

Technical Report No. 33

Evaluation, Validation, and Implementation of Alternative and Rapid Microbiological Methods

Revised 2026



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Table of Contents

1.0 Introduction	1
1.1 Purpose & Scope	1
2.0 Glossary	3
2.1 Abbreviations	8
3.0 Conventional Microbiology and the Move Toward Alternative/Rapid Microbiological Methods	9
3.1 Conventional Microbiological Methods	9
3.2 Introduction to Alternative/Rapid Microbiological Methods.....	11
3.3 Regulatory Perspectives	11
3.3.1 United States.....	12
3.3.2 Europe.....	13
3.3.3 Japan	14
3.3.4 Rest of World	15
3.3.5 Multi-Regional Regulatory Strategies for Alternative/Rapid Microbiological Methods.....	15
3.4 Business and Quality Considerations for Implementing Alternative/ Rapid Microbiological Methods	16
3.5 Risk Analysis.....	17
3.6 Automation of Conventional Methods.....	20
3.6.1 Performance Qualification.....	21
3.6.2 Time to Result	21
3.6.3 Method Suitability Test.....	22
3.7 Method Variability in Microbiology and Alternative/Rapid Microbiological Methods.....	22
3.7.1 Considerations for Detection and Enumeration Approaches	22
3.7.2 Considerations for Identification Approaches.....	23
3.8 Method Choice	24
3.8.1 Categorizing Technologies	25

4.0 Technology Review	26
4.1 Sample Effect Considerations	27
4.2 Qualitative Detection Principles and Approaches	27
4.2.1 Microcalorimetry	27
4.2.2 Respiration-Based Detection.....	27
4.2.3 Electrochemical Measurement	28
4.2.4 Single-Cell Detection/Method Development.....	28
4.2.5 Attachment Detection Technologies	28
4.2.6 Adenosine Triphosphate Bioluminescence	29
4.2.7 Alternative Biomarkers for Sterility Testing.....	29
4.2.8 Endotoxin Detection.....	29
4.2.9 Alternative Pyrogen Testing Monocyte Activation Test.....	30
4.3 Quantitative Measurement Principles and Approaches	30
4.3.1 Imaging and Optics.....	30
4.3.2 Fluorescent Probe Application	30
4.3.3 Autofluorescence	31
4.3.4 Flow Cytometry	31
4.3.5 Solid-Phase Cytometry.....	32
4.3.6 Hybrid Quantitative Identification Platforms	32
4.4 Identification Techniques, Principles, and Approaches.....	32
4.4.1 Hyperspectral Imaging	33
4.4.2 Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry.....	33
4.4.3 Fourier-Transform Infrared Spectroscopy	34
4.4.4 Raman Spectroscopy	34
4.4.5 Nucleic Acid Amplification Techniques	34
5.0 The Validation Process.....	37
5.1 Pre-Validation Activities	40
5.1.1 Proof of Concept.....	40
5.1.2 User Requirements Specifications	40
5.1.3 Assessment of Supplier Capabilities/Supplier Audit.....	41
5.1.4 Business Benefits or Return-on-Investment Considerations	42
5.1.5 Risk Assessment and Validation Planning	42
5.1.6 Design Qualification.....	42

5.1.7 The Validation Master Plan	43
5.2 Equipment and Software Qualification and Validation	43
5.2.1 Documentation of the Validation Protocol	44
5.2.2 Documentation of the Working Procedure	44
5.2.3 Computer System Validation and System Integration	44
5.2.4 Equipment Installation Qualification	45
5.2.5 Equipment Operational Qualification	45
5.2.6 Equipment Performance Qualification	45
5.3 Primary Validation, Method Suitability, and Comparability Testing	46
5.3.1 Primary Validation	46
5.3.2 Method Suitability	47
5.3.3 Comparability Testing in the Presence of the Test Sample	49
5.3.4 Unique Methods Requiring Additional or Modified Validation Strategies	50
5.4 Recommendations for Conducting Validation Studies	50
5.4.1 Specificity	51
5.4.2 Accuracy	52
5.4.3 Repeatability	54
5.4.4 Limit of Detection	55
5.4.5 Limit of Quantification	58
5.4.6 Linearity	59
5.4.7 Range	61
5.4.8 Ruggedness	61
5.4.9 Robustness	63
5.4.10 Method Suitability	64
5.4.11 Comparability	66
5.4.12 Guidance on Setting Minimum Incubation Times for Growth-Based Methods	69
5.4.13 Guidance on Changing Acceptance Criteria	70
5.4.14 Ongoing Maintenance and Periodic Reviews	72
5.4.15 Decommissioning of Equipment	73
5.5 Alternative/Rapid Microbiological Methods for Mycoplasma Detection	73
5.5.1 Primary Method Validation	73
5.5.2 Test for Interfering Substances/Method Suitability Testing	74
5.5.3 Mycoplasma Stocks and Standards	74

6.0 Implementation: Guidance on Secondary-Site Commissioning Versus Initial Validation	76
6.1 Guidance for the Transfer of an Alternative/Rapid Microbiological Method from a Primary Validation Site to a Secondary Validation Site	76
6.2 Equipment Installation and Operational Qualification at the Secondary Site	76
6.3 Equipment Performance Qualification at the Secondary Site	76
6.4 Implementation of the Alternative/Rapid Microbiological Method at the Secondary Site	77
7.0 The Role of Artificial Intelligence in Microbiological Testing	78
7.1 Use of Artificial Intelligence in Signal Detection	79
7.2 Use of Artificial Intelligence in Environmental Monitoring	81
8.0 References	87
9.0 Appendix 1: Validation of Alternative and Rapid Microbiological Methods: Statistical Analysis-Qualitative Methods.....	93
9.1 False-Positive Rates for Qualitative Alternative/Rapid Microbiological Methods.....	95
9.1.1 Estimation and Confidence Intervals.....	96
9.1.2 Sample Size Considerations When Estimating False-Positive Rates	97
9.2 Comparability on the Limit of Detection of Qualitative Alternative/Rapid Microbiological Methods	98
9.2.1 Demonstrating Comparability Using Ratios of Positive Rates	100
9.2.2 Justifying for Comparability with the Most Probable Number	101
9.2.3 Demonstrating Compatibility with the Generalized Most-Probable-Number Approach	103
9.2.4 Demonstrating Comparability Using Multiple Most Probable Number Experiments	105
9.2.5 Sample Size Considerations: Combining Multiple Species.....	107
9.3 Estimation of the Detection Limit	112
9.3.1 Logistic Regression Analysis.....	113
9.3.2 Most-Probable Limit Analysis	115
10.0 Appendix 2: Validation of Alternative and Rapid Microbiological Methods: Statistical Analysis-Quantitative Methods.....	116
10.1 Accuracy of Quantitative Microbiological Methods.....	119
10.1.1 Comparability of the Alternative/Rapid Microbiological Method for Each Concentration Against the Spiked Concentrations	121
10.1.2 Comparability of the Conventional Microbiological Method per Concentration Against the Spike Concentrations	127

10.1.3 Comparability of the Alternative/Rapid Microbiological Method with Respect to the Conventional Method per Concentration	129
10.2 Linearity and Accuracy	131
10.2.1 Weighted Linear Regression Analysis on the Counts.....	136
10.2.2 Poisson Regression Analysis on the Counts.....	137
10.2.3 Comparability on Accuracy of Alternative/Rapid Microbiological Methods Against Conventional Methods Using Concentration-Response Curves	138
10.3 Limit of Quantitation.....	145
10.3.1 Microbiological Methods with Positive Counts at Blank Samples.....	146
10.3.2 Microbiological Methods with Zero Counts at Blank Samples.....	147
10.4 Precision	148
10.4.1 Analysis of Variance for Precision	149
10.4.2 Generalized Linear Mixed Effects Model.....	152
10.4.3 Comparing Precision of the Alternative/Rapid Microbiological Method with the Conventional Method.....	153
11.0 Appendix 3: Validation of Alternative and Rapid Microbiological Methods: Additional Statistical Analysis-Non-CFU Signals	159
11.1 Qualitative Microbiological Methods with Non-CFU Signals.....	162
11.1.1 Limit of Detection when the Non-CFU Alternative/Rapid Microbiological Methods Does Not (or Hardly) Detects False Positives.....	162
11.1.2 Limit of Detection When the Non-CFU Alternative/Rapid Microbiological Method Demonstrates a Relevant False Positive Rate	164
11.1.3 Discussion on Fit for Purpose	166
11.2 Quantitative Microbiological Methods with Non-CFU Signals.....	169
11.2.1 Accuracy	170
11.2.2 Predicting the Counts of the Conventional Method Based on the Non-CFU Alternative/Rapid Microbiological Method Signal.....	174
11.2.3 Translating a CFU Upper Limit to an Upper Limit for the Non-CFU Alternative/Rapid Microbiological Method	175
12.0 Appendix References	179
Relevant Vendor and Supplier Resources.....	181

Figures and Tables Index

Table 3.5-1 Comparison of the Steps Taken in the Quality by Design and Analytical Quality by Design Processes (40)	18
Figure 3.6-1 Flow Chart for Equipment Qualification of Automated Colony-Counters and Method Validation	21
Table 5.0-1 Validation and Qualification Deliverables and Responsibilities.....	38
Table 5.4.8.3-1 Relative Standard Deviation for CFU Ranges	63
Table 7.2-1 Artificial Intelligence Approaches for Environmental Monitoring	82
Table 9.0-1 Nomenclature for Symbols Used in Section 9.0 (Appendix 1).....	94
Table 9.1-1 Presence/Absence Results on 20 Blank Samples.....	96
Table 9.1.2-1 Sample Size and 95% Upper Confidence Limit for Experiments with no Presence Tests	97
Table 9.2-1 Presence/Absence Results of Alternative/Rapid Microbiological Method and Conventional Methods on a Single Type of Microorganism	98
Figure 9.2-1 Probabilities for the Number of Microorganisms in Test Samples when the Average Number of Microorganisms in the Test Samples is Equal to 2.....	99
Figure 9.2.2-1 Visualization of Positive Rates as Function of the Average Number of Microorganisms in Test Samples	103
Table 9.2.4-1 Six Standard Most-Probable-Number Runs on a Single Microorganism Tested by Two Microbiological Methods.....	106
Table 9.2.4-2 Aggregated Data Per Concentration and Microbiological Method for the Multiple Most-Probable-Number Study.....	107
Table 9.2.5-1 Minimal Theoretical Sample Size per Microbiological Method for the gMPN Approach	108
Table 9.2.5-2 Present Results of the Alternative/Rapid Microbiological Method and Conventional Microbiological Method for Seven Different Species.....	109
Table 9.2.5-3 Most Probable Number Estimates and the gMPN Ratio Estimates for Multi-Species Experiment	110
Table 9.2.5-4 Critical Values Rejecting Homogeneity.....	111
Table 9.3-1 Number of Positives and Total Tested Test Samples for Different Concentrations.....	113
Figure 9.3.1-1 Visualization of the Positive Rate as Function of the Concentration (left) and Log Concentration (right).....	114
Table 10.0-1 Nomenclature for Symbols Used in Section 10.0 (Appendix 2).....	116
Figure 10.1-1 Visualization of Skewness for Counts at 10 CFU (left) and 50 CFU (right).....	120
Table 10.1-1 Counts for a Specific Microorganism for the Alternative/Rapid Microbiological Method and the Conventional Method at Different Concentration Levels	121
Table 10.1.1.1-1 Comparability of the Alternative/Rapid Microbiological Method on Recovery (%) per Concentration Level Using Different Data Analytic Approaches.....	123

Table 10.1.1.2-1 Output of Poisson Analysis for all Concentration Levels	124
Table 10.1.1.4-1 Sample Sizes for Non-Inferiority per Concentration at a Non-Inferiority Margin of 70% Using the Normal Distribution	126
Table 10.1.2-1 Non-Inferiority of the Conventional Method on Recovery per Concentration Level Using the Poisson Distribution	128
Table 10.1.3-1 Non-Inferiority of Alternative/Rapid Microbiological Method versus the Conventional Method per Concentration Level Using the Poisson Distribution	130
Table 10.1.3-2 Poisson Analysis Output.....	131
Figure 10.2-1 Visualization of the Estimated Concentration for an Unknown Sample Using a Linear Calibration Line for the Reference Sample in Analytical Measurement Systems.....	132
Figure 10.2-2 Visualization of Results from a Linear Concentration Response Curve (Green) and Results from a Non-linear Concentration Response Curve (Blue) Expressed in the Spiked Concentration and its Consequence for Accuracy.....	133
Figure 10.2-3A Visualization of the Mean Counts with Respect to the Spiked Concentration Levels.....	135
Figure 10.2-3B Visualization of the Mean Counts with Respect to the Spiked Concentration Levels.....	135
Table 10.2.1-1 Estimated Intercept and Slope Found with Weighted Linear Regression Analysis	136
Table 10.2.2-1 Estimates of the Parameters and 95% Confidence Intervals Using the Poisson Regression Model	138
Table 10.2.3-1 Counts of a Specific Microorganism for the Alternative/Rapid Microbiological Method and Conventional Method at Different Concentration Levels	139
Figure 10.2.3-1 Visualization of the Counts per Concentration and Connecting the Mean Counts	139
Table 10.2.3.1-1 Results Using Nonlinear Poisson Regression Analysis.....	141
Figure 10.2.3.1-1 Visualization of the Estimated Mitscherlich Functions with Respect to the Observed Counts	141
Table 10.2.3.1-2 Estimated Parameters.....	142
Figure 10.2.3.1-2 Visualization of the Estimated Recovery for the Log-Linear Concentration Response Curves of the Microbiological Methods	143
Figure 10.2.3-4 Visualization of the Estimated Mitscherlich Functions Using Weighted Non-Linear Regression with Respect to the Observed Counts	144
Table 10.2.3.2-1 Estimates of Parameters Using Weighted Nonlinear Regression.....	144
Figure 10.3-1 A Determination of the LoQ for Analytical Measurement Systems	145
Table 10.4-1 Precision Experiment for the Alternative/Rapid Microbiological Method Using Multiple Runs and Replicates	148
Figure 10.4-1 Visualization of the Relative Standard Deviation of a Poisson Distribution	149

Figure 10.4.1.2-1 Probability Plot of the Residuals (Analysis of the Counts (left); Analysis of the Log Counts (right)).....	151
Table 10.4.3-1 Precision Experiment for the CMM Using Multiple Runs and Replicates	153
Table 10.4.3-2 Minimum Sample Size for Two Variances.....	154
Table 10.4.3.2-1 Estimates and Standard Error for the Poisson Model.....	157
Table 11.0-1 Nomenclature for Symbols Used in Section 11.0 (Appendix 3).....	160
Table 11.1.1-1 Presence/Absence of 30 Samples for a Single Microorganism per Microbiological Method.....	163
Table 11.1.2-1 Testing of 30 Blank Samples and Spiked Samples on Both the Alternative/Rapid Microbiological Method on Absence/Presence for a Single Microorganism	164
Table 11.1.2-2 Positive Rates with Exact 95% Confidence Intervals	165
Figure 11.1.3-1 Agreement (%) Between the Alternative/Rapid Microbiological Method and the Conventional Microbiological Method as a Function of the Mean Number of Viable and Culturable Microorganisms.....	169
Table 11.2.1-1 Counts for a Specific Microorganism for the Alternative/Rapid Microbiological Method and Conventional Method at Different Concentration Levels	170
Figure 11.2.1-1 Visualization of the Enumerations of the Alternative/Rapid Microbiological Method and the Conventional Method for an Accuracy Experiment.....	171
Table 11.2.1-2 Recoveries of the Alternative/Rapid Microbiological Method versus Conventional Method per Concentration Level Using the Poisson Distribution.....	171
Table 11.2.1-3 Estimated Parameters of the Linear Concentration-Response Curve	173
Figure 11.2.1-2 Estimated Mean Difference Between the Two Methods with Lower 95% Confidence Limit	174
Figure 11.2.3-1A Probability Curves for the Alternative/Rapid Microbiological Methods for Testing a Single Sample Below the Translated Upper Confidence Limit ($U_{CMM} = 1$ (left); $U_{CMM} = 5$ (right)).....	177
Figure 11.2.3-1B Probability Curves for the Alternative/Rapid Microbiological Methods for Testing a Single Sample Below the Translated Upper Confidence Limit ($U_{CMM} = 10$ (left); $U_{CMM} = 100$ (right))	177

1.0 Introduction

Technical Report No. 33 (Revised 2026): Evaluation, Validation, and Implementation of Alternative and Rapid Microbiological Methods (TR 33) is intended to provide guidance for the successful evaluation, validation, and implementation of alternative/rapid microbiological methods (AMM/RMMs) used in the pharmaceutical, biotechnology, advanced therapy medicinal products (ATMPs), medical devices, and related industries. The use of these methods is aligned with the expectations for improving product and process quality, contamination control, and the rapid release of medically necessary drugs and devices. Applications may include, but are not limited to, sterility testing, microbiological examination of nonsterile products (enumeration and test for specified microorganisms), antimicrobial effectiveness testing, microbiological monitoring of cleanrooms and other controlled environments, analysis of pharmaceutical grade water, microbial characterization and identification, and microbiological in-process control testing.

The intended audience of TR 33 includes microbiologists and related individuals who are responsible for the validation and utilization of microbiological test methods in the microbiology testing laboratory, manufacturing environment, or other relevant testing settings. This technical report may also be of interest to suppliers of testing equipment, microbiology managers and supervisors, validation specialists, quality control (QC), compounders, and quality assurance personnel responsible for approving product release and validation protocols, manufacturing personnel, individuals who work on global compendial initiatives, and regulatory agencies. Additionally, **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) provides useful information to support statistical analysis of data from AMM/RMM validation and comparability studies.

1.1 Purpose & Scope

Microbiological testing plays an ever-increasing role in pharmaceutical and related industry laboratories. In response to this, a variety of alternative and rapid methodologies have emerged in recent years that automate conventional microbiological methods (CMM), make use of surrogate markers, or are based on wholly new technologies. These AMM/RMMs can offer significant improvements in terms of speed, accuracy, precision, and/or specificity, as compared with conventional microbiology test methodologies.

Most of the microbial testing performed today continues to rely on centuries-old CMMs that are based on the recovery and growth of microorganisms using solid or liquid microbiological growth media. This is true, in part, because these methods can be appropriate for their intended use and have a long history of application in both industrial and clinical settings. However, they are often limited by slow microbial growth rates, the unintended selectivity of microbiological cultures, and the inherent variability of microorganisms in their response to growth-based methods. Despite the limitations of CMMs, alternative, more rapid, and potentially superior microbiological methods have not been universally adopted within the pharmaceutical, biotechnology, ATMP, medical device, and related industries. This seemingly continues to be due in part to

a lack of understanding regarding the steps needed to demonstrate their equivalence or comparability to CMMs that would be acceptable to a firm's internal quality expectations as well as regulatory agency reviewers and inspectors. The original PDA TR 33 was published in 2002, and its first revision was released in 2013. Based on the industry, compendial, and regulatory developments since the 2013 revision, including the need for the rapid release of personalized medicines, such as ATMPs, sterile-compounded pharmaceutical preparations, radiopharmaceuticals, and the rapid release of therapeutic drug products needed in public health emergencies, the task force members believe that this revision of TR 33 is timely and will provide additional guidance to assist with the evaluation, validation and implementation of AMM/RMMs. While this document focuses on the manufacture of drug products, the validation approaches and principles defined herein can be translated to other product categories (e.g., consumer goods) where these AMM/RMMs may be applied.

Considerable guidance can be found regarding the validation of chemical methods, for example, *U.S. Pharmacopeia* (USP) General Chapter <1225> Validation of Compendial Methods, International Conference on Harmonisation (ICH) *Quality Guideline Q2(R2): Validation of Analytical Methods*, and U.S. Food and Drug Administration (FDA) *Guidance for Industry: Analytical Procedures and Methods Validation for Drugs and Biologics (1-3)*. These publications provide very specific instruction regarding the demonstration of alternative analytical chemistry methods and their equivalence to conventional chemical methods. Chapters introduced by the global compendia, including USP General Chapter <1223> Validation of Alternative Microbiological Methods, *European Pharmacopoeia* (Eur. Ph.) Chapter 5.1.6 Alternative Methods for Control of Microbiological Quality, and *Japanese Pharmacopoeia* (JP) General Information G4-6-170: Rapid Microbial Methods, provide guidance on the steps needed to validate an AMM/RMM (4-6). Due to the differences in recommended validation strategies within each guidance document and the potential variability in validation expectations across regulatory agencies, additional guidance that provides an understandable and holistic approach to the qualification and implementation of AMM/RMMs is desirable.

This technical report was developed as a collaborative effort among microbiologists and expert consultants who have practical experience validating and implementing AMM/RMMs within the pharmaceutical, biopharmaceutical, and medical device industries, AMM/RMM suppliers, and regulatory agency representatives. By providing updated and agreed-upon performance standards and industry best practices, the task force believes the validation and implementation of AMM/RMMs will be greatly accelerated.

2.0 Glossary

Definitions have been provided to help clarify the concepts discussed in this document. While some definitions used vary among companies, geographic location, etc., the definitions described below are consistently used within this technical report document. Where a definition is based on another published source, the source is cited.

Accuracy

A Closeness of the test results or measurement response obtained by the AMM/RMM to a known value obtained by the CMM or a reference material appropriate for the method, to be demonstrated across the dynamic (operational) range of the method. Accuracy is usually evaluated in quantitative methods.

Activation

Incubation of sample to be tested (typically) in liquid-nutrient media in order to (re)activate the metabolism of the viable microorganisms (contaminants), including spores that may be present in the sample. Proliferation of the viable microorganisms (contaminants) that may be present in the sample should be avoided because it may overestimate the cell number when conducting microbial enumeration.

Note: Also known as Activation Phase or Activation Step.

Alternative or Rapid Microbiological Methods (AMM/RMM)

Methods that differ from the CMM (e.g., USP General Notice 6.30). These AMM/RMMs use

technology that differs in measurement end points, automation, and software for managing testing processes and analyzing the resulting quantitative, qualitative, and microbial identification data. Such AMM/RMMs must undergo full validation and demonstrate comparable results to the CMMs within acceptable limits, established on a case-by-case basis.

Note: Within TR 33, the terms "alternative method," "rapid microbiological method," and "rapid method," are addressed as interchangeable terms and are represented by the acronym "AMM/RMM".

Analytical Quality by Design (AQbD)

Approach that aims to ensure the quality of a product by employing statistical, analytical, and risk-management methodology in the design, development, and manufacturing of medicines.

Artificial Intelligence (AI)

A scientific discipline established at the intersection of computer science, applied mathematics, and engineering, and dedicated to the development of systems capable of performing tasks traditionally requiring human intelligence. From a functional perspective, AI can be defined as the simulation or approximation of human intelligence in machines. AI encompasses machine-driven learning, reasoning, and perception, and is widely used across various industries, including finance and life sciences.

Automation

The use of technology to improve the efficiency, accuracy, and reproducibility of processes. By replacing manual work with automation, it increases productivity, reduces errors, and ensures more consistent results, while also speeding up turnaround times.

Biomarker

A measurable characteristic or molecule that indicates the presence, activity, or identity of microorganisms in a specific environment or sample.

Colony-Forming Unit (CFU)

One or more viable microorganisms that produce a visible, discrete growth entity on a semisolid, agar-based microbiological medium and under conditions of incubation and is used as the basis for quantitative enumeration (e.g., CFU/g or mL).

Comparability

Demonstration that the qualitative or quantitative detection of microorganisms, related analyte, or measurement response in the AMM/RMM is equivalent, non-inferior or not statistically different than the detection of microorganisms in the existing test method, depending on the statistical method used to compare the data.

Conventional Microbiological Method (CMM)

A classical or traditional growth-based method, such as enumeration on an agar plate or detection in a liquid broth when incubated for a specified time and temperature. These methods are used in USP <51>, <60>, <61>, <62>, <63>, and <71> which are harmonized with the Japanese Pharmacopoeia (JP) and European Pharmacopoeia (EP) pharmacopeial procedures.

Note: Throughout this document the term “conventional” is used and is understood to mean compendial, classic, existing, or traditional.

Equivalence

When the test results from two procedures are sufficiently similar for their intended purpose; demonstrating equivalence necessitates a predefined criterion for the level of similarity required. This evaluation generally involves a comparability testing study that produces comparative data, the test results of which are then subjected to statistical analysis.

Exclusivity

The capacity of an assay to differentiate a non-target microorganism from a target microorganisms.

False Negative

A test result that is incorrectly determined as negative (e.g., the absence of a viable microbial detection result when viable microorganisms are present).

False Positive

A test result that is incorrectly determined as positive (e.g., a viable microbial detection result when viable microorganisms are not present).

In-Line Measurements

Measurements taken within the process stream by placing measuring devices, typically sensors, directly in contact with or into the process stream. No portions of the material are therefore removed from the process stream.

Measurements must also be nondetrimental to the product. Measurements can be obtained in real time or close to real time.

Independent Samples (Determinations)

Samples selected from the same population or different populations that have no effect on one another, that is, no correlation exists between the samples.

Inclusivity

The ability of an assay to detect a target microorganism.

Intermediate Precision

Use of the AMM/RMM applied to different sample preparations tested in the same laboratory under conditions of maximum variability (e.g., different analysts, different reagents, different equipment (if available) and/or testing on different days).

Installation Qualification (IQ)

Provides documented evidence that the equipment is received as designed and specified, that it is properly and safely installed with the correct utilities in the selected environment, and that the environment is suitable for the operation and use of the equipment.

Laboratory Information Management System (LIMS)

Software-based solution with features that support a modern laboratory's operations by managing data related to sample experiments and instruments.

Limit of Detection (LoD)

The lowest concentration of microorganisms, related analyte, or measurement response in a test sample that can be reliably detected, but not necessarily quantified, under defined experimental conditions.

Note: Can be referred to as sensitivity.

Limit of Quantification (LoQ)

The lowest number of microorganisms, related analyte, or measurement response in a test sample that can be enumerated with acceptable accuracy and precision under defined experimental conditions.

Linearity

The ability to produce results that are proportional to the concentration of microorganisms present, related analyte, or measurement response in the sample within a given range.

Machine Learning (ML)

A specialized branch of artificial intelligence (AI) that utilizes algorithms and statistical

models to process data, generate insights, make predictions, and provide recommendations within complex datasets. In the pharmaceutical industry, ML identifies patterns, trends, and anomalies within complex datasets derived from drug discovery, clinical trials, manufacturing, regulatory compliance, pharmacovigilance, and patient outcomes.

Method Suitability

Demonstration of a lack of enhancement or inhibition by the product on the signal generated by the method.

Microbial Identification

The identification of a microbial isolate to genus or species level.

Next-Generation Sequencing (NGS)

A group of technologies that allows thousands to billions of DNA fragments to be simultaneously and independently sequenced.

Note: Also known as high-throughput or massive parallel sequencing.

Non-Inferiority

The hypothesis-testing framework used to demonstrate empirically that the AMM/RMM is not unacceptably worse on a specific feature (e.g., detection or decision-making) than the CMM by using a pre-specified non-inferiority margin.

On-Line Measurements

Sensor-based measurements made under real-time conditions by diverting a portion of the material from the process stream directly into a measuring device and the results are made available after a minimal time delay.

Depending on the nature of the test, for example, whether or not it is detrimental to the product, the diverted portion may be returned to the process stream; otherwise, it is discarded.

Note: Also referred to as “real-time measurements.”

Operational Qualification (OQ)

Provides documented verification that the equipment, as installed in the selected environment, performs effectively and reproducibly as intended throughout the anticipated or representative operational ranges, defined limits and tolerances. The OQ is also the focal point for the majority of the computer system, software and security validation activities.

Performance Qualification (PQ)

Provides documented evidence that the instrumentation, as installed, (and the method if applicable) consistently performs in accordance with predetermined criteria and thereby yields correct and appropriate results.

Precision

The degree of agreement among individual test results as demonstrated by repeatability, intermediate precision (ruggedness) or reproducibility.

Primary Validation

Validation usually performed by the supplier, typically in the absence of a test sample, demonstrating that the system or technology is adequate in detecting the intended target.

Process Analytical Technology (PAT)

A system that allows for the design, analysis, and control of manufacturing through real-time QC and performance attribute measurements.

Qualification

A generic term defining the activity of confirming the successful achievement of defined prerequisites required for validation.

Note: In TR 33, this term is applied to the training of personnel and the commissioning of process equipment. (See IQ, OQ, PQ.)

Quality

The degree to which a product, system, or process consistently fulfills established requirements and is suitable for its intended use.

Quality by Design (QbD)

Framework enabling the attainment of the desired state; a systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control based on sound science and quality risk management.

Quality Risk Management (QRM)

A systematic process for the assessment, control, communication, and review of risks to the quality of the drug product across the product lifecycle.

Range

The interval between the upper and lower levels of microorganisms, related analyte, or measurement response that have been demonstrated to be determined with accuracy, precision, and linearity.

Repeatability

Degree of agreement among individual test results when the procedure is applied repeatedly to multiple samplings under the same operating conditions.

Reproducibility

The precision among multiple laboratories (i.e., collaborative studies, usually applied to standardization of methodology).

Risk

The combination of the probability of the occurrence of harm and the severity of that harm.

Risk Assessment

A systematic process of organizing information to support a risk decision to be made within a risk management process. It consists of the identification of hazards and the analysis and evaluation of risks associated with exposure to those hazards.

Robustness

A method's capacity to remain unaffected by small but deliberate variations in method parameters while providing an indication of its reliability during normal usage.

Ruggedness

The degree of intermediate precision or reproducibility of test results obtained by assessing the same samples under a variety of normal test conditions.

Severity

A measure of the possible consequences of a hazard.

Specificity

The ability of an analytical procedure to accurately measure or detect the required range of microorganisms that may be present in the sample under test.

Superiority

The ability to outperform the conventional or existing method in terms of detection (i.e., presence or absence of the targeted microorganisms) or enumeration (i.e., microbial count of viable microorganisms) including a reduced time to results.

Note: The superiority may be so obvious that a statistical comparison would not be necessary.

Time to Result (TTR)

Time required to complete testing and obtain a result that can be used to support a conclusion regarding potential contamination of the sample test.

Validation

Demonstrating and documenting that the instrumentation as installed (and the method if applicable) functions effectively and is appropriate for its intended purpose. Formal validations are conducted prospectively following a written plan that includes justifiable acceptance criteria for the validation procedures.

Verification

Verification involves applying the required analytical performance characteristics outlined in the primary method validation to obtain reliable data for specific sample types, environments, or equipment, rather than duplicating the validation process.

2.1 Abbreviations

ANOVA	Analysis of Variance	MPL	Most Probable Limit
AMM*	Alternative Microbiological Method	MPN	Most Probable Number
ATMP	Advanced Therapy Medicinal Products	NADH	Nicotinamide Adenine Dinucleotide plus Hydrogen
ATP	Adenosine Triphosphate	NAT	Nucleic Acid Amplification Techniques
BFPC	Biofluorescent Particle Counters	OOT	Out-of-Trend
CFU	Colony-Forming Units	OQ	Operational Qualification
CMM	Conventional Microbiological Method	POC	Proof of Concept
EM	Environmental Monitoring	PCR	Polymerase Chain Reaction
FT-IR	Fourier-Transform Infrared Spectroscopy	PQ	Performance Qualification
GMP	Good Manufacturing Practice	QC	Quality Control
gMPN	Generalized Most Probable Number	QRM	Quality Risk Management
IQ	Installation Qualification	RMM*	Rapid Microbiological Method
LIMS	Laboratory Information Management System	TTR	Time to Result
LoD	Limit of Detection	UV	Ultraviolet
LoQ	Limit of Quantification	VBNC	Viable but Nonculturable
MALDI-TOF MS	Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry		

***Note:** In this document, the abbreviation AMM/RMM is used to indicate that both AMMs and RMMs are being discussed.

3.0 Conventional Microbiology and the Move Toward Alternative/Rapid Microbiological Methods

For decades, conventional techniques such as culture-based methods and microscopy have been used in pharmaceutical microbiological QC laboratories to assess the quality of pharmaceutical products. Many of these methods are still an essential component of regulatory guidance documents (e.g., pharmacopeias). However, technological advancements have ushered in a new era where AMM/RMMs, using alternative readout-technologies and molecular biology tools, are revolutionizing the way microbiologists assess the quality of pharmaceutical products. **Section 3.0** provides an overview of what aspects should be considered when implementing AMM/RMMs.

3.1 Conventional Microbiological Methods

Conventional pharmaceutical microbiology methods are comparatively simple growth-based techniques encompassing little technology and are limited in their scope of detection. These methods for the detection, enumeration, or identification of bacteria, yeast, and mold are applicable to sterility testing, antimicrobial or preservative effectiveness testing, environmental monitoring (EM), in-process control and/or raw material(s) bioburden monitoring, finished product enumeration and specified organisms, presence/absence testing including aseptic process simulation, bioburden testing of pharmaceutical-grade water and gases, and microbial identification. Biopharmaceutical production processes based on the use of cell cultures also require the industry to test for the presence of other adventitious microbiological agents such as viruses and mycoplasma in cell sources (cell banks), production-cell cultures, in-process samples, or finished products. Each of these growth-based techniques have primarily relied on basic microbiological media for culturing (e.g., bioburden, presence/absence testing relying on trypticase soy broth/agar for aerobic bacteria, Sabouraud's dextrose broth/agar for fungal growth, and fluid thioglycolate medium for anaerobic microorganisms, and Hayflick, Frey or Friis media for mycoplasma detection). These media have provided the basis for qualitative and quantitative microbial assays for microbial safety and quality product release, validation of in-process microbial tests, and sterilization validation for pharmaceuticals and medical devices in the 20th century. The accuracy and precision of these CMM are affected by the distribution of microorganisms in the test samples, cellular arrangement, available sample volume, the method itself including the media (supplier and/or quality of individual components), and the ability of microorganism growth.

Although the growth of microbial cells on agar surfaces or in liquid media provides the laboratory with critical information about the amount and type of organisms that may be present in a sample under evaluation, the time to result (TTR) is usually longer than desired. Days may elapse before microbial colonies are visually detected and, in many cases, confluent growth on agar plates prevents individual organisms from being isolated, necessitating their subculture onto additional agar media, delaying the TTR even further. This delay may hamper the industry in making timely forward-processing or good

manufacturing practice (GMP) release decisions, in timely investigations of events, and in support of confirming that manufacturing processes are in a microbiological state of control.

Additionally, environmental factors can affect the ability of microorganisms to replicate when cultured on microbiological media. These factors include stress due to nutrient deprivation or exposure to sublethal concentrations of antimicrobial agents such as preservatives, disinfectants, heat, or decontaminating gases. Furthermore, certain pharmaceutical manufacturing processes can impact microorganisms. As a result of any of these stressors, the physiologically injured microorganisms might not replicate when cultured on microbiological media, because the media and/or incubation parameters may not be optimal for the resuscitation and subsequent proliferation of organisms that may be present. The inability of stressed microorganisms, or even nonstressed microorganisms, to grow in microbiological media has also been termed viable but nonculturable (VBNC). Additionally, when microorganisms experience unfavorable growth conditions, it can induce some cells into dormancy, and these cells will not grow. VBNC and dormancy states are further discussed in the public literature (7, 8).

For these reasons, and to mitigate the limitations of CMMs as outlined in **Section 3.0**, the QC microbiological laboratory should look toward developing innovative approaches for the collection, detection, quantification, and identification of microorganisms to take advantage of emerging technologies.

From a quality risk management (QRM) perspective, the pharmaceutical/biopharmaceutical industry can benefit from implementing AMM/RMM testing to:

- Improve TTR for time-sensitive products (e.g., ATMPs, compounded drugs, nuclear medicine products, short-shelf-life or immediate-medical-need products, response to public health emergencies)
- Design and continuously improve robust processes that prevent contamination
- Ensure that a state of microbial control is maintained
- Develop strategies with earlier detection of contamination for faster investigation resolutions
- Monitor manufacturing environments, utilities, or processes to detect adverse microbiological trends or contamination in real time to avoid processing at risk
- Improve data integrity
- Evaluate appropriate sampling points for process improvement

Quality risk management principles are recommended by a number of quality and regulatory industry guides that have a direct impact on microbiological monitoring and control, for example, the European Commission's *Annex 1: Manufacture of Sterile Medicinal Products, EudraLex-Volume 4-EU Guidelines for Good Manufacturing Practice for Medicinal Products for Human and Veterinary Use (2022)* (EU GMP Annex 1) (9). Other examples are FDA's *Pharmaceutical cGMPs for the 21st Century—A Risk-Based Approach and Guidance for Industry: Process Analytical Technology, A Framework for Innovative Pharmaceutical Development, Manufacture, and Quality Assurance*, which implies that using a scientific framework to find ways of mitigating risk while facilitating continuous improvement and innovation in pharmaceutical manufacturing is a key public health objective (10, 11). Implementation of AMM/RMM, as described in USP General Chapter <1071> Rapid Microbial Tests for Release of Sterile Short-Life Products: A Risk-Based Approach, supports the medical and quality needs of time-critical patient therapies (12). These initiatives further promote the use of the latest scientific advances in manufacturing and technology, and this can apply to the implementation of AMM/RMMs and testing strategies.

3.2 Introduction to Alternative/Rapid Microbiological Methods

AMM/RMMs are based on a wide variety of scientific principles and can be used for a number of testing applications (13). Technologies can detect the presence of diverse types of microorganisms or a specific microbial species (qualitative), enumerate the number of microorganisms present in a sample (quantitative), or identify microbial cultures to the genus, species, and strain levels. The manner in which microorganisms are detected, quantified, or identified will depend on the specific technology, procedures, and/or instrumentation employed. Additionally, a number of these methods are considered to be more sensitive, accurate, precise, and reproducible, with improved data integrity, when compared with conventional, growth-based methods. Some methods are fully or semiautomated, offer increased sample throughput and less analyst hands-on time, provide significantly reduced TTR (e.g., from days to hours or minutes) and, for a few technologies, afford results in real-time. A more thorough review of technology platforms and the science behind these methods is provided in **Section 4.0**.

3.3 Regulatory Perspectives

AMM/RMMs have been understood, accepted, and encouraged by regulatory authorities in numerous regions across the globe (14). Various regulatory agencies, including the United States, Europe, and Japan, have developed written guidance on the validation and implementation of AMM/RMM. Regulators will generally accept a change in testing procedure if the change has been proven to be comparable to, superior to, and/or noninferior to the CMMs currently in place. While TR 33 contains recommendations for the validation studies to demonstrate acceptability of the AMM/RMM, the final decision on acceptance may be made by the affected regulatory agencies, for example, if the CMM is incorporated in a previously approved regulatory dossier. However, there are also instances where a formal regulatory submission may not be necessary, but validation would be expected and may be assessed during site inspection, especially for test samples that would not necessarily require a formal submission (e.g., in-process, raw material, water, EM samples) (15). In order to develop an appropriate strategy for the validation and implementation of these methods, it is important to fully understand the current regulatory expectations and specific details of the associated product dossier. Therefore, it is highly recommended that an open dialogue between the interested parties (i.e., the firm intending to implement the AMM/RMM and the relevant regulatory authority) be initiated early in the planning process. This dialogue can include discussions about the proposed method, impacted products, validation approach, and acceptance criteria, as well as regulatory submission requirements. Additionally, in current practice, this dialogue has been very helpful and has enabled the potential AMM/RMM users to move forward with greater assurance that they will be successful in gaining regulatory approval, when required.

A variety of different perspectives exist on AMM/RMM validation and submission strategies, depending on the regulatory body with which a firm's products are registered, and/or which local inspectorate is responsible for conducting GMP audits at a firm's manufacturing facilities. **Sections 3.3.1–3.3.5** summarize the most current regulatory expectations for validation, submission, and implementation; however, it is always recommended to monitor any regulatory updates or changes in this area, as appropriate.

Note: If and when a new technology becomes a conventional test, prior approval may not be required for the test. Firms should follow the directions in the conventional pharmacopeial chapters to demonstrate a test's suitability for use.

3.3.1 United States

The FDA has been accepting AMM/RMMs for several years, and this position is echoed in a variety of guidance documents and quality initiatives. For example, FDA's *Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing – Current Good Manufacturing Practice* recommends the use of rapid genotypic methods for microbial identification, as these methods have been shown to be more accurate and precise than biochemical and phenotypic techniques. The guidance also states that these methods are especially valuable for investigations into significant microbiological adverse events, such as sterility test failures and contaminated media fills. In addition, it confirms that other suitable microbiological tests (e.g., AMM/RMMs) can be considered for EM, in-process control testing, and finished-product release testing after it has been demonstrated that these new methods are comparable or better than CMM (16).

In 2012, FDA's Center for Biologics Evaluation and Research amended 21 CFR 610.12 *General Biological Products Standards: Sterility*, to allow the use of new sterility test methods that yield accurate and reliable test results (17). In 2020, the Center published the *Guidance for Industry: Chemistry, Manufacturing, and Control (CMC) Information for Human Gene Therapy Investigational New Drug Applications (INDs)*, where alternate sterility tests may be acceptable for certain types of products administered fresh or with limited hold time between final formulation and patient administration (18).

A separate FDA initiative known as the Strategic Plan for Regulatory Science (2011), calls for the development of sensitive, rapid, and high-throughput methods to detect, identify, and enumerate microbial contaminants and validate their utility in assessing product sterility (19).

The FDA has multiple pathways to facilitate formal meetings for drug products under the Prescription Drug User Fee Act (PDUFA), Generic Drug User Fee Act (GDUFA), Medical Device User Fee Act (MDUFA), or Biosimilar User Fee Act (BsUFA) (20-23). Another pathway is Type-C meetings which are commonly used for pre-submission alignment or marketed product changes related to ARMM to de-risk the validation strategy and obtain written feedback. Initial Targeted Engagement for Regulatory Advice on Center for Biologics Evaluation and Research/Center for Drug Evaluation and Research (CBER/CDER) Products (INTERACT) meetings are appropriate for pre-IND stage discussions on ARMM/RMM. To facilitate an effective meeting, include a clear statement of intent, a list of specific questions, an assessment of product/patient risk, a proposed validation strategy and a feasibility data set. One notable limitation of these meeting pathways is that they must be related to an application. Another useful pathway that exists in FDA for truly novel methods is the Emerging Technology Program, which is a collaborative program between the Agency and industry representatives to facilitate discussion of potential regulatory issues prior to implementation of a new technology (24).

Finally, the Agency recognizes that examining the suitability of an AMM/RMM may necessitate the use of the technology on production equipment and processes. The *Guidance for Industry PAT - A Framework for Innovative Pharmaceutical Development, Manufacturing, and Quality Assurance* provides advice to stakeholders on this topic, including that such activities can take place without prior notification to the Agency, that data collected using experimental tools should be considered as research data, and that studies conducted in production facilities be managed under the facility's quality system (11).

The FDA suggests that USP <1223>, Ph. Eur. Chapter 5.1.6, or PDA TR 33, be used as primary resources for the selection, validation, and implementation of AMM/RMMs (11). Additionally, many firms have successfully used the FDA Comparability Protocol (CP) as a means to manage the validation plan. Submitting a FDA CP for pre-approval review is another avenue to gather FDA feedback on a post-approval

change, though this pathway does not allow for discussion so it is useful when changing to a well-known ARMM/RMM method with a straightforward validation plan. Briefly, the CP is a regulatory submission (typically, a prior approval supplement) that contains a validation protocol for the AMM/RMM. The CP describes the proposed validation studies and the acceptance criteria to be met to demonstrate that the alternate method is acceptable. Once the CP has been approved, the applicant can use the protocol in the CP to validate the AMM/RMM for its intended use. The CP can be particularly useful for changes of a repetitive nature, such as the use of an AMM/RMM for multiple products or processes. Moreover, because the CP is reviewed by the FDA, the agency feedback can inform any required changes to the validation plan or approach prior to execution and therefore eliminating potential re-work. For more information on CPs, the guidance “*Comparability Protocols for Post approval Changes to the Chemistry, Manufacturing, and Controls Information in an NDA, ANDA, or BLA (2022)*” may be referenced (25).

3.3.2 Europe

Like their U.S. counterparts, European regulators have supported the validation and implementation of AMM/RMM technologies; however, there are subtle differences with respect to validation expectations and submission requirements.

Commission Regulation (EC) 1234/2008 went into effect in 2010 and applies to variations to a Marketing Authorization granted in a Mutual Recognition/Decentralized Procedure and to Community or Centralized Authorizations. In 2023, (EC) 2023/1182 went into effect, and the impact of this along with the issuance of ICH Q12 introduces a number of features aimed at reducing the workload for both competent authorities and applicants. One of the most important changes relating to AMM/RMMs makes it now possible to group variations under the same Marketing Authorization such that they can all be assessed at the same time. Furthermore, it is possible to combine the same variations or group of variations from different Marketing Authorizations and have all of these assessed at the same time under what is called a “Work Sharing Process” or “Common Assessment.” This could be the case for a single AMM/RMM technology being used for multiple products (26).

Next, there is an opportunity for scientific dialog with regulators through the European Medicines Agency (EMA) Scientific Advice and Protocol Assistance procedure. Here, a firm may ask for advice on its validation and implementation strategies. The Scientific Advice working party includes representatives from all European Union (EU) member states, and a written report is provided with the results of the advice process (27). For novel AMM/RMM technologies, EMA's Innovation Task Force (ITF) offers free, early informal dialogue to help identify the appropriate regulatory pathway and data requirements. For AMM/RMMs with broad applicability across multiple products, firms may also consider EMA's Qualification Advice procedure, which can even result in formal qualification of the methodology for industry-wide regulatory use.

Based on EC 1234/2008, the Commission published *Guidelines on the Details of the Various Categories of Variations...to the Terms of Marketing Authorizations for Medicinal Products for Human Use and Veterinary Medicinal Products* that implements the “post-approval change management protocol (PACMP)” (28). This voluntary process, which is very similar to FDA's CP, provides a strategy for managing the review of proposed validation plans prior to the start of testing (29).

In this two-step process, a change management testing protocol is first submitted as a Type 2 Variation. The protocol should include the overall testing strategy, such as the planned studies, acceptance criteria, and methods. Prior to submitting the PACMP, a firm may also discuss their testing strategies with the EMA

under the scientific advice procedure. Once the protocol is approved, the submitting company will perform the testing as specified in the protocol.

The second step of the PACMP process involves submitting the resulting data, assuming it has met the protocol's acceptance criteria. The decision to submit as a Type 1A or Type 1B variation is determined at the time of protocol review and approval. If the data is submitted under a Type 1A variation, the company can immediately implement the AMM/RMM; a Type 1B variation, on the other hand, requires a 30-day waiting period while the data is reviewed. These strategies are very similar to the FDA's Changes Being Effectuated-0 (CBE-0) and Changes Being Effectuated-30 (CBE-30) reduced reporting policies.

Chapter 5.1.6 of the European Pharmacopoeia (Ph. Eur.) addresses the validation of AMM/RMMs (5). Like TR 33, Ph. Eur. 5.1.6 provides a framework for the development of an appropriate validation plan, which can then be included in the PACMP as the basis for the studies that will be performed. Ph. Eur. 2.6.27 also provides good information as that chapter describes how AMM/RMMs can be used to demonstrate the quality of cell-based products (30).

The EMA recently extended the competencies of the process analytical technology (PAT) team, which is currently responsible for quality-by-design matters, to also cover all matters related to AMM/RMMs.

Additionally, the 2022 revision to the EU GMP Annex 1 recommends that the use of rapid microbial testing and monitoring systems should be considered to increase the protection of the product from potential extraneous sources of particulate and microbial contamination such as personnel, materials and the surrounding environment, and assist in the rapid detection of potential contaminants in the environment and product

As with all post-approval changes, discussing this with the EMA, relevant competent authorities, and/or the local inspectorate early in the implementation planning phase is recommended. This is especially relevant if a formal Type variation change may not be required, which will depend on the AMM/RMM's intended application, such as an in-process microbiology test that is not in a marketing authorization.

3.3.3 Japan

The Japanese Pharmaceuticals and Medical Devices Agency (PMDA) evaluates AMM/RMMs that will be used for product release, such as rapid sterility testing, but is generally accepting of such methods provided that they give results that are comparable to or better than CMMs. In Japan, when AMM/RMMs intended for use in product release decisions (product disposition)—such as rapid sterility testing—are employed, the relevant test method may be used in lieu of compendial testing for batch release, within the scope approved through an application for marketing authorization (or an application for a partial change to the approved matters) submitted to the regulatory authority.

However, if these methods are shown to give comparable or better results than CMMs, they are generally accepted by the PMDA.

In the Japanese Pharmacopoeia (18th Edition), information on these topics is listed in the category G4 Microorganism and GZ Others in the General Information category, including “Rapid Determination of Bacterial Count by Fluorescent Staining” <G4-8-152>, “Rapid Identification of Microorganisms by Genetic Analysis” <G4-7-160>, “Microbial Rapid Test Method” <G4-6-170>, “Alternative Methods for Endotoxin Testing and Measurement Reagent Using Recombinant Proteins” <G4-4-180>, and “Quality Control of Pharmaceutical Water” <GZ-2-172> (6).

3.3.4 Rest of World

The implementation of AMM/RMMs has been successful in non-U.S. and non-European regions, particularly in Japan, Australia, India, and China, but also in Mexico, Brazil, Central/South America and the Asia-Pacific region. As with U.S. and European agencies, it is recommended that an open dialogue with the affected regulators be initiated early in the implementation process. Because some of these agencies may not be familiar with the proposed technologies, having the vendor involved may also be recommended. Where some rest-of-the-world (ROW) countries may follow USP <1223> and/or Ph. Eur. Chapter 5.1.6, other countries may only follow their own local pharmacopeias. In either case, TR 33 can serve as a framework for discussion of the planned validation and implementation plans, and provides the following information regarding guidance given by authorities in these specific regions:

- The Australian Therapeutic Goods Administration utilizes relevant sections of Ph. Eur. 5.1.6, USP <1223>, the *British Pharmacopoeia*, ISO 17025, and TR 33 when working with companies wishing to validate and implement AMM/RMM technologies (31, 32).
- The *Chinese Pharmacopoeia* includes “Chapter 9201: Guidelines for the Validation of Alternative Microbial Detection Methods for Pharmaceuticals,” which, in general, allows AMM/RMMs to be used for final product testing where CMMs cannot meet QC requirements. Validation of AMM/RMMs is required, and the AMM/RMMs for final-product testing should give results that are comparable to or better than the CMM. Parallel testing can be performed to evaluate the comparability. Special considerations for the validation of AMM/RMMs are, for example, validation of each method parameter in the presence of the product, and validation of reproducibility in different labs, if possible. CMMs are used for post-market surveillance testing (33).
- The *Indian Pharmacopoeia* general chapter, “Approach to Alternative Microbiological Methods,” which was finalized in October 2025, describes methods that can be used as alternatives to conventional qualitative, quantitative, or identification tests. This chapter shares many scientific principles with USP <1223>, Ph. Eur. 5.1.6, and TR 33 (34).
 - In October 2023, the Indian Pharmacopoeia Commission officially joined the Pharmacopeial Discussion Group established in 1989 by the European Directorate for the Quality of Medicines and HealthCare, the United States Pharmacopeial Convention, and the Japanese Pharmacopoeia of the Ministry of Health, Labour and Welfare to harmonize pharmacopeial standards (excipient monographs and selected general chapters) in three major regions of the world. Their goal is to enhance global pharmaceutical standards, regulatory compliance, and international recognition of Indian pharmaceutical products.

3.3.5 Multi-Regional Regulatory Strategies for Alternative/Rapid Microbiological Methods

Companies that want to utilize an AMM/RMM for a product that is marketed in multiple countries require approval from each country’s regulatory authority. It is possible, however, that one regulatory authority may not approve the AMM/RMM, even though another regulatory authority has approved it. In this case, the company may need to establish an alternative product testing strategy for the country in which the AMM/RMM cannot be used, which may include using the CMM for that particular product (i.e., “dual testing strategy”). However, such dual testing strategies may introduce operational and compliance burdens including, parallel data trending, specification alignment, increased resource utilization, and potential challenges in product disposition decision-making. These implications should be considered during implementation planning and regulatory strategy development.

As stated in the USP “General Notice 6.30 Alternative and Harmonized Methods and Procedures,” if it appears there is a difference between the AMM/RMM and the CMM, or if there is a dispute as to whether a drug product complies with the monograph specifications, then the CMM is considered the referee test (35).

3.4 Business and Quality Considerations for Implementing Alternative/ Rapid Microbiological Methods

The industry can benefit from the application of modern process analytical tools including AMM/RMMs. The implementation of these technologies can provide a more in-depth understanding of their manufacturing processes in terms of microbial QC. As such, an opportunity exists to enhance process control, reduce variability, and increase manufacturing capacity and efficiencies.

To aid in the evaluation of which particular AMM/RMMs are appropriate for specific applications, firms should consider the business and economic, quality and regulatory, and technical considerations of the proposed method(s).

The business or cost requirements when implementing AMM/RMMs should be well understood. These considerations may have a significant impact on the decision to validate and implement new technologies for routine use. From a business impact perspective, a number of potential benefits may be achieved, including, but not limited to:

- Faster finished-product release times to reduce backlogs and inventory
- Reduction in risks associated with microbial control and current GMP decisions concerning forward- or continuous-processing of contaminated material that can have an impact on the acceptance and/or overall quality of the batch
- Elimination or reduction of offline assays by replacement with at-line, on-line, or in-line assays
- Increases in laboratory automation and reductions in manual testing, sample handling, and/or data management/data integrity, thereby reducing overhead and/or headcount for sampling and testing
- Faster response to contamination events or microbial data deviations, and the initiation of investigations
- Reduced repeat testing, lot rejection, reprocessing, and rework
- Reduction in plant downtime and investigations
- Reduced raw material, in-process and finished-goods inventory holdings

Many of these business considerations, in terms of potential cost savings and/or cost avoidances, may be analyzed using an appropriate financial or economic model, and the resulting information may be used to support, from an economic standpoint, the implementation of a new technology. However, the financial results of conducting such an exercise should not solely be used to make a final decision on whether or not to implement the technology. Rather, they should be considered in addition to all of the other relevant factors, including the quality and technical benefits, when implementing the technology.

There are a variety of financial models that can be used to assess whether an AMM/RMM will provide cost savings or cost avoidance when implemented. These include return-on-investment (ROI), Net Present Value and Payback Period, among others. For example, ROI is the ratio of money gained or lost on an investment relative to the amount of money invested. In this case, the investment is the implementation of an AMM/RMM. The Payback Period is the time required for the return on an investment to "repay" the sum of the original investment. A company's financial organization can assist in determining the most appropriate model to use depending on the technology, its applications, and other financial factors. In all cases, the operating costs associated with the CMM and the AMM/RMMs are identified, as well as the potential cost

savings/cost avoidances and the investment costs for validation and implementation of the AMM/RMM. These values are then used in the relevant financial model to calculate the potential financial benefit. In addition, soft savings, such as improvements in quality due to the use of the AMM/RMM, must be taken into consideration.

Examples of operating costs may include, but are not limited to, the following:

- Cost per test (e.g., consumables)
- Labor time and labor costs
- Equipment depreciation, calibration, qualification, and maintenance
- Laboratory overhead
- Data management, access, and storage
- Software updates and/or ongoing licensing/subscription fees
- Additional or dual testing (e.g., if the AMM/RMM is not approved in all countries)

Examples of investment costs may include, but are not limited to, the following:

- Capital costs for the new technology
- Installation costs, including laboratory modifications if necessary
- Costs for integration with electronic data management systems (e.g., laboratory information management system (LIMS))
- Training
- Validation
- Method transfer activities
- Regulatory filings and associated costs, where applicable

Finally, examples of potential cost savings/cost avoidances may include, but are not limited to the following:

- Better monitoring and control of the overall microbial quality of a manufacturing process and its associated product, thereby reducing risk to the patient
- Early identification of contamination in multistage processing, allowing process intervention to prevent manufacturing loss
- Greater understanding of manufacturing variability, enhancement of process knowledge, and contribution to continuous process and product improvement
- More accurate, sensitive, and reproducible monitoring, as well as enhanced trending of microbiological data
- Fewer transcription errors and improved documentation and traceability as a result of increased automation and data integrity

Additional information on how to develop a financial assessment may be found in the public literature (36).

3.5 Risk Analysis

The concepts of QRM and Analytical Quality by Design (AQbD) are useful to the discussion of evaluation, validation, and implementation of AMM/RMMs. The ICH defines risk for pharmaceuticals for human use as “the combination of the probability of occurrence of harm and the severity of that harm” (37). With an analytical method, risk is associated with categories like sample collection, sample preparation, test reagents, instrumentation, test environment, and personnel factors (38). These variables can be ranked according to the severity of the consequences on the analytical results, the probability of failure occurrence, and the difficulty of failure detection. Based on these rankings, risks are mitigated.

Two primary principles of QRM stated in ICH Q9(R1) are that the:

- Evaluation of the risk to quality should be based on scientific knowledge and ultimately linked to the protection of the patient
- Level of effort, formality, and documentation of the QRM process should be commensurate with the level of risk

An industry white paper thoroughly detailed how various AQbD tools contribute to developing robust analytical methods, whose validation status are continuously confirmed during a product lifecycle (39).

As quality by design (QbD) in pharmaceutical product development is more familiar than its application in assay development (i.e., AQbD), a useful comparison of the two processes is provided in Table 3.5-1.

Table 3.5-1 Comparison of the Steps Taken in the Quality by Design and Analytical Quality by Design Processes (40)

Quality by Design (QbD)	Analytical Quality by Design (AQbD)
Define quality target product profile	Define analytical target product profile
Establish critical quality product attributes	Establish critical quality method attributes
Conduct a risk assessment	Conduct a risk assessment
Establish the process space design	Establish the method operable design region
Develop a process control strategy	Develop a method control strategy
Implement lifecycle management	Implement lifecycle management

A QbD has been successfully applied to chemical assays and its principles can be adapted to microbiological methods through the AQbD framework (41). The six AQbD process steps in Table 3.5-1 are discussed more fully with examples relevant to modern microbiological methods here:

The steps are defined as:

1. Definition of the Analytical Target Profile with AMM/RMMs

Analytical target profile is a concept developed in analytical chemistry but is also applicable to microbiological methods. Microbiological methods are broadly classified as methods for microbial detection (e.g., sterility testing), microbial enumeration (e.g., microbial plate counts), limit tests (e.g., tests for specified or objectionable microorganisms), and microbial characterization and identification. The analytical targets of the conventional growth-based microbiological methods are viable microorganisms with the capability of growing in microbiological culture media, forming colonies on a plate or turbidity in a broth, that are detected and/or enumerated by visual inspection. With AMM/RMMs the analytical targets vary widely with the technology, the results are captured electronically, and they are often not comparable to the CMM based on the colony-forming unit (CFU) or turbidity. For example, the analytical target for bacteria and fungi in a bioluminescence method would be adenosine triphosphate (ATP) present in viable cells. The ATP content of microorganisms varies by the type, species, and growth-phase of the organism. Other targets include rRNA base sequences, respired CO₂, autofluorescence, vital stained nucleic acid, immunological reactions, enzymatic activity, and biomarkers.

2. Establish Critical Quality Method Attributes

Critical quality attributes (CQAs) are typically viewed as physical, chemical, or biological attributes that define the quality of a drug product. A major transition from microbiological attributes based on growth to the physicochemical attributes is occurring. CQAs for analytical methods, which can be called critical quality method attributes, address both method attributes and method parameters rather than product attributes. For example, a polymerase chain reaction (PCR) amplification sterility test method based on a nucleic acid base sequence, like 16S rRNA, depends on multiple operating parameters including nucleic acid extraction and capture, primer and probe reagents, target sequence, thermocycler performance, signal capture, and data reporting. Accordingly, these critical quality method attributes should be considered when validating this type of AMM/RMM.

3. Risk Management

A risk assessment is defined in the ICH Q9(R1) guideline as the identification of hazards and the analysis and evaluation of risks associated with exposure to those hazards. A risk assessment is an analytical component of the overall risk management process. Risk management is defined in the ICH Q9(R1) guideline as a systematic process for the assessment, control, communication, and review of risk to quality across the product lifecycle. Examples of risks that may be relevant to an AMM/RMM include false negative, false positive, and non-equivalent results. The greatest risk to a patient is a false negative result (i.e., a negative result when microorganisms are present in the test sample), which the industry must work to avoid. A false positive (i.e., a positive result when no viable microorganisms are present in the test sample), is a business risk; a good product could be rejected, which could affect the supply to the patient. In addition, the frequent lack of equivalency of a modern AMM/RMM to the CMM due to different signals makes the statistical comparison of the comparability difficult and complicates and delays the method validation and acceptance of AMM/RMM by the competent authorities.

4. Method Operable Design Region

Method operable design region is used to select the operational region for a routine analytical method, so critical method parameters and analyte sensitivity must be considered. For example, with a mycoplasma PCR screening method, the specificity of the targeted base sequence (method parameter) and a limit of detection (LoD) of 10 CFU/mL (analyte sensitivity) comparable to the growth-based CMM should be viewed as critical (42).

5. Method Control Strategy

The inclusion of negative and positive controls in an assay run is common with AMM/RMMs. They type of controls used will be specific to the technology or scientific principle of the method. Controls are useful in monitoring the performance of the assay.

6. Lifecycle Management

The current regulatory expectation is to assess the performance of an analytical method, make an improvement to the assay (if required), use change control to evaluate method changes and, if necessary, revalidate the method throughout the lifecycle from product introduction to product cancellation (43).

Generally, when a change is proposed, the potential risks associated with these changes should be identified; this is also the case when implementing an AMM/RMM. Therefore, a risk assessment should be performed prior to the start of any validation and implementation activities.

Identified risks may vary and may include:

- Inability to demonstrate equivalence in results between CMMs and AMM/RMMs
- Global regulatory acceptance
- Product sample requirements for evaluation
- Alternative signal to the conventional microbial growth-based result
- Potential for false-positive or false-negative results
- Computer system capabilities and securities
- Supplier issues

Risks may vary depending on the technology and the methodology it replaces, the nature of the measurements taken (qualitative, quantitative, or identification), the unit of measure (e.g., as compared with conventional microbiology measurements, such as the CFU or turbidity as an indicator for growth), and the particular product or process attribute being evaluated.

The risk assessment may be carried out by comparing the differences between each process step of the conventional method. See Gordon's *Validation of Milliflex® Quantum for Bioburden Testing of Pharmaceutical Product*, which provides additional guidance (44).

Risk analysis model tools, such as the Failure Modes and Effects Analysis (FMEA) or Hazard Analysis and Critical Control Points (HACCP), may be used to effectively decide which AMM/RMM to implement, to assist in the justification of technology implementation, or to better understand the impact of implementation on the product and the business. Additional information on these two methods of risk analysis as well as other risk analysis methods is available in PDA *Technical Report No. 44: Quality Risk Management for Aseptic Processes* and in World Health Organization *Annex 7: Application of Hazard Analysis and Critical Control Point (HACCP) Methodology to Pharmaceuticals* (45, 46).

3.6 Automation of Conventional Methods

Manual laboratory procedures can be automated, for example, Gram-staining, colony-counting, determining the inhibition zone, performing an antibiotics assay, or general sample preparation, such as dissolution or dilution of samples. Such automated and integrated systems offer many advantages to laboratories with very high sample throughput and recurring workflows (e.g., reduced sample-handling and variability from manual-handling). Whether they are worthwhile in individual cases can only be assessed on the basis of detailed cost-benefit evaluation (see **Section 3.4**).

Following this strategy, it is important to determine how the introduction of automation impacts the equipment qualification and method validation requirements (35). For example, there are a variety of methods available to detect the presence of viable cells. These methods may have applications in different tests (e.g., bioburden testing, sterility tests) but may not, in fact, replace critical aspects of the test entirely. When using imaging systems to detect and count the colonies, which in the narrower sense do not differ from the conventional pharmacopeial procedures (e.g., using membrane-filtration method, similar media, incubation temperature/conditions and time), it is probably sufficient to assess the accuracy and precision of the colony-counting system as part of the equipment qualification. It is advisable to carry out a risk assessment and to assess the possible effects on the validity of the determined colony number. It is probably not necessary to revalidate the automated colony-counting method in full, but a comparison of the accuracy and precision of the colony number determined by the automated colony-counter and the colony number determined by laboratory staff is advisable. The article "Systematic Approach for the Evaluation, Validation, and Implementation of Automated Colony Counting Systems" in the *PDA Journal of*

Pharmaceutical Science and Technology describes an equipment qualification for an automated colony-counting system that detects microcolonies based on their intrinsic autofluorescence and while allowing the incubation time to be reduced by 50% (47).

Figure 3.6-1 is a flow chart for an automated colony-counting system to decide whether a performance qualification (PQ) is sufficient (automation of an CMM) or if additional aspects should be assessed as part of a method validation (if changes to the CMM are introduced) (48).

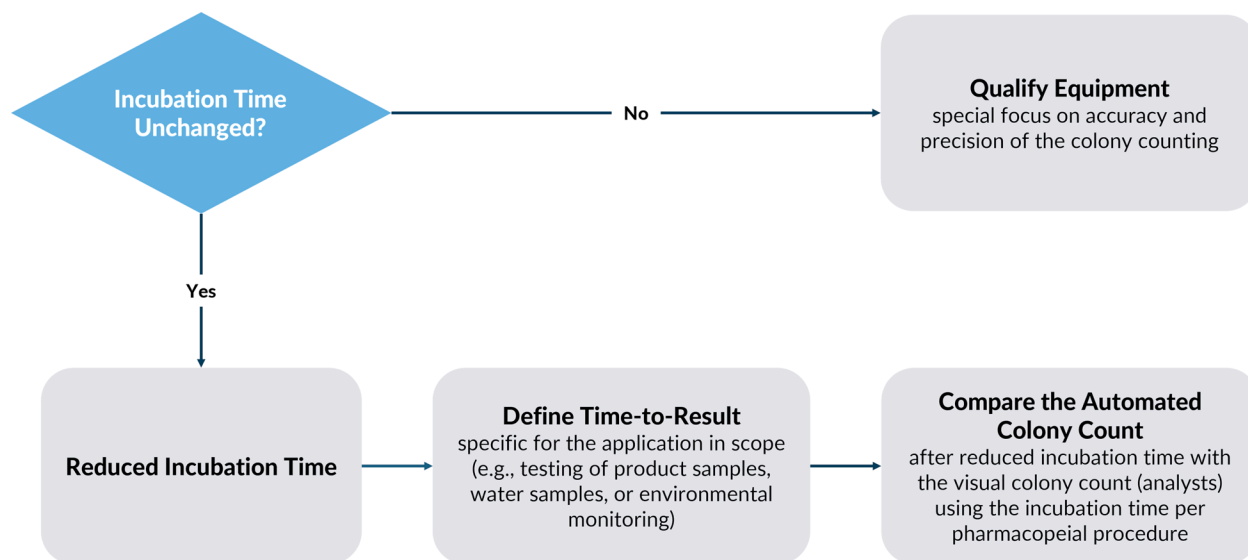


Figure 3.6-1 Flow Chart for Equipment Qualification of Automated Colony-Counters and Method Validation

The result of the decision process (see **Figure 3.6-1**) was that an extended equipment qualification should be performed if a shortened incubation time compared to the CMM is to be used.

In contrast, a colony-counter using the CMM with an incubation time described in guidance documents would merely compare the machine versus the visual count of the colonies. There is no need to establish the incubation time.

3.6.1 Performance Qualification

During the extended PQ, the accuracy and precision of the colony-counting device should be compared to the visual count on the same plate, rather than an assessment of the entire microbial analytical performance (47, 49). If the expected use of an automated colony-counting device is used to reduce the TTR, it is important to collect data before (e.g., development studies) or during the PQ to help determine the TTR.

3.6.2 Time to Result

The example for determining a TTR is based on risk assessments and on internal test methods for testing, for example, product or water samples (47). Testing of product samples was carried out according to pharmacopeial requirements with an incubation time of 3-5 days on tryptic soy agar. On the other hand, water samples based on the company's in-house flora, which includes slow-growing microorganisms, were incubated on Reasoner's 2A (R2A) agar with a prolonged incubation time of 7-8 days. These internal test methods were used as the basis for the experiments to define the sample-specific TTR.

The TTR may be defined as the time it takes for the automated colony-counting system to accurately count the slowest-growing organism on the solid media reaching a predefined recovery of the visual count of the same agar plate for the maximum incubation time described for the conventional readout (i.e., 5 days for testing of product samples, or 7-8 days for testing of water samples) (47, 49). The TTR may also be determined by statistically comparing microbial counts of relevant sampling sites (e.g., point-of-use water) between the automated colony counter and the CMM. In this case, the TTR in the automated colony-counter should demonstrate a statistical noninferior result as compared to the CMM. Applying the TTR, an assessment is needed as to whether the automated colony-counting system is “comparable” (Ph. Eur.) or “non-inferior” (USP) to the conventional visual colony-counting method with respect to the validation parameters, accuracy, and precision (see **Section 3.6.1**).

3.6.3 Method Suitability Test

Before routine testing can occur, method suitability must be demonstrated (30, 50). Method suitability demonstrates the recovery of microorganisms in the presence of the product to be tested.

Other modifications (e.g., different media types, different incubation temperatures) may require a method validation. The extent of this method validation should be defined based on a risk assessment (see **Section 3.5**).

3.7 Method Variability in Microbiology and Alternative/Rapid Microbiological Methods

Microbiological methods are subject to varying degrees of detection, and this is predominantly observed with methods that rely on the growth of microorganisms. Variability can be associated with media selection, incubation time, incubation temperature, incubation conditions (e.g., humidity, oxygen tension), intrusiveness of the sampling method, presence or absence of antimicrobial substances, or the actual method being used. Furthermore, microbial growth will be impacted by the physiological state of organisms intended to be detected, especially if the organisms are stressed, injured, dormant, or are VBNC.

3.7.1 Considerations for Detection and Enumeration Approaches

Method variability in CMM analysis is well documented and should be considered in assuring the microbial quality of a product (51). AMM/RMM approaches should be reviewed prior to validation as additional factors can influence the accuracy and sensitivity of microbial analysis, from sample preparation to detection, beyond those considered for CMMs. Reliable detection of microbial contamination at specified levels is essential, including the presence or absence of microbes and single-cell detection for sterility, or the definition of alert and action levels when used for process monitoring and control. An understanding of test material characteristics (e.g., solubility, viscosity) is a prerequisite to reliable analysis of the material and any required sample preparation steps to optimize the recovery and analysis of any microbes or associated chemical signals (e.g., endotoxins). Often, the recovery and suitability of AMM/RMMs can be addressed in the suitability of these materials in the pre-validation work with the supplier.

Key factors impacting the sensitivity and accuracy of AMM/RMMs include instrumentation limitations (e.g., offline verification of an in-line measurement), variability in sample preparation methods (e.g., dilution and filtration), the nature of the analyte (e.g., viscosity, solubility, microbial distribution, competing fluorescent chemistries, abiotic, and nonmicrobial biotic particles), the chosen test methodology (culture-based, molecular, chemical, immunologic), microbial viability, presence of VBNCs.

Using appropriate calibration standards, such as microbial count standards and surrogate chemical or particulate references as required by the measurement technology, for at-line and offline equipment is recommended to ensure consistency. Additionally, the inherent diversity of microorganisms within samples also adds to the variability. As the measurement field develops, there will be further challenges in ensuring the reliable operation of in-line technologies, for both the ongoing operation and the qualification and confirmation of detection within the process.

Sampling considerations for the incoming raw materials and in process samples are critical in ensuring quality production as is the sample plan applied to finished production, as the distribution of microorganisms can vary in unconventional materials, affecting detection accuracy. Statistical models like Poisson distribution are often employed to interpret microbial counts on the assumption that the microbes are uniformly dispersed within the test material. Alternative approaches for statistical models and considerations for the microbial distribution in food-based materials have also been proposed (52, 53).

Consistent preparation of bacterial inoculums is challenging, especially at low counts, and errors can arise during various stages of testing. While AMM/RMMs offer greater sensitivity than CMMs, they require meticulous sample preparation to accommodate the sample matrix and contamination distribution during validation and suitability testing.

Where the measured microbial indicator is dependent on microbial metabolic activity (e.g., ATP, metabolic conversion of fluorescent indicators), additional activation or incubation procedures may be required so stressed cells may recover or divide sufficiently for detection.

Unique validation strategies may be needed for AMM/RMMs, particularly with non-liquid samples and in line detection technologies. Additional vendor support or in-process confirmation of microbial detection is often crucial for end users in validating these technologies. A comprehensive understanding of method variability, along with careful planning in sampling, preparation, and validation, is essential for ensuring accurate detection and assessment of microbial contamination in various products.

3.7.2 Considerations for Identification Approaches

Various AMM/RMM identification methods are available for characterizing microorganisms in pharmaceutical products and raw materials. These serve as effective screening technologies to determine the absence of specified microorganisms, thereby necessitating validation according to established guidelines. Microbial characterization is crucial as it identifies potential impacts on products and production processes that may not be covered in compendial monographs.

These approaches are similar to conventional phenotypic and biochemical approaches in that the identification is calibrated to cultures of known identity. Qualification involves distinguishing between microbial classification, which groups organisms based on similarities, and identification, which specifies their genus or species. Identification methods include phenotypic approaches, such as colony morphology, Matrix-Assisted Laser Desorption/Ionization Time-of-Flight mass spectrometry (MALDI-TOF MS), and genotypic techniques like 16S rRNA sequencing. While considered the most-used reference methods, both may struggle to differentiate closely related species (e.g., *Bacillus spp.* or *Burkholderia spp.*), so require additional methods for accurate identification (e.g., multilocus sequencing technology). With the further development of next-generation sequencing, allowing whole-genome sequencing, and the use of spectral profiling (e.g., FT-IR, Raman) approaches, similar comparative qualification to reference cultures can be conducted.

New identification methods should be verified, focusing on accuracy and reproducibility (repeatability), either against CMMs through parallel testing (e.g., USP <1113> Microbial characterization, identification, and strain typing (54)), using known standard QC cultures, or agreement with reference laboratory results, with a goal of achieving over 90% accuracy and repeatability, defined as follows (55):

- Accuracy % = (Number of correct results/Total number of results) · 100
- Reproducibility (Repeatability) % = (Number of correct results in agreement/Total number of results) · 100

The Ph. Eur. 5.1.6 booklet provides an example of a validation approach for the 16S rDNA sequencing. The verification tests should include the assessment of (56):

- **Accuracy** (*Acceptance criterion:* Microorganisms are identified correctly to species level and above the pre-defined reporting level for sequence homology)
- **Specificity** (*Acceptance criterion:* the results of the specificity test using mixed cultures (quantified and titred based on CFU or using an analytical method to standardise the DNA titre) lead to valid identifications. These results must clearly demonstrate that the analysis of mixed cultures does not result in false identifications)
- **Robustness** (*Acceptance criterion:* Microorganisms are identified correctly to species level, with a pre-defined percentage match between the database and the test strains)

Where referential databases are used for identification, the profiles of all organisms may not be present due to the maturity of the identification technology, its use across industry, and its associated database. For known and culturable microorganisms, the databases can be updated for accuracy and completeness, ideally with multiple users verifying the accuracy of the input data as misidentifications can pose significant risks.

Those techniques that offer the detection of VBNC (e.g., microscopy and Raman, Cytometry) and are unable to refer to a cultured sample are limited to detection and enumeration, as the recovery of sufficient genetic material can limit the use of genotypic approaches for the confirmation of microbial identity.

For both culture and VBNC-sourced material, the purity of any observed or recovered organisms can be a problem as the sample may be composed of a mixture of microbes, and the method may bias the identification to the organism (e.g., efficiency of DNA extraction) (5).

Successful qualification hinges on a thorough comprehension of the system's capabilities, its limitations, and the implications of the results, enabling end users to make informed decisions when interpreting identification outcomes and managing any risk assessment for material dispensation.

For certain AMM/RMMs, the testing process may be destructive or may not permit recovery of a viable isolate for subsequent microbial identification. In such cases, firms may evaluate alternative strategies for the identification of contaminants, such as duplicate or parallel sampling.

3.8 Method Choice

Advances in specific measurement technologies are enabling the use of AMM/RMMs for targeted release and monitoring. These include technologies such as viability-based fluorescent staining and flow cytometry, which have enabled microbial detection at the single-cell level while certain hybrid platforms further integrate these with techniques like microscopy or spectral analysis to characterize individual targeted cells.

Other AMM/RMMs also offer distinct advantages. For example, ATP bioluminescence provides nearly instantaneous results for general hygiene or biomass monitoring and estimations, while nucleic acid

amplification techniques (NAT)/PCR deliver high taxonomic specificity and the ability to detect non-culturable microorganisms (57). In many cases, quantification and viability can be obtained in hours, with TTR dependent on the technology, its use, and sample preparation (often culture/enrichment). There are several in-line approaches, but a majority of these techniques apply to the at-line or off-line analysis of samples. Whether the full use of the method's capability is used for measurement purposes is at the discretion of the user, with the expectation that the scientific justification and validation of the selected measurement parameter is carried out. For example, the use of quantitative methods capable of enumerating viable particles in a sample for nonsterile applications can equally be applied to its use as a qualitative measure for the presence or absence of viable cells in sterile manufacture.

The development and use of AMM/RMMs for in-line or on-line measurement, to either monitor the sterility or the maintenance of sterile/hygienic manufacturing conditions, is currently limited to the quantification of biofluorescent particles which may consist of one or more cells, usually against an inert/transparent background chemistry (i.e., air or water). Additional processing and suitability considerations are often required for these technologies to be applied to more chemically complex matrices.

For AMM/RMMs, the TTR is defined by the time of detection for the most challenging organisms within the system (see **Section 3.6.2**). Unlike CMMs where TTR is traditionally based on fixed incubation periods for recovery (e.g., five days for products or seven to eight days for water), TTR for AMM/RMMs is determined by validation experiments. This involves demonstrating that the AMM/RMM provides a comparable or non-inferior level of detection within a shorter timeframe.

3.8.1 Categorizing Technologies

For the methods described in **Sections 4.2** and **4.3**, the aim is to detect or quantify the presence of even a single viable organism present in the process or test sample, with the potential to identify this organism where required (see **Section 4.4**). As a single technique may employ a combination of technologies (e.g., microscopy, activation, metabolized dyes) or targeted analytical techniques (e.g., FT-IR), each technology description has been categorized under the types of data output achieved: qualitative (presence/absence), quantitative (accurate enumeration), and identification, as defined in Ph. Eur. 5.1.6. This categorization does not imply that these methods cannot be used interchangeably across categories; for example, a method capable of single-cell discrimination and quantification can also serve for use as a presence/absence test, such as solid-phase cytometry.

Qualitative technologies will detect the presence of microorganisms (e.g., sterility test), the presence of a specific type of microorganism (e.g., specified organisms, *E. coli* or *Mycoplasma*), or related biomolecule (e.g., pyrogens), which can be semiquantitative by reference to relevant calibration curves (e.g., the monocyte activation test (MAT) for pyrogen detection (58)).

Quantitative technologies enumerate the number of microorganisms present in a test sample. The pharmacopeial microbial enumeration test is an example of a conventional growth-based method. In contrast, certain AMM/RMM technologies, such as solid-phase cytometry, provide enumeration to the single-cell level using cell- or viability-based staining approaches.

Identification technologies can identify microbes to the genus, species, subspecies, or strain levels generally requiring culture, with some systems capable of identification at the single-cell level (e.g., Confocal Raman spectroscopy). Some of these methods are fully automated strategies of conventional growth-based biochemical and carbohydrate methods, while others rely on novel scientific principles similar to chemical or analytical methods.

4.0 Technology Review

Compendial microbiological techniques are still widely and appropriately used to assess the sterility and microbiological quality of finished products and their associated processes (see **Section 3.1**). The move to biologically derived therapies, parametric release, and improved detection capability has led to a range of AMM/RMMs. With improvements in optics, digital imaging, and molecular/genetic, localized thermal, chemical, and electrophoretic techniques for measuring the presence, identity, and viability of microbes, a range of technologies have emerged or matured since the 2013 TR 33 revision. New technologies are now capable of either directly measuring a single cell, its activation, or alternatively, the early stages of cell division, active metabolism, or the presence of microbial by-products.

While technology dependent, AMM/RMMs may offer the promise of improved sensitivity, accuracy, automation, objectivity, precision, operational and data integrity, improved operational efficiency, and potentially shorter time to result. These new methods have not only increased the need for the modern industrial microbiologist to embrace digital, optical, and statistical expertise relative to interpretation and validation, but also include additional considerations in suitability testing and appropriate sample preparation of both biological and chemical samples for these new metrics.

The scope of use of AMM/RMMs can span in-line, on-line, at-line, and off-line use with data being either directly related to the release specification, for example, presence or absence of viable organisms for sterility or, for process monitoring purposes, rendering of viable cell levels in nonsterile applications or the verification of an aberrant result off-line.

More conventional, growth-dependent techniques have also benefited from these developments through either improved methods for detecting the conventional end point (e.g., the digital detection of microcolony formation) or by combining alternative technologies to increase method sensitivity and data integrity or to decrease TTR and analyst subjectivity (see **Section 3.8**). The application of automation has also benefited AMM/RMMs in a similar way to more conventional approaches, reducing errors in the procedure, improving the statistical robustness of the generated data, and increasing testing capacity while reducing the variability associated with direct handling of the samples in the laboratory (see **Section 3.6**).

The selection, validation, and use of AMM/RMMs is very process- and product-dependent and, ultimately, focused on providing the best assessment of product quality to minimize the risk of use for the safety of the patient or consumer. See **Section 4.1** for more information on sample effect considerations. Given the different product forms, processes, specifications (nonsterile, sterile), and controls used in pharmaceutical, medical device, and cell-and-gene-therapy manufacturing, the contents of **Section 4.0** can only be informative and guiding, not specific to product type or categories. Beyond the quality and safety requirements, the use of AMM/RMMs will be decided based on the release specification and a balance of improved risk management, production efficiency, validation cost, implementation, and regulatory compliance (59).

Section 4.0 introduces the technical principles and considerations for adopting these AMM/RMMs.

4.1 Sample Effect Considerations

For the validation and suitability qualification of any new method, or its application to the test product, additional consideration of what might affect the accuracy, sensitivity, and LoD for these new technologies is required as they are likely to differ from those used for conventional (growth-based) methods. The primary analytes for these methods range from particulate evaluation, fluorescence, spectral analysis of chemistry, and optical assessment and can often be combined in current AMM/RMMs (e.g., optic for counting and fluorescence for viability).

The impact of sample effects on AMM/RMMs needs to be evaluated, for example during method suitability studies (see **Section 5.3.2**). These sample effects can include the presence of background ATP or autofluorescence of particulates present in lab reagents, materials, or the sample matrix itself. Suppliers may provide guidance for overcoming known interference with their technology. This consideration may also require the evaluation of more conventional suitability criteria, such as the presence of any antimicrobials that previously affected culture techniques and cell growth but may now interfere with microbial metabolism and cell activation. Consideration of these factors and their impact on measurement will be system-dependent and may require additional reagent and sample processing (e.g., filtration to remove reagent particulates or the specific enrichment of any microbial content present relative to interfering moieties). To ensure method robustness, specific negative and positive controls must be established to mitigate these matrix effects, accompanied by explicit method suitability acceptance criteria that define permissible false-positive thresholds and signal-to-noise ratios.

4.2 Qualitative Detection Principles and Approaches

The primary objective of qualitative detection methods is to identify the presence of contaminating microorganisms that could render the material or product unusable.

4.2.1 Microcalorimetry

Microcalorimetry offers a non-destructive, direct inoculation method for continuous measurement of heat generated by metabolic activity. Microbial growth produces heat that can be detected by isothermal microcalorimetry with very high sensitivity. The signal is measured directly in the system without additional reagents. The sample is directly inoculated into a growth medium without additional manipulation and loaded into a vial for measurement. Microbial growth is determined from changes in heat flow over time, enabling earlier detection than turbidity-based methods. Continuous monitoring of heat production can detect and, under controlled conditions, quantify microbial growth/activity.

Beyond the research environment where conditions can be tightly controlled to enable quantification by comparison to reference growth curves-manufacturing-derived samples are most often assessed qualitatively (presence/absence). Heat output and signal dynamics can vary depending on organism physiology, stress state, and the sample matrix, making absolute quantification less reliable (60).

4.2.2 Respiration-Based Detection

Microorganisms growing in liquid culture produce carbon dioxide (CO₂), volatile organic compounds (VOCs) and other respiratory gases, along with various metabolites. These products can be detected through several methods, including monitoring specific analytes (e.g., pH), changes in pressure using transducers in a sealed container, or by analyzing the concentration of gases in the headspace of sealed containers when

using culture tunable diode laser absorption spectroscopy (TDLAS), all of which indicate microbial growth and a putative identification (e.g., indole) **(61)**.

The amount of CO₂ and other respiratory gases produced serve as a crucial indicator of organism viability. Test samples are added to media bottles containing a liquid emulsion or silicone sensor. During microbial growth, CO₂ diffuses from the medium into the sensor, where it interacts with hydrogen ions, leading to a decrease in pH and resulting in a color change of the sensor (e.g., from grey to yellow).

Additionally, respiratory gases can be sampled from the headspace and analyzed using gas sensors. The detection rate of CO₂ and respiratory gases is influenced by the initial concentration of microorganisms; a higher initial concentration results in a faster detection response. Moreover, the distinct profiles of respiratory gases produced by different microorganisms serve as unique metabolic fingerprints, facilitating high-resolution detection and early identification of specific microbial species **(62)**.

4.2.3 Electrochemical Measurement

Electrochemical methods measure changes in the electrical properties (conductivity) of microbiological media due to microbial metabolism. Liquid-growth media comprise relatively large uncharged or weakly charged molecules, and microorganisms growing in this media will break down the large molecules into smaller, more highly charged components (e.g., proteins into amino acids, fats into fatty acids, polysaccharides or sugars into lactic acid). These technologies can rapidly detect changes in measurable electrical threshold during microbial growth by monitoring the movement of ions between electrodes (conductance) or the storage of charge at the electrode surface (capacitance). This allows for faster detection of growing microorganisms compared to turbidity observations.

4.2.4 Single-Cell Detection/Method Development

Dielectrophoresis can be used as a selective separation method to capture microorganisms based on their size and dielectric properties. A combination of microfluidics and microelectronics is used to capture microorganisms followed by lab-on-a-chip quantification via optical or electrical sensing. The sample is pumped through the microfluidic chip with a system of electrodes. The electrodes in the chip generate an inhomogeneous electric field. As microorganisms enter the chip, the electric field selectively captures the microorganisms on the electrodes based on their frequency-dependent polarizability, which is then observed with a fluorescent microscope or measured via impedance changes.

4.2.5 Attachment Detection Technologies

Surface-based detection technologies respond to changes in conductance/resistivity or near-surface optical chemistry (e.g., guided mode resonance) due to microbial attachment and biofilm development **(63)**. Resistivity (ρ) is the inverse of conductivity ($\rho = 1 / \sigma$), which can be altered by inducing a small electrical current in the sensor to measure the changes in the electrical properties of process-fluid chemistry at the surface of a pipe. Microbial attachment, propagation, and the formation of extracellular matrix on the surface of the sensor indicates the presence of microbes.

Guided-mode resonance is an optical technique that uses a sensor or grating responsive to changes in the refractive index caused by a microbial attachment, providing qualitative data suited for process-monitoring and early warnings.

4.2.6 Adenosine Triphosphate Bioluminescence

ATP bioluminescence is the generation of light by a biological process. In the presence of the enzyme luciferase and the substrate luciferin, ATP is enzymatically broken down to produce photons of light. An instrument equipped with a photomultiplier tube can detect these photons. Because ATP is a key intracellular energy source in all metabolically active cells and is rapidly degraded upon loss of viability, measurement of ATP is a marker for viable organisms. To ensure analytical specificity, the method requires the selective removal of non-microbial extracellular ATP from the sample matrix prior to microbial lysis. Depending on the technology, some systems will detect the general presence of microorganisms by measuring the total relative light units from the test sample, while other systems can detect ATP bioluminescence from individual microcolonies following a shortened incubation period, thereby providing a quantitative assessment of the number of microorganisms from the original sample under evaluation.

The sensitivity and TTR of the ATP bioluminescence assay may also be improved with a two-phase reaction that begins by using an enzyme-catalyzed reaction to generate ATP to levels significantly higher than what is naturally contained in the microorganism. In the presence of microorganisms, specific microbial enzymes (adenylate kinase) can be used to convert the adenosine diphosphate (ADP) provided to the reaction into ATP and adenosine monophosphate. The enzymes are not consumed by the reaction; therefore, if ADP is present, ATP is continuously generated. The amplified ATP levels are then detected using the typical ATP bioluminescence reaction initially described in the prior paragraph.

4.2.7 Alternative Biomarkers for Sterility Testing

Existing definitions of biomarkers generally relate to the detection of host-related proteins and associated analytes in the clinical settings (64). Specifically in clinical microbiology, the use of biomarkers for detecting infectious diseases is becoming increasingly common. Notable examples of biomarkers for identifying life-threatening sepsis and systemic infections include C-reactive protein, procalcitonin, presepsin, interleukin-6, and CD64 (65).

In addition to the description provided regarding biomarkers in this section, researchers recently conducted a study to explore microbial biomarkers for sterility testing. They compared the exometabolomic profiles of uncontaminated and contaminated mesenchymal stromal cells and microbial culture supernatants using liquid chromatography-mass spectrometry. The study found that metabolically derived nicotinic acid was specifically present in bacterial-contaminated cell-therapy products with the ratio of nicotinic acid to nicotinamide in laboratory-contaminated cultures after a 24-hour incubation proved to be a useful biomarker for contamination in cell-therapy products (66). This study serves as a proof of concept (POC), demonstrating the potential of novel biomarkers being suitable determinants of microbial contamination.

4.2.8 Endotoxin Detection

The detection of bacterial endotoxins (i.e., lipopolysaccharide) has evolved with additional technologies such as:

- Cartridge- and microfluidic-based handheld systems that replicate quantitative, kinetic chromogenic methods with results in as quickly as 15 minutes.
- ELISA-based methods, where phage-derived protein binds lipopolysaccharide, in which the complex can be washed to remove interfering conditions.
- Recombinant Factor C/recombinant cascade reagent, in which synthetic but identical components in the conventional Limulus amoebocyte lysate bacterial endotoxins assay. The sensitivity of the Bacterial Endotoxin Test can be from 0.001 and 0.25 EU/mL.

4.2.9 Alternative Pyrogen Testing Monocyte Activation Test

The MAT was developed as an in-vitro assay alternative to the in-vivo rabbit pyrogen test. The test utilizes monocytic cell sources (e.g., whole blood or human peripheral blood mononuclear cells (PBMC), mononuclear cell lines, or recombinant cell lines) which, in the presence of bacterial endotoxins and other non-endotoxin pyrogens, will stimulate cytokine release, the latter of which can then be detected and quantified in an ELISA assay (58).

Several microfluidic or rapid-readout approaches exist that replace traditional plate-based ELISA in MAT workflows. Microfluidic immunoassay systems use cartridge-based nanochannels to perform automated sandwich immunoassays for cytokines like IL-6 or IL-1 β with much lower reagent volumes and faster read times. Reporter-gene assays instead detect NF- κ B activation in engineered monocytic cells using luminescent or fluorescent reporters, providing earlier pathway-level detection without antibody assays. Nucleic-acid-based MAT (NAT-MAT) measures cytokine mRNA induction using qPCR or digital PCR as a rapid transcriptional readout of monocyte activation. Emerging microfluidic and single-cell platforms further miniaturize these concepts, integrating immune stimulation and cytokine detection on chip to reduce assay time and sample volume while enabling higher-throughput pyrogen screening.

4.3 Quantitative Measurement Principles and Approaches

The primary objective of quantitative methods is to detect and quantify the presence of contaminating microorganisms, which can be used in risk assessments and may prohibit the use of the material or product.

4.3.1 Imaging and Optics

Technologies using an ultra-high definition camera or three-dimensional (3-D) optical scanning can be used to detect and quantify microcolonies. High-resolution images taken at frequent intervals (automated kinetic counting) distinguishes growing colonies from debris by analyzing multiple images over time, and according to kinetic counting suppliers (i.e., personal communication), this could lead to a potential reduction of 50% less incubation time relative to human-based observation of colony growth. This approach allows for precise quantification and standardization of counts, facilitating early notifications of limit excursions, such as exceeding alert or action levels, as well as identifying out-of-specification results as soon as they are detected. Additionally, 3-D optical scanning enables high-resolution detection of colony formation and early identification of microcolonies.

Light scattering is a phenomenon in which the propagation of light is disturbed by its interaction with particles. Instrumentation that utilizes Mie scatter (i.e., where the scattered light intensity is dependent upon the particle size in a certain size range) and fluorescence detection can provide information about the size and number of viable microorganisms in an air or water sample, with the increased use of machine learning (ML) and algorithms used to interpret microbial presence.

4.3.2 Fluorescent Probe Application

Viability-based technologies use metabolizable stains in combination with laser excitation for the detection and quantification of microorganisms, without the need for cellular growth. Viability-staining and laser-excitation can be used to detect and quantify single metabolizing cells and microcolonies. Using this approach, organisms that are stressed, injured, fastidious, or considered VBNCs may now be detected when these same organisms will not grow in or on conventional microbiological media. These types of

technologies can be used for a variety of applications (e.g., microscopy, solid-phase and flow cytometry) that require the detection and enumeration of microorganisms, such as bioburden and microbial-limits testing, EM, process-water analysis, and sterility testing.

While providing broader detection capabilities, the accuracy of these technologies is dependent on signal-to-noise ratios and the specificity of the stain for the target metabolic activity. These platforms are applicable to microscopy, solid-phase cytometry, and flow cytometry for tasks such as bioburden testing, environmental monitoring, process-water analysis, and sterility testing, provided they are validated against compendial methods for equivalent recovery.

4.3.3 Autofluorescence

Cellular autofluorescence is a property of all viable cells due to the presence of ubiquitous fluorescent biomolecules like NADH and riboflavin. Microbes emit autofluorescence when such endogenous fluorophores are excited by certain wavelengths of light. For example, microbes can emit autofluorescence in the yellow-green spectral region when illuminated with ultraviolet (UV) blue light due to the excitation of biomolecules including flavins, riboflavins and flavoproteins. An AMM/RMM example where autofluorescence is used is a digital imaging system that uses autofluorescence to enumerate micro-colonies. In such a system, test samples may be filtered, and the membrane is placed onto an agar surface and incubated. During this time, a light-emitting diode (LED) excites microcolonies to emit autofluorescence, which are then quantitated using a charge-coupled device (CCD) imaging system. Incubation can continue to allow for the recovery of larger colonies for subsequent analysis, such as microbial identification. Additionally, metabolizable fluorochromes can enhance signals from metabolically active colonies, improving method sensitivity.

4.3.4 Flow Cytometry

Flow cytometry quantifies microbes as they pass the light source and detector in a defined flow path. In-line execution can excite existing cell chemistry (e.g., amino acids, nicotinamide adenine dinucleotide plus hydrogen (NADH), and riboflavins) often referred to as biofluorescent particle counters (BFPCs) or use a microbe-specific fluorophore for direct staining or metabolic activation (67). The detection of the emitted fluorescence provides confirmation of the presence and enumeration of a biological particle or microbe in the flow. Given known flow rates, the number of fluorescent particles per volume can be estimated.

Fluorescence excitation is enabled using a laser or LED of a specific wavelength with detection of the emitted light measured via a wavelength-specific detector (e.g., photo multiplier tube or CCD). For in-line flow systems, the distance between the light source and detector is limited by the optics used and the transparency of the carrier fluid. With cell-based chemistry, the discrimination of microbes relative to other interferant biological materials present in the sample makes validation and a direct correlation to cell count challenging for intended use (e.g., process monitoring or trending).

The use of a fluorescent probe, whether it is a direct stain or a metabolized stain, often limits the volume of the test samples (in mL), and preincubation of the sample with the stain prior to measurement is necessary. Additional sample-processing, such as filtration, may also be required to remove interfering fluorescent substrates. The incubation of the sample with targeted stain, whether a direct (live/dead) stain or the use of a nonfluorescent precursor that is enzymatically cleaved by a viability-indicator enzyme (e.g., esterase present in an intact metabolizing cell) to fluorescently label the cell, improves the differentiation of microbes versus other interferant materials. This differentiation can be further enhanced by equipment designed with narrow

channels, such as microfluidics, which reduce the light-path length and allow for the testing of a wider variety of materials, including powders and opaque fluids.

4.3.5 Solid-Phase Cytometry

Solid-phase cytometry relies on capturing microorganisms on a solid phase (e.g., a membrane filter) and incubating the sample with either a direct (e.g., DNA-specific dyes) fluorophore, labelled antibodies, or metabolite-sensitive fluorophores to indicate microbial presence. Microorganisms retained on a membrane are exposed to an excitation light source (e.g., LED or laser) at a defined wavelength, inducing fluorescence from viable cells or labeled particles. The emitted fluorescence is subsequently detected through an imaging or scanning system, such as a CCD-based camera or a photomultiplier tube, or analyzed by artificial intelligence (AI), enabling enumeration of viable microorganisms and/or total particles through image analysis.

As with flow cytometry, extended contact and incubation times can be used to maximize the signal from these fluorophores, track metabolic activity, enhance cellular recovery for slow-metabolizing or injured cells, and further differentiation relative to nonfluorescent, inert particles present in the sample. Sample considerations include the removal of interferant chemistries (e.g., fluorescent chemistry, excessive particulates), which may obscure viable particles in the image, and the impact of fluid/carrier characteristics on the technique (e.g., the filterability of the sample) (49, 68).

4.3.6 Hybrid Quantitative Identification Platforms

Hybrid technologies now exist that combine spectroscopic and spatial signatures for the detection of microorganisms using transformer-based multimodal spectral imaging. This platform enables label-free, non-growth-based analysis of individual cells to the species level, providing both quantification and identification simultaneously.

4.4 Identification Techniques, Principles, and Approaches

Microbial identification is determining the identity of a pure culture to genus, and preferably to species or subspecies, if required, using phenotypic or genotypic methods. As all identification methods have limitations due to their underlying technology and the defined scope and level of verification of the databases using other methods, and as there is no compendial microbial identification method, there is no reference method for comparative purposes. Hence, the validation frameworks described in USP <1113> and Ph. Eur. 5.1.6 are appropriate for microbial identification validation. Performing a verification or fitness-for-use approach as described in USP <1113> and Clinical Laboratory Standard Institute (CLSI) M52 *Verification of Commercial Microbial Identification and Antimicrobial Susceptibility Testing Systems* is recommended (69).

This verification would involve identifying a number of laboratory cultures appropriate for intended use of the instrument, followed by parallel identification of a consecutive number of in-house isolates using the CMM and new AMM/RMM identification method. Any discrepant results would be resolved by an independent testing laboratory and justified, based on the limitation of the technology and database, and possible changes in microbial taxonomy.

Once identified, an organism can guide risk management regarding material disposition.

For most AMM/RMM approaches, identification often requires an additional laboratory processing step once contamination using qualitative or quantitative techniques has been confirmed. Many identification techniques require further activation, enrichment, and purification of the source contamination for accurate identification.

The discrimination of the identification technique requires evaluation against a range of organisms related and relevant to your process (70). Conventional biochemical identification techniques can still provide a quick means of providing insight into the type of organism; however, the expansion from biochemical, metabolism-based analysis to colorimetric, analytical-based (spectral analysis) and molecular approaches has increased the speed and accuracy with which the specific identification of a microorganism can be made. Beyond the techniques summarized in **Sections 4.4.1–4.4.5**, the application of well-developed analytical techniques (e.g., Raman, FT-IR, GC-MS) and proteomic-based techniques (e.g., MALDI-TOF MS) used in microbial identification are only limited by their ease of application, the simplicity of their execution, and their relation to established databases similar to how DNA/RNA-based identification techniques have developed.

4.4.1 Hyperspectral Imaging

Hyperspectral scanning is an imaging technique that can be used as a presumptive identification approach and can be combined with the enumeration of colonies/microcolonies immobilized on a surface (e.g., agar plate/filter). This imaging technique uses a hyperspectral camera to detect color variations in the colonies across wavelengths that can cover the visible and near infrared regions (400–1,000 nm).

This technique relies on comparison to a reference, such as an internal database, to confirm genus- and species-level identifications. However, it may face limitations due to the availability and number of captured wavelengths, especially where these wavelengths are fixed.

4.4.2 Matrix-Assisted Laser Desorption/Ionization Time-of-Flight Mass Spectrometry

Mass spectrometry systems, such as the MALDI-TOF MS, are used for microbial identification. Detection of specific microorganisms is an option, too, but the main application of MALDI-TOF MS is the identification of microorganisms. MALDI-TOF MS is an analytical technique in which analytes are separated and detected based on their mass-to-charge (m/z) ratio. Sample preparation typically involves one of three methods:

- **Direct Transfer Method:** A small amount of a pure culture is applied directly to a target plate, followed by coating with a matrix solution (usually cinnamic acid). This method is recommended for bacterial identification.
- **Extended Direct Transfer Method:** Formic acid is added to the sample on the target plate prior to coating with the matrix. This optimized procedure is recommended for the identification of bacterial isolates.
- **Formic-Acid Extraction Method:** The isolate to be analyzed is purified in several steps by the addition of ethanol, formic acid, and acetonitrile before being applied to the target and coated with matrix. This procedure is recommended for the analysis of molds and yeasts.

After sample preparations, the procedure for each method is the same; during the crystallization of the matrix, the microorganism is embedded into the crystal structure. In the MALDI-TOF MS instrument, the sample spot is hit with a pulsed beam from a solid-state laser under a high vacuum. This vaporizes the

matrix explosively and entrains the components of the microorganism (i.e., proteins) to establish the mass to charge ratio of the components.

The high energy of the laser favors the formation of ions, which are then accelerated in an electric field due to their charge. Subsequently, in a voltage-free field (i.e., flight tube), the separation of the analyte components takes place depending on their mass. The mass-to-charge ratio of individual particles of the microorganism determines the time of flight, that is, the time required for the particle to reach the detector after evaporation. Characteristic mass spectra result from the determination of the time of flight. Thus, for most bacteria, yeasts, or fungi—considering the fingerprint range between 4,000-12,000 Da—partly species-specific spectra can be distinguished. An automated comparison of the spectra generated by MALDI-TOF MS with a reference database is then carried out using software designed for this purpose.

4.4.3 Fourier-Transform Infrared Spectroscopy

Fourier-transform infrared spectroscopy (FT-IR) is an analytical technique used to capture high-resolution infrared spectra emitted or absorbed from a microbial cell. The sample is irradiated with infrared radiation, and the resulting spectrum is a plot of the intensity of the transmitted or reflected light as a function of wavelength. Comparing the microbial spectral profile to a database of known organisms allows for the identification and characterization of the organism being examined and is applicable to bacteria, fungi, and viruses.

4.4.4 Raman Spectroscopy

Raman spectroscopy is an established analytical method based on the Raman-scattering properties of a material under evaluation. When laser light interacts with the sample, molecules are excited, resulting in a photon-energy shift related to molecular vibrations and rotations. The collection of the Raman-scattered light results in a unique spectral fingerprint, which is characteristic of the composition and molecular structure of a microorganism when immobilized on a surface (e.g., a filter). Comparative software and curated reference databases facilitate accurate identification in less than 15 minutes.

Because this is a chemistry-based technique, additional considerations are required for sample analysis, including selection of the appropriate immobilization background (e.g., Raman-negative filter/solid phase) and the subsequent isolation and removal of interferant chemistry from the targeted organism. This is important as these sample analysis considerations may impact the quality of the Raman spectrum.

4.4.5 Nucleic Acid Amplification Techniques

NAT employ a variety of scientific principles including, but not limited to, DNA-based PCR (e.g., touch-down PCR, real-time PCR, digital drop, and digital PCR), RNA-based reverse-transcriptase amplification, 16S rRNA typing, gene sequencing, and other novel techniques.

Many of these methods will detect the presence of a specific microorganism, such as an “objectionable” or pharmacopoeia-specified organism, or can provide a microbial identification, in some instances, to the strain or subspecies level. Additional methods can be used to estimate the number of viable microorganisms in a sample based on the number of amplification cycles or copy numbers required to reach a baseline or threshold level. Because there are many methods and systems based on NAT, it is not possible to cover all the scientific principles in TR 33; therefore, a brief example of key methods is provided, and additional information may be found in the literature and in online resources (36, 71-74).

It is important to distinguish microbial taxonomy from routine identification used in pharmaceutical microbiology laboratories. While taxonomy focuses on phylogenetic classification based on conserved gene regions (e.g., 16S rRNA or ITS), such sequence lengths may be insufficient to resolve closely related species. As a result, higher-resolution technologies, including whole-genome sequencing, have increasingly been adopted where greater discriminatory power is required (55).

4.4.5.1 Polymerase Chain Reaction

In a conventional PCR reaction, DNA is extracted from microorganisms (e.g., from an isolated colony on an agar plate) and heated to separate the double strands. DNA primers are then added that will bind to unique target sequences on the template DNA. The primer is elongated when a heat-stable DNA polymerase and nucleotide bases are added. The result is two new copies of the template DNA. This PCR process is then repeated, resulting in millions of copies of the target DNA. After the amplification step, the amplified nucleic acid targets must be analyzed using several techniques: amplicon detection/analysis in gel electrophoresis, detection by hybridization techniques (e.g., fluorescent-in-situ-hybridization or microarray-based techniques), or DNA-sequencing.

Real-time PCR uses unspecific nucleic-acid intercalating fluorescence dyes or target-specific probes labelled to detect the amplified nucleic acid after each amplification cycle but also allows a quantification of the target at timepoint t_0 . There are many different types of probes that bind to double-stranded DNA or to specific sequences as they amplify and accumulate in the test system. The increase in fluorescent signals is then detected. As an example, if a sample contains a DNA target sequence associated with a particular microorganism, following PCR amplification of that target, the fluorescence signal from the probe will be detected, and the system will provide a positive response for that microorganism. If the target is not present in the original sample, then no amplification will occur, and no fluorescence will be detected (above the threshold or background level). If the method uses real-time quantitative PCR, the DNA amplification reaction is measured as it occurs. Typically, fully automated real-time PCR systems are used, which reduces the hands-on time.

Digital PCR systems separate the sample to be tested into thousands of partitions using either nano well plates (with 20,000 partitions of 45 μ L, 28,000 partitions of 30 μ L, or even 100,000 partitions of 15 μ L) or into droplets (digital droplet PCR). In these little partitions, the amplification using fluorescent probes takes place. After the amplification step, the nano well plate partitions or the droplets will individually be assessed for fluorescence. Partitions or droplets containing amplicons are detected by fluorescence and scored as positive. Partitions or droplets without amplicons are scored negative. A Poisson statistical analysis of the number of positive and negative partitions or droplets allows a quantitation of the target sequence.

4.4.5.2 Reverse Transcriptase Polymerase Chain Reaction

Reverse transcriptase PCR (RT-PCR) uses RNA, instead of DNA, as a starting template for the PCR reaction. In this process, RNA is extracted from the cell, and the enzyme reverse transcriptase will create a complementary strand of copy DNA (cDNA). Ribonuclease H (RNase H) treatment will then remove the original single strand of RNA. A second primer and DNA polymerase is then used to create double-stranded cDNA, which will be used in the conventional PCR reaction previously described in **Section 4.4.5.1**. RT-PCR has some advantages over conventional PCR, such as a lower risk of contamination from nonviable cell DNA or residual DNA from the sample or work environment. RT-PCR is currently being used for the detection of specific types of microorganisms, such as viable, proliferating microorganisms, as well as the estimation of viable cell count.

4.4.5.3 Gene-Sequencing

Gene-sequencing is used for the identification of a wide variety of microorganisms, including bacteria, yeast, mold, and Mycoplasma. The scientific principle involves sequencing each nucleotide base of a specific DNA target after PCR amplification. Typically, the first 500 base pairs (bp) of the 16S rRNA gene are used for the identification of bacteria, although the entire, approximately 1,500 bp long 16S rRNA gene has also been employed for greater accuracy. Sometimes, the 23S rDNA gene is sequenced to identify bacteria. Fungi, yeast, and molds are identified using the D2 region of a large subunit ribosomal RNA gene sequence. DNA is first extracted from a pure culture of cells and then amplified via PCR in four separate reactions—one reaction for each of the four deoxynucleotide bases: adenine (A), thymine (T), guanine (G), and cytosine (C). However, a mixture of these standard nucleotides and dideoxynucleotides (ddNTP) are used, where the latter nucleotides lack a 3'-hydroxyl (-OH) group on their deoxyribose sugar.

When a ddNTP is randomly incorporated during the amplification reaction, elongation of the PCR primer is terminated. This provides DNA fragments of varying lengths. Because each ddNTP is labelled with a different fluorescent dye, a series of fluorescently labelled copies of the amplified sequence, each terminating at a different base, is formed. These copies differ in molecular weight and can be separated and detected (based on their fluorescence) during gel electrophoresis of the reaction mixes. By simultaneously analyzing each of the four reaction mixes, representing the four deoxynucleotide bases, software within the gene-sequencer will reconstruct the linear arrangement of these bases in the sequence being analyzed. The resulting sequence is then compared with a library of known microorganism sequences and, if a sequence match is found, a genus and species identification are provided.

4.4.5.4 Next-Generation Sequencing

Next-generation-sequencing (NGS), also known as whole-genome-sequencing or high-throughput-sequencing, are novel technologies in nucleic acid analysis. In contrast to previously known sequencing methods, several hundred million fragments can be sequenced simultaneously in a sample and compared against a known library. NGS involves integration of fluorescently tagged deoxyribonucleotide triphosphates (dNTPs) into a DNA template strand during cycles of DNA synthesis. Fluorophore excitation at the addition of each nucleotide is used to identify the nucleotides during each procedure.

While NGS offers increased resolution and broad detection capability, interpretation of results from complex or mixed microbial populations may be challenging. Because NGS-based approaches sequence all nucleic acids present in the sample including background flora, non-viable organisms, and host-derived DNA, careful consideration must be given to data analysis strategies, relevance of detected organisms, and alignment with the intended application.

5.0 The Validation Process

There are multiple definitions of validation within the pharmaceutical industry. For example, ICH Q2(R2) states that “The objective of validation of an analytical procedure is to demonstrate that it is suitable for its intended purpose” (2). In modern regulatory practice, validation is understood as a lifecycle activity that extends beyond an initial validation study and includes continued assurance of method performance during routine use.

The validation of AMM/RMM should therefore be conducted within this lifecycle framework. In addition to demonstrating suitability for intended purpose, validation activities should ensure that the method, equipment and instruments, associated software, supporting systems, and test procedures or process workflows remain capable of consistently delivering reliable results throughout their operational life. Accordingly, this technical report provides guidance on a holistic approach to the validation and implementation of AMM/RMMs and their associated systems.

Many different approaches have been used successfully for the validation of AMM/RMM, many of which have been accepted by regulatory authorities. While examples of specific validation strategies are provided here in **Section 5.0**, it may be necessary to modify or customize these strategies to address a specific technology, method, or application.

For the purposes of validation, AMM/RMM may be classified as qualitative (e.g., presence/absence assays) or quantitative (e.g., enumeration assays), and the applicable validation criteria will differ accordingly. However, minor differences exist among current guidance documents, USP <1223>, and Ph. Eur. 5.1.6 including terminologies, procedures employed during the validation process, data interpretation, acceptance criteria, and the use of statistics (56). Therefore, the purpose of this current version of TR 33 is to help potential users of AMM/RMMs to appropriately select and understand the performance parameters requiring validation for any new method. The goal is for one strategy to be acceptable in all geographic regions, by all regulatory authorities and, most importantly, to be scientifically defensible and justified.

It is also strongly recommended that the appropriate regulatory agencies be contacted and involved early in the decision-making process for both the validation activities and the implementation of an AMM/RMM that will be used during the manufacture of pharmaceutical products.

Table 5.0-1 outlines the main validation activities and qualification deliverables and the portion of **Section 5.0** where each is discussed.

Table 5.0-1 Validation and Qualification Deliverables and Responsibilities

Validation/Qualification Deliverables	Responsibility		TR 33 Section
	Supplier	End User	
Proof of Concept (POC) Studies	Execute upon request of end user	Review and assess supplier's primary validation study or Execute	5.1.2
User Requirement Specifications (URS)	Support end user	Execute	5.1.3
Supplier Audit, Business Case, and Risk Assessment	Support end user	Execute	5.1.4 – 5.1.6 & 3.5
Design Qualification (DQ)	Execute and Support end user	Review and assess supplier URS/DQ or Execute, in case it does not exist for the intended use	5.1.7
Validation Master Plan and Associated Documentation	N/A	Execute	5.1.8 5.2.1 – 5.2.2
Equipment IQ/OQ/PQ, including System Integration and Computer System Validation (CSV)	Execute IQ/OQ/PQ and Support system integration and CSV	Review and assess IQ/OQ/PQ protocols, data and reports ¹ and Execute system integration and CSV	5.2.3 – 5.2.6

Validation/Qualification Deliverables	Responsibility		TR 33 Section
	Supplier	End User	
Primary Validation	Execute	Review and assess supplier primary validation or Execute, in case the method is other than the one validated by the supplier or if no primary validation exists	5.3 – 5.4
Evaluation of Validation Parameters in the Absence of Product or a Test Sample	N/A	Execute	5.3.1
Method Suitability Testing	N/A	Execute	5.3.2
Comparability Testing in the Presence of Product or a Test Sample	N/A	Execute	5.3.3
Ongoing Maintenance and Periodic Reviews	Execute maintenance	Introduce change control process and maintenance program	5.4.14
¹ Based on the assessment, the end user may decide to perform additional studies			

5.1 Pre-Validation Activities

A variety of activities may be successfully completed prior to initiating validation of the system, as some of these tasks are critical to the subsequent validation activities required. Completion of these tasks can aid in completing a successful validation and, hopefully, reduce the number of deviations and amount of retesting that may become necessary during the validation activities.

Suppliers are responsible for providing end users a clear description of their technology, the scientific principle for microbial detection, the intended applications for its use, the materials and equipment needed for the test method, and expected data output or signal when detecting, counting, or identifying microorganisms. If an end user requires information on test-sample suitability prior to the start of validation or the selection of an AMM/RMM, the end user can work with suppliers to perform feasibility or POC studies to generate this information.

5.1.1 Proof of Concept

The AMM/RMM selected for use should initially be evaluated to ensure confirmation of POC, feasibility, or principle (i.e., assessing whether the method and accompanying system is compatible with the intended product or sample matrix). This POC phase is most appropriate if the method supplier has no supporting data on similar products, sample matrices, or types of microorganisms that the end user will routinely test; therefore, POC testing, if required, should be conducted prior to making the final decision to purchase the equipment or instrumentation. POC would also allow for mitigation of potential issues with the instruments and method customization for the intended use. This activity can be conducted by the end user or the supplier of the system or method. For example, the products or other sample matrices for evaluation during POC testing and, when appropriate, the number and types of microorganisms chosen to challenge the new method should be carefully selected in order to ensure that the method is compatible with the product matrix of the sample(s) to be tested, and the resulting data provides a compelling indication that the validation of the intended method and accompanying system will have a high probability of success.

5.1.2 User Requirements Specifications

The URS is a key document that explicitly describes the characteristics of the method that will be required for routine use. As such, the content of the URS may well determine the success or failure of the method selection process. The specification is typically prepared by the end user; however, it is important to seek input from other internal stakeholders, such as Regulatory Affairs, Quality, Information Technology, and other relevant validation groups, as well as potential suppliers of the method. Many AMM/RMM suppliers possess their own system specifications, often referred to as an External Specification or Supplier URS. Careful review and comparison of the supplier's URS and the end user's URS may help identify criteria not included in one or the other that may be critical performance characteristics to be considered during the system selection process. This may include not only the microbiological aspects of the method and accompanying system, but also throughput, automation, environmental requirements, supplier expectations and communications, and computer system capabilities (e.g., LIMS interface, data management). In cases where the end user requires a very specific workflow (e.g., based on sample matrices and/or applications), the URS may serve as a starting point for additional performance characteristics that the end user and the supplier may need to consider and/or develop.

The URS will also be the starting point for establishing the validation criteria that will be tested by the end user. Therefore, the URS should describe the functions the method and accompanying system must be

capable of meeting that will be very specific to the end user's needs and the materials to be assessed. Essentially, the requirements specified in the URS will directly influence the entire strategy for validation and acceptance criteria.

5.1.3 Assessment of Supplier Capabilities/Supplier Audit

The ability of a potential supplier to meet the specified requirements (e.g., as outlined in the User Requirements Specification (URS)) is very important (see **Section 5.1.3**). Within the framework of the URS, suppliers for AMM/RMM technologies should be scrutinized for their ability to provide high-quality instrumentation, reagents and consumables, software, and technical support. Suppliers should have an appropriate quality system in place for designing, manufacturing, testing, and releasing equipment, software, reagents, and consumables throughout the technology lifecycle. Additionally, suppliers may provide technical documentation, training, validation support, troubleshooting, calibration services, preventive maintenance programs, and/or field service support. To ensure that the suppliers meet the user's internal quality requirements, as well as good practice guidelines, the end user should evaluate each supplier to ensure their ability to meet these expectations. Supplier assessments or audits may be conducted through a review of relevant documentation provided by the supplier and/or a physical audit at the supplier's manufacturing, design and development, and testing facilities.

Some of the assessment areas that should be the focus of an audit include, but are not limited to:

- Quality assurance procedures and standards, including change control and ISO certification
- Results from other audits (regulatory or other users)
- Financial stability
- Availability to provide instrumentation
- Ability to provide an uninterrupted supply of reagents and other consumables
- Ability to respond to field issues of a technical nature
- Ability to provide equipment maintenance
- Training and validation support and associated documentation
- Supplier's internal validation procedures and data
- Management of software updates, including the impact on validation activities

Additional information that may support the justification of implementing the supplier's technology may be requested, such as peer-reviewed publications, user manuals, and other relevant documentation.

Some suppliers may also have submitted additional technology performance information to FDA in the form of a Master File. The contents of the Master File are confidential and available only to the FDA. Master Files typically include proprietary information specific to the technology or data from validation studies. For example, certain types of validation studies, such as ruggedness and robustness, may be performed by the supplier and included in a Master File. End users may request a letter of authorization from the supplier to allow the FDA to reference the supplier's Master File to supplement their own submissions to the FDA, when appropriate (75). Other regulatory authorities, such as the EMA and other agencies, may not have a Master File process in place for AMM/RMM method suppliers. In these cases, it may be appropriate to obtain validation data directly from the suppliers and include those in the end user's validation package (75).

5.1.4 Business Benefits or Return-on-Investment Considerations

Based on the POC evaluation and other pre-validation activities, a thorough analysis should be conducted to determine the technical appropriateness of the AMM/RMM for its intended use. Additionally, a ROI analysis may also be helpful to support or justify the business case for purchasing the equipment and its subsequent validation and implementation for routine use (see **Section 3.4**). The final instrument selection and purchase should, of course, take place before initiating validation.

5.1.5 Risk Assessment and Validation Planning

In keeping with the principles of QRM, the first deliverable of any validation exercise should be a documented risk assessment (see **Section 3.5**). Risk assessments should follow established QRM principles, be structured and documented, and be commensurate with the intended use and impact of the AMM/RMM. Critical method steps and parameters, technical and scientific risks on method performance, and the resulting data should be considered in this assessment. The outcomes for the risk assessment should identify critical functions and potential failure modes, which in turn can be used to define the scope and depth of validation activities required.

The next phase is the development of an overall strategy, a validation plan. The end user is responsible for ensuring that the validation plan is appropriate and correctly documented. Depending upon an end user's requirements, the validation plan may or may not include pre-validation activities (see **Section 5.1**).

When it is determined that the new method requires validation, then an approved validation plan should be followed that will govern the process from beginning to end and will detail precisely what activities are necessary to produce an appropriately validated system. Another key component of the validation plan is the definition of system validation responsibilities, such as the identification of the individuals, organizations, or departments responsible for performing, reviewing, and approving the work. The validation plan should also specify how deviations from the approved testing strategy are handled, documented, reviewed, and approved. There are also situations where validation responsibilities may fall on both the end user and the supplier of the new method.

5.1.6 Design Qualification

Design qualification (DQ) is the documented review and verification that the proposed design of the equipment or system is suitable for its intended purpose, including meeting GMP compliance. Therefore, the DQ is most critical if no commercial off-the-shelf equipment is available for the intended application and must be specifically designed for that purpose. DQ would typically consider material, instrument, and test specifications and capabilities and maintenance and cleaning possibilities. However, since most AMM/RMM systems are commercial off-the-shelf equipment, the DQ serves to verify that the equipment specifications will meet the requirements identified in the URS. The DQ may also take data from POC testing into consideration, when appropriate. While this is termed "qualification" and is part of the qualification process, the DQ activity typically occurs prior to the purchase and validation of the method and accompanying system.

The DQ can be performed by the end user or the supplier; however, the end user is responsible for verifying that the method and accompanying system meet the requirements specified in the URS. The methods for accepting the DQ and an instrument's suitability for use will be determined by the nature of the instrument, the complexity of the proposed application, the complexity of the software used for instrument operation

and data analysis, and the prior history with the supplier. Supplier audits, review of documentation provided by the supplier, and/or direct examination of the system can satisfy the DQ requirement.

5.1.7 The Validation Master Plan

The Validation Master Plan should include a documented roadmap including development studies of all the performance functions and requirements for the method and accompanying system, and identify what will be tested to ensure that the method and accompanying system performs as specified in the URS. The Validation Master Plan may include microbiological and performance characteristics such as system and method functionality, configuration, input/outputs, environmental conditions, utilities, computer and communication architecture, interfaces, data management, and security. The specific test scripts where each performance function or requirement will be evaluated and verified against preestablished acceptance criteria are usually contained within the relevant phases of the Validation Master Plan, namely, the IQ, OQ, PQ, and Method Validation, including Method Suitability and Comparability sections. The Validation Master Plan may be written by the end user, the supplier, or both parties and may also be incorporated into other relevant documents depending on a company's validation requirements or preferences.

5.2 Equipment and Software Qualification and Validation

The functionality of an AMM/RMM involves the entire system; therefore, prior to validating the AMM/RMM, the testing equipment or instrumentation should be qualified, and the associated software/computer system should be validated. For this process, an adaptation to the Analytical Equipment Qualification Model may be used (76).

The steps described in **Section 5.2** provide a useful framework that can be applied to the equipment qualification and method validation of a complete system, that is, all the components of the new test method including any instrumentation, software, firmware, and reagents. This will guide the end user through the process steps involved both in the decision-making and the practical work required for implementing a new AMM/RMM analytical instrument and associated method(s).

Note: The framework provided here reflects a highly detailed approach, and some end users may not consider some of the deliverables a requirement for commercial off-the-shelf equipment (e.g., conducting a design qualification). As such, the overall validation strategy for a particular method should be evaluated in a risk assessment and documented in the validation plan (see **Section 5.2.1**).

Not all the activities in each section need to be carried out in serial order; as such, parallel path activities can occur (e.g., combining the installation qualification (IQ) and operational qualification (OQ), if appropriate). However, some activities should not begin until previous activities have been completed, for example, the PQ should be initiated only after the OQ acceptance criteria have been met, and this phase of the validation plan has been reviewed and approved.

Finally, it may not be possible to verify every feature associated with every piece of equipment. Therefore, a decision should be made regarding the relevance of testing those features of the system (e.g., during the IQ and OQ) that will not be used during the routine use of the new test method. One way to address this type of concern is to obtain a certificate of conformance for those features within the instrument received from the supplier of the system. Alternately, the validation plan should specifically exclude those features and the reason they will not be tested. For example, a system may offer the option to document assay parameters via a thermal printer or a regular printer. Depending on which option an end user decides to pursue, the other

may not be applicable and, therefore, not in scope for testing. Similarly, if a computer function will not be used routinely, then that computer function may not need to be tested during computerized system validation.

5.2.1 Documentation of the Validation Protocol

The execution of the validation plan, test results, and verification that acceptance criteria have been met must be adequately documented. Accordingly, relevant worksheets or checklists may be incorporated into each of the validation plan phases (i.e., IQ, OQ, PQ).

5.2.2 Documentation of the Working Procedure

Written procedures should be developed that specify the appropriate use of equipment or instrumentation, method workflow, and other work instructions to allow for the proper and consistent execution of the validation plan. These procedures may be captured in qualification protocols or standard operating procedures, depending on the end user's policies. Certain procedures may need to be in place and approved for certain phases of the validation plan. The need for effective instructions is important so that personnel can understand exactly how to perform the new test and operate and maintain the associated instrumentation.

Additionally, the analysts who will conduct the validation and/or operate the system should be appropriately qualified for these purposes. Therefore, training and developmental studies should be completed before the validation activities begin; such training may be conducted in-house (usually during the initial commissioning of the equipment) and/or at the supplier's own training facility, when available.

Appropriate documentation of personnel training is important to verify compliance with regulatory requirements.

5.2.3 Computer System Validation and System Integration

For equipment or instruments that contain a computerized system, a computerized system validation must be included during the equipment qualification. As per *FDA Guidance for Industry and Food and Drug Administrative Staff: Computer Software Assurance for Production and Quality System Software*, software validation is:

...confirmation by examination and provision of objective evidence that software specifications conform to user needs and intended uses, and that the particular requirements implemented through software can be consistently fulfilled (77).

System integration usually refers to information technology (IT), or computer systems, and involves bringing together all of the component subsystems into a single operating system and ensuring that all of the components of the system function appropriately. If the AMM/RMM technology requires connection to a separate data management storage and retrieval system, such as a LIMS or similar platform, system integration and validation between those various components may be required. In cases where the LIMS supplier assumes responsibility for the interface, the instrument supplier may provide only the instrument output in a predefined, documented format, while the LIMS supplier develops, configures, and validates the interface logic independently.

5.2.4 Equipment Installation Qualification

IQ studies should establish that the equipment is properly and safely installed with the correct utilities in an appropriate laboratory or, in some cases, in a manufacturing environment. A significant part of IQ is a verification that all new equipment was received and meets the design specifications for the equipment ordered. Any exceptions to the original specifications should be documented, showing the corrected specification and approvals. Also, it should note that an IQ is instrument-specific and portions of an IQ may need to be repeated if the equipment is moved within the laboratory or to another user site. An exception to this rule may be for instrumentation designed to be portable.

IQ studies should be performed in accordance with an approved protocol. Examples of the fundamental types of information to be included in an IQ document include system descriptions, utility requirements, operating environmental conditions, safety features, calibration requirements, software to be installed, and supporting documentation (e.g., technical manuals, blueprints, drawings).

Computerized or microprocessor-controlled systems should also document important features such as dip-switch settings, cabling connections, microprocessor chips used, computer configuration, any special features of the equipment required, printer connections, buffers, files, and memory requirements. In addition, the software required and its appropriate version numbers should be documented, including any operating systems used by the computer.

The IQ may be carried out by the supplier during the initial installation or system integration of the equipment and witnessed by the end user. However, depending on the extent of the supplier's IQ, as compared with receiving the end user's validation requirements, a separate and more extensive IQ may also be performed by the end user.

5.2.5 Equipment Operational Qualification

The OQ verifies and documents that relevant system parts or functionalities (and, if applicable, associated software) work within predetermined limits when operated in accordance with its operational procedures. Typical OQ parameters may, for instance, be verification of specified heating or cooling rates and adequate performance of probes, sensors, and optical systems or proper functioning of the user interface.

Furthermore, there is an expectation that any system that will be used to generate electronic records or electronic signatures is appropriately validated for this use, for example, to generate records that are accurate and reliable and can be appropriately maintained and accessed in compliance with the expectations in *21 CFR Part 11: Electronic Records; Electronic Signatures*. Other examples of computer testing that may be performed during the OQ include, but are not limited to, administrator control and operator access, user ID and password setup, user and system lockout, data integrity, audit trails, report generation, system integration, data transfer and server communication, data backup and recovery, database management and integrity, and interference (radio frequency, electromagnetic or wireless). In addition to 21 CFR Part 11, guidance and requirements regarding the validation of computer systems may be found in EU *Annex 11: Computerised Systems* and ISPE *Good Automated Manufacturing Practice (GAMP) (78-80)*.

5.2.6 Equipment Performance Qualification

PQ provides documented confirmation that the equipment, as installed and operated in accordance with operational procedures, consistently performs in accordance with predetermined criteria to yield appropriate

results. This may be demonstrated, for instance, through application of an experimental setup that may encompass an appropriate selection and detection limit of test microorganisms.

Qualification can also be demonstrated by running assay controls depending on the technology (e.g., nucleic acid standards such as exogenous and endogenous positive controls, no template, no amplification controls). For the purposes of PQ, the assay controls may be considered representative samples of actual product testing for the method. The parameters selected should be based on the method's detection system and suitability acceptance criteria. Normally, the supplier of the equipment will perform these types of studies to generate "primary validation" data to demonstrate that the equipment is capable of detecting, enumerating, and identifying microorganisms, depending on the intended application(s).

Microorganisms suspended in a suitable diluent are used during primary validation studies and are not actual test samples or product, the latter of which are used during method suitability and comparability testing.

An end user may choose to perform additional testing to verify or supplement supplier-provided primary validation data. It is the end user's responsibility to determine when this type of additional testing is warranted, for example, if an end user plans to use the method with different organisms of interest than the supplier's testing, if different incubation conditions (time/temperature) are desired, or if the end user's application is different from its original design or intended use (using a quantitative method as a presence/absence method).

5.3 Primary Validation, Method Suitability, and Comparability Testing

The validation process for AMM/RMMs should result in documented evidence that the equipment or instrumentation, as installed and operated in accordance with established sample preparation and workflow procedures, consistently performs in accordance with predetermined criteria and thereby yields correct and appropriate results.

Validation is performed in phases by both suppliers of AMM/RMM and stakeholders who will implement the methods for routine use. These phases include primary validation by the supplier, method suitability in the presence of the test sample, and comparability in the presence of the test sample. Each of these phases are summarized in **Sections 5.3.1-5.3.3**.

Note: When a previously validated AMM/RMM is to be replaced by a new generation of the method or by another AMM/RMM, which impacts the original validated state of the AMM/RMM or the previously validated AMM/RMM's comparability to the original CMM, then the new AMM/RMM to be implemented must be validated against the CMM, not against the established AMM/RMM.

5.3.1 Primary Validation

Primary validation is usually performed by the supplier, customarily in the absence of a test sample; however, some suppliers may include common or universal test sample matrices to demonstrate test sample suitability when introduced into the instrument or workflow. Suppliers should also provide the end user with primary validation test data demonstrating the system and method that will detect, enumerate, and/or identify microorganisms.

Primary validation should demonstrate that the workflow is adequate in obtaining the intended target to be detected by the technology. This may include, for example, procedures for extracting cellular targets from viable microorganisms, establishing minimum incubation times for enrichment steps, or activation

conditions for the uptake of viability stains. The use of surrogate targets, such as purified nucleic acids, ATP, or standardized fluorescent beads, may also be used to demonstrate that the technology is capable of detecting the intended target, as long as the target is appropriate for the technology.

The end user is responsible for reviewing and verifying the primary validation data to determine if the information is sufficient to support the end user's validation plan.

Normally, the end user is not required to repeat a supplier's primary validation studies unless one or more of the following apply:

- Test method is employed differently than what is defined by the supplier
- Equipment parameters are altered by the end user
- Panel of organisms used by the supplier are not representative of the organisms relevant to the end user's requirements
- Supplier did not conduct any primary validation studies

In these instances, the end user would need to perform the primary validation using relevant microorganisms and in the absence of product or actual test samples. End users may consider using surrogate analytes, as appropriate for the technology, to demonstrate that alterations to the AMM/RMM still detect the intended cellular target, as long as the method will continue to capture the target from viable cells for analysis.

When performing primary validation, the supplier should select the appropriate validation criteria to evaluate (e.g., pertaining to a qualitative or quantitative method). However, because some AMM/RMM applications may overlap (e.g., a method may offer both quantitative and qualitative capabilities), suppliers should assess relevant validation criteria to demonstrate all of the intended applications.

For qualitative applications, the supplier should assess specificity (using an appropriate number of relevant organisms), LoD, ruggedness (e.g., intermediate precision), and robustness.

For quantitative applications, the supplier should assess specificity (using an appropriate number of relevant organisms), accuracy, repeatability (within-run variability), limit of quantification (LoQ), range, linearity, ruggedness (e.g., intermediate precision), and robustness.

The supplier can also evaluate relevant validation criteria, in both the AMM/RMM and a CMM, which can demonstrate, at a minimum, comparability. This information may provide the end user with a level of confidence that the AMM/RMM will be an appropriate alternative to a CMM intended to be replaced by the AMM/RMM.

For AMM/RMMs that rely on the growth of microorganisms, the supplier should establish minimum incubation times when performing studies against the relevant validation criteria.

5.3.2 Method Suitability

Method suitability testing is a product or test sample-specific validation that should be performed in the presence of the test sample using the intended sample preparation steps and workflow for the AMM/RMM. Each novel test sample or product formulation should be assessed for suitability (i.e., compatibility with the workflow and instrumentation) unless otherwise justified via a risk assessment (e.g., when a prior study was performed on a similar sample matrix, or a formulation with a higher concentration of a sample component, such as an API). Where applicable, method suitability should account for product batch-to-batch variability. The number of distinct batches evaluated should be scientifically justified, as differences in raw materials, manufacturing conditions, or product age may influence microbial recovery, signal interference, or matrix

effects. In general, evaluation of three (3) independent batches is commonly considered appropriate, if available, unless otherwise justified based on risk assessment and product knowledge.

Validated methods that are addressed in a compendial test chapter may necessitate specific method suitability studies to be performed. End users should consult with the relevant pharmacopeias for additional guidance and testing requirements.

Method suitability testing should be performed on a sample matrix prior to conducting intended use or comparability studies using a relevant panel of microorganisms that have been chosen based on the type of test being validated (e.g., compendial strains), frequency and type of previously recovered organisms from the test sample (using the CMM), process, or the manufacturing environment or organisms that may present a risk to product, process, and/or the patient). These studies demonstrate the suitability of the method to detect and/or quantify microorganisms in the presence of the test sample.

A justification may be used to support not evaluating facility or environmental isolates or organisms naturally occurring in the test sample if these are similar to compendial indicator organisms.

Alternatively, a separate method suitability study may not be required if the sample matrix is evaluated during intended use or comparability testing, as long as a relevant panel of microorganisms is and the concentration of the challenge organisms is within a numerical range usually used during method suitability testing (e.g., less than 100 CFU, but not necessarily at the AMM/RMM's purported LoD or LoQ).

Method suitability testing should be performed during transfer of the method to a different facility to demonstrate that the facility can perform the method appropriately. Additional method suitability studies may be required, especially if local facility isolates pose a risk to the product, processes, or patient, and those same isolates were not assessed by the initial test validation site. A risk assessment should be performed to determine whether additional studies by the secondary facility are warranted (e.g., whether local isolates not represented in the initial validation studies should be considered or to perform a limited method suitability test to verify the method transfer was executed appropriately) (see **Section 6.4**).

Method suitability testing should demonstrate an absence of unacceptable enhancement of signals or background noise (e.g., false positives) or the inhibition, masking, occlusion, or interference in detecting viable microorganisms (e.g., false negatives). These studies are usually performed by the end user, but may be contracted to an external party, such as the AMM/RMM supplier or a contract facility. For AMM/RMMs that cannot be directly challenged by the end user (e.g., air-monitoring biofluorescent particle counters), method suitability may be carried out under actual-use conditions, such as in a controlled environment where the challenge levels are consistent, or by the supplier who has the capability of performing standardized testing using a panel of microorganisms.

False-positive testing is performed in the presence of the test sample but in the absence of challenge organisms.

False-negative testing is performed using a relevant panel of challenge organisms under actual-use conditions.

Qualitative methods should demonstrate a positive response in the presence of challenge microorganisms.

Quantitative methods should recover at least 50% of the challenge microorganisms inoculated into the test sample when compared with a control sample (e.g., a suitable diluent).

For AMM/RMMs that rely on the growth of microorganisms, the minimum incubation times established during primary validation should be verified; otherwise, incubation times may need to be revised based on the results obtained during method suitability or comparability testing (whichever incubation time is greater) in the presence of the test sample.

5.3.3 Comparability Testing in the Presence of the Test Sample

Comparability testing is usually performed by the end user in the presence of the test sample using the AMM/RMM and the CMM. The intent of comparability testing is to demonstrate that the detection of microorganisms in a qualitative method or the enumeration of recovered microorganisms in a quantitative AMM/RMM is comparable, not statistically different than or superior to, the detection of microorganisms in the CMM, depending on the statistical method used to compare the data (81). The validation criteria assessed during comparability testing will be based on the AMM/RMM's application (e.g., LoD and specificity for qualitative methods, accuracy, repeatability, LoQ and specificity for quantitative methods).

The choice of organisms used during comparability studies should be clearly justified by the end user and based on a sound scientific rationale. This may be based on the frequency and type of previously recovered organisms from the test sample (using the CMM), the type of test being validated (e.g., compendial strains for pharmacopeial procedures monograph tests), environmental or facility isolates, organisms that are relevant to challenging the AMM/RMM and the different workflows that may be required (e.g., different media for growth-based methods), and/or organisms that may present a risk to the product, the process or the patient.

A justification may be used to support not evaluating facility or environmental isolates or organisms naturally occurring in the test sample, if these are similar to compendial indicator organisms.

It may be appropriate to justify reduced testing during comparability studies (e.g., when the availability of test sample is limited). In this instance, the validation of an AMM/RMM to detect microorganisms in the presence of the test sample would only require a comparison with the CMM at the detection limit of the AMM/RMM in lieu of executing comparability studies covering all validation parameters. However, if method suitability and full comparability tests are able to be performed on the test sample, this alternative testing strategy would not be necessary.

Using an appropriate panel of microorganisms, an assessment against relevant validation criteria is conducted in the presence of the test sample and only in the AMM/RMM. The selection of validation criteria to be assessed can mirror what is performed during the primary validation for qualitative and quantitative methods.

When a prior method-suitability test has been performed on the test sample using a relevant panel of microorganisms, the end user may justify selecting a limited number of these same challenge organisms for comparability studies (e.g., all the method-suitability test organisms do not need to be evaluated during comparability studies). However, if a prior method-suitability test was never performed or the data generated during comparability studies will also be used to support method suitability, the end user should consider an appropriate panel of relevant microorganisms to support these studies.

The actual challenge level(s) for these studies are determined by the end user and may be based on the intended test procedure or the required level of microorganism detection or recovery. For example, testing at the AMM/RMM's purported LoD or LoQ may require a challenge level of 1–5 microbial cells or, alternatively, a series of dilutions into the fractional range could be evaluated.

When the test sample is expected to contain naturally occurring bioburden, these organisms may also be considered, given the target of the test.

Comparability testing for qualitative applications should be performed in the presence of the test sample and should address specificity and LoD in the AMM/RMM and the method to which it is being compared.

Comparability testing for quantitative applications should be performed in the presence of the test sample and should address specificity. The challenge concentration(s) employed should be relevant to the level of microorganisms required to be recovered and quantified, which could be at or near the LoQ and/or at a particular specification level.

For qualitative and quantitative applications, ruggedness (intermediate precision) may also be assessed. Based on the AMM/RMM, however, depending on the type of product or test sample and the manufacturing process, the results from ruggedness studies performed by the supplier during primary validation (in the absence of product) may be used.

For AMM/RMMs that rely on the growth of microorganisms, the minimum incubation times established during primary validation and/or method suitability testing should be verified; otherwise, incubation times may need to be revised based on the results obtained during comparability testing (e.g., whichever incubation time is greater).

When a test sample has successfully met the acceptance criteria for comparability, it may not be necessary to repeat comparability testing for new samples that have a similar formulation or matrix. However, new test samples that are fundamentally different in composition, may introduce background noise or interference, or may affect the AMM/RMM to detect microorganisms may require comparability testing. An example may be the difference between a human cell-based therapy versus an oligopeptide-based product.

5.3.4 Unique Methods Requiring Additional or Modified Validation Strategies

The method validation testing strategies discussed in TR 33 primarily use standardized microbial cultures in liquid suspension. The possibility exists, however, that some AMM/RMM detection systems may need to consider different approaches, especially if a liquid suspension cannot physically be introduced into the detection system for analysis. Technologies that use airborne, aerosol, or other nonliquid-based samples may fall into this category. Therefore, expanding or adapting these validation strategies may be appropriate, as long as the testing methods are scientifically justified. Because the end user may not possess the expertise or specialized equipment to conduct such studies, the vendor may need to play a greater role in supporting the end user during the validation of these types of technologies.

5.4 Recommendations for Conducting Validation Studies

Sections 5.4.1-5.4.11 provide recommendations for each of the validation criteria, including testing procedures, acceptance criteria, and statistical analyses for quantitative and qualitative AMM/RMMs.

Validation testing may also be designed and executed where the data from one validation criteria study may be used for several other validation criteria requirements. For example, the data derived during the test for linearity may also be used for the test for accuracy.

Note: These recommendations are not all-inclusive, and suppliers and stakeholders may find alternative strategies (e.g., revised, additional, or reduced testing based on risk assessments) that are also satisfactory

for use and accepted by regulators. In this instance, stakeholders are encouraged to discuss their validation strategies with relevant regulatory authorities prior to initiating their proposed validation plan.

5.4.1 Specificity

See the definition of specificity in **Section 2.0**.

Specificity is demonstrated when a relevant panel of microorganisms is used to challenge the AMM/RMM during the assessment of other validation criteria (e.g., accuracy, repeatability, LoD, LoQ), and when conducting method suitability and comparability testing. As such, there is no requirement to perform a separate or dedicated specificity study.

5.4.1.1 Specificity Testing Procedure

An AMM/RMM is generally expected to detect and/or enumerate a wide range of microorganisms unless the AMM/RMM is specifically designed to detect a single target of interest (e.g., *Staphylococcus aureus*).

When selecting a panel of microorganisms, consideration should be given to the type of test being validated (e.g., strains required for a CMM), limitations to the AMM/RMM under evaluation, the frequency and type of organisms previously recovered from the test sample (using the CMM), the manufacturing process and environment, and the organisms that may present a risk to the product quality, the process, or the patient.

Accordingly, microorganisms can include a variety of Gram-negative and Gram-positive rods, Gram-positive sporulating bacteria, Gram-positive cocci, yeast, and mold.

For growth-based methods, the selection of organisms may need to demonstrate microbial detection in different media and incubation conditions, based on the technology being assessed.

The concentration of microorganisms used will be dictated by the validation criteria being assessed.

For AMM/RMMs that will detect the presence/absence of specific target microorganisms, a relevant panel should include the target microorganism and appropriate nontarget, unrelated, and closely related species to demonstrate inclusivity and exclusivity, respectively.

An appropriate low concentration for inclusivity and an appropriate high concentration for exclusivity testing must be used.

Organisms may be commercially sourced or established within the end user's laboratory.

Pure cultures of a specific microorganism are generally utilized; however, if an AMM/RMM has multiplexing or simultaneous detection capabilities, the use of mixed cultures may be considered to assess this capability.

Specificity may also include demonstrating that nonviable cells are not detected by the AMM/RMM, particularly for nongrowth-based technologies where detection of nonviable cells could be interpreted as a false-positive result depending on the intended use. These types of studies may be most relevant during the establishment of method suitability. End users should consider practical and appropriate procedures for preparing nonviable cells (e.g., temperature treatment, exposure to biocidal agents).

The use of stressed organisms is not recommended, as there are no currently accepted global standards for generating cultures in a stressed state.

5.4.1.2 Specificity Statistical Analysis

The selection of appropriate statistical analyses will be dependent on the validation criteria the microorganisms are being used to assess.

5.4.1.3 Specificity Acceptance Criteria

Acceptance criteria will be based on the requirements of the validation criteria being evaluated.

5.4.2 Accuracy

See the definition of accuracy in **Section 2.0**.

5.4.2.1 Accuracy Testing Procedure

Separate dilutions of a relevant panel of microorganisms must be prepared within the practical range of the quantitative test or to the enumeration levels required for the intended application. The number of separate dilutions prepared should be justified. Examples of what may be considered include:

- The end user's microbial specifications, based on compendial requirements or internal acceptance levels
- The suppliers purported lower-to-upper quantitation range
- A practical range as defined in published literature
- If one dilution is used, it should be in the middle of the practical range of the test. If the range of the test is different for specific organisms (e.g., bacteria versus fungi), then that dilution may be used for each
- If the practical range of the test is broad, more than one dilution should be considered

See **Section 5.4.1** for examples of what relevant microorganisms should be considered.

An appropriate number of replicates per test organism should be used (e.g., a minimum of five (5) replicates per organism) to meet the requirements for the intended data analysis and/or test power. See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance.

A greater number of replicates or test runs may be used when combining accuracy with other validation tests (e.g., precision, range, linearity). Accuracy may also be combined with ruggedness (intermediate precision) by testing on different days, with different analysts, and/or using different instruments.

For each organism, the same number of cells or volume from each dilution into the AMM/RMM and the CMM in parallel must be challenged. Parallel testing using the same suspension is preferable; however, if parallel testing is not performed, the experiment and statistical analysis need to address the possible differences in spikes across suspensions. Parallel testing may suggest a statistically "paired" analysis (e.g., paired *t*-test), because the samples tested with the AMM/RMM and CMM come from the same suspension, but such statistical approaches only apply when the enumeration of both methods are conducted on the same samples, not the same suspension.

The recovery count from each replicate must be determined and the mean recovery for each organism and in each method calculated.

5.4.2.2 Accuracy Statistical Analysis

A test to demonstrate equivalence, comparability, or statistical difference between the recovered counts in both methods should be used, for example:

A two one-sided equivalence test (TOST) that uses a proper 90% confidence interval on the difference in or ratio between the method's expected values (or mean count) can demonstrate statistical equivalence between the two methods, given a suitable equivalence margin. However, if the AMM/RMM is more sensitive than the CMM and may generate statistically higher counts, a noninferiority test that uses a proper one-sided 95% confidence limit on the difference in or ratio between the expected values should be used. Depending on the probability distribution assumption for the underlying enumerations, the following approaches can be used when the AMM/RMM is compared to the CMM for one microorganism using one concentration:

- If the counts can be assumed to be Poisson- or Negative-Binomial-distributed, closed-form asymptotic confidence limits can be used on the ratio of the expected values of the two methods using Poisson or Negative-Binomial regression analysis (and the logarithmic link function).
- If no parametric distribution can be assumed (on the original or transformed count data), then nonparametric confidence limits can be considered (e.g., those that are developed for comparing differences in medians), but nonparametric approaches usually have poorer power and require a larger number of replicates than parametric approaches.
- Making use of confidence limits based on regression-modeling approaches (linear regression or nonlinear modeling approaches (Poisson regression)) for the estimation of concentration-response curves when multiple concentrations are considered simultaneously may also be considered.
- If mean values are considered and a normal distribution can be assumed, approximately (possibly after suitable transformation) normal or conventional confidence limits can be used for TOST (either from a dependent- or independent-samples *t*-test for paired or independent samples, respectively). However, it is important to confirm that the number of test runs is of adequate size to ensure the results are representative of the true condition of the data populations and, hence, will provide more accurate or reliable test results.

The data from different challenge-organism replicate tests may need to be pooled to achieve the desired test power. If this is done, the results from individual challenge organisms should be reviewed to determine whether the pooled results are acceptable (e.g., the results for a specific organism are not masked by pooling all of the replicates across multiple-challenge organisms). Accordingly, a justification for the statistical method used (e.g., pooling versus individual organism analyses) should be performed.

See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance on the selection of an appropriate statistical analysis method based on the desired test power and number of replicates.

5.4.2.3 Accuracy Acceptance Criteria

Specific acceptance criteria will depend on the statistical method used to compare the data from the two methods. For example, the mean recovery obtained in the AMM/RMM should be noninferior to, or comparable to, the mean recovery obtained in the CMM. If the ratio in the expected counts is being compared between the two methods, a comparability margin of 0.7 is often considered. It is common to denote this as $\Delta = 0.3$, but this value is not the comparability margin itself, it is one minus the comparability margin. In case of equivalence, $\Delta = 0.3$ means that the 90% confidence interval on the ratio of expected counts falls within the interval $[1 - \Delta, 1 + \Delta] = [0.7, 1.3]$.

See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance on acceptance criteria based on the statistical method used.

5.4.3 Repeatability

See the definition of repeatability in **Section 2.0**.

The repeatability (or within-run variability) of an AMM/RMM is the degree of agreement among individual test results when the procedure is applied repeatedly to multiple samplings of the same suspension of microorganisms. Repeatability is usually evaluated in quantitative methods.

5.4.3.1 Repeatability Testing Procedure

Repeatability (also known as within-run variability) testing is performed on a set of samples collected from the same suspension (replicates) by a single analyst using the same equipment in the same laboratory. Intermediate precision (also known as ruggedness) is performed on a set of samples under different conditions that would naturally vary during routine use of the AMM/RMM method (e.g., multiple suspensions, analysts, days, different instruments). **Section 5.4.3.1** focuses on repeatability while **Section 5.4.8** will address intermediate precision (ruggedness).

At least one dilution should be prepared of a relevant panel of microorganisms within the practical range of the quantitative test or what enumeration levels are required for the intended application. The number of separate dilutions prepared should be justified. Examples of what may be considered include:

- The end user's microbial specifications, based on compendial requirements or internal acceptance levels.
- The suppliers purported lower-to-upper quantitation range.
- A practical range as defined in published literature.
- If one dilution is used, this should be in the middle of the practical range of the test. If the range of the test is different for specific organisms (e.g., bacteria versus fungi), then a particular dilution may be used for each.
- If the practical range of the test is large, then more than one dilution should be considered.

See **Section 5.4.1** for examples of what relevant microorganisms should be considered.

An appropriate number of replicates per test organism should be used, but no fewer than five (5) replicates per organism (in alignment with analytical assays), to meet the requirements for the intended data analysis and/or test power. A relevant number of replicates or test runs may be used when combining repeatability with other validation tests, for example, accuracy, range, or linearity). See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance.

Repeatability may preferably be combined with ruggedness (intermediate precision) by testing on different days with different analysts and/or using different instruments.

For each organism, the same number of cells or volume from each dilution into the AMM/RMM should be challenged and the CMM should be challenged in parallel. Parallel testing using the same suspension is preferable; however, each aliquot from the same suspension will represent an independent sample. If the same aliquot can be analyzed in both methods, then these would be treated as dependent samples. The manner in which aliquots are used will dictate the appropriate statistical test to utilize, as described in **Section 5.4.3.2**.

5.4.3.2 Repeatability Statistical Analysis

Repeatability is best expressed in terms of a relative standard deviation (%) or coefficient of variation per suspension, either with respect to the spiked level or with respect to the mean count, together with an approximate 95% confidence interval for variances using, for instance, the chi-square distribution (not addressing the uncertainty in the denominator). Repeatability is best calculated for mean counts larger than 5 CFU. Due to the variability in counts between test samples as a result of the sampling of test samples, the repeatability is expected to be at least in the range of the reciprocal of the square root of the mean count. A McKay approximation may possibly be used when the distribution of the counts is approximately normal (82).

A comparison with the CMM can be conducted by demonstrating equal variance between the recovered counts in both methods. To demonstrate superiority of the AMM/RMM compared to the CMM, Levene's test (preferably Brown and Forsythe's version using medians instead of means) can be applied (83). When a comparability analysis is desired, an upper 95% confidence limit can be applied on the difference in the logarithmically transformed variances (e.g., AMM/RMM less than CMM).

Depending on the statistical test used, the data from different challenge-organism replicate tests could possibly be pooled to achieve the desired test power. If this is done, the results from individual challenge organisms should be reviewed to determine whether pooling is acceptable (e.g., the results for a specific organism are not masked by the pooling all of the replicates across multiple-challenge organisms). Accordingly, a justification for the statistical method used (e.g., pooling versus individual organism analyses) should be documented.

See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance on the selection of an appropriate statistical analysis method based on the desired test power and number of replicates.

Note: For the lower concentration of organisms used during these studies (e.g., <10 CFU), the repeatability is anticipated to be lower due to the relatively higher variability in counts at these levels. However, this would be expected in both the AMM/RMM and the CMM.

5.4.3.3 Repeatability Acceptance Criteria

Specific acceptance criteria will depend on the statistical method used to compare the data from the two methods. For example, the variance obtained in the AMM/RMM should not be statistically different, is noninferior to, or is comparable to the variance obtained in the CMM. A comparability margin of 0.5 on the ratio (CMM/AMM) or 0.693 in the natural-log scale on relative standard deviations is common practice.

See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance on acceptance criteria based on the statistical method used.

5.4.4 Limit of Detection

See the definition of LoD in **Section 2.0**.

The LoD is usually evaluated through qualitative or presence/absence methods.

5.4.4.1 Limit of Detection Testing Procedure

The intent of the LoD validation criterion is to establish the lowest level at which positive samples are detected (commonly at 95% probability (i.e., LoD₉₅)) for the AMM/RMM and the CMM, and to

demonstrate the AMM/RMM LoD is comparable from the CMM LoD. The LoD may also be used to justify the challenge level of target organisms during inclusivity testing.

A relevant panel of microorganisms is used when performing LoD studies. See **Section 5.4.1** on specificity for examples of what relevant microorganisms should be considered.

The LoD₉₅ should be established as follows:

1. An LoD established by a single point (e.g., the lowest level detected, such as a single CFU) has limited utility because the certainty aspect (e.g., a 95% confidence level) may not be precisely determined. Modeling approaches based on empirical data, such as probit analysis, allow the calculation of an LoD at a desired level of certainty (e.g., 95%) and is recommended when establishing the LoD for an AMM/RMM.
2. For each microorganism, a series of serial dilutions into the fractional range should be prepared, with the first dilution having a microbial concentration expected to give all positive results. The dilutions should include a level that represents the purported LoD from the AMM/RMM supplier. For example, dilutions of 10, 1, and 0.1 CFU, or 5, 0.5, and 0.05 CFU, or 2, 1.2, 0.7, 0.4, and 0.25 CFUs may be considered. For precise estimation of the LoD, it is better to use five concentrations.
3. For an appropriate test power, each dilution should be composed of at least 20 to 40 replicates. Data from multiple challenge organisms can be pooled to reduce the replicates per dilution when the statistical analysis approach incorporates the different microorganisms.
4. The same number of cells or volume from each prepared suspension should be challenged into the AMM/RMM and the CMM in parallel. If parallel testing cannot be performed (i.e., different suspensions used in each method), then differences in spiked levels between the suspensions should be accounted for during the subsequent analyses.
5. The number of positive and negative detection-events must be determined for each microorganism replicate in the AMM/RMM and in the CMM.
6. If the fraction of positive replicates equals 1 or 0 for two dilution concentrations, other appropriate dilution levels should be selected.

The AMM/RMM LoD should be compared to the CMM LoD as follows:

1. The data generated when establishing the LoD₉₅ can be used to compare the LoD in the AMM/RMM and in the CMM.
2. Comparisons of LoDs between methods can be conducted for a single suspension when the challenge level can be chosen, such that the same (or similar) number of positives and negatives are present in both methods. The stakeholder should choose an appropriate challenge level for the intended application, and the data of multiple organisms may be pooled in the analysis.
3. A most probable number (MPN) method may also be used if the number of replicates and the dilution concentration used are consistent with the requirements in the FDA MPN tables (**84**). The stakeholder should determine the number of replicates per dilution concentration that is practical to perform the analysis. Most importantly, the number of MPN runs should be sufficient to adequately apply the intended statistical analysis. Calculate the MPN and the upper and lower 95% confidence intervals for each challenge organism and for each method. Refer to the FDA MPN table relevant to the number of replicates used and instructions for selecting dilutions for the calculations. Dilutions other than those prescribed in the FDA method may also be considered, although the FDA tables should not be used to calculate the final MPN (e.g., using spike levels of 4, 2, 1 CFU rather than the FDA-recommended 10-fold dilutions, such as 10, 1, 0.1 CFU).

4. Parallel testing using the same suspension is preferable; however, if parallel testing is not performed, a statistical analysis must be used that could address the potential differences in spiked levels between suspensions and multiple suspensions for each method.

5.4.4.2 Limit of Detection Statistical Analysis

When determining the LoD_{95} in each method, a probit/logit analysis is performed using the positive and negative results for the dilution levels assessed. This may also be performed using natural logarithms of the dilution levels.

The data from all microorganism dilutions and replicates are used to achieve the desired test power, but pooling data from multiple organisms requires a specialized analysis approach. However, the results from individual challenge organisms should be reviewed to determine whether the pooled results are acceptable (e.g., the results for a specific organism are not masked by pooling all the replicates across multiple challenge organisms). Poolability can be evaluated with hypothesis tests, outlier tests, or goodness-of-fit measures (see **Section 9.0** (Appendix 1)).

If the model fit is sufficient according to the chi-square goodness-of-fit test, the analyte concentration at which the estimated positive rate equals 95% is the estimated mean number of organisms in the test samples that provides a positive result with 95% confidence. The LoD_{95} , which is the exact lowest number of organisms that leads to a detection probability of 95%, is typically below this estimated mean number of organisms. If the model fit is insufficient, additional replicates or dilutions must be analyzed.

The method of Most Probable Limit (MPL) can be used when the AMM/RMM is expected to provide a LoD of one or two CFUs to demonstrate an almost-perfect method of detection, implying that a direct comparison with the CMM is not needed.

When comparing the AMM/RMM and the CMM $LoDs$, and depending on the statistical test used, the data from different challenge organisms replicate tests may need to be pooled to achieve the desired test power. If this is done, the results from individual challenge organisms should be reviewed to determine whether the pooled results are acceptable (e.g., the results for a specific organism are not masked by pooling all of the replicates across multiple-challenge organisms). Accordingly, a justification for the statistical method used (e.g., pooling versus individual organism analyses) should be presented.

Two statistical approaches provide a reliable comparison: one which uses the well-known standard MPN approach repeatedly (i.e., multiple executions of the MPN experiment with possibly adapted spike levels for the three dilutions), while a generalized MPN (gMPN) approach can be used for a single dilution. Both approaches are suitable to incorporate or pool data from multiple microorganisms.

If an MPN approach is used for a single microorganism, it is suggested to calculate confidence limits on the mean difference in the repeated log-transformed MPNs to demonstrate comparability between the methods (similar to accuracy for quantitative methods). In case multiple microorganisms are being pooled, the repeated log-transformed MPNs per microorganism are being used in a two-way fixed-effects analysis of variance (ANOVA) model to estimate the mean differences in MPNs across microorganisms between both methods.

An alternative, more-specialized analysis approach exists, referred to as the gMPN approach, that can compare the detection proportions (i.e., the probability of detecting samples with exactly one microorganism) between the two microbiological methods for individual microorganisms and for pooled data from multiple microorganisms. When the total number of test samples is identical, the gMPN approach,

which uses a single dilution, is more powerful than the MPN approach, but the MPN approach is more robust to spiking errors due to the use of multiple dilutions. The gMPN approach is a less straightforward analysis than the MPN approach.

See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance on the selection of an appropriate statistical analysis method based on the desired test power and number of replicates.

5.4.4.3 Limit of Detection Acceptance Criteria

The LoD₉₅ values for each method should be reported.

When comparing the LoDs, specific acceptance criteria will depend on the statistical method used to compare the data from the two methods. For the comparison of MPNs and detection proportions, a comparability margin of 0.7, commonly denoted as $\Delta = 0.3$, is common practice. This value is not the comparability margin itself, it is one (1) minus the comparability margin. In case of equivalence, $\Delta = 0.3$ means that the 90% confidence interval on the ratio of expected counts falls within the interval $[1 - \Delta, 1 + \Delta] = [0.7, 1.3]$.

See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance on acceptance criteria based on the statistical method utilized.

5.4.5 Limit of Quantification

See the definition of LoQ in **Section 2.0**.

LoQ is usually evaluated using quantitative methods.

5.4.5.1 Limit of Quantification Testing Procedure

LoQ can be determined from the accuracy or linearity studies.

Several serial dilutions, for which one should be close to the purported LoQ, should be prepared across the desired range of the assay in order to ensure that microorganisms are enumerated close to the true quantification limit.

An appropriate panel of microorganisms should be utilized, similar to what would be used during accuracy testing. Ideally, the organisms selected would be expected to be quantified by the AMM/RMM at very low concentrations.

The number of replicates used per dilution should be appropriate to reduce the potential for variability at very low challenge concentrations.

Alternatively, the concentration of microorganisms used can also be defined differently, for example, in a certificate of analysis from a supplier for commercially prepared microbial challenges, as long as these are very accurate at low concentrations.

The procedures used during accuracy or linearity studies can be followed for reporting recovered microorganisms. If recovered counts are less than the challenge concentration, the recovered counts should be reported as less than the dilution level and not used to calculate the true LoQ (e.g., if the lowest dilution is expected to contain 10 CFU, any count recovered that is less than 10 should be reported as “<10”).

5.4.5.2 Limit of Quantification Statistical Analysis

The lowest concentration in the practical range of the assay that meets the acceptance criteria for accuracy and/or linearity is considered to be the quantitation limit of the method.

Data analysis may be based on the fact that microorganisms in an inoculation suspension approximately follow a Poisson distribution.

The most appropriate concentration-response curve (possibly linear or proportional to the spikes) using Poisson (or linear) regression should be determined, and the lowest concentration level for which the 99% lower confidence limit is above zero on all higher concentrations should be calculated.

This lowest level is considered the LoQ if accuracy (and possibly repeatability) has been demonstrated at this concentration or lower. If not, the LoQ becomes the lowest concentration at which accuracy has been demonstrated.

5.4.5.3 Limit of Quantification Acceptance Criteria

There is no acceptance criterion for LoQ testing, unless a certain level for the LoQ is required for the purpose of the AMM/RMM. In that case, the estimated LoQ in the AMM/RMM should be at or below the target or intended LoQ.

5.4.6 Linearity

See the definition of linearity in **Section 2.0**.

Linearity is usually evaluated using quantitative methods.

5.4.6.1 Linearity Testing Procedure

Linearity is a parameter that must be evaluated in relation to accuracy, in particular when accuracy is conducted at several concentrations separately (without assuming a concentration-response curve). Satisfying linearity would then guarantee that accuracy also holds for each concentration in between any two concentrations for which accuracy was demonstrated.

When accuracy is evaluated using concentration-response curves, the test for linearity is less important and may even be unnecessary.

To be able to investigate linearity, a minimum of three concentrations (across the range of the method) is required, but more concentrations are preferred.

Linearity can be visualized by examining either the actual counts obtained in the AMM/RMM or by examining the relative counts with respect to the (true) amount of microorganisms in the dilution (e.g., using the spiked concentration level or the expected or mean counts of the CMM). When performed properly, both approaches would lead to the same conclusions on linearity.

5.4.6.2 Linearity of the Actual Counts Obtained in the Alternative/Rapid Microbiological Method

The expected or mean counts of the AMM/RMM is a straight line in the number of microorganisms *with a nonzero slope*. The ideal linear function is a straight line through the origin, in which case linearity is

referred to as *proportionality*. Under proportionality, the slope represents the *recovery* of the AMM/RMM across the investigated range of concentration levels.

5.4.6.3 Linearity of the Relative Counts of Alternative/Rapid Microbiological Method

The relative counts of the AMM/RMM (e.g., calculated from the dilutions using an initial spiked concentration) is a straight line. The ideal straight line is a horizontal line with a slope equal to zero, since the average relative count over all spiked concentrations is an estimate for the recovery of the AMM/RMM.

5.4.6.4 Linearity Statistical Analysis

Linearity of the actual counts or the relative count of the AMM/RMM is best evaluated with an appropriate regression analysis (e.g., standard linear regression, weighted linear regression, Poisson regression), considering the following options:

- Use the *p*-value of the slope to test that the slope is nonzero. A nonzero slope means that the response (counts or relative counts) is dependent on the number of microorganisms. For a regression analysis of the AMM/RMM counts, a nonzero slope is required while, for a regression analysis of the relative counts, a zero slope is preferable.
- Use the *p*-value of the intercept to test that the intercept is nonzero. For a regression analysis of the AMM/RMM counts, an intercept that is not distinguishable from zero supports proportionality (i.e., the slope then represents the mean recovery). For a regression analysis of the relative counts (or recoveries), the intercept represents the estimated mean recovery (where the slope should not differ significantly from zero) and should be significantly different from zero.
- Apply a goodness-of-fit test to evaluate the appropriateness of linearity by comparing the regression line with a more extensive model. For instance, add a quadratic term to the regression analysis or compare the regression line with the saturated model (a model that estimates the mean at each spiked concentration). The goodness-of-fit test should not be statistically significant.
- The R^2 value is only an indirect measure of linearity. The larger the R^2 value, the less likely linearity is violated, but it is no guarantee that linearity is true since it also depends on the range of spike levels being tested.

Transformation of the counts (e.g., log 10) and/or the concentration level may be a suitable approach for the purpose of demonstrating linearity, but the interpretation of the parameters of the regression line will change.

The use of more complex statistical analyses, where the counts of the AMM/RMM and CMM are jointly analyzed, are also appropriate. See **Section 10.0** (Appendix 2) for additional information.

5.4.6.5 Linearity Acceptance Criteria

The criteria on the model parameters are described within the statistical analysis (nonzero intercept and slope), and the acceptance criteria depend on what type of outcome (absolute counts, relative counts, or transformation) is being studied. A lack-of-fit of the regression line requires a deeper investigation of accuracy. For a regression analysis of the AMM/RMM counts as a function of the number of microorganisms (e.g., the concentration level or the expected or mean counts of the CMM), it is recommended to also require a high R^2 value (at least 90%) to show that the AMM/RMM can appropriately predict the number of microorganisms in a dilution appropriately.

5.4.7 Range

See the definition of range in **Section 2.0**.

Range is usually evaluated using quantitative methods.

5.4.7.1 Range Testing Procedure

Suppliers and stakeholders can use the same data generated during accuracy, repeatability, and linearity studies, as long as the concentrations tested are consistent with the intended range of the assay.

5.4.7.2 Range Statistical Analysis

There is no statistical analysis associated with demonstrating range.

5.4.7.3 Range Acceptance Criteria

The information that accuracy, repeatability, and linearity are demonstrated within the intended range of the assay should be reported.

5.4.8 Ruggedness

See the definition of ruggedness in **Section 2.0**.

Test conditions to be examined in a ruggedness study include the use of different analysts, different instruments, different lots of reagent, or on different days. When performing a ruggedness study, intermediate precision is performed within the same laboratory, and reproducibility is performed between laboratories. Ruggedness can also be considered the intrinsic resistance to the influences exerted by operational and environmental variables on the results of the method. Ruggedness is usually evaluated using both quantitative and qualitative methods.

Ruggedness is usually performed by the supplier, and the data may be accepted by the end user to support the validation program and regulatory submissions. The end user should perform a risk assessment of the supplier's data to identify any ruggedness gaps for their intended use of the AMM/RMM and determine if additional studies are needed. Ruggedness testing may need to be repeated by the end user if there are modifications to the instrument or workflow made by the end user that could render testing by the supplier obsolete.

5.4.8.1 Ruggedness Testing Procedure

Ruggedness is performed on a set of samples that would naturally vary with routine practice, including, but possibly not limited to different analysts, instruments, lots of reagents, and days.

There is no need to repeat all challenge organisms for ruggedness; it should be performed using the most relevant challenge organisms that are most difficult to enumerate.

Concentrations should be considered similarly as done with repeatability (see **Section 5.4.3**).

For a comparison of ruggedness with the CMM, it is preferred that both microbiological methods enumerate samples from the same suspension, and the results are used for a comparability test.

The appropriate number of runs and replicates per run should be calculated to meet the requirements for the intended data analysis and/or test power, but it should never be below the six runs and three replicates per run.

Data can be pooled from different microorganisms when the variability in detection or enumerations is (almost) independent of the type of microorganisms.

If the variety of testing conditions are already included during the testing of certain other validation parameters, such as accuracy, then a separate ruggedness study may not be needed if the ruggedness can be appropriately determined from the data of this other validation parameter.

5.4.8.2 Ruggedness Statistical Analysis

For the estimation of ruggedness for quantitative methods, for each microorganism, concentration level, and microbiological method (when both microbiological methods are included in the study), the random effects ANOVA (where run is the random factor) is used in case the expected counts (or concentration levels) are above five (5) CFUs.

From the ANOVA, collect the variance components of the within and between-run variability and the mean counts. Calculate the ruggedness or intermediate precision as a relative standard deviation (expressed in percentages) and conduct the following steps:

1. Add up the two variance components.
2. Take the square root of the sum of variance components.
3. Divide the square root by the mean count.
4. Multiply by 100%.
5. Calculate a 95% confidence interval on the sum of the variance components using the Satterthwaite approach; take the square root, divide by the mean count, and multiply by 100%.
6. In case a comparison with the quantitative CMM is needed, use the ruggedness or intermediate precision estimates and their standard errors to calculate a 95% lower confidence limit on the ratio of the estimates (CMM divided by the AMM/RMM) per microorganism and concentration level. This may be evaluated, possibly in the logarithmic scale.

For the estimation of ruggedness for qualitative methods, demonstrate that the detection of microorganisms under the different conditions is independent of or almost unaffected by these different conditions. Logistic regression approaches should be used for each microorganism and concentration level to demonstrate that a lack-of-effect of these conditions exists (although a combined analysis of microorganisms and concentrations is allowed).

Demonstrating the lack-of-effect can be done for the AMM/RMM separately using comparability or conventional hypothesis testing approaches on the coefficients of the logistic regression.

Demonstrating the lack-of-effect in comparison to the CMM should demonstrate that comparability with the CMM is independent of the testing conditions, that is, it should demonstrate that there are no interaction effects present.

More sophisticated statistical analyses (such as a generalized linear mixed model) may be applied as an alternative to ANOVA (when counts are less than 5 CFU or when normality of the counts are violated) or logistic regression to estimate ruggedness and conduct comparability between the AMM/RMM and CMM.

5.4.8.3 Ruggedness Acceptance Criteria

Absolute criteria on ruggedness depends on the purpose of the method. For quantitative methods, the set of criteria in **Table 5.4.8.3-1** may be useful for the estimate of ruggedness.

Table 5.4.8.3-1 Relative Standard Deviation for CFU Ranges

Mean Count Range (CFU)	Relative Standard Deviation
5 – 9	45%
10 – 16	35%
17 – 45	25%
46 – 100	15%
101 – 400	10%

For a comparison of two quantitative microbiological methods, a comparability margin of 0.5 on the ratio of CMM/AMM or 0.693 in the natural-log scale on the relative standard deviations is common practice. For qualitative methods, a comparability margin of 0.7 on the detection probability is common practice.

5.4.9 Robustness

See the definition of robustness in **Section 2.0**.

Robustness is usually performed by the supplier, and the data may be accepted by the end user to support the validation program and regulatory submissions. The end user should perform a risk assessment of the supplier's data to identify any robustness gaps in their intended use of the AMM/RMM and determine if additional studies are needed. Robustness may need to be repeated by the end user if there are modifications made to the instrument or workflow that could render testing by the supplier obsolete. Robustness is usually evaluated using both quantitative and qualitative methods (see **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3)).

5.4.9.1 Robustness Testing Procedure

The supplier should determine the impact of variations on critical method and system parameters. These may include, but are not limited to, reagent concentrations/volume, environmental conditions (e.g., temperature, humidity), instrument operational limits, physical conditions (e.g., vibration), extraction methods (where applicable), sample volumes, and incubation parameters. This information provides an indication of the limitations of the test method and associated instrumentation.

A suspension of relevant microorganisms should be prepared, and the recovery (detection or enumeration, depending on the test method) should be evaluated against relevant validation criteria (e.g., accuracy, repeatability, LoD, LoQ, linearity) after changes to critical parameters have been made.

For qualitative methods, the supplier can perform LoD testing; for quantitative methods, the supplier can perform accuracy and precision testing. The supplier can also perform a study for a specific robustness evaluation without having to do a full validation criteria experiment. For example, recovery counts would be

within 70% or with no statistical difference in accuracy but with the critical parameter in question; if not, then the final parameter should be used for full accuracy studies.

5.4.9.2 Robustness Statistical Analysis

Similar statistical analyses and data evaluation approaches for validation criteria (e.g., accuracy, repeatability, LoD, LoQ, linearity) may be applied when assessing robustness.

5.4.9.3 Robustness Acceptance Criteria

Acceptance criteria for validation criteria (e.g., accuracy, repeatability, LoD, LoQ, linearity) may be applied when assessing robustness. Other criteria may be used; for example, for nucleic acid amplification technologies, demonstrating the threshold cycle or crossing point values are within acceptable ranges. For each test condition, a range should be demonstrated within which the AMM/RMM operates in a robust manner. The results should be used to define adequate precautions or limitations when the method is used routinely.

5.4.10 Method Suitability

See the definition of method suitability in **Section 2.0**.

To demonstrate that an AMM/RMM is compatible with specific product formulation or a sample matrix that will be routinely assayed, each material should be evaluated for suitability. The potential to produce interfering or abnormal results, such as false positives or false negatives needs to be understood. This may also include evaluating whether cellular debris, dead microorganisms, or other viable entities, such as mammalian cell cultures, have any impact on the ability of the AMM/RMM and accompanying system to operate as it is intended. Accordingly, method suitability is evaluated in both quantitative and qualitative methods.

Although it is possible that certain classes of materials will produce similar results, the end user is responsible for ensuring that all test samples will be compatible with the new method (see **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3)).

5.4.10.1 Method Suitability Testing Procedure

The test sample should be the same as what would be used during routine testing. In some cases, the stakeholder may consider using the most challenging matrices, in terms of formulation, potency, and/or quantity required for testing, to avoid having to repeat method suitability for similar formulations that would not require analysis (e.g., same product classes, matrix with the same formulation composition but using a smaller quantity, similar formulations with the exception of a lower concentration of API or excipient).

Note: The end-user should determine any impact that final packaging may have on the assay prior to performing any product grouping approach. For example, modifying final package material may impact the number of background particles detected in some non-growth-based AMM/RMMs.

The test sample should undergo the same specific workflow for the AMM/RMM under evaluation (e.g., nucleic acid extraction and amplification steps, organism capture, sample concentration, addition of reagents).

The number of distinct test samples or product batches used to demonstrate method suitability should be justified; for example, three (3) separate batches are commonly used, if available.

- **False-positive testing:** Challenge the test sample, which does not contain any challenge microorganisms, into the AMM/RMM using the same amount of material and workflow the stakeholder intends to use for routine testing. A negative control in the absence of the test sample should also be challenged in the AMM/RMM.
- **False-negative testing:** Using a relevant panel of microorganisms (e.g., similar to what was used for other validation criteria-testing, including specificity), inoculate <100 CFU into the same amount of material the stakeholder intends to use for routine testing. If the workflow or sample preparation steps could affect the viability of microorganisms during these studies, the stakeholder should inoculate the test sample prior to performing these steps. A positive control in the absence of the test sample, but with microorganisms, should also be challenged in the AMM/RMM. Surrogate markers specific to the AMM/RMM may also be added to the test samples; however, the stakeholder should demonstrate that these markers would be obtained from viable organisms during sample preparation or when using the intended method workflow and be used at a concentration that would correlate with a <100 CFU viable cell concentration. Although a <100 CFU challenge is specified, this does not necessarily mean the challenge should be at the LoD or LoQ level.

Stakeholders who perform validation for the intended-use studies (i.e., where microorganisms are challenged in actual test samples and assessed against validation criteria but not compared with the CMM), method suitability will need to be demonstrated in support of these studies.

For organism-specific detection methods, the test sample should be spiked with the intended target organism (or surrogate target material) for inclusivity testing. The test sample should be spiked with nontarget or closely related target organisms (or surrogate target material) for exclusivity testing.

Additional studies to assess potential interfering materials may also need to be considered. For example, certain AMM/RMMs that rely on the detection of microorganisms via biofluorescence may illicit a false-positive response in the presence of extrinsic particles, disinfectants, or other substances that could fluoresce in the AMM/RMM instrument. When applicable, the stakeholder should identify these types of substances and challenge the AMM/RMM to determine the potential for false-positive detection events. Similarly, if dead cells could illicit a positive detection event in the AMM/RMM, stakeholders may want to consider challenging the AMM/RMM with dead or recently killed cells to understand whether the resulting signals could be interpreted as a false-positive event.

5.4.10.2 Method Suitability Statistical Analysis

There is no statistical analysis associated with demonstrating method suitability.

5.4.10.3 Method Suitability Acceptance Criteria

There should be no false positives unless the stakeholder determines some level of positives (or background interference) is acceptable for routine use. For example, a baseline measurement in the presence of the test sample (but no microorganisms) may be determined. The baseline measurement may then be used to determine when a true-positive detection event is observed (e.g., when the result is above an acceptable signal-to-noise ratio).

To minimize the occurrence of false negatives, the detection of microorganisms in the presence of the test sample, based on a predefined positive threshold, should be observed for qualitative methods. Similarly, at least a 50% recovery (or a factor of 2) for AMM/RMMs that can be directly and quantitatively compared with the positive control (organisms or surrogate material in the absence of the test sample) should be observed. For quantitative AMM/RMM signals that cannot be directly compared with a CMM (e.g., plate

count), alternative strategies may be considered. For additional guidance, see comparability testing in **Section 5.4.11.1**.

Inclusivity testing should show a positive detection event in the presence of the target organism; exclusivity testing should show a negative detection event in the presence of nontarget or closely related organisms.

If potentially interfering materials are shown to provide a false-positive response, the stakeholder should ensure these are absent during routine testing of the sample. Alternatively, strategies for minimizing exposure of the AMM/RMM to interfering materials should be considered. For example, when an AMM/RMM is used for EM and certain disinfectants are known to cause a false-positive event, sampling should commence only after the disinfection activity has ended and residual disinfectant has been cleared from the sampling area.

5.4.11 Comparability

See the definition of comparability in **Section 2.0**.

The signal from a growth-based CMM will usually be reported as a CFU on solid agar or on a membrane and as visual turbidity in liquid media. Examples of methods that produce a signal other than the CFU or turbidity are fluorescent units from a viability-staining and laser-excitation method, relative light units or microcolony detection counts from ATP measurements, a positive response from a nucleic acid amplification technique, or intrinsic, autofluorescent bioparticle counts. While many AMM/RMMs analyze individual samples at one time, resulting in a single-data signal, other methods have the capability to produce large amounts of data over a longer sampling period, even under continuous sampling conditions.

As such, the manner in which comparability testing is performed will depend on a number of factors, including whether the signals from the AMM/RMM and the CMM can be directly or indirectly compared. For example, AMM/RMMs that produce a CFU signal can be directly compared with conventional CFU-based methods using similar validation criteria as that used during primary validation. However, AMM/RMMs that produce a non-CFU signal may or may not be directly compared with a CFU, and this may be reflected in the manner in which the data is obtained, the samples analyzed in both methods, and other factors. **Section 5.4.11** provides guidance on options for comparing the same or different signals and addressing the need to separate out nonmicrobiological detection events such as true false positives.

Ideally, test samples should not cause interference, false-positive or false-negative outcomes. This can be assessed during method suitability studies, preferably prior to conducting comparability testing (see **Section 5.4.10**).

When the AMM/RMM and CMM use the same microorganism-related signal, comparability should typically focus on noninferiority using similar validation criteria as that used during primary validation.

When the AMM/RMM produces a non-CFU signal and the CMM produces a CFU, comparability should focus on noninferiority or superiority, as it is commonly accepted that non-CFU or nongrowth-based signals can provide greater recovery or detection events due to the ability to recovery stressed, injured, or VBNC organisms. This aligns with the decision-equivalence concept described in USP <1223>.

5.4.11.1 Comparability Testing Procedure

The amount of test sample used and the number of independent determinations for comparability studies should be justified and explained. For example, at least three (3) lots of product, if available, or a practical

number of tests performed over some period of time may be considered. When a statistical analysis is performed on the resulting data, the number of independent determinations should be adequate for the intended analysis (e.g., to meet a certain test power). Depending on the type of study, the test plan should consider what has already been performed during prior studies (e.g., primary validation) to potentially reduce the amount of testing and test material required to demonstrate comparability.

If a consistent level of natural bioburden is present in the test sample, then the methods should be directly challenged with the test samples. If the samples are not expected to contain microorganisms, however, or if the natural bioburden is not consistent or adequately controlled, laboratory studies using inoculated challenge organisms, similar to validation criteria-testing in absence of the test sample, may be performed.

When using challenge organisms, an adequate panel and concentration of relevant strains appropriate for the AMM/RMM being evaluated should be selected. For example, growth-based AMM/RMMs should consider the use of slow-growing microorganisms and/or strains specific to testing the required media employed in the AMM/RMM procedure; nongrowth-based methods may consider more difficult types of organisms to detect, such as spores. If there is a low availability of test sample, the end user may justify performing comparability testing with a limited number of organism strains. Alternatively, the end user may justify using a similar sample matrix to test all of the relevant challenge organisms.

Alternative challenge material may be considered when viable microorganisms would not provide a set of results that can be compared with the AMM/RMM. Examples may include purified nucleic acid, cellular components such as ATP, or standardized biofluorescent beads. When alternative challenge material is used, it must be appropriate for the AMM/RMM under evaluation.

Some AMM/RMMs may produce data representative of multiple or continuous sampling events, making it more challenging to compare the data with a CMM that produces only a single test-sample result. For these reasons, the stakeholder may not be able to numerically correlate non-CFU signals to the CFU. The goal is then to determine ranges of CFU signals for the CMM for given levels of non-CFU signals. **Section 5.4.11.2** provides guidance on how to demonstrate comparability and how to make a quality decision about the test sample in this regard.

5.4.11.2 Comparability Statistical Analysis

Statistical analyses should be commensurate with the intent of the study and the expected results (comparability, equivalence, or superiority). If the AMM/RMM is using a non-CFU signal, it could be necessary to obtain either a prediction model from non-CFU-to-CFU signals or otherwise obtain an upper limit on the CFU signal, given the non-CFU signal.

5.4.11.2.1 Comparability of Qualitative Methods

Because the data will always be presented as a positive or negative result, the data from different methods can be directly compared, regardless of the detection-event signal. As such, comparability may be demonstrated in similar ways LoD and specificity are assessed in the absence of test sample (i.e., using an appropriate conventional hypothesis or a comparability statistical test).

5.4.11.2.2 Comparability of Quantitative Methods

When individual test samples can be evaluated in the AMM/RMM and numerical values are obtained in each method, comparability can be conducted using similar testing procedures and statistical analyses to those for accuracy, repeatability, and specificity during primary validation or validation-for-intended-use to

demonstrate comparability or superiority, depending on the comparison of signals (85). Obtaining a prediction model using some form of regression analysis to be able to predict CFU-signals from non-CFU signals or to obtain a natural range of CFU signals may be beneficial (see **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3). A prediction model requires a specific study with different CFU concentrations and multiple samples per concentration.

For AMM/RMMs that do not analyze individual samples but operate continuously or cannot be challenged with known microorganisms (i.e., using artificially inoculated test samples), other strategies for demonstrating comparability to a CMM can be employed. Examples of this include BFPCs which are used for environmental or water monitoring. For AMM/RMM such as these, the following options may be considered:

- Initial studies using only the AMM/RMM may help to define the normal or expected baseline levels for the AMM/RMM in the presence of the test sample or the test environment. The environmental conditions of the area under investigation should be well defined and be expected to represent routine operations. Activities that are not considered routine when monitoring should be identified and excluded from baseline studies. Excursions from baseline levels may help define what an out-of-expectation (OOE) result is in the AMM/RMM. This is done by calculating the upper control limits for the normal signals (e.g., two or three times the acceptable signal-to-noise ratio, as described in ICH Q2(R2) and USP <1225>).
- Parallel testing of the test sample or in the test environment with the CMM and the AMM/RMM could be performed; this is most meaningful with quantitative methods when the microbiological methods are testing separate samples. The stakeholder should determine an appropriate number of test samples or a predefined testing period to obtain meaningful data (e.g., nonzero detection events or numerical values) to determine comparability. When non-CFU continuous-monitoring methods are compared with conventional growth-based episodic or single-sample methods, the number of test runs for the CMM may need to be increased. For example, a new active air sampler may need to be initiated when a prior sampler has finished its sampling; alternatively, a new agar plate is added to a slit-to-agar sampler to simulate “continuous” conventional sampling.
- In instances where positive detection events are not expected or observed (e.g., Grade A environments), comparability studies may need to be evaluated where microbial recovery is expected (e.g., lower-classified cleanroom environments). The main goal is to determine the distribution or range of CFUs for the CMM based on the results of the AMM/RMM. These distributions can be used to determine acceptable ranges for the signal of the AMM/RMM. The results from both methods may be compared in the following manner:
 - If signals are consistent, use regression analysis (linear, nonlinear, weighted, or Poisson regression) to determine a calibration model. This approach may require analyzing microbial concentrations across the practical range of the test instead of performing parallel testing on a sample.
 - Compare the ratio of recovered counts between the two datasets. Demonstrate comparability if the same microorganism-related signals are being used for the AMM/RMM and the CMM, or superiority when the AMM/RMM is using a more-sensitive microorganism-related signal and the CMM provides a CFU signal.
 - Compare the frequency of detectable events in both methods. The stakeholder should define what a detectable event or OOE event is in the AMM/RMM, and what CFU levels in the CMM are relevant for comparability studies (which may include alert or action levels in the CMM). The AMM/RMM should detect at least the same frequency of detection events (e.g., deviations from expected baseline values in the AMM/RMM) as the CMM produces

- in OOE events. Demonstrate this using statistical tests for the comparison of proportions (either noninferiority or superiority).
- Quantify the distribution of CFU counts in the CMM for each level of AMM/RMM counts. Based on the OOE CFU counts, an upper level on AMM/RMM counts can be translated. See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 2) for additional guidance.

5.4.11.3 Comparability Acceptance Criteria

Similar acceptance criteria should be used as was used during the assessment of validation criteria in the primary validation when similar statistical approaches are being used. When a noninferiority statistical method is used, the recovery of organisms in the AMM/RMM should be noninferior to the recovery of organisms in the CMM. When a superiority statistical method is used, criteria for conventional hypothesis testing can be applied. Criteria for prediction models and setting upper-control limits should be based on the uncertainty of estimating the range of CFU levels, but this is usually translated to a proper number of test samples, batches, and/or concentration levels. See **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) for additional guidance.

5.4.12 Guidance on Setting Minimum Incubation Times for Growth-Based Methods

Growth-based methods require an appropriate incubation time to ensure the recovery of microorganisms in the presence of the test sample and under the conditions of the test method. Minimum incubation times may be established from the longest TTR for challenge microorganisms determined during primary validation, method suitability, and/or comparability studies.

Adding a safety factor to the longest TTR should be considered. For example, rounding up to the nearest whole day can be a relatively simple way to add such a safety factor. For example, if the longest TTR for a slow-growing microorganism is 4.6 days, then the minimum incubation time for the routine test can be 5.0 days.

When appropriate, a larger safety factor may be considered, for example, an extra full day may be added to the longest TTR. Using the same 4.6 day length of the longest TTR for a slow-growing microorganism, the minimum incubation time for the routine test could be 5.6 days.

Both of these options are relatively simple to implement, and the stakeholder can justify a minimum TTR that is suitable for the test method and routine-use requirements.

Another approach in setting a safety factor is based on the time needed for a ten-fold increase in the amount of incubation time for the slowest-growing organism used during validation studies. This can be calculated by multiplying the estimated generation time in the test media by $\log_2(10)$ as illustrated in the following equation:

$$T = t_{ttr} + \log_2(10) \cdot G$$

Where:

- T = incubation time for microbial detection in the test sample to be examined (hours (h))
- t_{ttr} = longest TTR determined during primary validation, method suitability, and/or comparability studies (h)
- G = generation time of slowest-growing microorganism (h), which may be determined by conducting separate studies or based on published literature

Using the same example, if the slowest microorganism has a TTR of 4.6 d (110.40 h) and has a generation time of 1 h, the minimum time of incubation would be $110.40 + 3.32 \times 1 = 113.72$ h (or 4.7 d). Similarly, if the slowest microorganism has a TTR of 4.6 d (110.40 h) but has a longer generation time of 5 h, the minimum time of incubation would be $110.40 + 3.32 \times 5 = 127$ h (or 5.3 d).

Stakeholders who want to calculate their own generation times can inoculate relevant organisms in the alternative growth-based method and use the following equation:

$$G = \frac{t}{3.3 \times \log_{10} \left(\frac{N}{N_0} \right)}$$

Where:

- t = time interval (h)
- N = number of cells/CFU at the end of the time interval
- N_0 = number of cells/CFU at the beginning of the time interval

Note: Depending on the output signal measured, other formulas may apply.

5.4.13 Guidance on Changing Acceptance Criteria

Though growth-based, CMM provide results in terms of a CFU, which could arise from a single cell or multiple cells. Some AMM/RMMs will provide results or signals that are different from the CFU (e.g., a viable cell count, a spectral analysis, relative light units, fluorescent units, etc.). As such, these AMM/RMMs could potentially produce superior results (e.g., higher quantitative counts) when compared with growth-based methods. Although strategies exist for demonstrating comparability between a non-CFU method and a CFU method, data derived from non-CFU methods may not align with conventional acceptance criteria, such as alert and action levels or specifications, because the number or rate of recovery in these methods may be higher (see **Section 5.4.10**). Accordingly, the implementation of these non-CFU methods may require the establishment of new acceptance criteria.

This is not a new concept. In a 2006 non-peer reviewed publication, the author's concluded that new microbiological methods rely on a completely different body of information—some may be direct measurements, some indirect—and that previous acceptance criteria may not be applicable. Therefore, implementation of newly developed, or more rapid microbiology methods may also require the establishment of new acceptance criteria (86).

The 2022 revision of EU GMP Annex 1 allows AMM/RMMs, instead of growth-based, CMMs, to be used for EM as long as they are appropriately validated and meet the intent of providing information during steps when potential for contamination of the product exists. More importantly, throughout the document, the microbiology limits are based on the CFU and, if new technologies are used that present results in a manner different from the CFU, the manufacturer should scientifically justify the limits applied and, where possible, correlate them to the CFU (9).

Similarly, the 2023 Swissmedicines Inspectorate publication Interpretation of GMP Annex 1 2022 (Rev. 1) addresses whether it is possible to fully replace microbiological monitoring using settle plates and volumetric air-sampling systems by other integrated sampling and testing systems, such as AMM/RMM. The document states that the equivalence of methods should be demonstrated, and the effectiveness of the chosen method should be proven. Additionally, for the use of real-time viable particle-counting and given the nonequivalence of autofluorescent unit versus CFUs and current GMP pharmacopeial limits are in CFUs, the company needs to collect data on their process for the real-time viable particle-counting to compare it to the standard EM data (87). However, consideration should be given to the environments in which these studies are performed, as some samples will provide microbial recoveries (e.g., Grade C and D, or controlled, non-classified areas) while other environments may not (e.g., Grade A and B areas), in which case alternative strategies may be used, including trending of zero to non-zero results. Additional guidance is provided in **Section 11.0** (Appendix 3).

AMM/RMMs that do not provide results as a CFU signal, are not based on the growth of microorganisms, or are more sensitive than growth-based CMMs, may provide results that are numerically superior (higher) than the CMM being compared. For these reasons, acceptance criteria historically based on the CFU, such as EM alert and action levels or microbial specifications, may need to be changed to reflect the recovered counts using the non-CFU method.

Establishing a numerical relationship between a non-CFU signal method to a CFU method may be possible but only if the data can be correlated mathematically. For example, if the non-CFU signal is always five times greater than the CFU method numerical values, then a 5X correlation factor may be used to establish new acceptance levels. However, when a direct correlation between numerical values cannot be demonstrated across the practical range of the test, across different sample matrices, different microorganisms, or across different environmental areas, a general correlation may not be possible. In this case, setting new acceptance levels for the AMM/RMM should be considered.

Understanding the AMM/RMM's baseline measurements, similar to those described for comparability studies, may be useful in establishing new acceptance levels, as recoveries considered to be "normal" would not necessarily be treated as an OOE detection event. Where numerical baseline measurements indicate the normal background noise in the absence of a test sample, these measurements may be subtracted from the test sample measurements, when justified.

The most conservative approach is to continue to use the existing CFU levels and extend this to data derived from using the AMM/RMM. For example, if an AMM/RMM demonstrates a normal baseline for

continuous EM as 0 CFU/m³, then excursions to this baseline may indicate an OOE event. Separately, if an AMM/RMM indicates occasional recovery events over the course of some monitored period of time, then these data should be used to establish the normally expected baseline. Using this information can help to establish new acceptance criteria that can be related to the historical control measures using a conventional CFU method. In this instance, the results from the AMM/RMM are used to make a quality decision about the test result. This aligns with the decision-equivalence concept described in USP <1223>.

Stakeholders may consider the following examples of how to establish new acceptance criteria:

- Set the highest AMM/RMM recovery counts as an action level when you also show the corresponding CFU count is at an acceptable level.
- Use 97.5% and 99.5% percentiles or 2 and 3 standard deviations from the mean-AMM/RMM counts for alert and action levels, respectively, based on the AMM/RMM historical data (i.e., baseline). This may also confirm correspondence to CFU acceptable levels.
- Apply trending rules to determine when an out-of-trend (OOT) result is observed, when compared with historical (baseline) data that may include numerical values for recovered counts and/or the frequency of recovery or nonzero events over some period of time. This may be most appropriate for a large number of test samples or monitored environments that are expected to have no or very-low detection events. This may also be applied to higher detection events as long as the baseline is appropriately defined. OOT events may include the following:
 - One or more points outside the established alert or control limits
 - Six points in a row that are steadily increasing
 - Fourteen points in a row that are alternating up and down
 - An unusual or nonrandom pattern in the data, when compared with historical data
 - Any combination of the above recommendations

Discussions with relevant regulatory agencies or regulatory submissions may also be required prior to any changes being finalized or implemented.

Ultimately, any changes to established specifications must be related to fitness for use and be suitable for the application of interest. Additionally, final specifications should be scientifically sound, justified, and based on an appropriate risk analysis.

5.4.14 Ongoing Maintenance and Periodic Reviews

Following validation of the new microbiological method and accompanying system, appropriate procedures should be established to maintain the system in a validated state; for example, standard operating procedures should be implemented, the method and accompanying system should be included in a change control program, and all instrumentation or equipment should be maintained in good working condition.

Once the new method and accompanying system has completed validation, subsequent change controls should assess the impact of any change on the validation status of the system. Additionally, a formal mechanism should be put in place to periodically review the method and the accompanying system's performance, as well as the overall validation program in relation to current good practices.

A special area of concern is the preventive maintenance program, frequently handled by the supplier of specialized equipment. Many programs include updating system software with the most current software versions, extension of databases, and periodic calibration checks. It is important to ensure that appropriate requalification testing is performed before placing the system back into use, that the end user is cognizant of

good practice requirements, and that they perform the maintenance accurately in addition to documenting the activities that were performed. The supplier should also provide information on the potential impact of repairs or spare-part changes, or drift in instrument performance, in order to define requalification or verification requirements.

Finally, the results or outputs of this phase should feed back into the first phase of the validation process, namely, the risk assessment. This will ensure that the overall operation of the new method and accompanying system does not introduce additional quality risks associated with the lifecycle of the products and sample matrices that will be tested routinely.

5.4.15 Decommissioning of Equipment

Decommissioning is part of the equipment's lifecycle, and it must follow the same change-control rules to ensure that the qualified and calibrated state is maintained until the final retirement. Decommissioning activities include change documentation, documentation that the equipment is in a qualified and calibrated status until its final use, termination of service or maintenance contracts (if applicable), archival of all equipment-specific documentation, electronic files and the ability to retrieve them, decontamination prior to retirement (if needed), update of the equipment inventory, and description of what happened to the equipment—whether it is stored, given away, sold, or scrapped.

5.5 Alternative/Rapid Microbiological Methods for Mycoplasma Detection

PDA *Technical Report No. 50: Alternative Methods for Mycoplasma Testing* (TR 50) describes mycoplasmas as “an unusual group of bacteria distinguished by the absence of a cell wall, a small genome size and low G + C content” and provides a table of the most significant genera, which includes pathogenic, saprophytic and commensal species (**88, 89**).

TR 50 provides a comprehensive review of the various methods available for mycoplasma testing as well as describing what methods to use in evaluating the appropriateness of these methods.

Many of the recommendations provided in this technical report may be used in support of validating an alternative or rapid mycoplasma method; however, reviewing the recommendations provided in TR 50 for a thorough understanding of what is expected is recommended. Furthermore, Eur. Ph. Chapter 2.6.7 Mycoplasmas may provide additional guidance as it provides a detailed protocol for the primary method validation and the product-specific tests for interfering substances as well as a section on NAT that has been extensively revised to reflect the state of the art in science and technology, as of October 2025.

5.5.1 Primary Method Validation

The parameters of the LoD, specificity, robustness, and comparability should be assessed during the primary method validation as discussed in **Sections 5.4.1-5.4.11**.

5.5.1.1 Limit of Detection

For the assessment of the LoD, the lowest amount of target nucleic acid in a sample that can be detected, the positive cut-off point is determined. The positive cut-off point is the minimum number of target sequences per sample-volume tested that can be detected in 95 percent of the test runs. To determine the positive cut-off point, at least three (3) independent 10-fold dilution series of characterized and calibrated working

strains or standards should be tested. This design, with a sufficient number of replicates at each dilution, gives a total number of 24 test results for each dilution which enables a statistical analysis of the results.

The specificity of a NAT, the ability to detect unambiguously the mycoplasma target nucleic acid, depends on the primer and probe sequences and the stringency of the NAT conditions. Hence, the specificity experiments should include a range of bacterial species other than mycoplasmas, that is, microorganisms with a close phylogenetic relationship to mycoplasmas, such as other Gram-positive bacteria like *Clostridium*, *Lactobacillus*, *Streptococcus*, or others, to evaluate the possibility for cross-detection of the non-mycoplasma bacteria.

5.5.1.2 Robustness

The robustness of the NAT, the capacity of the NAT to remain unaffected by small, but deliberate variations in the method parameters, is assessed for example, by using different lots of the PCR-reagents, using different thermal-cycler devices, or testing of samples on different days by different analysts, to examine the variation between test runs. In addition, robustness of the alternative NAT can be evaluated by collaborative studies.

5.5.1.3 Comparability/Equivalency Study

To demonstrate comparability/equivalency between the culture methods, the indicator-cell culture method, and the NAT-based mycoplasma detection assay, cell-culture fluid should be spiked with relevant mycoplasma strain stocks. Based on this initially spiked sample (e.g., 100 CFU/mL), a serial tenfold dilution should be prepared using unspiked cell-culture fluid as diluent leading to samples spiked with 100 CFU/mL, 10 CFU/mL, 1 CFU/mL, 0.1 CFU/mL and, if needed, 0.01 CFU/mL. All dilutions are tested using all three different mycoplasma detection methods in parallel without freezing the samples.

Acceptance criteria for the comparability study:

- The NAT must be able to detect at least 10 CFU/mL or less than 100 GC/mL if the NAT is planned as a replacement for the culture method.
- The NAT must be able to detect at least 100 CFU/mL or less than 1000 GC/mL if the NAT is planned as a replacement for the indicator-cell culture method.
- The NAT must be able to detect at least 10 CFU/mL or less than 100 GC/mL if the NAT is planned as a replacement for both the culture method and the indicator-cell culture method.

5.5.2 Test for Interfering Substances/Method Suitability Testing

For each new product or cell source, a test for interfering substances must be performed to demonstrate that the sample tested does not interfere with the NAT and/or the enrichment step (if applicable). Typically, samples from three different lots of the product to be assessed or batches of the cell-culture harvests (if available) are spiked with a maximum of 10 CFU/mL or 100 GC/mL and tested by the validated NAT to demonstrate sufficient recovery (e.g., 95%) of the target nucleic acid in the presence of the product to be tested.

5.5.3 Mycoplasma Stocks and Standards

For the primary method validation and the product-specific test for interfering substances, use of a broad range of mycoplasma, including challenging organisms (based on risk assessment), is recommended.

The selection process should consider the following:

- Manufacturing process (such as use of mammalian, avian, or insect cell expression systems)
- Use of mammalian or plant-derived media components
- Product category (biopharmaceutical, ATMPs, or live vaccines) and the occurrence of contamination (based on literature or their own experience)

Ph. Eur. Chapter 2.6.7 provides a suggestion of mycoplasma strains for validation testing based on their frequency of occurrence as contaminants phylogenetic relationship.

The mycoplasma strains can be viable cultures, a preparation of nonviable whole organism derived from these cultures, nucleic acid standards (e.g., the WHO International Standard for mycoplasma DNA for nucleic acid amplification technique-based assays designed for generic mycoplasma detection), or assay control plasmids. Positive control cultures should not be more than 15 passages from isolation, enumerated (harvested during the exponential phase of growth to guarantee a high viability of the culture and titered in CFU prior to freezing), and characterized using both the supernatant and cellular fractions, for genomic copies (GC), that is, their GC/CFU ratio. A GC/CFU ratio of the reference preparations of less than 10 is normally accepted by the relevant authorities; higher values must be justified.

The selection of the mycoplasma standard or reference material should be appropriate to the detection technology employed and should reflect the complete analytical workflow, including any pre-treatment or pre-processing steps (e.g., filtration, enrichment, nucleic acid extraction, lysis, concentration, or inactivation procedures). For culture-based methods, viable organisms are generally required to evaluate growth characteristics and recovery. For NAT or other molecular-based assays, nucleic acid standards or appropriately characterized inactivated preparations may be suitable. When pre-processing steps are part of the validated method, the chosen standard should allow assessment of the entire method performance, not solely the detection step.

6.0 Implementation: Guidance on Secondary-Site Commissioning Versus Initial Validation

Section 6.0 provides an overview and points to consider for transferring a validated AMM/RMM from a primary validation site to a secondary site. For the purposes of this document, the “primary validation site” is defined as the site performing the initial validation of the AMM/RMM, including equipment qualification and method validation. The “secondary site” is defined as the site to which the method is transferred.

6.1 Guidance for the Transfer of an Alternative/Rapid Microbiological Method from a Primary Validation Site to a Secondary Validation Site

The first time an AMM/RMM is qualified, it is expected that a comprehensive validation plan, as that recommended in TR 33, is appropriately performed. When the AMM/RMM is transferred from a primary validation site to a secondary site (e.g., testing or manufacturing facility), a copy of the original validation package should be provided to the secondary site. Each new installation must be separately evaluated to determine the extent of additional IQ, OQ, PQ, or method validation testing that may need to be performed. As part of the installation, qualification, and method transfer activities at the secondary site, all necessary training will be conducted, relevant procedures will be written and approved, and maintenance and calibration programs will be developed prior to performing the analyst qualification and subsequent routine use.

6.2 Equipment Installation and Operational Qualification at the Secondary Site

When an identical technology is installed at the secondary site, it is expected that a standard equipment IQ and OQ will be performed following the original qualification package (see **Section 5.2.8** and **Section 5.2.9**). However, if the AMM/RMM is used at a secondary site for a different purpose or in a different manner (e.g., standalone or LIMS integrated) than at the primary site, additional IQ, OQ and method validation may be needed. There may also exist environmental or physical conditions related to the secondary location that could warrant additional qualification or validation considerations, such as extremes in altitude, relative humidity, or temperature.

6.3 Equipment Performance Qualification at the Secondary Site

Because an exhaustive microbiological testing plan will be completed during method validation at the primary site, when a like-for-like instrument (e.g., the exact same model and components with the exact same version numbers for all software, microprocessors, and computers) is installed, it may not be necessary for the secondary site to repeat this testing in its entirety. However, it is recommended that a reduced

microbiological challenge from the original PQ, in addition to representative isolates recovered at the secondary site, be performed to demonstrate that the system is operating as intended. A review of locally recovered microbial isolates should indicate if the reference strains used during the qualification performed at the primary site are representative of the isolates recovered at the secondary site. If they are not, then the qualification plan should include a list of microbial isolates to be evaluated and what qualification requirements they will be tested against. Additionally, it may be appropriate to determine if the original qualification will meet current GMPs and internal requirements, taking into consideration the date the original qualification was completed.

6.4 Implementation of the Alternative/Rapid Microbiological Method at the Secondary Site

Method suitability in the presence of product must be repeated at the secondary site to demonstrate that the method will provide the expected results when performed by the analysts at the secondary site. As part of the method suitability test, local isolates from the secondary site (if it is a secondary manufacturing site), especially those that were not tested at the primary site, must be included. See **Section 5.3.2** for further information on method suitability.

Once the AMM/RMM has been successfully installed and qualified for use at a secondary site and required regulatory approval is obtained, the system may be used routinely.

In the event the secondary site intends to qualify the same instrument for use with a new product or process material that was not assessed by the primary site, a risk assessment should be performed to determine if a full-method validation, including method suitability testing and comparability testing, is required to be performed (see **Section 5.2.1**). **Section 5.3** discusses the use of test plans and acceptance criteria.

7.0 The Role of Artificial Intelligence in Microbiological Testing

AI is a scientific discipline established at the intersection of computer science, applied mathematics, and engineering, dedicated to the development of systems capable of performing tasks traditionally requiring human intelligence. From a functional perspective, AI can be defined as the simulation or approximation of human intelligence in machines **(90)**. Although AI incorporates ML and statistical methodologies, it remains an independent and multidisciplinary field based on algorithm design, optimization, and logic-based modeling **(91)**.

In pharmaceutical manufacturing, AI is applied to support process control, enhance decision-making, and strengthen quality operations. Within microbiological testing, AI contributes to improving the precision, consistency, and efficiency of testing instruments, data analysis, and trending.

Applications of AI in microbiological testing include:

- Within microbiological testing instruments, where it assists in interpreting signals for the enumeration, detection, and identification of microorganisms, particularly in AMM/RMMs and imaging-based technologies.
- Outside the testing instruments, where it is used to analyze microbiological data (enumeration, detection, and identification results) and to assist in decision-making, including trend analysis, deviation detection, and contamination risk assessment.
- In EM programs, where AI models process historical and real-time data to identify patterns associated with microbial excursions and guide data-driven contamination control strategies.

From a regulatory perspective, the FDA has initiated several efforts to clarify expectations for the use of AI in pharmaceutical development and manufacturing. In 2023, the FDA published the discussion paper *Artificial Intelligence in Drug Manufacturing* which outlined potential regulatory considerations and invited stakeholder feedback on how existing regulatory frameworks and current GMP requirements could apply to AI-based models used in pharmaceutical manufacturing. This discussion paper served as an early step toward developing a regulatory approach to AI-enabled technologies. Building on these efforts, the FDA released the draft guidance *Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products* in 2025, which proposes a risk-based credibility assessment framework for AI models used to generate data or information submitted in regulatory applications. In parallel, the EMA published the draft guidance *Annex 22: Artificial Intelligence*, which addresses the use of static AI systems in GMP-related processes where outputs may affect product quality, patient safety, or data integrity. The document emphasizes clearly defined intended use, validation using independent datasets, explainability, human oversight, and lifecycle monitoring to ensure that AI systems remain controlled, auditable, and compliant within the GMP environment **(92-94)**.

These published guidance documents reflect a growing recognition of the need to define clear expectations for the validation, governance, and application of AI technologies within pharmaceutical development and

manufacturing. In this context, microbiological testing represents a critical area where AI can be responsibly adopted in alignment with both scientific standards and evolving regulatory guidance. The integration of AI into microbiological testing aligns with established scientific and regulatory frameworks, contributing to improved data management, enhanced process understanding, and more robust microbial control.

7.1 Use of Artificial Intelligence in Signal Detection

Suppliers of microbiological methods have incorporated AI into the microbiological system to generate a “reportable value” (e.g., microbial count, absent/present result, or identification) directly from raw instrument data such as images or scatterplots.

For example, search algorithms are being used to screen databases with DNA patterns of known microorganisms to map the observed DNA pattern from a test sample to one of the known DNA patterns in a data library. Another example is the flow cytometry technology that provides two signals: light-scattering and fluorescence from the individual microbial cells moving through the activators and detectors. The detected signals are presented on a screen as a scatterplot that is used to define the enumeration of microorganisms.

Traditional microbiological quality control methods rely on manual interpretation of culture plates, colony morphology, or other analog signals generated during microbial detection assays. In many rapid microbiological methods, signal interpretation has historically been performed using deterministic, rule-based algorithms embedded in analytical instruments. In contrast, AI models learn patterns directly from large datasets rather than relying solely on predefined rules, enabling the identification of complex relationships and subtle variations in digital microbiological data. When properly trained and validated, these models support more consistent and scalable analysis of signals generated by modern microbiological technologies.

Many rapid microbiological methods produce digital image data that are analyzed computationally. In image-based detection systems, microbial colonies or cells are represented as arrays of pixels describing variations in light intensity or color. Computer-vision algorithms analyze these pixel patterns to recognize shapes, textures, and spatial features associated with microbial cells or colonies. Because microbial structures exhibit characteristic morphological patterns, AI-based image analysis improves the consistency and throughput of microbial detection compared with manual counting, particularly when large image datasets are analyzed (95, 96).

Other rapid microbiological technologies generate multidimensional optical or molecular signals that also benefit from AI-assisted interpretation. Flow cytometry-based microbial detection systems, for example, measure light-scattering and fluorescence signals emitted by individual cells as they pass through optical detectors. Machine-learning models analyze these signal patterns to differentiate microbial cells from background particles and support automated enumeration of microbial populations (97, 98). Similarly, genomic identification methods compare DNA sequences obtained from a test sample with reference databases of known microorganisms; machine-learning approaches enhance these analyses by identifying complex sequence patterns and improving classification performance within large genomic datasets (99).

Among computer-vision techniques used in microbiological image analysis, object-detection models such as the “You Only Look Once” (YOLO) architecture are widely applied. YOLO analyzes an image by dividing it into regions and simultaneously predicting the location and classification of objects within each region, enabling rapid identification of microbial cells or colonies in complex images. Because the model evaluates

the entire image in a single computational step, it efficiently recognizes biological structures and spatial patterns within pixel-based microscopy images **(100)**.

For the validation of AMM/RMM, it is important to distinguish between two types of AI model approaches that may be implemented within the instrument's software: static (locked) and dynamic (adaptive) AI models.

Static AI models are trained once on a fixed dataset and remain unchanged during the operational use of the microbiological method unless the software and its embedded models are formally updated. These models are typically built using algorithms trained on extensive datasets to distinguish between “noise” and true microbial signals. The performance and reliability of a static model depend on the quality, representativeness, and diversity of the data used during its initial training, which remains the responsibility of the supplier. Because these models do not modify their parameters during routine use, their validation follows the established experimentation and data analysis recommendations provided in this technical report. If the instrument software is updated and the AI model is retrained, for example, by incorporating new data to enhance performance, the changes must be assessed, and revalidation of the microbiological method may be required to ensure continued compliance and reliability **(93, 101)**.

Dynamic AI models, in contrast, are designed to continuously learn and adapt by updating their parameters as new data is generated during routine use of the microbiological method. This ongoing learning process can potentially enhance the model's ability to accurately detect microbial signals over time, especially as the dataset expands beyond the initial training population. While dynamic models offer the advantage of adaptability and potential performance improvements, they also introduce additional validation challenges. Continuous updates may lead to model drift or unintended shifts in performance, particularly affecting the detection of less-represented or newly emerging microbial subpopulations **(93)**. Consequently, microbiological methods employing dynamic AI models require ongoing monitoring to confirm that model performance is improving or remaining stable and not deteriorating in critical areas.

Actual AI development routinely incorporates explainability techniques to ensure that model predictions remain interpretable and transparent, especially in regulated environments. Methods such as the Shapley Additive Explanations and Local Interpretable Model-Agnostic Explanations help clarify how complex models reach their conclusions by identifying the contribution of each input variable to the final prediction **(102)**. Additionally, frameworks like the Open Neural Network Exchange provide a standardized format for representing AI models across platforms, supporting both model portability and explainability **(103)**. These tools are particularly valuable for dynamic AI models in microbiological testing, ensuring reliable performance monitoring and regulatory transparency.

The current guidance provided in this document can be applied to determine the performance of a microbiological method employing AI models at a given point in time. However, dynamic AI models additionally require regular verification of their operational behavior and performance trends. Defining standardized procedures for ongoing monitoring, acceptance criteria, and revalidation for dynamic AI models remains an active area of research, and specific regulatory expectations for this type of application are still evolving **(93, 104)**. It should be noted that certain regulatory bodies have adopted a more conservative preliminary position regarding dynamic AI models in GMP-critical applications; notably, the draft EU GMP Annex 22: Artificial Intelligence (EMA, 2025) currently limits its scope to static, deterministic models and recommends that dynamic models should not be used in critical GMP applications. Regulatory expectations in this area, however, continue to evolve, and other frameworks, such as the FDA draft guidance on the use of AI to support regulatory decision-making **(85)** and the FDA Total

Product Lifecycle approach for AI/ML-based Software as a Medical Device, do not preclude the use of dynamic models provided that appropriate validation, monitoring, and lifecycle controls are in place.

7.2 Use of Artificial Intelligence in Environmental Monitoring

AI is also being used on sets of data (or reportable values) that are generated by the microbiological method to obtain a better understanding of the observed data and to assist with the decision-making process. An active application area is EM, where it is important to know when a production or testing environment is starting to show a number of microorganisms that is unlikely to occur under normal use of the environment. An upward change in the number of microorganisms may increase the risk of product contamination. Indeed, Section 2.1 of EU GMP Annex 1 states that:

Facility, equipment and process should be appropriately designed, qualified and/or validated and where applicable, subjected to ongoing verification according to the relevant sections of the Good Manufacturing Practices (GMP) guidelines. The use of appropriate technologies (e.g. Restricted Access Barriers Systems (RABS), isolators, robotic systems, rapid/alternative methods and continuous monitoring systems) should be considered to increase the protection of the product from potential extraneous sources of endotoxin/pyrogen, particulate and microbial contamination such as personnel, materials and the surrounding environment, and assist in the rapid detection of potential contaminants in the environment and the product.

EM is part of the field of statistical process control, where three main activities are being implemented (105):

1. Monitoring the process (here, the environment)
2. Detecting deviations from what is considered in-control
3. Taking corrective action in response to an alarm signal to bring the environment back in control

Thus, part of EM can be viewed as a rule-based decision system for detecting an abnormal situation, and specific AI techniques for anomaly detection can be used, such as:

- Isolation Forests (effective for outlier detection in microbial count trends)
- Autoencoders (unsupervised learning for anomaly detection in complex datasets)
- Bayesian Networks (probabilistic models for uncertainty management in EM)

The primary AI approaches for EM, including their use cases, data requirements, and validation considerations, are summarized in **Table 7.2-1**.

Table 7.2-1 Artificial Intelligence Approaches for Environmental Monitoring

Summary of AI Approaches for Environmental Monitoring	
Supervised Learning (e.g., Random Forest, Gradient Boosting)	
Typical EM Use Case	Predicting out-of-trend (OOT)/out-of-action (OOA) events, prioritizing investigations, multivariable risk scoring from historical EM metadata.
When Most Appropriate	When a site has sufficient labeled historical events and stable definitions of excursions/deviations.
Main Advantages	Good predictive performance on complex multivariable data; tree-based models are often favored in manufacturing QA because they handle high-dimensional sensor/process data well; some interpretability is possible through feature importance.
Main Trade-Offs	Requires enough representative labeled data; can overfit site-specific patterns; false positives may rise if the event class is rare or process conditions shift.
Validation Considerations	Use temporally separated training/test data or cross-validation; assess precision/recall, not accuracy alone; perform external/site-specific revalidation and lifecycle review because pharmaceutical models require risk-based validation and maintenance.
Peer Reviewed Reference Examples	<i>Explainable Machine Learning for the Regulatory Environment: A Case Study in Micro-droplet Printing</i> by D. Ryan, E. Harris, and G. O'Connor (2024) <i>Machine Learning Algorithms for Manufacturing Quality Assurance: A Systematic Review of Performance Metrics and Systematic Review of Performance Metrics and Applications</i> by A. Karim Kausik, A. Bin Rashid, R. Baki, and M. Maktum (2025)

Summary of AI Approaches for Environmental Monitoring	
Unsupervised Learning (e.g., k-means, DBSCAN, clustering-based anomaly detection)	
Typical EM Use Case	Detecting novel anomalies, shifts in contamination patterns, spatial or temporal clustering of atypical EM behavior when labels are limited.
When Most Appropriate	When labeled OOT events are scarce, inconsistent, or unavailable; useful for exploratory surveillance and early signal detection.
Main Advantages	Does not require labeled deviations; can identify previously unrecognized patterns or rare states; useful when excursions are infrequent.
Main Trade-Offs	More sensitive to preprocessing and parameter choices; cluster meaning is not always obvious; higher review burden because anomalies still need SME interpretation. DBSCAN is attractive when noise/outliers are expected, whereas k-means is simpler but requires choosing cluster number and assumes more compact cluster structure.
Validation Considerations	Predefine anomaly-review rules; test robustness to hyperparameters and time windows; compare outputs against known historical excursions/trends and expert assessment before routine deployment.
Peer Reviewed Reference Examples	<i>Detection and Diagnosis of Process Fault Using Unsupervised Learning Methods and Unlabeled Data</i> by A. Rahoma, S. Imtiaz, S. Ahmed, and F. Khan (2023).

Summary of AI Approaches for Environmental Monitoring	
Hybrid/Staged Strategy (e.g., unsupervised detection followed by supervised classification or transfer learning)	
Typical EM Use Case	Early detection of atypical states followed by risk ranking or classification; adapting models across lines, rooms, or equipment with data shifts.
When Most Appropriate	When sites have mixed data maturity: many unlabeled records, few confirmed events, or changing operating conditions between campaigns/areas.
Main Advantages	Combines strengths of both approaches: unsupervised methods discover new patterns, while supervised or transfer components improve prioritization and robustness under data shift. Hybrid methods are also discussed as promising for better prediction in bioprocess settings.
Main Trade-Offs	More complex to implement and govern; more moving parts means more documentation, change control, and validation effort.
Validation Considerations	Validate each stage separately and end-to-end; assess transferability across equipment/contexts; include model maintenance, drift monitoring, and retraining triggers in the lifecycle plan.
Peer Reviewed Reference Examples	<i>Anomaly Detection for Drug Product Manufacturing Considering Data Limitations and Shifts: A Case Study on Industrial Freeze-dryers</i> by G. Lombardini et. al. (2025) <i>Explainable Machine Learning for the Regulatory Environment: A Case Study in Micro-droplet Printing</i> by D. Ryan, E. Harris, and G. O'Connor (2024) <i>Review on Machine Learning-based Bioprocess Optimization, Monitoring, and Control Systems</i> by P. Pratim Mondal et. al. (2023)

Conventionally, EM data is collected through various sampling methods, including active air, surface contact plates, settle plates, and personnel monitoring. The collected data is typically evaluated using control charts based on the Poisson or negative-binomial distribution, which incorporate statistically derived limits reflecting the natural variation inherent in production or testing environments due to common causes. When, at a specific point in time, the enumeration exceeds the action limit (generating an out-of-control (OOC) signal) or shows an adverse trend (OOT signal), for example, through a higher frequency of alert limit excursions or a systematic increase in counts, it may indicate a change in the state of the controlled environment (106). However, AI models can offer a more flexible and predictive approach by analyzing complex, high-dimensional, and time-dependent patterns in EM data that conventional Poisson-based control charts may not detect.

The cause of a potential change in the microbial burden depends on the characteristics of the environment under investigation. For example, OOC or OOT signals in a cleanroom may be caused by such factors as the opening of a barrier system, a HEPA-filter integrity failure, or a reduction in laminar airflow velocity. In pharmaceutical water systems, potential causes may include diminished UV lamp performance, biofilm formation, or leaks in the distribution system since the last sanitization.

To enhance the early detection and characterization of such deviations, AI techniques are increasingly being applied in EM data analysis. A practical way to present these methods is to distinguish them by data availability and intended use. Supervised models, such as Random Forest and Gradient Boosting, are most appropriate when large, well-labeled historical EM datasets are available for excursion prediction or risk scoring (107). In contrast, unsupervised approaches like k-means or Density-Based Spatial Clustering of Applications with Noise (DBSCAN) are more suitable when labeled OOT events are scarce and the goal is to detect unrecognized anomalies (108). Hybrid strategies may be preferable for mature EM programs as they combine exploratory detection with targeted classification, though they require more rigorous lifecycle validation.

Moreover, recurrent neural networks, particularly Long Short-Term Memory (LSTM) models, are well suited for analyzing sequential EM datasets, enabling the prediction of future excursions or trends by capturing temporal dependencies and latent trends in microbial counts. These AI-driven methods support proactive contamination control by identifying potential system disturbances before traditional control charts would signal an alert. Integrating these AI techniques into EM programs offers the potential to move from reactive deviation management to predictive risk-based control strategies. Their application enhances sensitivity to subtle environmental changes while reducing false positives caused by expected background variability, thus improving contamination risk management and overall operational robustness.

Many companies are considering or are transitioning to microbial particle-counters for EM that generate counts in an almost continuous manner, providing real-time and high-frequency data. High-frequency data is often considerably more erratic than data obtained from conventional approaches, which is typically collected at much lower frequencies (e.g., daily). In continuous monitoring, small and transient changes to the environment immediately affect the datastream and would not be captured when data is accumulated over longer intervals as such fluctuations tend to average out. As a result, these minor, short-term aberrations complicate the application of conventional control charts on continuous datastreams, as they may generate frequent OOT or OOC signals. Control charts typically assume a Poisson or negative-binomial distribution for the observed data that remains constant over time while the environment is in control. However, normal operational activities often introduce additional peaks and fluctuations that violate this assumption of a stable, time-independent distribution.

To address these challenges, AI algorithms specifically designed for automatic anomaly detection in high-frequency time-series data are increasingly applied in industrial and pharmaceutical monitoring environments. One widely used method is the Isolation Forest technique, which isolates anomalies by randomly selecting features and splitting the dataset into a tree-like structure. Since outliers differ markedly from normal patterns, they require fewer splits to be isolated, making this approach highly efficient for real-time anomaly detection in continuous monitoring systems (109). Another effective method involves LSTM Autoencoders, which are neural-network architectures trained to learn and reconstruct normal sequential patterns. When unexpected anomalies occur, these models fail to reconstruct the sequence accurately, resulting in elevated reconstruction errors that act as reliable signals for deviations. This approach is particularly suitable for detecting subtle or emergent shifts in continuous industrial process-monitoring, environmental-contamination data, network security, and biomedical signals (110).

One approach for monitoring environments with high-frequency data is to aggregate the data to a larger period of time (total or average number of counts per day) and then use conventional control charts. The small aberrations will be averaged out over longer periods, assuming a constant distribution is more likely. However, aggregation would not maximize the available information. Instead, ML methods may be applied to the high-frequency datastream to learn what patterns of data are considered in-control (i.e., normal variation even if it would lead to signals in conventional control charts) and what patterns of data are considered OOC (i.e., signals beyond the normal aberrations or outlier results). The most suitable ML approaches for EM are approaches that could detect outliers in data sets. The reason is that separating in-control from OOC situations is very similar to detecting outliers or erratic data patterns.

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9.0 Appendix 1: Validation of Alternative and Rapid Microbiological Methods: Statistical Analysis-Qualitative Methods

The content provided in **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) has been written in a manner that a microbiologist or an analyst with similar training would be able to understand the fundamental (statistical) concepts described. However, stakeholders may want to engage the expertise of a statistician to ensure the recommendations in these sections are appropriately executed.

Section 9.0 (Appendix 1) and **Section 10.0** (Appendix 2) provide several acceptable statistical analyses for the validation of alternative qualitative and quantitative microbiological methods, respectively, while **Section 11.0** (Appendix 3) focuses on additional considerations for non-colony forming units (CFU) qualitative and quantitative alternative/rapid microbiological methods (AMM/RMMs). The statistical analyses recommended in the body of *PDA Technical Report No. 33 (Revised 2026): Evaluation, Validation, and Implementation of Alternative and Rapid Microbiological Methods* are restated with example data to illustrate how the analyses can be used. Although the example data is artificial, the data is based on real validation experiments previously conducted by the authors. Since these methods are presented only as examples, the stakeholder is responsible for selecting the most appropriate statistical analyses to use based on the data to be analyzed and the validation criteria being assessed. It is recommended to align the validation approach with the firm's quality management system (QMS).

The choice of statistical analysis depends on the type of experiment that is being performed and on the type of data analyzed (i.e., binary or positive/negative results versus recovered counts). In each example, multiple options for a statistical analysis based on the observed data will be presented, as the option selected will depend on the statistical assumptions being made to describe the characteristics of the data. As such, **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) will recommend several alternative statistical analyses for a given dataset while explaining the advantages and/or disadvantages of each.

Furthermore, **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) recommend statistical analyses that are feasible for use without specialized software specifically designed for microbiological methods as much as possible. Should specific calculations be needed that are not readily available through user-friendly, general-purpose statistical software (e.g., Minitab®, SPSS, STATA), **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) provides mathematical formulas that can be used manually, imported into an appropriate spreadsheet with calculation capabilities (e.g., Microsoft Excel), conducted with programmable statistical software (e.g., SAS®, R), or implemented in electronic systems using attributable, legible, contemporaneous, original, and accurate (ALCOA) principles. For these reasons, the text may appear significantly technical in nature.

Finally, where possible, **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) discusses sample size in relation to the specific validation criteria and statistical analyses. Certain statistical analyses require a relevant number of data points to be valid for validation studies, which may present a challenge for stakeholders if large amounts of test sample are not readily available. Consequently, **Section 9.0** (Appendix 1) through **Section 11.0** (Appendix 3) discusses the importance of test power and provides examples of how to pool the data from multiple organism experiments to increase the total sample size (where needed). Because the results associated with pooling data may come at the expense of additional statistical assumptions, the limitations of this practice are discussed.

Section 9.0 (Appendix 1) contains many formulas, **Table 9.0-1** provides a key of all the symbols used in the formulas and their meaning.

Note: The notation $\log$ will be used for the natural logarithm in **Section 9.0** (Appendix 1).

Table 9.0-1 Nomenclature for Symbols Used in Section 9.0 (Appendix 1)

Symbol	Meaning
Δ	The non-inferiority margin that is selected a priori to do a non-inferiority test. In some other texts, the notation δ is used.
λ	The mean number of microorganisms or the bacterial density in a set of test samples or in a suspension. When we apply the MPN method for the AMM/RMM and CMM we may denote λ_A and λ_C as theoretical bacterial densities for AMM/RMM and CMM, respectively. When we apply logistic regression, λ_{LOD} represents the mean number of microorganisms in the test samples that provide a positive rate of 95%.
p	The proportion of independent test samples that were tested positively for the presence of viable microorganisms with the microbiological method, also called the positive rate. p_A and p_C are the positive rates for the AMM/RMM and the CMM, respectively. When the test samples are only blank samples, the positive rate is referred to as the false positive rate.
θ	The detection proportion for the microbiological method, which indicates the probability for a presence test for test samples with exactly one microorganism present in the test sample. θ_A and θ_C indicate the detection proportions for the AMM/RMM and CMM, respectively.
R	The ratio of a statistic for the AMM/RMM and CMM and is viewed as a measure for accuracy or recovery. The ratio R_{pr} is the ratio of positive rates p_A/p_C . R_{gMPN} is the ratio of detection proportions θ_A/θ_C . R_{MPN} is the ratio of bacterial densities λ_A/λ_C . R_{Pool} is a pooled ratio taken from multiple ratio estimates.
L	The limit of detection (LoD).
α	The intercept for a logistic regression analysis on the positive rate with log concentration as the independent variable for a microbiological method.
β	The slope for a logistic regression analysis on the positive rate with log concentration as the independent variable for a microbiological method.

Symbol	Meaning
LCL	The lower confidence limit. Often there will be an index X attached to it, i.e., LCL_X , to indicate for what (validation) parameter or statistic the confidence limit is calculated.
UCL	The upper confidence limit. Often there will be an index X attached to it, i.e., UCL_X , to indicate for what (validation) parameter or statistic the confidence limit is calculated.
n	The number of independent test samples for a single concentration tested with the microbiological method. n_A and n_C are the number of test samples for AMM/RMM and CMM, respectively.
m	The number of species tested for a specific validation parameter.
$\bar{x}$	The average log-transformed MPN estimates from multiple MPN runs. $\bar{x}_A$ and $\bar{x}_C$ are the average log-transformed MPN estimates for the AMM/RMM and CMM, respectively.
$\bar{X}_{\log(R)}$	The average of the log-transformed ratios R .
$S_{\log(R)}$	The standard deviation of the log-transformed ratios R .
χ_p^2	Pearson's chi-square statistic calculated for contingency tables.
C_m	The t -value that depends on the number of species m and the confidence level of 90%.
P_m	The t -value that depends on the number of species m and the confidence level of 95%.

9.1 False-Positive Rates for Qualitative Alternative/Rapid Microbiological Methods

One aspect of method suitability studies is to demonstrate that false positives or background interference signals are not detected in an AMM/RMM when the test sample is known not to contain any viable microorganisms. However, some level of false positives may be acceptable, depending on how this information is used (e.g., when establishing a baseline level or an acceptable signal to noise ratio) (see **Section 5.3.2**). The false-positive rate can be quantified by testing several independent blank samples (i.e., samples that are not expected to contain any viable microorganisms) and by observing the number of positive and negative detection events. **Table 9.1-1** provides artificial data for illustrating how to quantify the false-positive rate. The data show that of twenty (20) independent blank samples tested for the presence of a viable microorganism detection event in a single AMM/RMM, only one sample (#6) tested positive.

Table 9.1-1 Presence/Absence Results on 20 Blank Samples

Sample	Presence	Sample	Presence
1	0	11	0
2	0	12	0
3	0	13	0
4	0	14	0
5	0	15	0
6	1	16	0
7	0	17	0
8	0	18	0
9	0	19	0
10	0	20	0

9.1.1 Estimation and Confidence Intervals

Using the data in **Table 9.1-1**, the false-positive rate of the AMM/RMM, denoted by p_A , is estimated as 0.05 or 5% (i.e., 1 positive/20 samples). This false-positive rate is based on only one experiment using 20 test samples. However, if the same experiment is repeated multiple times, it may result in a different, but valid, false-positive rate for each repeated experiment. A 95% confidence interval can quantify the range of valid estimates without having to repeat the experiments. In other words, the interpretation of the interval is a range of false positives that captures the true false positive rate of the AMM/RMM with 95% confidence. The smaller the experiment or the higher the confidence level, the wider the confidence interval. If the goal was to obtain the absolute true false-positive rate with 100% confidence, an unrealistically large experiment would need to be conducted. This is why the use of an appropriate confidence interval, such as 95%, is more practical and acceptable.

Many different methods exist for constructing a confidence interval for analyzing proportions (*1*). The simplest 95% confidence interval is the asymptotic or Wald confidence interval, which is based on a normal approximation and calculated using the following formula:

$$p_A \pm 1.96 \sqrt{\frac{p_A(1 - p_A)}{n}}$$

Where:

n = number of samples tested (from **Table 9.1-1**, $n = 20$)

Using the formula above, the asymptotic 95% confidence interval for the false-positive rate based on the 20 samples is then estimated as (0, 0.146). Because the lower value of the interval was estimated below zero, the result was truncated to zero, since negative false-positive rates do not exist. The asymptotic 95% confidence interval indicates that the true false-positive rate of the AMM/RMM, when 20 samples are tested, lies between 0% and 14.6%, with 95% confidence.

One limitation in using the asymptotic confidence interval is that the number of samples tested should be above, for example, 50, and the number of events (here defined by the number of positive test samples) should not be close to/not approaching zero, for example, above 5. Because the example in **Table 9.1-1** used 20 samples and the number of positive test samples was one, it would be more appropriate to use an alternative confidence interval known as the exact or Clopper-Pearson confidence interval (*I*), which is better suited for a smaller number of samples and low number of events. This interval is based on a binomial distribution (instead of a normal distribution for the asymptotic confidence interval), but the mathematical expression is far more complicated.

General-purpose statistical software packages can easily perform the exact confidence interval. Using the data in **Table 9.1-1**, the confidence interval becomes (0.001, 0.249). Thus, the exact 95% confidence interval implies that the true false-positive rate of the AMM/RMM lies between 0.1% and 24.9%, with 95% confidence. Due to the limitations of the asymptotic confidence interval, using the exact confidence interval when estimating a false-positive rate is recommended.

9.1.2 Sample Size Considerations When Estimating False-Positive Rates

If the length of the confidence interval for the false-positive rate is too large and a smaller confidence interval is needed, the sample size must be larger. **Table 9.1.2-1** shows the minimum sample size with an upper 95% confidence limit for a false-positive rate when the experiment does not contain any positive results.

Table 9.1.2-1 Sample Size and 95% Upper Confidence Limit for Experiments with no Presence Tests

n	UCL	n	UCL	n	UCL	n	UCL	n	UCL
5	45.1	30	9.50	55	5.30	80	3.68	120	2.47
10	25.9	35	8.20	60	4.87	85	3.46	150	1.98
15	18.1	40	7.22	65	4.50	90	3.17	200	1.49
20	13.9	45	6.44	70	4.19	95	3.10	250	1.19
25	11.3	50	5.82	75	3.92	100	2.95	300	0.99

To demonstrate that the false-positive rate is less than 1%, with 95% confidence, 300 blank samples would need to be tested, and all of the results must be negative. Alternatively, if 30 test samples tested negative, the false positive rate would not be higher than 9.5% with 95% confidence. Given a maximum allowable false positive rate *p*, for all experiments with no positive test results, the sample size can be calculated as:

$$n \geq \frac{\log(0.05)}{\log(1 - p)}$$

While given the sample size n , the maximum allowable false positive rate is given by $1 - (0.05)^{1/n}$, with 95% confidence.

9.2 Comparability on the Limit of Detection of Qualitative Alternative/ Rapid Microbiological Methods

For a comparability evaluation of a qualitative AMM/RMM, the detection of microorganisms in the AMM/RMM must be compared with the conventional microbiological method (CMM). Such a comparison does not require an estimation of the limit of detection (LoD); however, this appendix provides guidance on how to perform this analysis. Indeed, by demonstrating the AMM/RMM provides similar detection results as the CMM, the conclusion that the AMM/RMM has a LoD that is not inferior to the LoD of the CMM can be reached, even without specifying what the LoD is in each method.

Table 9.2-1 illustrates an example dataset that can be used to demonstrate comparability of a qualitative AMM/RMM. An experiment with a single suspension of microorganisms was used in which the challenge concentration was adjusted to provide a positive rate of approximately 50%, as recommended in USP <1223>. The AMM/RMM and the CMM were each allocated 40 random test samples (i.e., 80 samples in total) and these were assayed for the presence or absence of microorganisms.

Table 9.2-1 Presence/Absence Results of Alternative/Rapid Microbiological Method and Conventional Methods on a Single Type of Microorganism

Microbiological Method	Test Result		
	Presence	Absence	Total
AMM/RMM	25	15	40
CMM	28	12	40
Total	53	27	80

These samples are representative of independent test samples, even though the samples are taken as separate aliquots from the same microbial suspension, because each sample cannot be tested in both methods. The data is considered paired if both microbiological methods measure the exact same test sample. Additionally, there is a probability for a test sample to contain a specific number of microorganisms, and test samples may differ in the number of microorganisms (irrespective of whether the sample is enumerated or tested for absence/presence with the microbiological method). For example, **Figure 9.2-1** illustrates the probabilities (the vertical axis) for some number of microorganisms (the horizontal axis) in the test sample when the mean number of microorganisms in the test sample (often denoted with λ) is equal to 2 and the number of microorganism in the test samples follows a Poisson distribution. **Figure 9.2-1** shows that almost 55% of all the test samples will contain just one or two microorganisms, 13.5% of the test samples do not contain any microorganisms, and more than 5% of the test samples have five or more microorganisms in the test samples.

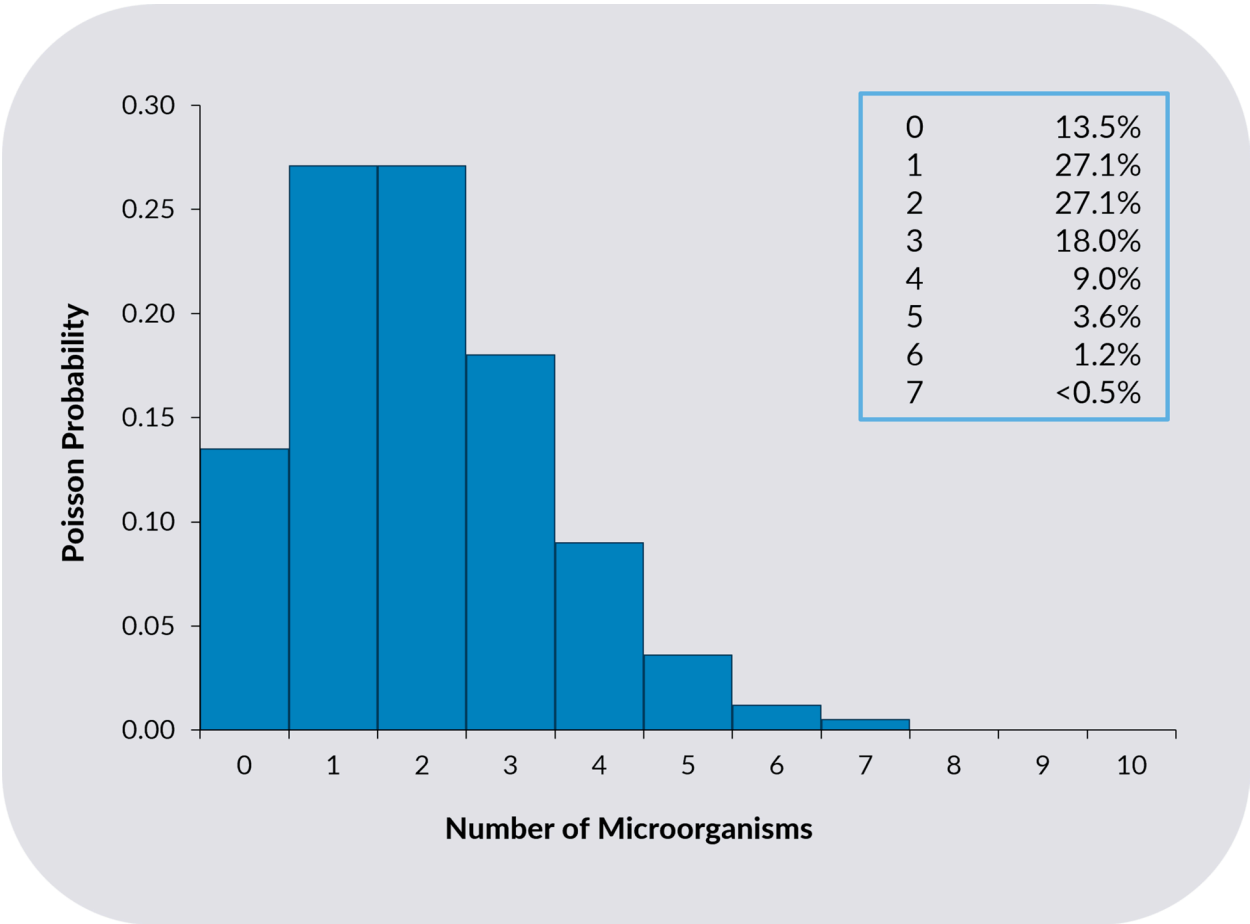


Figure 9.2-1 Probabilities for the Number of Microorganisms in Test Samples when the Average Number of Microorganisms in the Test Samples is Equal to 2

It is important to emphasize that a random allocation of test samples to the AMM/RMM and CMM is essential to guarantee that both microbiological methods would test samples with different numbers of microorganisms in the same ratios. For example, if 22 of the 80 test samples in **Table 9.2-1** contains a single microorganism, it can be assumed that both the AMM/RMM and CMM would analyze 11 test samples with a single microorganism when random allocation is applied. An imbalance in these ratios could severely affect the comparison of detecting microorganisms between the AMM/RMM and CMM, because it would be easier to detect test samples that contain more microorganisms. Thus, if the AMM/RMM is testing samples with higher numbers of microorganisms than the CMM, the test results could lead to a conclusion that the AMM/RMM is superior in detecting microorganisms than the CMM, whereas this was actually just a result of an incorrectly executed experiment (e.g., of not applying random allocation). Conversely, if the CMM received test samples with a higher number of microorganisms, the noninferiority in the AMM/RMM could not be demonstrated. Since the exact number of microorganisms in the test samples is unknown, the test samples cannot be perfectly allocated with the same number of microorganisms to both microbiological methods and, as a result, the outcome relies on the randomization described above for all experiments. Consequently, it is important to ensure the homogeneity of microbial suspensions as best as possible.

9.2.1 Demonstrating Comparability Using Ratios of Positive Rates

Using the data summarized in **Table 9-2.1**, the positive rates for the AMM/RMM and CMM can be calculated, together with their 95% exact confidence interval, as described in **Section 9.1.1**. The proportion and confidence interval for the AMM/RMM (p_A) is estimated at 0.625 (0.458, 0.773); the proportion and confidence interval for the CMM (p_C) is estimated at 0.700 (0.535, 0.834).

Considering these results, two conclusions are clear. First, the positive rate for the CMM is statistically larger than the planned or intended 50% positive detection rate, since the confidence interval falls above the value 0.5. Thus, the goal of achieving an approximate 50% positive rate was rejected with the collected data, indicating that the spike used in the suspension was possibly higher than originally designed. Second, the positive rate for the CMM appears somewhat larger than the positive rate for the AMM/RMM. Testing the null hypothesis that the microbiological methods have the same positive rates ($H_0: p_A = p_C$) can be performed with the Pearson's chi-square test, which is available in many general-purpose statistical software packages. The data can be entered in a 2x2 contingency table as presence and absence values for each method.

Using the data in **Table 9.2-1**, the Pearson's chi-square statistic is equal to 0.503, and the two-tailed p -value is equal to 0.478, indicating that the null hypothesis, $H_0: p_A = p_C$, cannot be rejected. Because the p -value is >0.05 , the association between the number of positives between the AMM/RMM and the CMM is considered to be not statistically significant. Another option for testing the null hypothesis $H_0: p_A = p_C$, is to use the Fisher's exact test, which is more accurate for small sample sizes (e.g., <50) or when the counts in the contingency table are low (e.g., <5). Using the same data in **Table 9.2-1**, the two-tailed p -value is equal to 0.637, also indicating that the null hypothesis cannot be rejected. Although both tests seem to provide a good result (equality of the positive rates for AMM/RMM and CMM cannot be rejected), some care is needed, because sample size plays an important role in accepting the conclusion when using traditional hypothesis tests. The smaller the number of test samples, the less likely the null hypothesis of equal positive rates will be rejected even if a large difference in positive rates is present. On the other hand, if many more samples had been tested, only a small irrelevant difference in positive rates would already be rejected.

Although the Pearson's chi-square or Fisher's exact test provide valuable information, in particular when the null hypothesis is being rejected, performing a non-inferiority test is preferred, where the primary goal is to demonstrate that the AMM/RMM is "not worse" than the CMM. A key element of a non-inferiority test is the use of a prespecified "non-inferiority margin," which defines the largest acceptable difference or ratio in positive rates between the AMM/RMM and the CMM. The non-inferiority hypothesis is then formulated as

$$H_0: \frac{p_A}{p_C} < \Delta$$

Where:

Δ = non-inferiority margin (or δ in some statistical applications) that must be stated at the beginning of the experiment

p_A = positive rate for the AMM/RMM

p_C = positive rate for the CMM

Using this ratio of positive rates approach is most appropriate when the positive rates are expected to be low (e.g., <50%). Furthermore, a Δ equal to 0.7 is commonly used, unless a stricter margin is desired (e.g., $\Delta = 0.8$). The null hypothesis for non-inferiority indicates that the AMM/RMM is inferior to the CMM and that this hypothesis should be rejected with the observed data to demonstrate non-inferiority. To execute the non-inferiority test, a 90% confidence interval on the ratio of positive rates should be calculated. This approach is often referred to as the Two-One-Sided Test (TOST), where an upper (UCL) and a lower (LCL) 95% confidence limit is calculated. However, for non-inferiority, only the lower 95% confidence limit (which would be provided by the 90% confidence interval) needs to be calculated. When the lower 95% confidence limit is above the non-inferiority margin, the null hypothesis of inferiority is rejected.

Similar to the confidence interval utilized when estimating false-positive rates, there are different ways of constructing a confidence interval on the ratio of positive rates, and this may be related to the statistical software package being used. For a qualitative microbiological method, USP <1223> refers to a confidence interval developed by Farrington and Manning in 1990 (2). The Farrington and Manning approach is programmed in the software package SAS® using the FREQ procedure, however, this may not be available in all general-purpose software packages. In Minitab, the 2-Sample Equivalency Test for numerical data may be used, as long as the two positive rates are away from the boundaries (e.g., $0.2 < \text{positive rate} < 0.8$).

Alternative approaches for confidence intervals on ratios of proportions may also be used. For example, in clinical trials the ratio of proportions is referred to as relative risk, and several calculation approaches exist to determine non-inferiority as alternatives to the Farrington and Manning approach. These calculation approaches are essentially different ways of constructing the confidence interval on the relative risk, and there is no approach superior to all other approaches. These approaches can be calculated from a logistic regression approach essentially using an asymptotic confidence interval, a Bayesian approach where prior information can be used, or a direct approach making use of explicit formulas (3-6).

Applying the Farrington and Manning approach to the data in **Table 9.2-1**, the ratio of positive rates is estimated with $R_{pr} = p_A/p_C = 0.893$, and its lower 95% confidence limit is determined at 0.676. Since this lower 95% confidence limit of 0.676 is below the margin 0.7, the null hypothesis of inferiority **cannot** be rejected. Even though the null hypothesis of equal positive rates could not be rejected with Pearson's chi-square or Fisher's exact tests, there is not enough evidence to show that the AMM/RMM is non-inferior to the CMM for non-inferiority margins of at least 0.7.

9.2.2 Justifying for Comparability with the Most Probable Number

It was demonstrated in 2022 that the comparison of positive rates may be unreliable if the experiment is not properly executed (2). For example, if higher numbers of microorganisms are spiked in a test suspension, a greater number of positive test responses will be observed in both methods, regardless of each method's detection limits. Conversely, if very low numbers are spiked in a test suspension, and there are differences in the detection limits for each method, the analysis may lead to incorrect results. To illustrate this, assuming the average number of microorganisms in test samples is four (4), and the Poisson probability distribution are calculated similarly as that shown in **Figure 9.2.2-1**, only 7.3% of all test samples would contain exactly one organism and 76.2% of all test samples would contain three or more microorganisms. If it is further assumed that the CMM has a detection limit of 1 (resulting in a probability of 0.95 of a positive test when a single microorganism is present) and the AMM/RMM has a detection limit of 4 (resulting in an inferior probability of $0.527 = 1 - (1 - 0.95)^{1/4}$ for a positive test when a single microorganism is present)*, a comparison of positive rates would not be able to detect a difference in recovery when the average spike in the test samples is four or more, while the ratio of probabilities for samples with a single microorganism is

equal to 0.555 ($= 0.527/0.95$) and much lower than the non-inferiority margin of 0.70. For this situation with higher number of microorganisms in test samples, the experiment may result in 39 positives for the CMM and 35 positives for the AMM/RMM (still assuming 40 samples are tested in both methods as was done in **Table 9.2-1**). Conducting a non-inferiority test with positive rates as in **Section 9.2.1** leads to a rejection of the null hypothesis of inferiority, since the results will be equal to $R_{pr} = p_A/p_C = 0.897$ (0.781, 0.997)

and the lower 95% confidence limit is far above the non-inferiority margin of 0.70. This is counterintuitive compared to the ratio of 0.555 for test samples with single microorganisms and to the conclusion of not rejecting inferiority in **Section 9.2.1**, where results were closer to each other (28 positives for the CMM and 25 positives for the AMM/RMM) when samples containing, on average, fewer organisms (around two) were observed. Therefore, the conclusion derived with the ratio of positive rates when positive rates are close to the boundary of one would be incorrect.

***Note:** The formula used to calculate the inferiority probability in this example is:

$$\theta \leq 1 - \exp\left(\frac{\log(0.05)}{\text{LoD}}\right)$$

Where:

θ = detection proportion

Alternatively, if the detection proportion is known, the LoD can be calculated using this formula:

$$\text{LoD} \geq \frac{\log 0.05}{\log(1 - \theta)}$$

To graphically illustrate how the positive rates would be affected by the change in the average number of microorganisms, **Figure 9.2.2-1** shows the positive rates for a CMM with a detection limit of one, and an AMM/RMM with a detection limit of 4. The positive rates and *their ratio* converge to one when the average number of microorganisms in a test sample increases. Thus, no matter how large the difference in the detection limit, the ratio of positive rates will pass the inferiority margin when enough organisms are spiked in the test samples. When the average number of microorganisms goes to zero, the positive rates of the AMM/RMM and CMM also go to zero. However, mathematical peculiarities occur when the *ratio* of these positive rates (as a consequence of dividing by zero) are examined. Consequently, the ratio of positive rates converges to the *true accuracy* for testing test samples with only single microorganisms when the average number of microorganisms converges to zero (under certain mathematical assumptions). Thus, only when the observed positive rate for the CMM in the experiment is low, for example, below 50%, would the non-inferiority approach discussed in **Section 9.2.1**, when comparing positive rates, be appropriate to use (2).

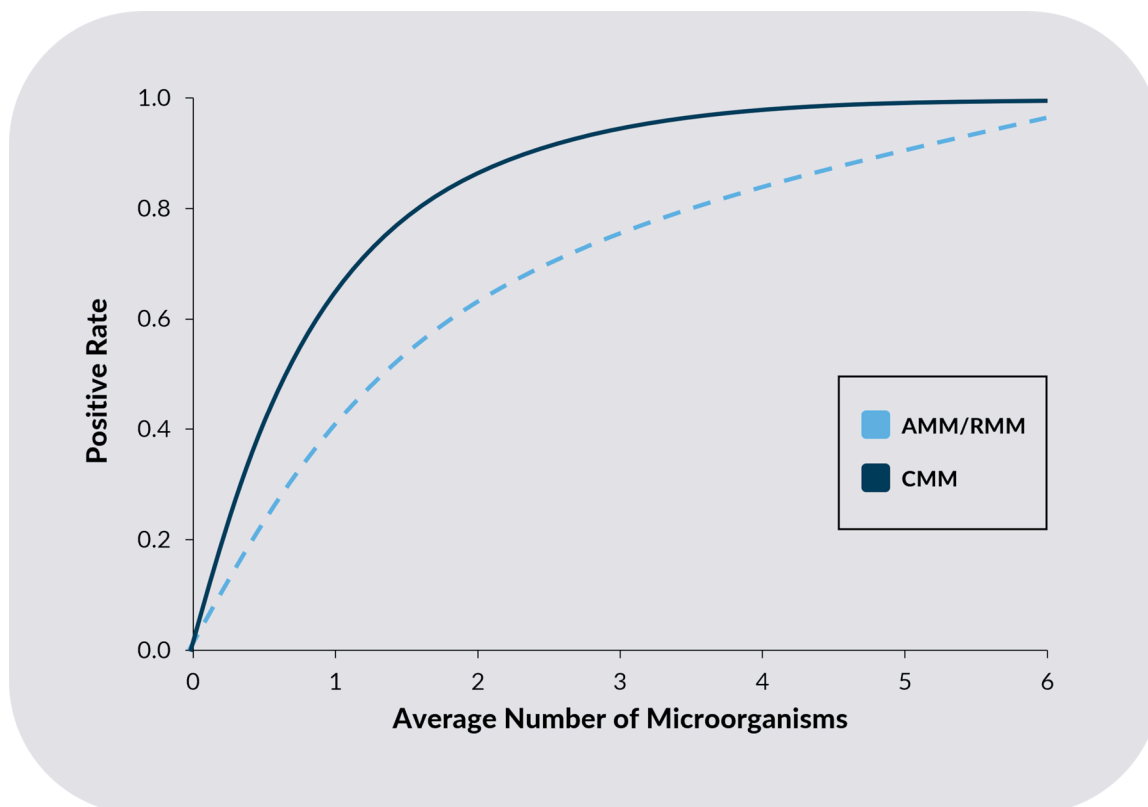


Figure 9.2.2-1 Visualization of Positive Rates as Function of the Average Number of Microorganisms in Test Samples

A more reliable approach less affected by the spiked concentration of microorganisms, is to calculate the most probable number (MPN) for both the AMM/RMM and CMM. This would require an experiment, where multiple MPN values derived from the presence/absence data across multiple challenge concentrations would be obtained for both microbiological methods. A non-inferiority analysis based on the *t*-test could then be applied (see **Section 9.2.3**). Alternatively, literature has provided the so-called generalized MPN (gMPN) method that could be conducted on a single concentration without repeating the experiment, such that the data in **Table 9.2-1** could be analyzed. This method is currently only programmed in the specialized statistical software TriMSA (7). As such, the mathematical details for this type of analysis are described in **Section 9.2.3** so stakeholders can use this approach without the specialized software.

9.2.3 Demonstrating Compatibility with the Generalized Most-Probable-Number Approach

The MPN is based on the premise that the probability of a positive test result is equal to $1 - \exp\{-\lambda\}$, with λ being the average number of microorganisms in the test samples. When a number of test samples are taken from a single suspension and tested for presence/absence with the AMM/RMM, the parameter λ can be estimated by:

$$\lambda = -\log(1 - p_A)$$

Where:

p_A = positive rate for the AMM/RMM

The MPN is then calculated by:

$$\text{MPN} = \frac{V\lambda}{v}$$

Where:

V = volume of the suspension from which the test samples were taken

v = volume of the test sample (8)

The gMPN assumes that the probability of a positive test result is equal to $1 - \exp\{-\theta\lambda\}$, with θ the probability of a positive test for a test sample with exactly one microorganism. The MPN approach assumes this probability is equal to one. The parameter θ is referred to as the *detection proportion*. Non-inferiority of the AMM/RMM with respect to the CMM is then conducted on the detection proportions of the AMM/RMM and CMM, that is, the null hypothesis of inferiority is:

$$H_0: \frac{\theta_A}{\theta_C} < \Delta$$

Where:

θ_A = detection proportion of the AMM/RMM

θ_C = detection proportion of the CMM

Δ = non-inferiority margin

The ratio $R_{\text{gMPN}} = \theta_A/\theta_C$ of detection proportions can be viewed as a kind of “accuracy” or “recovery”. Since the detection proportion is the probability that test samples with a single microorganism are tested positively, the ratio would indicate how much of the positive-tested samples of the CMM would be recovered with the AMM/RMM. A ratio less than one would indicate that not all of the positively tested samples of the CMM are being recovered (8, 9).

The ratio $R_{\text{gMPN}} = \theta_A/\theta_C$ of detection proportions, when the test samples are all collected from a single suspension (like in **Table 9.2-1**), can be estimated by:

$$R_{\text{gMPN}} = \frac{\log(1 - p_A)}{\log(1 - p_C)}$$

Where:

p_A = positive rate for the AMM/RMM

p_C = positive rate for the CMM

Its asymptotic lower 95% confidence limit (an exact confidence interval does not exist) can then be calculated by:

$$LCL_{gMPN} = R_{gMPN} \cdot \exp \left(-1.645 \sqrt{\frac{p_A}{n(1 - p_A)[\log(1 - p_A)]^2} + \frac{p_C}{n(1 - p_C)[\log(1 - p_C)]^2}} \right)$$

Where:

R_{gMPN} = ratio of detection proportions

p_A = positive rate for the AMM/RMM

p_C = positive rate for the CMM

n = number of test samples offered to each microbiological method

$\exp$ = inverse of the natural logarithm

Applying the gMPN non-inferiority approach to the data obtained in **Table 9.2-1**, obtains $R_{gMPN} = 0.815$, with a lower 95% confidence limit equal to $LCL_{gMPN} = 0.506$.

To illustrate that the gMPN is not affected by the average number of microorganisms in the test samples, the gMPN approach was also applied to the data with 35 positives for the AMM/RMM and 39 positives for the CMM, where the AMM/RMM is inferior to the CMM for a non-inferiority margin of $\Delta = 0.70$. This data would fit with the situation where the detection proportions are equal to $\theta_C = 0.95$ (an LoD equal to one) and $\theta_A = 0.527$ (an LoD equal to 4) for the CMM and AMM/RMM, respectively, that is, a theoretical ratio of 0.555 indicating the AMM/RMM is inferior to the CMM. The gMPN provides an estimate of $R_{gMPN} = 0.564$ close to the true ratio, with a lower 95% confidence limit equal to $LCL_{gMPN} = 0.325$. The gMPN approach does not reject the null hypothesis of inferiority as it should, implying that the AMM/RMM is inferior.

9.2.4 Demonstrating Comparability Using Multiple Most Probable Number Experiments

To determine the non-inferiority of the AMM/RMM with respect to the CMM on detecting microorganisms, multiple standard MPN experiments can be applied. The multiple MPN results obtained from both microbiological methods can then be compared to demonstrate non-inferiority. **Table 9.2.4-1** provides the results of six (6) MPN runs in both microbiological methods where the same suspension was used to make three (3) 10-fold dilutions in each of the runs. Three (3) replicates per dilution are analyzed. The two (2) MPN values per run, derived from FDA tables, are considered independent (similar to the discussion for the data of **Table 9.2-1**), and a two-sample t -test can be used to determine non-inferiority. However, if each run had come from a separate suspension (e.g., different suspension preparations, tested on different days or by different analysts), the two MPN values per run would be considered paired, and a paired t -test should be applied. The last row in **Table 9.2.4-1** presents the MPN results derived from the FDA MPN tables (**10**).

Table 9.2.4-1 Six Standard Most-Probable-Number Runs on a Single Microorganism Tested by Two Microbiological Methods

Dilution	Run 1		Run 2		Run 3		Run 4		Run 5		Run 6	
	CMM	AMM/RMM	CMM	AMM/RMM	CMM	AMM/RMM	CMM	AMM/RMM	CMM	AMM/RMM	CMM	AMM/RMM
1.00	3	3	3	3	3	3	3	3	3	3	3	3
0.10	2	3	2	2	1	3	1	2	2	2	2	1
0.01	0	1	1	0	1	0	1	0	0	2	2	1
MPN	9.3	46.0	15.0	9.3	7.5	24.0	7.5	9.3	9.3	21.0	21.0	7.5

The MPN values can be considered estimates of the bacterial density of the suspension. Although the suspension contained just one true number of organisms, both microbiological methods should provide similar estimates when the methods would detect microorganisms in the same way. If the microbiological methods were different, it could be conceptually formulated that the AMM/RMM would estimate bacterial density λ_A , and the CMM would estimate λ_C . Non-inferiority would then be formulated by:

$$H_0: \frac{\lambda_A}{\lambda_C} < \Delta$$

Where:

Δ = non-inferiority margin (equal to 0.7 in this example)

λ_A = bacterial density for the AMM/RMM

λ_C = bacterial density for the CMM

A natural logarithmic transformation of the MPN values, that is, $x = \log(MPN)$, is preferred, as this would reduce skewness in the MPN values (i.e., thicker right tail) and make the data more normally distributed. The non-inferiority hypothesis would also be simplified, since it would become a difference in log-transformed bacterial densities, that is, $H_0: \log(\lambda_A) - \log(\lambda_C) < \log(\Delta)$, instead of a ratio in bacterial densities. For the transformed MPNs, then the TOST could be used that results from the t -test (available in almost any general-purpose statistics software), since the difference $\log(\lambda_A) - \log(\lambda_C)$ is estimated with the difference in averages of the log-transformed MPNs, and a confidence interval can be calculated with the t -distribution.

Applying the t -test to the logarithmically transformed MPNs leads to an average (standard deviation) $\bar{x}_C = 2.374$ (0.415) for the CMM and $\bar{x}_A = 2.754$ (0.709) for the AMM/RMM. Traditional hypothesis-testing with the two-sample t -test does not reject the null hypothesis, $H_0: \log(\lambda_A) = \log(\lambda_C)$, for equality in log-transformed bacterial densities (p -value = 0.289). The difference in means is now estimated at 0.381 with a 90% confidence interval equal to (-0.227, 0.989). Since the lower limit of the 90% confidence interval is larger than $\log(0.7) \approx -0.3567$, the null hypothesis of inferiority is rejected, and the AMM/RMM is considered non-inferior to the CMM. The mean difference between the two methods in the logarithmic scale

can be exponentiated (reverted to the original scale where the mean difference becomes a ratio) to obtain an accuracy estimate $R_{MPN} = \lambda_A/\lambda_C$. Furthermore, if the traditional null hypothesis of $H_0: \log(\lambda_A) = \log(\lambda_C)$ had been rejected and the AMM/RMM recovers more microorganisms than the CMM (i.e., the AMM/RMM MPN is higher than the CMM MPN), the AMM/RMM can be claimed superior, although this is not required.

Although the comparison of multiple MPNs between the AMM/RMM and CMM is an appropriate analysis approach, an alternative analysis on the aggregated data (ignoring the run structure) is also possible. **Table 9.2.4-2** presents the aggregated data according to a concentration and microbiological method from the data in **Table 9.2.4-1**. This approach is somewhat more powerful than the comparison of multiple MPNs, but is more difficult to analyze with general-purpose statistical software (2). There is no closed mathematical formula for the ratio $R_{gMPN} = \theta_A/\theta_C$ when multiple concentrations are involved. The analysis is programmed in the specialized software TriMSA, and it can be programmed in the SAS® software or in the open-access software [R]. Applying the gMPN approach to the aggregated data in **Table 9.2.4-2**, the ratio is estimated at $R_{gMPN} = 1.338$ with a 90% confidence interval equal to (0.726, 2.466). Thus, the gMPN approach also shows that the AMM/RMM is non-inferior to the CMM at a non-inferiority margin $\Delta = 0.70$.

Table 9.2.4-2 Aggregated Data Per Concentration and Microbiological Method for the Multiple Most-Probable-Number Study

Dilution	CMM		AMM/RMM	
	Count	Total	Count	Total
1.00	18	18	18	18
0.10	10	18	13	18
0.01	5	18	4	18

9.2.5 Sample Size Considerations: Combining Multiple Species

Demonstrating non-inferiority in detecting microorganisms between the AMM/RMM and CMM requires a substantial number of test samples per microbiological method. For traditional hypothesis testing, where the null hypothesis is that the detection limits are identical, at least 200 test samples per microbiological method are needed (2,9).

For the gMPN approach, the minimal sample sizes are provided in **Table 9.2.5-1**. These sample sizes do not yet incorporate the uncertainty that comes from estimating the parameters, implying that (substantial) larger sample sizes are needed.

Table 9.2.5-1 Minimal Theoretical Sample Size per Microbiological Method for the gMPN Approach

θ_C	$\frac{\theta_A}{\theta_C}$	Non-Inferiority Margin 0.7					Non-Inferiority Margin 0.8				
		Average number of microorganisms (λ)					Average number of microorganisms (λ)				
		1.5	2.0	2.5	3.0	3.5	1.5	2.0	2.5	3.0	3.5
0.35	0.90	224	184	162	148	140	1017	835	734	673	636
0.55	0.90	167	145	136	133	135	757	660	617	604	611
0.75	0.90	144	134	134	141	154	655	608	609	640	699
0.95	0.90	135	134	145	165	195	613	610	657	747	886
0.35	1.00	107	89	78	72	69	273	225	199	184	175
0.55	1.00	81	71	67	66	68	205	181	170	168	172
0.75	1.00	70	66	67	72	80	179	169	171	183	202
0.95	1.00	67	68	74	86	104	170	172	188	218	264
0.35	1.10	65	54	48	44	42	130	108	96	89	85
0.55	1.10	49	44	42	42	43	99	88	83	83	89
0.75	1.10	43	41	43	46	52	87	83	85	93	104
0.95	1.10	42	43	48	57	70	83	86	96	114	141

The sample size depends on the detection proportion of the CMM (θ_C), the true accuracy (θ_A/θ_C), the average number of microorganisms (λ) in the test samples, and the non-inferiority margin (Δ). When the AMM/RMM is superior to the CMM ($\theta_A/\theta_C > 1$), the minimal number of test samples in **Table 9.2.5-1** may be reasonable in practice when the non-inferiority margin can be taken equal to 0.7. However, in most other cases, an unrealistically large sample size is needed. Indeed, in case the true accuracy (θ_A/θ_C) is close to the non-inferiority margin, the sample size can increase to hundreds of test samples per microbiological method. Moreover, when spiking a suspension is imprecise, the required sample size must be robust against deviations in the average number of microorganisms. In most settings, where the true accuracy may not be known and spiking can vary substantially, a sample size of 200 samples (that was also needed for traditional hypothesis) may be needed, even when the non-inferiority margin can be taken equal to 0.7. Since testing 200 test samples for a single species is often too much of a burden, creative approaches should be considered.

One approach is to combine the results from multiple species to obtain a total sample size of at least 200 test samples per microbiological method for a single dilution. However, to maintain a reliable estimate per species, using at least 30 test samples per species is recommended, otherwise the risk is too high to obtain positive rates at the boundary (i.e., equal to zero or one), and the width of the confidence interval becomes very large. For example, if six species are being investigated on the LoD, a total of 34 test samples per species and microbiological method is needed, to make the total sample size larger than 200 and the sample size per species larger than 30. When meeting these sample-size recommendations is not possible (e.g., when enough test material is not available), it does not mean the studies cannot be conducted. Rather, stakeholders will need to accept that the confidence interval may be too wide to demonstrate 70% non-inferiority, unless the AMM/RMM is superior to the CMM.

Combining the results from different species would then be necessary to achieve enough power for non-inferiority testing, but some statistical considerations should be mentioned, because a mistake in the analysis could lead to incorrect conclusions. The simplest approach is to analyze the total data as one set, ignoring the structure obtained by the species. An analysis as presented in **Section 9.2.2** would then be appropriate or, alternatively, the analysis in **Section 9.2.1** if the conditions of the positive rate for the CMM is satisfied. However, ignoring the species in the analysis is only acceptable when the ratios $R_{\text{GMPN}} = \theta_A/\theta_C$ for the different species would be very similar across the species. Without this assumption, the AMM/RMM can be inferior for one species, but this may not be detected because the data is masked by the data of the other species where non-inferiority may be the truth. A somewhat more cumbersome alternative approach is making use of the analysis shown in **Section 9.2.2** for each species separately and then pooling the estimated ratios. Both approaches are illustrated in **Table 9.2.5-2** using presence/absence data.

Table 9.2.5-2 Present Results of the Alternative/Rapid Microbiological Method and Conventional Microbiological Method for Seven Different Species

Species	CMM		AMM/RMM	
	Count	Total	Count	Total
B. subtilis	23	30	22	30
C. sporogenes	27	30	26	30
C. albicans	20	29	22	30
S. aureus	25	30	24	30
A. brasiliensis	21	30	19	30
P. aeruginosa	15	30	8	30
E.coli	25	28	26	29
Total	156	207	147	209

Ignoring the structure of species, just the numbers at the last row of **Table 9.2.5-2** can be used for analysis. The positive rates are then estimated by $p_A = 0.703$ (0.636, 0.764) and $p_C = 0.753$ (0.689, 0.811) for the AMM/RMM and CMM, respectively, using the exact approach for confidence intervals. The positive rate for the CMM is too high for a reliable analysis with the positive rates directly (see **Section 9.2.3**). With the gMPN analysis of **Section 9.2.3**, a ratio of $R_{\text{gMPN}} = \theta_A/\theta_C = 0.867$ can be obtained with a lower 95% confidence limit of $LCL_{\text{gMPN}} = 0.708$. Thus, the gMPN analysis on all counts together shows that the AMM/RMM is non-inferior to the CMM for a non-inferiority margin of $\Delta = 0.7$. However, if each species were analyzed separately with the gMPN analysis, the results in **Table 9.2.5-3** (number of microorganisms in 1 mL) would be obtained. For each species, the null hypothesis of inferiority at the non-inferiority margin of $\Delta = 0.7$ cannot be rejected. This is not unexpected, since 30 test samples (or less) is too low a number for non-inferiority testing per species (**Table 9.2.5-1** demonstrates this even if the AMM/RMM would be slightly superior to the CMM). Moreover, the ratio for *P. aeruginosa* is different from the other six species, which all vary around 0.90. This somewhat lower ratio would not be detected by a gMPN analysis of the total counts.

Table 9.2.5-3 Most Probable Number Estimates and the gMPN Ratio Estimates for Multi-Species Experiment

Species	MPN AMM	MPN CMM	R_{gMPN}	LCL_{gMPN}	$\log(R_{\text{gMPN}})$
<i>B. subtilis</i>	1.32	1.46	0.908	0.534	-0.09624
<i>C. sporogenes</i>	2.02	2.30	0.875	0.507	-0.13346
<i>C. albicans</i>	1.32	1.17	1.130	0.657	0.12190
<i>S. aureus</i>	1.61	1.79	0.898	0.529	-0.10731
<i>A. brasiliensis</i>	1.00	1.20	0.833	0.482	-0.18233
<i>P. aeruginosa</i>	0.31	0.69	0.447	0.216	-0.80417
<i>E. coli</i>	2.27	2.23	1.016	0.578	0.01559

The individual ratios for the species in **Table 9.2.5-3** can be combined or pooled using different meta-analyses methods but here, one of the simplest methods though perhaps not the most powerful technique, has been used (**II**). All the meta-analyses methods for pooling a certain statistic across different groups, also called a “measure of effect,” assume that the statistic is normally distributed. To best satisfy this assumption, the log transformation of the ratios R_{gMPN} in **Table 9.2.5-3** is recommended as the measure of effect for accuracy that is pooled. The log transformed ratios are also listed in **Table 9.2.5-3**.

The simplest meta-analysis method is to calculate the arithmetic average and the standard deviation of the log-transformed ratios and then calculate a standard 90% confidence interval for normally distributed data

(available in any general-purpose statistical software package). Since this interval is in the log-transformed scale, it must be transformed back to the ratio scale. Thus, for the confidence interval, the following must be calculated:

$$\exp \left\{ \bar{X}_{\log(R)} \pm C_m \frac{S_{\log(R)}}{\sqrt{m}} \right\}$$

Where:

$\bar{X}_{\log(R)}$ = average value of the log transformed ratios of detection proportions (see below)

$S_{\log(R)}$ = standard deviation of the log transformed ratio of detection proportions (see below)

R_{gMPN}^k = ratio of detection proportions for species k (see below)

C_m = t -value (see **Table 9.2.5-4**) that depends on the number of species and the confidence level chosen (here 90%)

m = number of species being pooled

$\exp$ = inverse of the natural logarithm

$$\bar{X}_{\log(R)} = \sum_{k=1}^m \left[\frac{\log(R_{\text{gMPN}}^k)}{m} \right] \text{ and } S_{\log(R)}^2 = \sum_{k=1}^m \left[\frac{(\log(R_{\text{gMPN}}^k) - \bar{X}_{\log(R)})^2}{(m-1)} \right]$$

Using the data in **Table 9.2.5-3**, the numbers can be calculated as $\bar{X}_{\log(R)} = -0.16943$, $S_{\log(R)} = 0.29796$, and $C_m = 1.94318$. The pooled ratio and its 90% confidence interval are then estimated at $R_{\text{Pool}} = 0.844$ (0.678, 1.051).

Table 9.2.5-4 Critical Values Rejecting Homogeneity

Number of Species	C_m	P_m
3	2.91999	4.30265
4	2.35336	3.18245
5	2.13185	2.77645
6	2.01505	2.57058
7	1.94318	2.44691
8	1.89458	2.36462
9	1.85955	2.30600
10	1.83311	2.26216

In this pooled analysis, the null hypothesis of inferiority at margin $\Delta = 0.70$ cannot be rejected because the lower limit is below this value. The reason is that the result of *P. aeruginosa* is less masked in the pooled analysis, since the result on *P. aeruginosa* pulls the pooled ratio down and increases the variability. Removing the results of *P. aeruginosa*, which seems to be an outlier, the pooled ratio with its 90% confidence interval is equal to $R_{\text{pooled}} = 0.938$ (0.856, 1.029). Thus, non-inferiority is attained for the species *B. subtilis*, *C. sporogenes*, *C. albicans*, *S. aureus*, *A. brasiliensis*, and *E. coli*; but, this cannot be demonstrated for *P. aeruginosa* with the collected data.

To demonstrate that the result for *P. aeruginosa* is an outlier, 95% prediction limits for the ratios can be calculated. A prediction limit is calculated similar to the confidence limit:

$$\exp \left\{ \bar{X}_{\log(R)} \pm P_m \cdot S_{\log(R)} \cdot \sqrt{\left(1 + \frac{1}{m}\right)} \right\}$$

Where:

$\bar{X}_{\log(R)}$ = average value of the log transformed ratios of detection proportions

$S_{\log(R)}$ = standard deviation of the log transformed ratios of detection proportions

P_m = *t*-value (listed in **Table 9.2.5-4**) that depends on the number of species and the choice of confidence level (here selected equal to 95%)

m = number of species being pooled

$\exp$ = inverse of the natural logarithm

The prediction limits should be calculated without the extreme value (minimum or maximum ratio) to judge if the extreme value falls within the prediction interval. If the extreme value does not fall inside the prediction interval, the result is considered unexpected and could be qualified as an outlier. Thus, after excluding the results of *P. aeruginosa* in the analysis, the prediction interval for a new ratio R_{gMPN} is determined at (0.688, 1.280). The result of 0.447 for *P. aeruginosa* is clearly outside the prediction interval and can therefore be viewed as an outlier. This confirms the conclusion that the detection of *P. aeruginosa* with the AMM/RMM, compared to the detection of *P. aeruginosa* with the CMM, cannot be viewed similarly to the other species.

9.3 Estimation of the Detection Limit

Table 9.3-1 contains the results of an experiment set up to determine the LoD of the AMM/RMM, without making a direct comparison with the CMM. The data was collected for one type of microorganism or species with one suspension created and spiked with a known number of microorganisms that is close to the AMM/RMM's purported LoD (e.g., <10 cells). Five (5) additional suspensions, using different concentrations from the initial suspension, were created. From each suspension, twenty (20) samples were drawn and tested on the presence/absence with the AMM/RMM. The concentrations listed in **Table 9.3-1** represent the expected average number of microorganisms present in the test sample.

Different statistical approaches exist for estimating the LoD based on the data in **Table 9.3-1**. The logistic regression approach with different risk curves and the most probable limit (MPL) is described here (12-15).

Table 9.3-1 Number of Positives and Total Tested Test Samples for Different Concentrations

Concentration	Counts	Total	Positive Rate
2.2059	18	20	0.90
1.1691	15	20	0.75
0.6177	8	20	0.40
0.3309	5	20	0.25
0.1765	2	20	0.10

9.3.1 Logistic Regression Analysis

The logistic regression approach models the positive rate as a function of concentration. Often the following relationship is proposed (12, 13):

$$\log\left(\frac{p_A}{(1 - p_A)}\right) = \alpha + \beta \cdot \log(\lambda)$$

Where:

- p_A = positive rate for the AMM/RMM
- λ = average number or concentration of microorganisms in the test samples
- α = intercept
- β = slope indicating how quickly the positive rate changes with the changes in log concentration $\log(\lambda)$

The value $\log(p_A/(1 - p_A))$ is called the “logit” of positive rate p_A . The value $\exp\{-\alpha/\beta\}$ is the concentration at which a positive rate of 50% is obtained.

This model assumes that the probability of a presence test for a blank test sample is equal to zero, since it forces the positive rate to zero when no microorganisms are present in the test sample. Additionally, the positive rate as a function of the log concentration has a nice S-shape (blue curve on the right of **Figure 9.3.1-1**) but, as the function of the concentration, the positive rate quickly rises to a high probability and then flattens (blue curve on the left in **Figure 9.3.1-1**). This latter curve may be considered as a function of the negative rate (i.e., one minus the positive rate), because the slow increase with decreasing concentration and rapid increase when the concentration reaches zero seems like an exponential function. This alternative model may be formulated as:

$$\log(1 - p_A) = -\theta \cdot \lambda$$

Where:

- p_A = positive rate for the AMM/RMM
- θ = detection proportion
- λ = average number or concentration of microorganisms in the test samples

In this expression we only have a single parameter θ instead of the intercept α and slope β for the logistic regression analysis. In this exponential model, the positive rate would then be formulated as $p_A = 1 - \exp\{-\theta\lambda\}$, which resembles the function for the gMPN approach*. The red curves in **Figure 9.3.1-1** show this alternative model and seems very close to the two-parameter logistic curve.

***Note:** This gMPN-related curve has been shown in **Figure 9.2.2-1** as well.

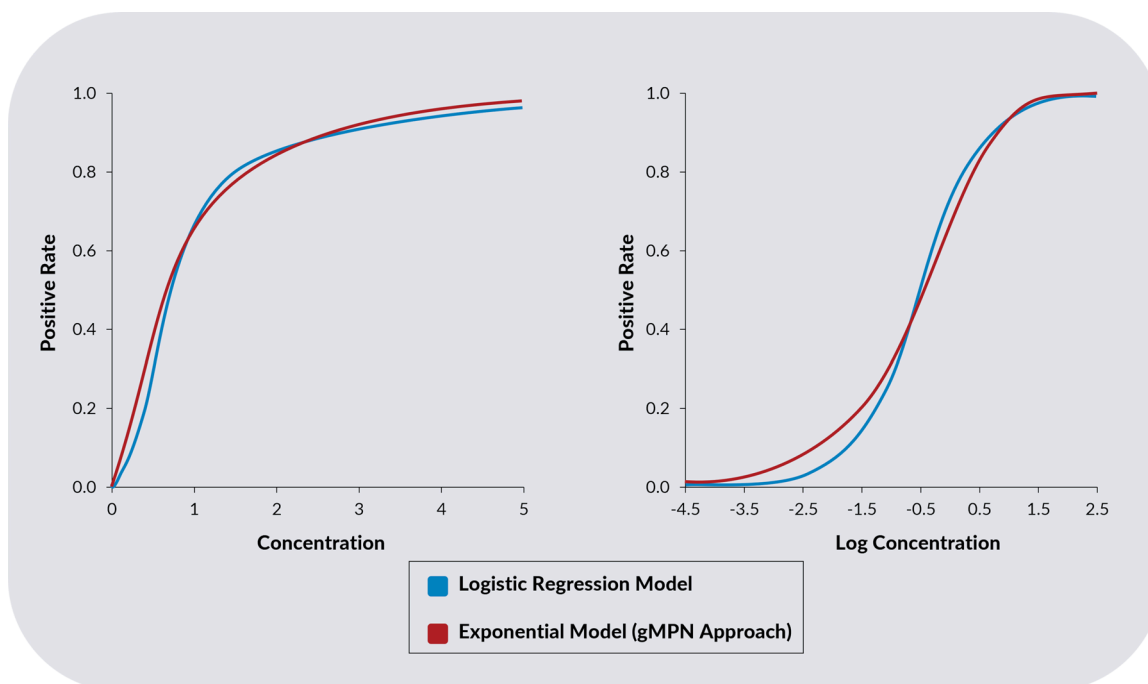


Figure 9.3.1-1 Visualization of the Positive Rate as Function of the Concentration (left) and Log Concentration (right)

Both curves in **Figure 9.3.1-1** can be estimated on multi-concentration data using logistic regression analysis programmed in general-purpose statistical software but fitting the curve related to the gMPN approach requires a deeper understanding of the options for the logistic regression function. Thus, the two-parameter logistic function is more straightforward in practice and, therefore, easier to fit with general-purpose statistical software. Nevertheless, the gMPN-related curve has only one parameter and is therefore a simpler model, and it has a direct connection to the MPN approach, making it easier to determine the LoD.

Fitting both curves to the data in **Table 9.3-1** results in the following parameter estimation with its asymptotic 95% confidence (an exact confidence interval does not exist):

Logistic curve: $\alpha = 0.707$ (0.140, 1.275) and $\beta = 1.746$ (1.081, 2.412)

gMPN-related curve: $\theta = 0.972$ (0.673, 1.000)

The curves with the estimated parameters were already visualized in **Figure 9.3.1-1**. If the LoD is the smallest number of organisms in a test sample that can be detected with at least 95% probability, the fitted curves can be used to establish these limits.

The average concentration in test samples that can be detected with exactly 95% probability are given by:

$$\text{Logistic curve: } \lambda_{\text{LoD}} = \exp\left(\frac{-\alpha}{\beta}\right) \cdot (19)^{\frac{1}{\beta}} = 3.60 \text{ (1.04, 6.16)}$$

$$\text{gMPN-related curve: } \lambda_{\text{LoD}} = \frac{-\log(0.05)}{\theta} = 3.08 \text{ (2.13, 4.03)}$$

These limits are not the LoD, but an average concentration in test samples, which is typically an overestimation of the LoD. An illustration of this is to assume that test samples with an average concentration of 3.6 CFU's per test sample have been created. If, additionally, the distribution of the number of microorganisms in the test samples equals a Poisson distribution (with mean 3.6), 2.73% of the test samples will be blank test samples and 97.27% of the test samples contain one or more microorganisms. Something similar has already been visualized in **Figure 9.2-1**, but with an average number of 2 CFUs per test sample. If, in addition, the AMM/RMM does not produce any false positives, all blank samples will be tested as absent, which is approximately 2.7%. To obtain a 95% probability for all test samples, the AMM/RMM should produce a presence test for almost all test samples that do contain at least one microorganism. Thus, the $\lambda_{\text{LoD}} = 3.60$ implies that the LoD is most likely equal to one. This conclusion is supported by the gMPN-related curve, because the parameter θ represents the detection probability, which is the probability of a positive test for a test sample containing a single microorganism. The parameter θ is estimated at $\theta = 0.972$, which is above 95%, indicating that the LoD is one. This can be confirmed by calculating the cumulative 5% (0.05) probability for a Poisson distribution plot with a mean value = 3.60 (i.e., adding the individual probabilities and reporting the first number in the plot that passes the cumulative 5% threshold).

9.3.2 Most-Probable Limit Analysis

The analyses so far have made use of statistical models where presence tests occur with a certain probability. Although these models are reasonable statistical models, the mechanism of detection may also be deterministic. This means that all test samples with a number of microorganisms below the LoD will be tested negatively with 100% certainty, while test samples containing a number of microorganisms equal to or greater than the LoD will test positive with 100% certainty. This model has been described in literature and has been used for estimation of the LoD of AMM/RMMs. It is referred to as the MPL and extends the MPN approach in a different way. The MPN approach assumes that the LoD is equal to one and then provides an estimate of the number of organisms in a suspension using a serial-dilution experiment. The MPL approach assumes that the LoD is equal to an unknown value L and estimates both the LoD and the number of microorganisms in the suspension (**14, 15**). The MPL approach is programmed in the specialized software TriMSA and can be programmed in SAS[®] and [R]. Applying the MPL approach to the data in **Table 9.2.5-4**, an estimate of the LoD of $L = 1$ with 95% confidence interval of [1, 2] and an estimate of the number of microorganisms in the initial suspension of MPN = 87 with 95% confidence interval of [56, 126] is obtained. If the LoD was equal to two, the MPN estimate would become MPN = 197 [160, 235]. The MPL approach applied to the data in **Table 9.2.5-4**, shows that the LoD is either one or two.

10.0 Appendix 2: Validation of Alternative and Rapid Microbiological Methods: Statistical Analysis-Quantitative Methods

As mentioned at the beginning of **Section 9.0** (Appendix 1), that section and **Section 10.0** (Appendix 2) provide several acceptable statistical analyses for the validation of alternative qualitative and quantitative microbiological methods, respectively, while **Section 11.0** (Appendix 3) focuses on additional considerations for non-CFU AMM/RMMs. The statistical analyses recommended in the body of *PDA Technical Report No. 33 (Revised 2026): Evaluation, Validation, and Implementation of Alternative and Rapid Microbiological Methods* will be restated with example data to illustrate how the analyses can be used. Although the example data is artificial, the data is based on real validation experiments previously conducted by the authors. While these methods are presented as examples, the stakeholder is responsible for selecting the most appropriate statistical analyses to use based on the data to be analyzed and the validation criteria being assessed.

Section 10.0 (Appendix 2) contains many formulas, **Table 10.0-1** provides a key of all the symbols used in the formulas and their meaning.

Note: The notation $\log$ will be used for the natural logarithm in **Section 10.0** (Appendix 2).

Table 10.0-1 Nomenclature for Symbols Used in Section 10.0 (Appendix 2)

Symbol	Meaning
Δ	The non-inferiority margin that is selected a priori to do a non-inferiority test. In some other texts the notation δ is used.
Φ	The standard normal distribution function.
λ	The mean number of microorganisms or the bacterial density in a set of test samples or in a suspension. We also use λ for a spiked level of microorganisms in an accuracy, linearity, or precision experiment. In some settings we may use an index to illustrate the purpose of the concentration level, e.g., λ_{LoQ} is the concentration at which the microbiological method produces an average count at the upper tail for blank samples.
θ	The parameter used to quantify the accuracy (often in terms of recovery) of the microbiological method. θ_A and θ_C are sometimes used as accuracy for the AMM/RMM and CMM against the spiked concentration level, while θ_A can also be the accuracy of the AMM/RMM against the CMM.

Symbol	Meaning
n	The number of samples tested typically for a single concentration with the microbiological method. n_A and n_C are the number of samples tested with the AMM/RMM and CMM, respectively. n can play the role of all samples tested, i.e., $n = n_A + n_C$ or represent n_A or n_C if we only do an analysis for a single microbiological method.
m	The number of runs in a precision experiment (per microbiological method).
y_i	A count for test sample i with the microbiological method. In some cases, the index i is eliminated to make notations clearer, but it still represents a count for a test sample. In other cases, multiple indices may be used to indicate test samples for different subgroups, e.g., y_{ij} is the count of test sample j for group (e.g., run) i .
$\bar{y}$	The mean count for a set of test samples at the same concentration level for the microbiological method, i.e., $\bar{y} = \sum_{i=1}^n (y_i/n)$. The means $\bar{y}_A$ and $\bar{y}_C$ are the mean counts for test samples tested with the AMM/RMM and CMM, respectively.
r_i	The recovery for test sample i , where the recovery is equal to the observed count y_i for the sample divided by the spiked concentration λ for the sample.
$\bar{r}_{\log}$	The mean value of the (natural) log-transformed recoveries, i.e., $\bar{r}_{\log} = \sum_{i=1}^n \log(r_i) / n$ and with n the number of samples tested.
z_i	Used as latent variable in a generalized linear mixed effects model for group i (e.g., run in a precision experiment).
β_0	The intercept in a (linear) concentration-recovery curve and the slope in a (linear or log-linear) concentration-response curve.
β_1	The slope in a concentration-recovery curve, but plays the role of intercept in the non-linear Mitscherlich function.
β_2	The “slope” in the non-linear Mitscherlich function, i.e., the coefficient of the power of the concentration level.
β_3	The power parameter in the non-linear Mitscherlich function.
α	The intercept in a (linear) concentration-response curve or in a Poisson regression analysis. α_A and α_C are the intercepts in the (linear) concentration-response curve obtained with the AMM/RMM and CMM, respectively.
β	The slope in a (linear) concentration-response curve or in a Poisson regression analysis. β_A and β_C are the slopes in the linear concentration-response curve obtained with the AMM/RMM and CMM, respectively.
γ	The coefficient for a quadratic concentration-response curve.
ε	The residual in a regression equation. It represents the unexplained variability from the regression equation that was fitted through the observed data (counts or recoveries). It is typically assumed that the residual follows a normal distribution with a variance that may depend on the concentration level.
μ	The notation for a mean count in the ANOVA approach for precision.

Symbol	Meaning
σ^2	A variance component and indicates the variation in counts. It is used for the variance of the residual and for variation sources in an ANOVA. We may use indices to indicate for what part of variation the variance belongs. For instance, it could represent a variance for the residuals at a specific concentration level, e.g., σ_0^2 is the residual variance for blank samples, or it could represent the variance within (σ_W^2) and between runs (σ_B^2).
τ^2	A variance component used to quantify the variability of the latent variable in a generalized linear mixed effects model that quantifies the between run variability in the logarithmic scale of a precision experiment.
η^2	The variance of the estimator for the total variability calculated from a one-way random effects ANOVA. η_A^2 and η_C^2 are the variances for AMM/RMM and CMM, respectively.
$m(\lambda)$	The concentration-response curve for the microbiological method in (spiked) concentration level λ . $m_A(\lambda)$ and $m_C(\lambda)$ are the concentration-response curves for the AMM/RMM and CMM, respectively.
$\tau(\lambda)$	The standard deviation of the log-transformed estimated concentration-response curve for the microbiological method at concentration level λ . $\tau_A(\lambda)$ and $\tau_C(\lambda)$ are the standard deviations for the log-transformed estimated concentration-response curves for the AMM/RMM and CMM, respectively.
SE	The abbreviation of standard error and represents the standard deviation of an estimate of a parameter. In some cases, an index is used to indicate the parameter of interest, e.g., SE_α the standard deviation of the estimate of the intercept α in a regression equation.
LCL	The lower confidence limit. Often there will be an index X attached to it, i.e., LCL_X , to indicate for what (validation) parameter or statistic the confidence limit is calculated.
UCL	The upper confidence limit. Often there will be an index X attached to it, i.e., UCL_X , to indicate for what (validation) parameter or statistic the confidence limit is calculated.
UQ	The abbreviation for upper quantile and represents a value in the upper tail of the distribution of observations, in particular the upper quantile for the distribution of counts of blank test samples.
MS	The abbreviation of mean squares used in ANOVA. We may use indices to identify for which factor the mean squares is calculated, e.g., MS_W and MS_B are the mean squares within and between the runs of a one-way ANOVA.
RSD	The abbreviation of relative standard deviation for enumeration of a microbiological method. We will use an index to indicate what relative standard deviation we are referring to, e.g., RSD_R and RSD_{IP} are the relative standard deviations for repeatability and intermediate precision of a microbiological method. To be able to indicate the measures of precision for AMM/RMM or CMM, we may write RSD_R^A or RSD_R^C for repeatability and RSD_{IP}^A or RSD_{IP}^C for intermediate precision. In case we want to express the relative standard deviation in percentages we use $RSD(\%)$.

10.1 Accuracy of Quantitative Microbiological Methods

When the quantitative or enumeration data of an AMM/RMM is analyzed, different ways of investigating the accuracy of the AMM/RMM exist. The three different approaches discussed in this section are typically related to the type of data-analytic assumptions one is willing to make.

The first assumption is related to the spiked concentration levels used in the accuracy experiment. An important question about the spiked concentration levels is whether they can be used as a known and precise concentration level, like that done for analytical measurement systems. If the spiked concentration levels are considered precise, the recovery of the microbiological method(s) could be investigated at each concentration level with respect to the spiked concentration levels (see **Section 10.1.1**). The recovery is here defined as the ratio between the observed count and the spiked concentration level. This approach is typical for analytical measurement systems, and also suitable for microbiological methods. However, in case the spiked concentration levels are considered less precise, a comparison in counts between the two microbiological methods may be more appropriate (also **Section 10.1.1**).

The second assumption is related to the count distribution for the replicated results. Often, it is appropriate to assume that these counts are from a Poisson distribution (see **Section 10.1.1**). Indeed, if samples from a larger suspension were collected and the microorganisms are randomly distributed throughout the full suspension, the number of organisms in the test samples may be considered Poisson-distributed (8). However, in practice, the replicated counts may sometimes show more variation than can be expected from the Poisson distribution (e.g., when microorganisms are clustered in the suspension). For Poisson data, the mean should be approximately equal to the variance. When more variation is present, this feature is referred to as *overdispersion*. In such cases, the use of a Poisson distribution should be avoided and an overdispersed Poisson distribution should be used on the counts directly, like the negative binomial distribution or, alternatively, a normal distribution typically on the logarithmically transformed counts should be used (see **Section 10.1.1**).

The logarithmic transformation on the counts is beneficial to correct for the skewness in the counts and make the transformed counts more normal. Counts may often show positive skewness, in particular at lower concentration levels where the tail of the distribution of the counts is longer on the right side of the distribution than on the left side of the distribution. At higher concentration levels, replicated counts (at the same concentration level) often show more symmetry in the distribution of the counts. **Figure 10.1-1** provides examples of two histograms with five hundred (500) replicated counts spiked at concentration levels 10 CFU and 50 CFU, respectively. The right tail of the histogram at 10 CFU (left histogram) is larger than the left tail, while the histogram at 50 CFU (right histogram) does not show this skewness.

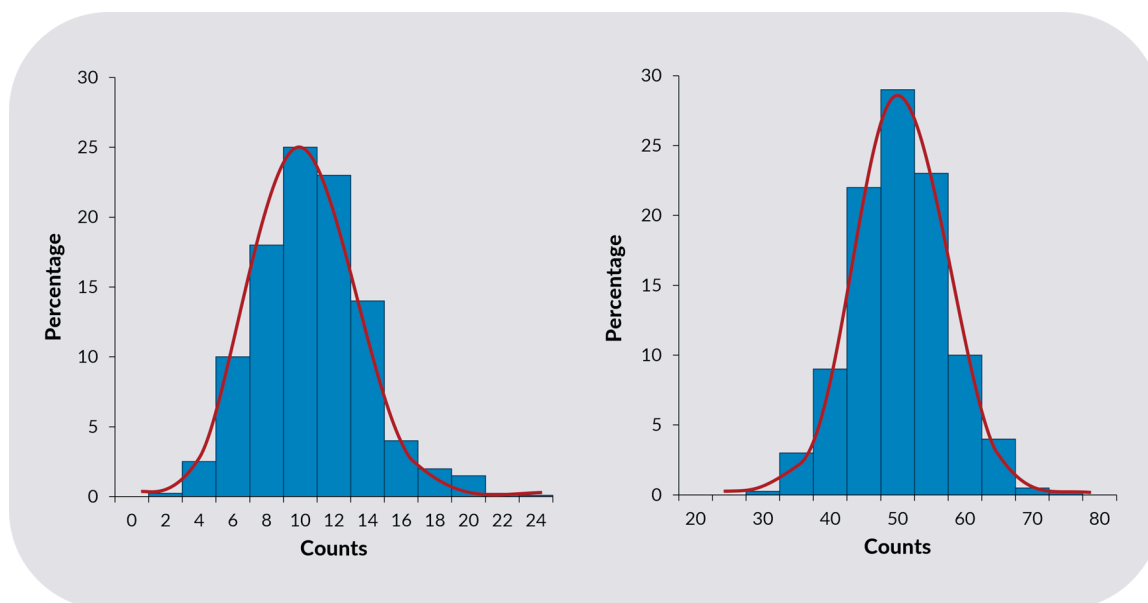


Figure 10.1-1 Visualization of Skewness for Counts at 10 CFU (left) and 50 CFU (right)

The third and final assumption is related to how the count data from all spiked concentrations could be analyzed simultaneously. In some cases, it may be beneficial to build a regression equation between the counts and the spiked concentrations to establish accuracy. Such an approach is also necessary for the validation parameter linearity. An advantage of using regression models is that they can determine the level of accuracy at concentration levels other than the spiked concentrations levels used in the experiment. Moreover, the regression models can be used to determine the set of concentration levels for which non-inferiority is appropriate between the two microbiological methods (see **Section 10.2**). However, this benefit would only be useful when the spiked levels are considered known and precise.

Some of the proposed analyses can be demonstrated by using the data of an accuracy experiment provided in **Table 10.1-1**. The experiment consists of six concentration levels (5, 25, 50, 100, 150, 250 CFU) for a specific microorganism for both the AMM/RMM and CMM. At each spiked concentration, five replicates were considered. **Table 10.1-1** also contains the mean count and the variance in counts for each concentration level and microbiological method.

Table 10.1-1 Counts for a Specific Microorganism for the Alternative/Rapid Microbiological Method and the Conventional Method at Different Concentration Levels

Replicate	AMM/RMM						CMM					
	5	25	50	100	150	250	5	25	50	100	150	250
1	6	20	42	72	138	213	6	19	47	91	118	201
2	3	14	52	77	120	197	4	20	42	86	130	194
3	4	22	34	83	110	209	6	24	46	103	139	203
4	8	24	39	80	119	193	8	21	53	84	112	209
5	3	19	44	69	121	163	5	25	36	69	136	200
Mean	4.8	19.8	42.2	76.2	121.6	195.0	5.8	21.8	44.8	86.6	127.0	201.4
Variance	4.7	14.2	44.2	32.7	103.3	388.0	2.2	6.7	39.7	151.3	135.0	29.3

10.1.1 Comparability of the Alternative/Rapid Microbiological Method for Each Concentration Against the Spiked Concentrations

Section 10.1.1 will demonstrate the normal and the Poisson distribution approach for accuracy of the AMM/RMM against the spiked concentration levels. These approaches make sense when the spiked concentrations have been determined reliably. Non-inferiority of the accuracy of the AMM/RMM is determined by investigating the *recoveries*. The recovery results are the observed counts divided by the spiked level at which the count was obtained and then multiplied by 100% to calculate the so-called “observed recoveries.” The ideal outcome of the accuracy experiment is to obtain 100% recovery for each sample.

10.1.1.1 Using the Normal Distribution Approach

The normal distribution approach makes sense when either the recoveries or the logarithmically transformed recoveries follow a normal distribution. The mean recovery per spiked concentration, together with its 90% confidence intervals (calculated directly from the replicated recoveries at the spiked concentration or through the inverse logarithm of the logarithmically transformed recoveries), is then established using the normal distribution. To establish non-inferiority at a spiked concentration, the lower 95% confidence limit should remain above the non-inferiority margin (e.g., 70%). The approach is illustrated here, in **Section 10.1.1.1**, on the logarithmically transformed recoveries. A standard fixed-effects ANOVA approach, where the recoveries from multiple spiked concentrations are used simultaneously, is *not recommended* for the accuracy of quantitative microbiological methods, although it is commonly used for analytical measurement systems. The reason is that the standard fixed-effects, one-way ANOVA approach assumes that the within-concentration variability across all spiked concentration levels is constant, that is, *homoscedastic* within-concentration variances, an aspect that is typically violated for microbiological methods. Microbiological

methods need to accommodate the presence of what is called heteroscedasticity—the within-concentration variance of the recoveries becomes smaller when the spiked concentration is increasing.

Using the normal distribution per spiked concentration may be appropriate when the recoveries or logarithmically transformed recoveries follow a normal distribution; when the counts (on which the recoveries are calculated) within each spiked concentration level shows overdispersion; and when the counts are non-zero. When assessing counts, it is often expected that some right skewness may be present (more variability to the right than to the left), and the normality for the recoveries may not hold. Therefore, using the normal distribution approach on the logarithmically transformed recoveries is recommended. Due to the presence of heteroscedasticity, the confidence interval per spiked concentration level is best calculated using the variability per concentration level. This analysis can be easily done using any general-purpose statistical software package. Following this, the calculated 90% confidence intervals in the log scale must be transformed back to the original scale, though this back-transformation to the original scale may need to be conducted manually. Although software packages can easily provide the results, the formulae are provided here:

The mean recovery for one of the spiked concentrations is obtained by:

$$\theta_A = \exp\{\bar{r}_{\log}\} = \exp\left\{\sum_{i=1}^{n_A} \left[\frac{\log(r_i)}{n_A}\right]\right\}$$

Where:

r_i = the recovery (i.e., the count divided by the spiked concentration level)

$\bar{r}_{\log}$ = mean of the logarithmically transformed recoveries

n_A = number of counts for the specific spiked concentration of the AMM/RMM

The lower 95% confidence limit is obtained by:

$$LCL = \theta_A \cdot \exp\left\{t_{n_A-1}^{-1}(0.95) \cdot \frac{s_{\log}}{\sqrt{n_A}}\right\},$$

with $s_{\log}$ the standard deviation on the log transformed recoveries defined by its variance:

$$s_{\log}^2 = \sum_{i=1}^{n_A} \left[\frac{(\log(r_i) - \bar{r}_{\log})^2}{(n_A - 1)}\right]$$

and with $t_{n-1}^{-1}(0.95)$ the 95% upper quantile of the t -distribution with $n_A - 1$ degrees of freedom. See the constants C_m in **Table 9.2.5-4**, where the number of species m in **Table 9.2.5-4** plays the role of n_A . In case sample size n_A is large, the value $t_{n_A-1}^{-1}(0.95)$ is approximately equal to 1.645.

The results of the approach with the normal distribution on the logarithmically transformed recoveries are presented in **Table 10.1.1.1-1** under the column heading “Normal Distribution.” The results show that non-inferiority (when comparing the observed counts to the spiked or expected counts) for a non-inferiority margin of 70% is achieved for the concentration levels 50, 100, 150, and 250 CFUs. Although the mean recovery is above 70% for the lower two concentrations (5 and 25 CFU), the amount of variability in relation to the number of replicates is not small enough. In this instance, if more test samples had been tested at these two lower concentrations, non-inferiority may have been achieved as well, but with the

current data collected, inferiority at 70% recovery cannot be rejected. Conversely, when more replicates are conducted, the mean recovery may change as well, and it would not be certain that non-inferiority can be reached at those lower concentrations.

Table 10.1.1.1-1 Comparability of the Alternative/Rapid Microbiological Method on Recovery (%) per Concentration Level Using Different Data Analytic Approaches

Spike	Normal Distribution		One-Way ANOVA		Poisson Distribution	
	Mean % Recovery	LCL	Mean % Recovery	LCL	Mean % Recovery	LCL
5	88.8	58.7	88.8	75.3	96.0	68.6
25	77.9	64.1	77.9	66.1	79.2	67.1
50	83.6	72.0	83.6	70.9	84.4	75.4
100	76.0	70.7	76.0	64.5	76.2	70.0
150	80.8	74.8	80.8	68.5	81.1	75.8
250	77.7	70.2	77.7	65.8	78.0	74.0

Table 10.1.1.1-1 also contains the results from a standard fixed-effects, one-way ANOVA analysis on the logarithmically transformed recoveries, where the variability within concentration is pooled, to show that such an analysis approach is less appropriate. For the one-way ANOVA approach, the non-inferiority is achieved at 5 CFU and 50 CFU, but not at the largest three concentrations where the normal distribution approach showed non-inferiority. The pooled variance for the within-concentration variability in the ANOVA approach is too small for the lower concentrations and too large for the higher concentrations, making the results from the one-way ANOVA analysis unreliable.

10.1.1.2 Using the Poisson Distribution Approach

An alternative approach, which would make use of the Poisson distribution, is to first calculate the mean counts and a lower 95% confidence limit for the mean counts using the Poisson distribution, and then changing these values to recoveries by dividing the mean count and its lower 95% confidence limit by the spiked concentration levels. The asymptotic or approximate lower 95% lower confidence limit on the mean count that makes use of the Poisson distribution is equal to:

$$LCL = \bar{y}_A \cdot \exp \left\{ -1.645 \cdot \sqrt{\frac{1}{(n_A \cdot \bar{y}_A)}} \right\}$$

Where:

$\bar{y}_A$ = mean count at a single concentration

n_A = number of counts for the AMM/RMM at the same concentration

To determine recoveries with the Poisson approach, the mean count $\bar{y}_A$ and the *LCL* are both divided by the spiked concentration level.

This analysis can easily be programmed with MS Excel but can also be done using any general-purpose statistical software where a Poisson regression analysis is available. When using statistical software, the counts of the AMM/RMM are the dependent variable, and the spiked concentration levels are the independent variable that should be accounted for in the analysis as the categorical variable. The default link function, which is the logarithmic link function, should be used. Statistical software requires a procedure that can fit a generalized linear model. In SAS®, this is done with the procedure GENMOD, but in Minitab the operation is accomplished via the “Stat > Regression > Poisson Regression > Fit Poisson Model” Function. Note that the confidence limits are approximate or asymptotic and not exact confidence limits.*

***Note:** For data analysts experienced in Poisson regression, the recoveries could also be obtained directly from the Poisson regression analysis by including what is called an “offset” parameter. If the logarithmically transformed spiked concentration level is chosen as the offset, the mean counts are expressed as a ratio of the spiked concentration level, which are then the mean recoveries.

The output of a Poisson analysis as described above is provided in **Table 10.1.1.2-1**. It provides an intercept which represents the expected count at (in this case) the highest concentration level, but on the logarithmic scale. Transforming this back to the original scale, an expected count of 195 is obtained, which is the average count provided in **Table 10.1.1-1** for the 250 CFU concentration for the AMM/RMM. The other coefficients represent the difference in expected counts between the highest concentration and the other concentrations in the logarithmic scale. Thus, the difference in counts between the concentrations of 5 and 250 CFUs is -3.7044 in the log scale, which means that the mean count at 5 CFU is equal to 2.46% of the mean count in the 250 CFU concentration. This can also be determined from **Table 10.1.1-1**, because 2.46% is equal to $100\% \times 4.8/195$. The *p*-value for this difference (which is <0.001) indicates that the equality in the mean count between the two spiked concentrations is rejected at the significance level of $\alpha = 0.05$.

Table 10.1.1.2-1 Output of Poisson Analysis for all Concentration Levels

Effect	Estimate	SE	95% CI	Chi-Square	p-Value
Intercept	5.2730	0.0320	(5.2102 ,5.3358)	27109.4	<0.001
Conc. 5	-3.7044	0.2066	(-4.1093, -3.2994)	321.43	<0.001
Conc. 25	-2.2873	0.1055	(-2.4941, -2.0806)	470.21	<0.001
Conc. 50	-1.5306	0.0759	(-1.6794, -1.3818)	406.36	<0.001
Conc. 100	-0.9396	0.0604	(-1.0581, -0.8212)	241.88	<0.001
Conc. 150	-0.4723	0.0517	(-0.5735, -0.3710)	83.52	<0.001
Conc. 250	0.0000	0.0000	(0.0000, 0.0000)	-	<0.001

The recovery estimates from this Poisson analysis can be calculated with general-purpose statistical software, but the instructions will depend on the package used. For instance, in SAS®, the option

“LSMEANS” may be used with the procedure instructed to use exponentiated means with a significance level of 0.1. On the other hand, in Minitab, after the Poisson regression analysis was conducted, the Predict option from the Poisson regression analysis must first be used to obtain predicted counts and lower 95% confidence limits at each spiked concentration level. Then these predicted values and lower limits must be divided by the spiked concentration level. The results are presented in **Table 10.1.1-1**.

When comparing the results from the normal and Poisson distribution approaches, the mean recoveries deviate from the normal distribution approach (and the fixed-effects, one-way ANOVA approach) on the logarithmically transformed recoveries because the Poisson distribution is based on the arithmetic mean of the recoveries, while the normal distribution is based on the geometric mean of the observed recoveries.

Nevertheless, the lower confidence limits for the normal distribution and the Poisson distribution are relatively close, and they lead to the same conclusion on non-inferiority. Using the results from **Table 10.1.1-1**, non-inferiority at the margin of 70% is achieved for concentrations of 50, 100, 150, and 250 CFUs, but not for 5 CFU and 25 CFU, because the LCL is lower than 70%.

10.1.1.3 Lack-of-Fit Analyses

As discussed in **Section 10.1.1.1**, the one-way ANOVA method is not reliable, and either the Poisson distribution or the normal distribution approach should be considered for non-inferiority testing. While it may not be easy to decide between these two analyses, the Poisson distribution would be preferred over the normal distribution approach when the number of microorganisms in the test samples follow the Poisson distribution. But often there may be violations (e.g., clustering of microorganisms), and these violations would typically increase the variability, so the confidence limits from the Poisson distribution will then be too narrow. Thus, when the Poisson distribution is violated, due to overdispersion, it is better to use an alternative distribution than the Poisson distribution, and the normal distribution approach may be a suitable alternative.

Demonstrating that the distribution of the counts violates the Poisson distribution is not straightforward. Alternatively, checking the residuals for the normal distribution on the recoveries or the log-transformed recoveries is not recommended due to the heterogeneity in variability across spiked concentrations. Moreover, an evaluation of normality per concentration is not appropriate either due to the small number of replicates.

One way of investigating the violation of the Poisson distribution is to evaluate overdispersion. This lack-of-fit test may be done qualitatively, by investigating the ratio of the variance/mean per concentration. This ratio should be approximately one (1) when no overdispersion is present, while the presence of overdispersion would make this ratio larger than one. **Table 10.1.1-1** also shows the variances per concentration, and the only concentration that is questionable is the highest concentration for the AMM/RMM, since the ratio is around the value of two (2). This larger variation is most likely due to the outlying value of 163 counts for replicate #5. Thus, taking all of the concentrations into consideration, this qualitative analysis does not indicate strong evidence for overdispersion.

A more formal analysis is to rerun the Poisson regression analysis with a negative binomial distribution. The negative binomial distribution can be viewed as a Poisson distribution with additional variability; it estimates an additional overdispersion parameter. Thus, comparing these two regression models on their fit-to-the-count data investigates whether a violation of the Poisson distribution is related to overdispersion. The two regression models can be compared using Akaike’s corrected information criterion (AICC). Essentially, AICC describes the trade-off between the goodness of fit of a statistical model and the

simplicity of the model, and it is typically useful in comparing different statistical models. The AICC value on its own does not provide relevant information rather it is typically provided by the general-purpose software in the output of the Poisson and negative-binomial regression analyses. Thus, for statistical software programs with the capability of performing both a Poisson and negative-binomial regression analysis (like SAS®), the results of the AMM/RMM in **Table 10.1.1-1** shows an AICC of 208.2 for the Poisson regression analysis and 211.6 for the negative-binomial regression analysis. Since the Poisson regression analysis has a smaller AICC, the Poisson model is considered more appropriate than the negative-binomial.

Some statistical programs cannot extend the Poisson distribution to the negative-binomial distribution in the regression analysis for counts, but they may provide generic tests for the lack-of-fit of the Poisson distribution, like the deviance goodness-of-fit test or the Pearson chi-square test. The deviance test compares the “selected model” to the data against the “saturated Poisson model” (when a Poisson distribution is used). The saturated Poisson model for the current data is the model in which it is assumed that the count at each concentration comes from the Poisson distribution. The Pearson goodness-of-fit test investigates how well the predicted counts resemble the observed counts. These goodness-of-fit tests come with a *p*-value. For the AMM/RMM counts in **Table 10.1.1-1**, the *p*-value of the deviance test is equal to 0.452 and, for the Pearson’s goodness-of-fit test, the *p*-value is equal to 0.459. The results show that a violation of the Poisson distribution was not demonstrated at a significance level of $\alpha = 0.05$.

10.1.1.4 Sample Size Considerations

An appropriate number of replicates or sample size for each concentration can be determined when the logarithmically transformed counts follow the normal distribution with a variance that is equal to the mean. The sample size depends on the concentration level, and it is typically larger when the concentration is lower. **Table 10.1.1.4-1** shows the sample sizes for different concentration levels under different truths (i.e., the true recovery rate) when the non-inferiority margin is equal to 70% (for a significance level of $\alpha = 0.05$ and a power of $1 - \beta = 0.8$). Here the truth is determined by the true recovery being 100%, 90%, or 80%. The lower the true recovery of the AMM/RMM, the more samples must be determined to demonstrate it is away from the 70% margin. Never using less than three replicates is recommended, so the lowest sample size has been set to three (3).

Table 10.1.1.4-1 Sample Sizes for Non-Inferiority per Concentration at a Non-Inferiority Margin of 70% Using the Normal Distribution

True Recovery	Spiked Concentration Level											
	5	10	15	20	25	50	75	100	150	200	250	300
100%	20	10	7	5	4	3	3	3	3	3	3	3
90%	42	21	14	11	9	5	3	3	3	3	3	3
80%	157	79	53	40	32	16	11	8	6	4	4	3

To calculate the values in **Table 10.1.1.4-1** or to calculate other settings, the smallest sample size n_A (>2) that provides a value of 0.8 for the following probability is the sample size that can be used:

$$\text{Power} = 1 - \Phi \left(\Phi^{-1}(0.95) + \frac{\log(\Delta) - \log(\theta)}{\sqrt{\frac{\lambda^{-1} + (\theta\lambda)^{-1}}{n_A}}} \right)$$

Where:

- Φ = cumulative standard normal distribution function
- Φ^{-1} = quantile of the standard normal distribution function
- n_A = number of test samples
- θ = true recovery (not expressed in percentages)
- λ = spiked concentration level
- Δ = non-inferiority margin (e.g., 0.7)

10.1.2 Comparability of the Conventional Microbiological Method per Concentration Against the Spike Concentrations

When a Poisson distribution (and normal distribution) approach is used for accuracy, the analysis on the recoveries of the AMM/RMM with respect to the spiked concentration levels in **Section 10.1.1** demonstrates that inferiority is not rejected for the concentrations of 5 CFU and 25 CFU. However, the CMM also shows somewhat lower recoveries compared to what has been considered the true spiked concentration level, except for the lowest concentration level. Hypothetically, we expect the recovery for the CMM to be 100% for each spiked concentration level when using the conventional/compendial method. In practice, it is understood there will always be some variability in microbiological testing, and it may be appropriate to investigate the accuracy of the CMM against the spiked concentration levels.

Using the Poisson distribution on the counts of the CMM, the mean recoveries with their lower 95% confidence limits are presented in **Table 10.1.2-1**. For the CMM, inferiority is rejected for all concentrations at the non-inferiority margin of 70%, but the estimated recoveries are still lower than 100% (when compared with the spike levels) for the concentrations 25 CFU and beyond. Testing the traditional null hypothesis H_0 : Recovery = 100% for the CMM on each spiked concentration level separately using the Poisson regression analysis, shows that the null hypothesis is rejected for the three concentration levels 100, 150, and 250 CFUs, see the column “ p -Value*” in **Table 10.1.2-1**.

***Note:** To obtain this p -value, the following p -value could be programmed (into MS Excel, for instance) using the following formula:

$$p = \begin{cases} 2 * \left(1 - \Phi \left(\frac{\log \left(\frac{\bar{y}_C}{\lambda} \right)}{\sqrt{\frac{1}{n_C \bar{y}_C}}} \right) \right) & \text{if } \bar{y}_C > \lambda \\ 2 * \Phi \left(\frac{\log \left(\frac{\bar{y}_C}{\lambda} \right)}{\sqrt{\frac{1}{n_C \bar{y}_C}}} \right) & \text{if } \bar{y}_C \leq \lambda \end{cases}$$

Where:

$\bar{y}_C$ = mean count of the CMM at the spiked concentration

λ = spiked concentration level

n_C = number of replicates

Table 10.1.2-1 Non-Inferiority of the Conventional Method on Recovery per Concentration Level Using the Poisson Distribution

Spike	Mean % Recovery	LCL	p-Value
5	116.0	85.5	0.424
25	87.2	74.5	0.153
50	89.6	80.3	0.100
100	86.6	80.0	0.003
150	84.7	79.3	<0.001
250	80.6	76.5	<0.001

Alternatively, these p -values may be obtained from the Poisson regression analysis where the logarithmically transformed concentration is used as an offset parameter and where an LSMEANS statement is used to obtain the predicted recoveries for each spiked concentration (this is possible with SAS[®] but not with Minitab). Using this offset parameter normalizes the counts to rates expressed as ratios of the spiked concentration.

When both the CMM and the AMM/RMM show lower recoveries than 100%, it may be a sign that the spiked concentrations are not very reliable (i.e., there is too much variability). Thus, it may be better to compare the AMM/RMM against the CMM to see if the AMM/RMM is non-inferior to the CMM when it comes to enumeration. Clearly, for accuracy, it would be best to compare the AMM/RMM against the CMM, but when the CMM shows a recovery of 100% for each spiked concentration level, a comparison of the AMM/RMM against the spiked concentration levels could be more powerful since only the variability in counts of the AMM/RMM are incorporated in the non-inferiority test.

10.1.3 Comparability of the Alternative/Rapid Microbiological Method with Respect to the Conventional Method per Concentration

A comparison can be conducted using a Poisson regression analysis again using the logarithmic link function, where the variables “method” and “concentration” are included as categorical variables and where the interaction between method and concentration level is considered. The inclusion of the interaction term makes it possible to investigate differences between the two methods for each spiked concentration separately. Differences between the mean counts of the AMM/RMM and CMM per concentration can only be obtained with general-purpose statistical software that can calculate least-squares means (as already applied in **Section 10.1.1** for Poisson regression). SAS® and [R] have this option, but Minitab does not have this option for Poisson regression. When the logarithmic link function is applied, differences in means between the AMM/RMM and CMM calculated with the least-squares means are determined in the logarithmic scale and they must be exponentiated. These exponentiated mean differences can then be viewed as a recovery of the AMM/RMM compared to the CMM (instead of a recovery against the spiked concentrations, as previously described for each method in **Section 10.1.2**). **Table 10.1.3-1** presents the recoveries of the AMM/RMM against the CMM and the lower 95% confidence limit. Although they were obtained with the Poisson regression analysis of the counts of AMM/RMM and CMM, the recoveries are just the ratios of the two means of the two microbiological methods multiplied by 100%. For example, using the data in **Table 10.1-1**, the recovery of 82.8% for concentration 5 CFU is obtained by multiplying the ratio 4.8/5.8 by 100%. The calculation of the lower 95% confidence limit is more difficult, but can be calculated by the following expression:

$$\left(\frac{\bar{x}_A}{\bar{x}_C}\right) \cdot \exp \left\{ -1.645 \cdot \sqrt{\frac{1}{(n_A \cdot \bar{x}_A)} + \frac{1}{(n_C \cdot \bar{x}_C)}} \right\}$$

Where:

$\bar{x}_A$ = average count for the AMM/RMM

$\bar{x}_C$ = average count for the CMM

n_A = number of samples tested with AMM/RMM

n_C = number of samples tested with CMM

The results show that the null hypothesis of inferiority of the AMM/RMM compared to the CMM is rejected for all spiked concentrations in the experiment, when a non-inferiority margin of 70% is being used, except for the spiked concentration at 5 CFU.

Looking at the recoveries of AMM/RMM versus CMM in **Table 10.1.3-1**, all the values are close to each other (around 90%) which may suggest that the recovery is constant across the spiked concentrations and a pooled recovery can also be obtained. Pooling has the advantage that the power for the non-inferiority test can be increased, because all counts are now used to calculate one recovery value. The lack of sample size for non-inferiority at the lower spiked concentrations is compensated for by the other concentrations, if it can be assumed that the recovery is constant across spiked concentrations (see **Table 10.1.1-2**). However, pooling may have a disadvantage in that it can mask inferior results for one (1) concentration when all of the other concentrations non-inferior results (similar to the discussion of pooling accuracy from multiple microorganisms in **Section 9.0**). The lower confidence limit at the lower spiked concentration may just indicate that the recovery is much lower than the non-inferiority margin, and pooling may be considered as “testing into compliance.” Thus, pooling should not be conducted automatically and should be substantiated by microbiological arguments and additional statistical analysis.

Table 10.1.3-1 Non-Inferiority of Alternative/Rapid Microbiological Method versus the Conventional Method per Concentration Level Using the Poisson Distribution

Spike	Mean % Recovery	LCL
5	82.8	52.6
25	90.8	72.3
50	94.2	80.5
100	88.0	78.4
150	95.8	87.2
250	96.8	89.9

To demonstrate the estimation of the pooled recovery, a Poisson regression analysis can be used again, where concentration and method can be used in the analysis as categorical variables, but now *not* incorporating the interaction term. The interaction term makes the recoveries different across the spiked concentration levels, but now the goal is to obtain a single-recovery estimate. The output of this Poisson analysis is provided in **Table 10.2.3-2**.

Using the results from the output, the pooled percent recovery is then estimated at:

94.3% ($= 100 \cdot \exp(-0.0587)$) with a 95% lower confidence limit (or the lower boundary of the 90% confidence interval) equal to 89.9% ($= 100 \cdot \exp(-0.1066)$).

Thus, when the recovery of the AMM/RMM compared to the CMM is assumed to be homogeneous across spiked concentration levels, inferiority is rejected for all concentrations levels between 5 CFU and 250 CFU at the non-inferiority margin of 70%. Thus, an investigation of constant accuracy across the concentration levels may be beneficial to achieve a wider range of concentrations for which accuracy would be satisfied. Note that the *p*-value of 0.0434 in the output of the Poisson analysis demonstrates that the AMM/RMM does enumerate fewer microorganisms than the CMM (the null hypothesis of equal means is rejected at significance level 0.05), but the accuracy is still far above the non-inferiority margin.

Table 10.1.3-2 Poisson Analysis Output

Effect	Estimate	SE	90% CI	Chi-Square	p-Value
Intercept	5.3182	0.0265	(5.2746, 5.3618)	40193.0	<0.001
Method: AMM/RMM	-0.0587	0.0291	(-0.1066, -0.0109)	4.08	0.0434
Method: CMM	0	0	(0.0000, 0.0000)	-	-
Conc. 5	-3.6216	0.1392	(-3.8505, -3.3926)	677.03	<0.001
Conc. 25	-2.2534	0.0729	(-2.374, -2.1344)	956.66	<0.001
Conc. 50	-1.5168	0.0529	(-1.6036, -1.4294)	820.37	<0.001
Conc. 100	-0.8899	0.0416	(-0.9584, -0.8214)	456.96	<0.001
Conc. 150	-0.4666	0.0362	(-0.5261, -0.4071)	166.30	<0.001
Conc. 250	0.0000	0.0000	(0.0000, 0.0000)	-	-

Checking the assumption of a constant difference between the AMM/RMM and CMM statistically, the AICC can be used to determine which Poisson regression model describes the data best, that is, the Poisson model with a recovery per spiked concentration or the Poisson model with a constant recovery across spiked concentrations. The AICC value for the Poisson model with a constant recovery across the spiked concentrations is equal to 400.0, while the AICC for the Poisson model with a recovery estimate per spiked concentration is 412.8. Thus, the AICC (smaller is better) suggests that the recovery is constant across the concentration levels. If no other substantiated knowledge contradicts this result, the pooled analysis is appropriate to use, and a calculation per concentration is not necessary. Then, non-inferiority can be claimed for all concentrations levels, including the 5 CFU spike.

10.2 Linearity and Accuracy

For analytical measurement systems, linearity is an important validation parameter, because the analytical measurement system makes use of a reference sample in routine analysis to determine the concentration value of an unknown sample. The reference and unknown sample are being tested in one analytical run and, for each run, a run-specific single-point calibration line is created that goes through the origin and through the response of the reference sample (the green line and dot in **Figure 10.2-1**). Since the true concentration value of the reference sample is known (x_R), the observed concentration value of the unknown sample is obtained by the concentration of the calibration line (x_U) that matches the value of the line to the response of the unknown sample (the red arrows and dot in **Figure 10.2-1**). To be able to use such a procedure in routine analysis for analytical measurement systems, it is imperative that the concentration-response curve is linear and goes through the origin. Thus, for single-point calibration models for analytical measurement systems what is needed is *proportionality*.

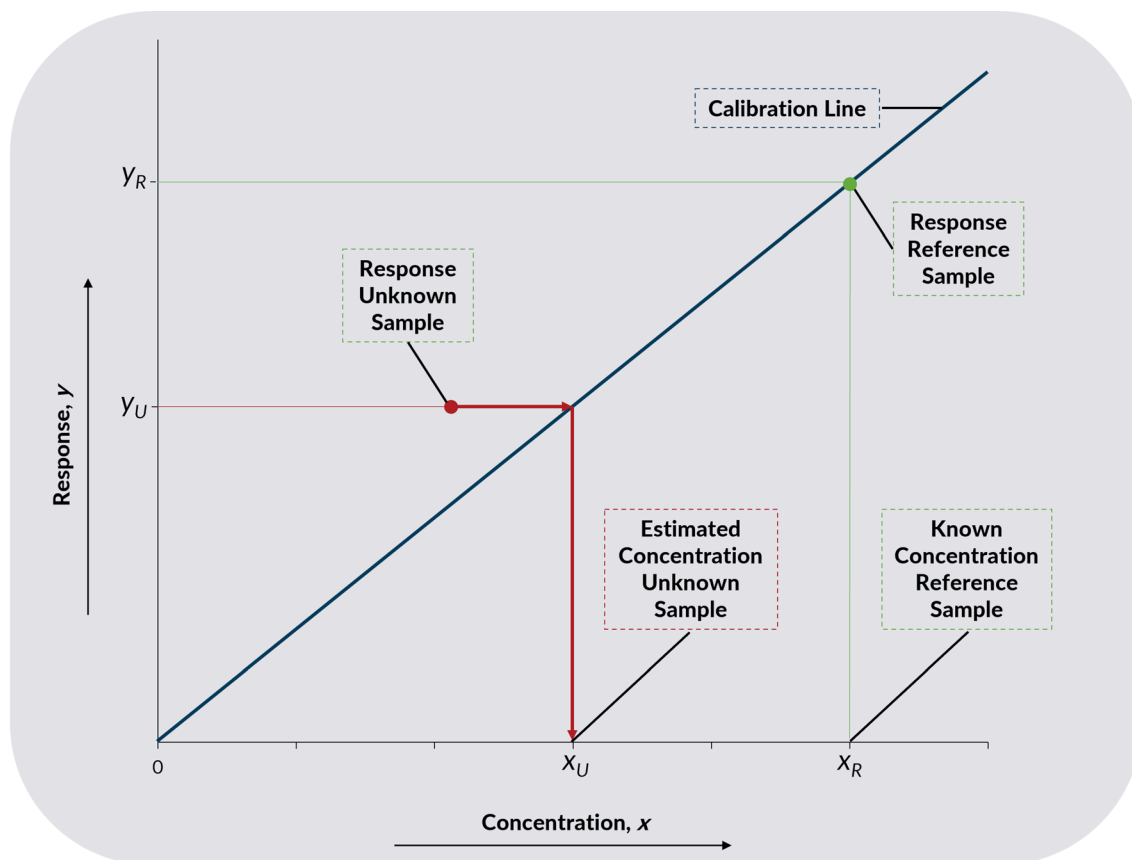


Figure 10.2-1 Visualization of the Estimated Concentration for an Unknown Sample Using a Linear Calibration Line for the Reference Sample in Analytical Measurement Systems

In microbiology, linearity has a different purpose, and should support the validation parameter accuracy as conducted in **Section 10.1**. If accuracy is achieved at each concentration level that has been tested in an accuracy experiment (**Sections 10.1.1, 10.1.2, and 10.1.3**), that accuracy only applies for the tested concentration levels and cannot automatically be guaranteed (and may fail) at non-tested concentration levels within the full interval. In order to extend the accuracy at specific concentration levels to all non-tested concentration levels within the spiked interval, a linear or, more precisely, a monotonic concentration-recovery curve is required. This operation is a smooth concentration curve that is nondecreasing or nonincreasing over the interval of interest where the recovery is defined as the mean count of the AMM/RMM divided by the true concentration level (often the spiked concentration level or a precise estimate of the concentration level with the CMM) at which the mean count is observed. A linear or monotonic concentration-recovery curve guarantees that the worst bias or lack of accuracy is attained at the two outer concentration levels in the experiment (the green dots and green line in **Figure 10.2-2**). The ideal recovery for the AMM/RMM is 100%, or more if non-CFU signals are generated in the AMM/RMM. Therefore, the worst accuracy for the green recovery results in **Figure 10.2-2** occurs at the last (or sixth) spiked concentration, as it is the lowest recovery across the range of concentration levels in the interval of the experiment (the green dots and green line in **Figure 10.2-2**).

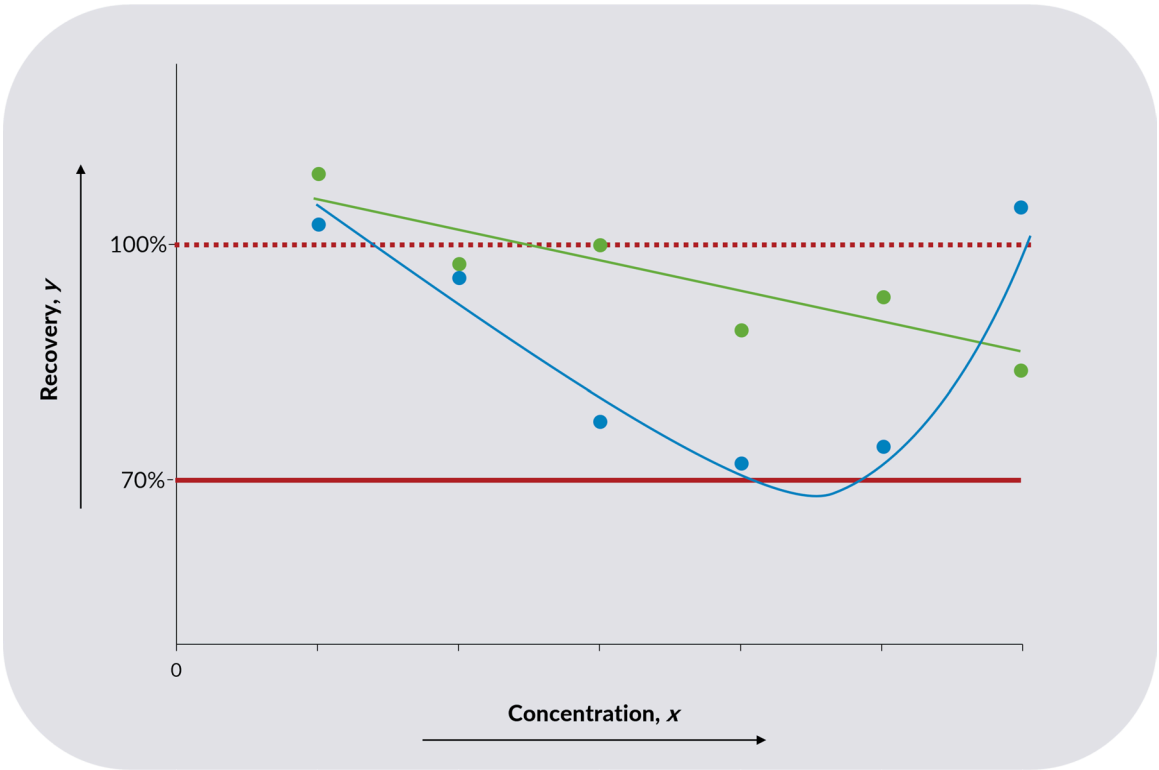


Figure 10.2-2 Visualization of Results from a Linear Concentration Response Curve (Green) and Results from a Non-linear Concentration Response Curve (Blue) Expressed in the Spiked Concentration and its Consequence for Accuracy

The blue dots in **Figure 10.2-2**, on the other hand, show that accuracy may potentially hold for each spiked concentration level in the experiment separately. In this case, the blue dots (the mean recoveries) are all away from the non-inferiority margin of 70%, even though some results are still close to this boundary. Following the expected recovery of the blue nonlinear concentration-recovery curve in **Figure 10.2-2**, however, the accuracy would most likely fail at specific untested concentration levels between the fourth and fifth spiked concentration levels. Indeed, the expected recovery reaches below the non-inferiority margin of 70%. Thus, without linearity or monotonicity of the concentration-recovery curve, there is no guarantee that accuracy holds across the full range of spiked concentration levels even if it has been demonstrated that non-inferiority holds at all the spiked concentration levels in the accuracy experiment.

To determine linearity for the concentration-recovery curve, the following mathematical formula is used:

$$\frac{\mathbb{E}(y|\lambda)}{\lambda} = \beta_0 + \beta_1\lambda$$

Where:

- $E(y|\lambda)$ = expected concentration of the AMM/RMM of an arbitrary test sample that is tested at the true concentration level λ
- λ = true concentration level
- $\mathbb{E}(y|\lambda)/\lambda$ = would represent the recovery (not expressed in percentages)
- β_0 = intercept of the linear concentration-recovery curve
- β_1 = slope of the linear concentration-recovery curve

The expected concentration $\mathbb{E}(y|\lambda)$ would be estimated with the mean counts at the spiked concentration level observed in the experiment, and the true concentration level would either be estimated by a precise estimate from the CMM* or taken by the spiked concentration level. If a stricter assumption is made and it is assumed that the slope β_1 is equal to zero, the concentration-recovery curve becomes a horizontal line, and the intercept represents the expected recovery for all concentration levels. This was demonstrated in **Section 10.1.3** when the recovery of the AMM/RMM against the CMM was constant. Under this stricter assumption, the concentration-response curve for the AMM/RMM is then proportional to the true concentration level λ , that is, $\mathbb{E}(y|\lambda) = \beta_0\lambda$ as was discussed regarding the single-point calibration model of an analytical measurement system. Thus, if proportionality can be shown for the concentration-response curve, this demonstrates a more defined linearity needed to support that accuracy has been demonstrated across the full interval of spike levels.

***Note:** In many linearity experiments, the number of replicates taken at each concentration for the CMM is often too small to be able to obtain a precise estimate of the true concentration (too much variability), and a regression analysis where the mean counts of the CMM are used as independent variables and the mean counts of the AMM/RMM are considered the dependent variable may lead to incorrect estimates of the intercept and slope of the regression line.

In microbiology, however, it is common to demonstrate a linear concentration-response curve, that is, $\mathbb{E}(y|\lambda) = \alpha + \beta_0\lambda$, where the mean counts of the AMM/RMM are modeled as a function of the true concentration. Unfortunately, this concentration-response curve does not lead to a linear concentration-recovery curve, since the concentration-recovery curve becomes $\mathbb{E}(y|\lambda)/\lambda = \beta_0 + \alpha/\lambda$ which is no longer a linear function of the true concentration λ . Demonstrating a linear concentration-response curve is still appropriate, however, because the concentration-recovery function remains monotonic (i.e., it is a non-increasing function when α is positive), which is necessary for the concentration-recovery curve to support the validation parameter accuracy. Additionally, demonstrating a log-linear concentration-response curve (i.e., $\log(\mathbb{E}(y|\lambda)) = \alpha + \beta_0\log(\lambda)$ with $\log$ the natural logarithm) also leads to a monotonic concentration-recovery curve: $\mathbb{E}(y|\lambda)/\lambda = \exp\{\alpha\} \cdot \lambda^{\beta_0-1}$.

Thus, when it can be shown that the concentration response curve is linear or even log-linear, there is no need to evaluate the concentration-recovery curve due to the explanation given here. However, when other forms of concentration response curves, which are discussed in the text above, are identified, the concentration-recovery curve may need to be studied directly to be able to support the validation parameter accuracy (see **Section 10.2.3**).

Using the data from **Table 10.1-1** to illustrate the evaluation of concentration-response and concentration-recovery curves, visualizing the mean counts against the spiked concentration levels in **Figure 10.2-3A** and **Figure 10.2-3B** will help evaluate the possible choices in curves. The mean counts have been linearly connected without building a regression equation at this point. Qualitatively, the concentration-response curves (top two figures) show what would likely be a linear function of the spiked concentration level, but the concentration-recovery curves may deviate from linearity, mostly because the results at the lowest concentration level seem to be somewhat higher than what can be expected from linearity of the other concentrations. However, the variation in the replicated recoveries at the lowest concentration level is also relatively large, which makes this determination difficult. Finally, it can also be seen that the variability in the replicated counts is nonconstant across the spiked concentrations levels (heteroscedasticity), and this should be addressed when an appropriate regression equation is being constructed from the observed data.

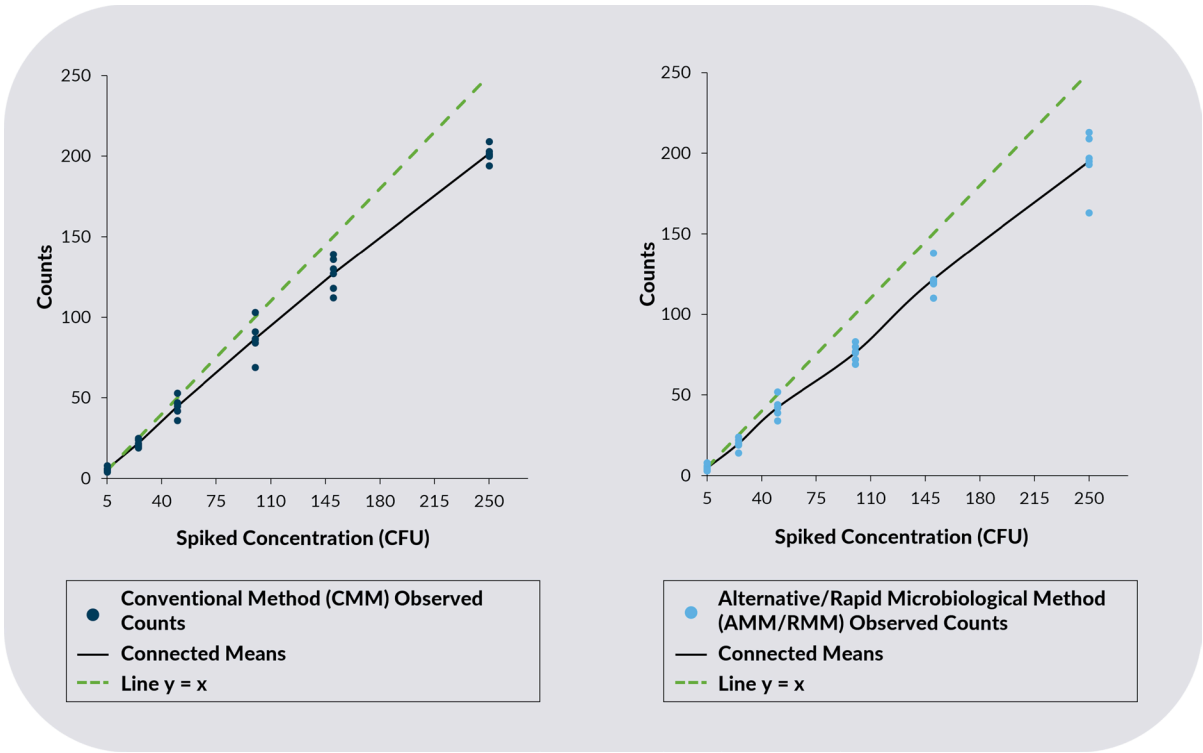


Figure 10.2-3A Visualization of the Mean Counts with Respect to the Spiked Concentration Levels

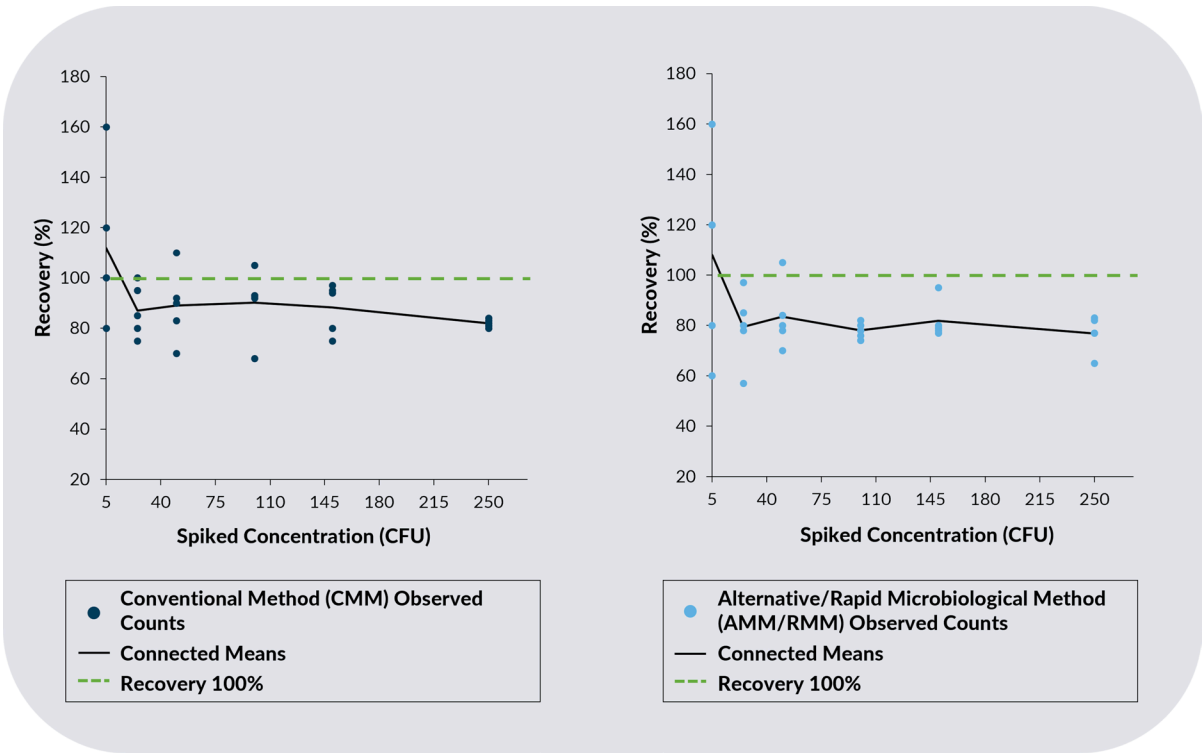


Figure 10.2-3B Visualization of the Mean Counts with Respect to the Spiked Concentration Levels

10.2.1 Weighted Linear Regression Analysis on the Counts

To create a concentration-response curve, it must first be assumed that the counts can be approximately modelled by the following linear regression equation:

$$y = \alpha + \beta\lambda + \varepsilon$$

Where:

- y = a count that is observed at spiked concentration level λ
- α = intercept of the line
- β = slope of the regression line
- λ = spiked concentration level
- ε = residual having a normal distribution with mean zero and variance $\sigma^2(\lambda)$

Here, it is assumed that the variance in the replicated counts at spiked concentration level λ also depends on the spiked concentration (as seen from the data). If the residual variance varies with the concentration, the regression analysis should weight the observed value with the inverse variance. This type of analyses is programmed in any general-purpose software package.

Table 10.2.1-1 shows the estimates of the intercept and slope with a 95% confidence interval, if this weighted linear regression analysis is applied to the counts of the AMM/RMM and CMM separately.

Table 10.2.1-1 Estimated Intercept and Slope Found with Weighted Linear Regression Analysis

Parameter	AMM/RMM	CMM
Intercept α (95% Confidence Interval)	0.829 (-0.975, 2.634)	1.922 (0.721, 3.122)
Slope β (95% Confidence Interval)	0.778 (0.740, 0.816)	0.803 (0.783, 0.822)

Based on the estimated results, it follows that the intercept of the regression line for the AMM/RMM is not statistically different from zero (0) (the 95% confidence interval contains the value zero (0)). If linearity is satisfied for the AMM/RMM, that is, no better relations describe the data, the AMM/RMM satisfies proportionality, implying a constant concentration-recovery curve with a recovery close to 77.8% (i.e., the slope multiplied with 100, for every concentration (see **Section 10.2** and **Section 10.2.2**). On the other hand, the 95% confidence interval for the intercept of the CMM positively indicates that the intercept deviates from zero (since the number zero (0) is not contained in the 95% confidence interval), implying that the CMM has a nonzero background enumeration level of 1.922 CFUs for blank samples if linearity holds true. In other words, based on the estimated intercept, the CMM would generate false positives when test samples do not contain microorganisms.

To investigate the linearity, the explained variation (R^2) with the fitted regression line is equal to 98.4 for the AMM/RMM and 99.6 for the CMM. These values can be obtained from the output of general-purpose software packages when the weighted linear regression analyses are performed. These high values for the R^2 are often considered as evidence for satisfying linearity, since there seems hardly any variation left that could be explained by possible nonlinearity. While there is some truth in this, a high R^2 can also be

achieved when a wide range of concentrations is used and linearity is violated. A more formal way of investigating linearity is to test for a deviation from linearity, where a quadratic term for concentration level is included by using the regression equation $y = \alpha + \beta\lambda + \gamma\lambda^2 + \varepsilon$, and then testing whether the coefficient γ is equal to zero. When this extended regression model is applied, the p -values of 0.821 and 0.189 for the AMM/RMM and CMM, respectively, can be obtained, indicating that there is no evidence for nonlinearity for both microbiological methods. Thus, the conclusion is that both microbiological methods satisfy linearity.

The conclusion of linearity for the AMM/RMM with the attained non-inferiority of accuracy for concentration levels 50, 100, 150, and 250 CFU with the normal distribution in **Table 10.1.1-1** implies that non-inferiority is attained for any concentration level between 50 and 250 CFU.

10.2.2 Poisson Regression Analysis on the Counts

An alternative analysis on the count data that would also deal with the heteroscedasticity is Poisson regression. Instead of assuming normality for the counts, the counts are assumed to be Poisson-distributed. Furthermore, Poisson regression typically uses a logarithmic link function, which would guarantee that the intercept can never become negative. This is different in a weighted linear regression analysis, where the intercept can become negative. Additionally, Poisson regression does not require weights, since the Poisson-distribution dictates that the variance in the replicates equal the mean of the replicates. (Poisson regression analysis is available in many general-purpose statistical software packages.)

In Poisson regression (with the default log-link function), the concentration-response curve is assumed to take the following form:

$$\mathbb{E}(y|\lambda) = \exp\{\alpha + \beta \cdot \psi(\lambda)\}$$

Where:

$\mathbb{E}(y|\lambda)$ = expected concentration of the AMM/RMM of an arbitrary test sample that is tested at the true concentration level λ

λ = spiked concentration level

$\psi(\lambda)$ = transformed concentration level

α = intercept

β = slope

For validation of microbiological methods, the two typical transformations are the logarithmic transformation ($\psi(\lambda) = \log(\lambda)$) and no transformation (i.e., the identity function $\psi(\lambda) = \lambda$). For the logarithmic transformation, the concentration-response curve is forced through the origin (when the slope $\beta > 0$), implying no false-positive rate or background noise. Moreover, when the slope is close to one ($\beta \approx 1$), the value $100\% \cdot \exp\{\alpha\}$ represents the recovery of the microbiological method with respect to all spiked concentration levels. When the slope is larger than one ($\beta > 1$), the microbiological method has difficulty in enumerating microorganisms at a lower concentration, but it improves the enumeration when concentrations are getting higher. When the slope is less than one ($\beta < 1$), the microbiological method has more difficulty in enumerating microorganism at higher concentration, often due to saturation problems. If the identity function is taken, the value $\exp\{\alpha\}$ represents the mean count or background noise at blank

samples, which is always positive, although this does not guarantee a more sensitive enumeration at the lower concentration.

Fitting the Poisson regression model to the data of **Table 10.1-1** for both microbiological methods separately, using the logarithm of the concentration level (i.e., $\psi(\lambda) = \log(\lambda)$), leads to the estimates of the parameters and their 95% confidence intervals shown in **Table 10.2.2-1**.

Table 10.2.2-1 Estimates of the Parameters and 95% Confidence Intervals Using the Poisson Regression Model

Parameter	AMM/RMM	CMM
Intercept α (95% Confidence Interval)	-0.0999 (-0.3898, 0.1901)	0.1386 (-0.1358, 0.4131)
Slope β (95% Confidence Interval)	0.9732 (0.9152, 1.0312)	0.9368 (0.8817, 0.9919)

Both microbiological methods show a slope that is less than one, indicating some saturation at higher concentration levels. The CMM has a slope that deviates from one, since one is not contained in the 95% confidence interval. However, the slope for AMM/RMM cannot be demonstrated to be different from one, since the value one is contained in the 95% confidence interval, indicating that the AMM/RMM may satisfy proportionality. Assuming the assumption of proportionality holds, the recovery of the AMM/RMM against the spiked concentration levels is approximately equal to 90.5% ($= 100 \cdot \exp(-0.0999)$) for all concentration levels within 5 to 250 CFUs, although a better estimate is obtained when the slope is fixed to the value one in the Poisson regression analysis.

10.2.3 Comparability on Accuracy of Alternative/Rapid Microbiological Methods Against Conventional Methods Using Concentration-Response Curves

The analyses of accuracy in **Section 10.1** treated the spiked concentration levels as a categorical variable, while the spiked concentration levels were treated as a continuous variable for the linearity assessment in **Section 10.2.1** and **Section 10.2.2**. The two analyses on accuracy and linearity together can then be used to make a statement about the accuracy for concentration levels other than those tested in the experiment. An alternative approach is to model accuracy as a function of the concentration level and to establish a 90% confidence interval at each concentration that can be compared with the non-inferiority margin. **Section 10.2.3** discusses the data analysis approach that models the concentration-response curves for the AMM/RMM and CMM separately and then uses these curves to calculate the accuracy of the AMM/RMM with respect to the CMM.

Note: This type of data analysis is rather technical and may require the help of a statistician to be able to program it in a programmable statistical software package such as SAS®, R, or Python; the software TriMSA has already implemented such an analysis. To aid readers, **Section 10.3.3** provides some formulae that could be implemented in their preferred software package using the data shown in **Table 10.2.3-1**.

Table 10.2.3-1 Counts of a Specific Microorganism for the Alternative/Rapid Microbiological Method and Conventional Method at Different Concentration Levels

Observation	AMM/RMM						CMM					
	5	25	50	100	150	250	5	25	50	100	150	250
1	7	31	48	70	119	210	3	18	36	77	92	215
2	4	22	59	74	102	194	1	18	32	73	103	205
3	4	33	39	80	93	206	3	22	35	88	111	217
4	9	36	45	77	101	190	4	20	41	71	87	224
5	4	29	49	66	103	160	2	24	27	57	109	215
Mean	5.6	30.2	48.0	73.4	103.6	192.0	2.6	20.4	34.2	73.2	100.4	215.2

The data from **Table 10.2.3-1** will be used to illustrate this approach of calculating accuracy through concentration-response curves. It is assumed that the counts for both the AMM/RMM and CMM are either normally or otherwise Poisson-distributed. The counts have been visualized in **Figure 10.2.3-1**, where the mean counts per spiked concentration are connected. The figure shows that the concentration-response curve is nonlinear in spiked concentration.

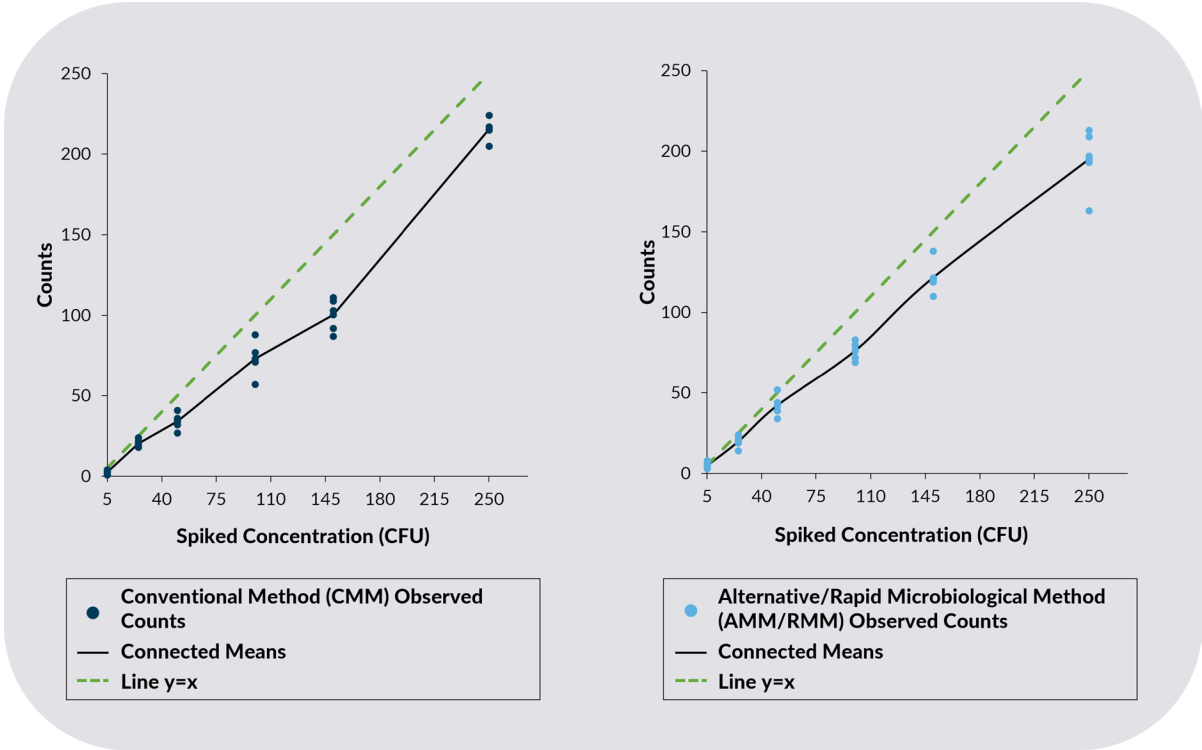


Figure 10.2.3-1 Visualization of the Counts per Concentration and Connecting the Mean Counts

For both microbiological methods, the Mitscherlich function is appropriate to use to extend the concentration-response curves to nonlinear relationships in spiked concentrations:

$$m(\lambda) \equiv E(y|\lambda) = \beta_1 + \beta_2 \exp\{\beta_3 \log(\lambda)\}$$

Where:

$E(y|\lambda)$ = expected concentration

λ = concentration level

β_1 = intercept

β_2 = slope

β_3 = power parameter

The Mitscherlich function is a so-called nonlinear regression model and requires specific regression analysis tools to estimate this function. It is relatively flexible in modeling concentration-response curves and has broad applicability. When the intercept is zero ($\beta_1 = 0$), the Mitscherlich function becomes log-linear:

$$\log(E(y|\lambda)) = \log(\beta_2) + \beta_3 \log(\lambda)$$

When the power parameter is equal to one ($\beta_3 = 1$), the Mitscherlich function becomes a linear function in the concentration λ :

$$E(y|\lambda) = \beta_1 + \beta_2 \lambda$$

Finally, when both the intercept is zero and the power parameter is equal to one, the relation between the mean count and concentration level becomes proportional. Thus, under certain conditions, the nonlinear Mitscherlich function becomes a (log-) linear function, and it can also take the shape of a proportional linear model.

When the counts are Poisson-distributed, a nonlinear Poisson regression analysis must be applied. Most general-purpose statistical software cannot manage the nonlinear aspect, and specialized software is required. The Mitscherlich function with Poisson-distributed counts can be run in SAS®, R, and TriMSA. If the counts are assumed to be normally distributed, the Mitscherlich function should be estimated with a *weighted nonlinear* regression analysis. The weights are needed to address the heteroscedasticity in the counts across concentrations (which is not needed with a Poisson regression analysis), and the nonlinearity is needed to fit the Mitscherlich function. Such regression analysis is available in SAS®, R, and Minitab. While the results of both regression analyses can be obtained, calculating the accuracy of the AMM/RMM against CMM is even more complicated with the weighted nonlinear regression analysis than with the nonlinear Poisson regression analysis, because a model for the variance of the counts across the spiked concentrations must be determined. In the Poisson analysis, this variance function is determined by the Poisson-distribution. This technical report illustrates the accuracy for the nonlinear Poisson regression analysis in **Section 10.2.3.1** and the weighted nonlinear regression analysis in **Section 10.2.3.2**.

10.2.3.1 Using Nonlinear Poisson Regression Analysis

Fitting the Mitscherlich function to the data in **Table 10.2.3-1** for both microbiological methods separately, results in the estimates of the parameters of the Mitscherlich function and its 95% confidence intervals shown in **Table 10.2.3.1-1**.

Table 10.2.3.1-1 Results Using Nonlinear Poisson Regression Analysis

Parameter	AMM/RMM	CMM
Intercept β_1 (95% Confidence Interval)	0.5780 (-0.37396, 4.8957)	0.7096 (-1.6797, 3.0989)
Slope β_2 (95% Confidence Interval)	1.4908 (0.6423, 2.3393)	0.4150 (0.1952, 0.6348)
Power β_3 (95% Confidence Interval)	0.8671 (0.7611, 0.9731)	1.1217 (1.0213, 1.2222)

The fitted Mitscherlich functions have been plotted to the observed counts in **Figure 10.2.3.1-1**.

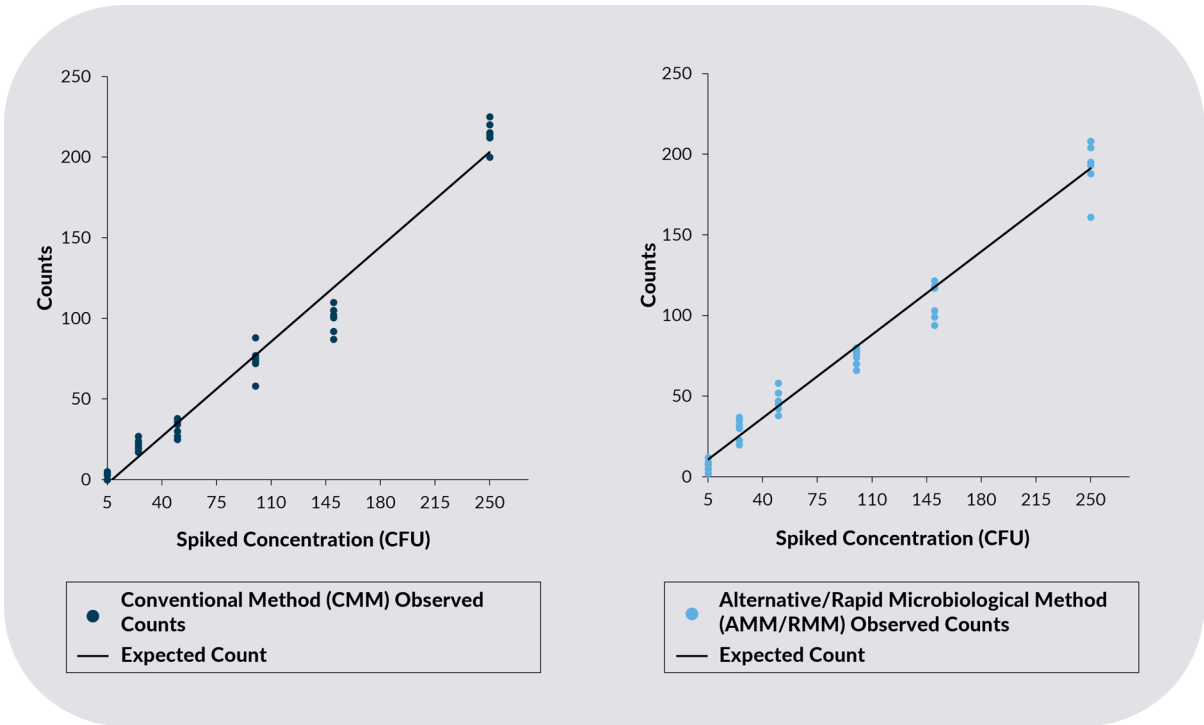


Figure 10.2.3.1-1 Visualization of the Estimated Mitscherlich Functions with Respect to the Observed Counts

The two microbiological methods show a different concentration-response curve. The CMM has some difficulty enumerating microorganisms at lower concentrations but improves when the concentration increases. This can be seen from a slope less than one and a power parameter larger than one. The AMM/RMM shows an opposite pattern, where enumeration is good at lower concentrations but worsens when concentrations increase. This pattern reflects a slope larger than one and a power parameter smaller than one.

Both power parameter estimates deviate from one significantly (the value one does not fall in the 95% confidence interval), but the intercepts may be equal to zero, which may suggest a log-linear model. This model can be fitted with the same procedure, by setting the intercept to zero and locking this as the value so that it will no longer be estimated and, therefore, remains equal to zero. The estimated parameters for the log-linear model are shown in **Table 10.2.3.1-2**.

Table 10.2.3.1-2 Estimated Parameters

Parameter	AMM/RMM	CMM
Intercept β_1	0	0
Slope β_2 (95% Confidence Interval)	1.5924 (1.1470, 2.0378)	0.4693 (0.3125, 0.6262)
Power β_3 (95% Confidence Interval)	0.8552 (0.7986, 0.9118)	1.0991 (1.0329, 1.1653)

By comparing Akaike's corrected information criterion for the log-linear and the Mitscherlich function for both microbiological methods separately, the log-linear model appears to be a better fit (AMM/RMM: Mitscherlich 224.8 versus log-linear 222.4; CMM: Mitscherlich 208.9 versus log-linear 206.8). Thus, using the log-linear concentration-response curves can be continued.

To determine non-inferiority in enumeration between the AMM/RMM and CMM, the recovery or ratio of the two estimated concentration-response curves can be calculated for each concentration in the experimental range [5, 250]: $\hat{m}_A(\lambda)/\hat{m}_C(\lambda)$. For this ratio, a lower 95% confidence limit can be calculated and then compared to the lower limit with the non-inferiority margin.

The lower 95% confidence limit is best calculated in the logarithmic scale:

$$\left[\frac{\hat{m}_A(\lambda)}{\hat{m}_C(\lambda)} \right] \cdot \exp \left\{ -1.645 \sqrt{\hat{\tau}_A^2(\lambda) + \hat{\tau}_C^2(\lambda)} \right\},$$

with $\tau_A^2(\lambda) = \text{VAR}(\log(\hat{m}_A(\lambda)))$ and $\tau_C^2(\lambda) = \text{VAR}(\log(\hat{m}_C(\lambda)))$ the variance of the estimated log transformed Mitscherlich function of the AMM/RMM and CMM, respectively.

The variance of the estimated logarithmically transformed Mitscherlich function is quite elaborative and can be found in Heidari et al. (2022) (16). For the log-linear model, the variance $\tau^2(\lambda)$ is somewhat less elaborate and is equal to:

$$\tau^2(\lambda) = \text{VAR}(\log(\hat{\beta}_2)) + [\log(\lambda)]^2 \text{VAR}(\hat{\beta}_3) + 2 \log(\lambda) \text{COV}(\log(\hat{\beta}_2), \hat{\beta}_3).$$

To obtain this variance, it is easier to model the slope parameter in the log scale (as is provided in TriMSA). The estimated recovery (solid line) and the lower 95% confidence limit (dotted line) is shown in **Figure 10.2.3.1-2** (left figure). The procedure NLMIXED in SAS® for the analysis and the visualization was used for **Figure 10.2.3.1-2**, but TriMSA can provide the same results. The data show that the AMM/RMM is non-inferior to the CMM (at non-inferiority margin of 70%), but the recovery of the AMM/RMM is much better at the lower end of the concentration range than at the upper range. This is due to the enumeration pattern seen from the two microbiological methods separately. Finally, the lower 95% confidence limit at 250 CFU is estimated at 82.7%, so non-inferiority would not have been rejected at 80% non-inferiority margin either.

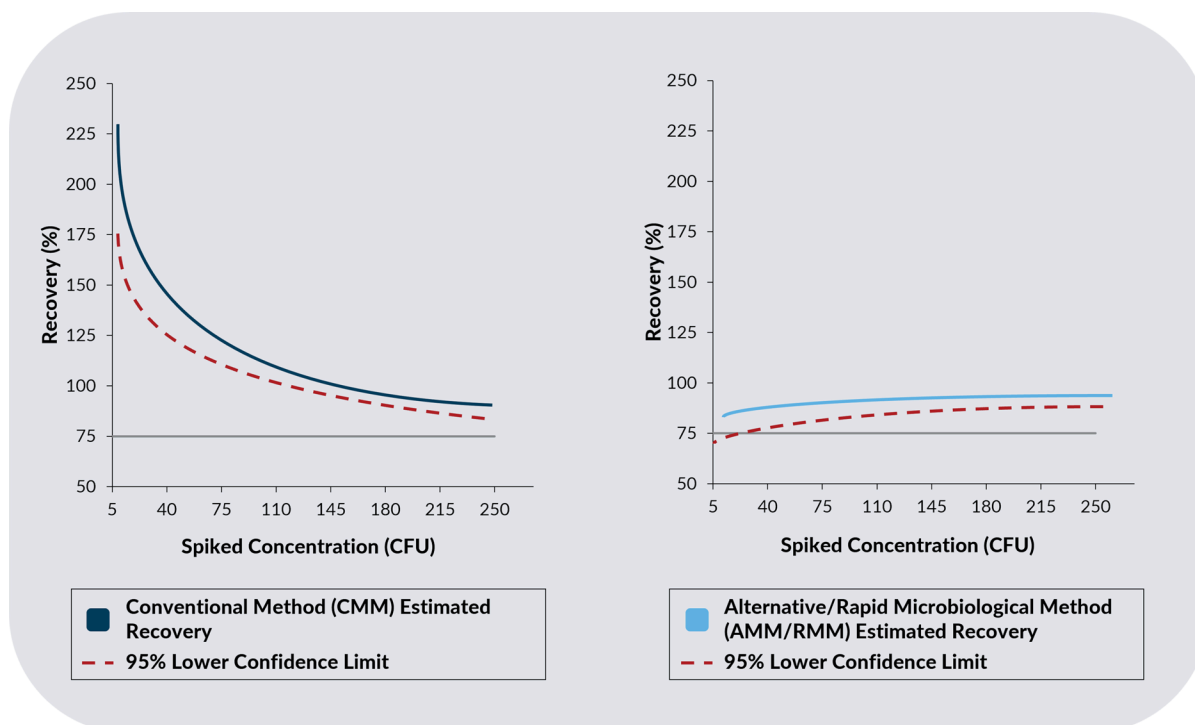


Figure 10.2.3.1-2 Visualization of the Estimated Recovery for the Log-Linear Concentration Response Curves of the Microbiological Methods

This model-based analysis for accuracy could sometimes help reduce variation compared to a per concentration analysis and, therefore, can be more efficient than the analyses between the AMM/RMM and CMM with concentration used as categorical variable. The data in **Table 10.1-1** is used to illustrate this. The analysis in **Section 10.1.2** with the results presented in **Table 10.1.3-1** shows that non-inferiority between AMM/RMM and CMM occurs at 25 CFU for the first time. However, since proportionality for both the AMM/RMM and CMM on the concentration-response curves was demonstrated, consistent accuracy across the full range of 5 to 250 CFU was also demonstrated. Therefore, non-inferiority could be extended across the full experimental range. But if proportionality did not hold for at least one of the microbiological methods, non-inferiority below 25 CFU could not be claimed (see **Table 10.1.3-1**). If the log-linear models were the appropriate models for the concentration-response curves and the model-based approach for accuracy is applied, it would show that non-inferiority would begin from 8.4 CFU (see **Figure 10.2.3.1-2**), which is below the 25 CFU.

10.2.3.2 Using Weighted Nonlinear Regression Analysis

Using the data in **Table 10.2.3-1** for both microbiological methods separately and using weighted nonlinear regression the estimated results of the parameters of the Mitscherlich function and its 95% confidence intervals are shown in **Table 10.2.3.2-1**.

Table 10.2.3.2-1 Estimates of Parameters Using Weighted Nonlinear Regression

Parameter	AMM/RMM	CMM
Intercept β_1 (95% Confidence Interval)	0.0270 (-5.1512, 5.2052)	0.2723 (-1.7477, 2.2923)
Slope β_2 (95% Confidence Interval)	1.6468 (0.3034, 2.9938)	0.4686 (0.2295, 0.7078)
Power β_3 (95% Confidence Interval)	0.8371 (0.6783, 0.9958)	1.1065 (1.0130, 1.2001)

Here, the weights have been selected equal to one over the variance of the counts for each spiked concentration, the same as was done for weighted linear regression in **Section 10.2.1**. **Figure 10.2.3-4** provides a visualization of the data and the fitted Mitscherlich function.

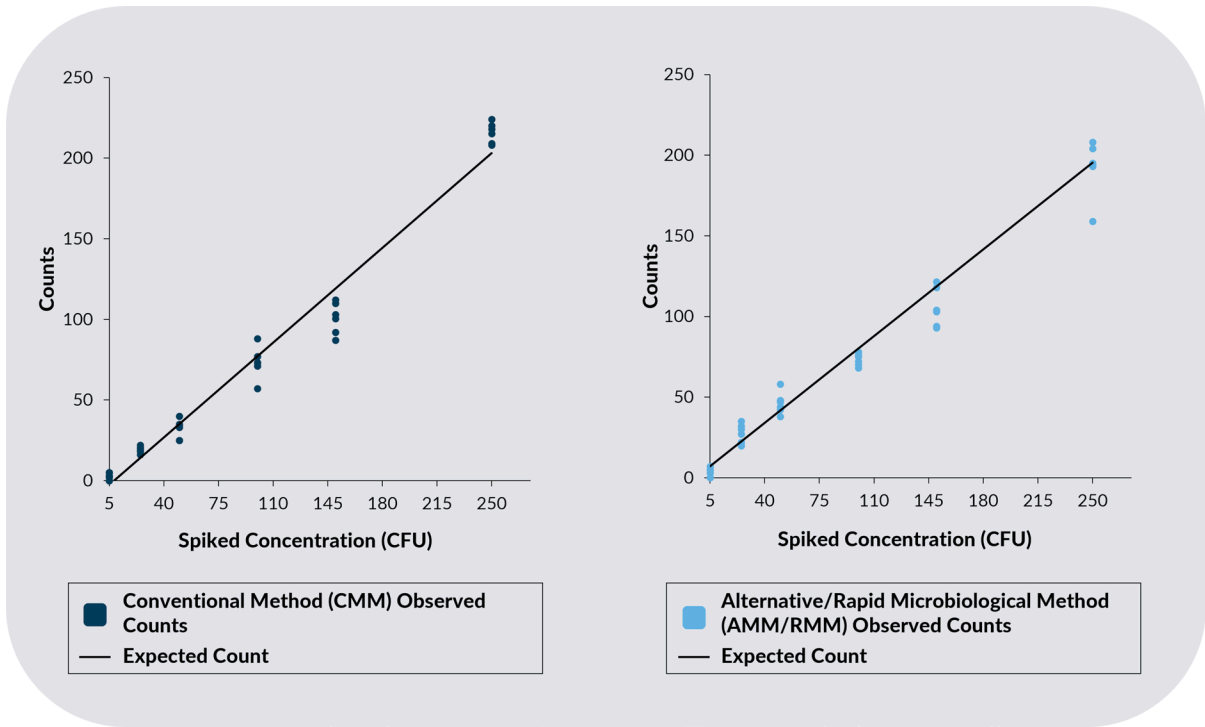


Figure 10.2.3-4 Visualization of the Estimated Mitscherlich Functions Using Weighted Non-Linear Regression with Respect to the Observed Counts

The coefficients are rather different from the coefficients estimated by the nonlinear Poisson regression, particularly for the intercept and slope, but the conclusion about the enumeration patterns do not change. The CMM still has more enumeration issues at lower concentration levels, while the AMM/RMM has more enumeration issues at higher concentration levels. Also, the conclusions that the intercept may be equal to zero and that the power parameters deviate from one are the same as with the nonlinear Poisson regression analysis.

Furthermore, the predicted curves are not so different from the nonlinear Poisson regression analysis in **Figure 10.2.3-2**, even though the coefficients of the Mitscherlich function are somewhat different. However, visually, the nonlinear Poisson regression analysis seems to fit slightly better than the weighted nonlinear regression analysis and, therefore, may be preferred over the weighted nonlinear regression.

10.3 Limit of Quantitation

Determining the concentration of a specific chemical component with analytical measurement systems is always affected by random errors, even if the chemical component is not present in the sample. Only measurements that exceed a specific threshold, that is, the limit of quantitation (LoQ), can be considered a reliable quantification of the concentration. For analytical measurement systems, the LoQ is often established in two steps. In the first step, an upper quantile for the noise of a blank sample is calculated. Then, in the second step, the upper quantile is interpolated on the concentration-response curve to determine the concentration level that would provide, on average, a result above the noise of a blank sample. The LoQ is often quantified during a linearity experiment for the measurement system where the upper quantile is calculated from the unexplained variability of the concentration-response curve and the normal or log-normal distribution. For the analytical measurement methods, the LoQ is often set at 10 standard deviations away from zero when the normal distribution is used; but for count data, this seems excessive due to the skewness of the distribution of the counts at lower concentration levels. Instead, an upper quantile at the level of 0.001 has been chosen. **Figure 10.3-1** shows the calculation approach.

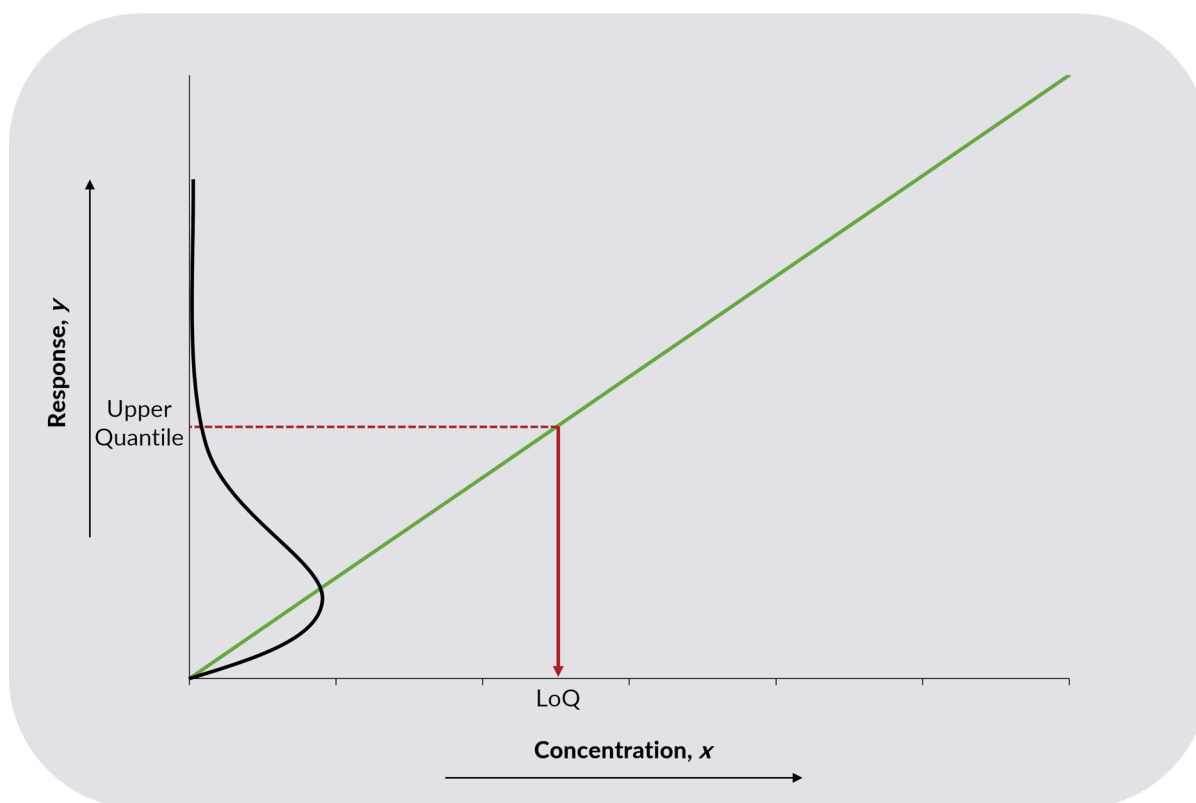


Figure 10.3-1 A Determination of the LoQ for Analytical Measurement Systems

This calculation approach applied to analytical measurements makes sense for microbiological methods if the method generates background noise or false positives for blank samples. This approach will be illustrated in **Section 10.3.1**. If the microbiological method does not produce any positive counts for blank samples, the LoQ cannot be calculated using the analytical measurement method, because there is no noise present at blank samples. An alternative approach is suggested in **Section 10.3.2**.

10.3.1 Microbiological Methods with Positive Counts at Blank Samples

The data in **Table 10.1-1** of the AMM/RMM will be used to illustrate the calculation approach used for analytical methods and follows the weighted regression analysis that was applied in **Section 10.2.1**. Most of the analysis can be done with general-purpose statistical software, but will likely also require some additional manual calculations. The coefficients for the linear concentration-response curves were estimated at $\alpha = 0.829$ (intercept) and $\beta = 0.778$ (slope). The standard error of the intercept was determined as $SE_{\alpha} = 0.881$, which can be found from the regression analysis output that was described, but not reported in **Section 10.2.1**. The residual variance σ^2 , which can normally be found under the “Mean Squared” or “Adjusted Mean Squared Error” function (depending on the software package being used), for the residual error in the ANOVA table produced by the weighted regression analysis, was estimated at $\hat{\sigma}^2 = 0.946$. This residual variance is a “corrected variance” which is corrected for the weights used in the regression analysis. To get the variance around the concentration-response curve at each level of the concentrations, the estimated residual variance of 0.946 should be multiplied with the sample variance that was calculated at each of the concentrations.

To obtain the variance for an individual count at a blank sample, the sample variance for blank samples must be known; but a blank concentration was not included in this experiment (nor was it required) and, therefore, an observed sample variance is not provided. Therefore, to predict the variance at concentration level zero, a (standard) linear regression analysis must be conducted on the sample standard deviations calculated for the six concentrations (see **Table 10.1-1**). The linear regression analysis on the sample standard deviations leads to an intercept of 1.590 and a slope of 0.067, which is obtainable with any general-purpose statistical software package. This estimated intercept is the predicted sample standard deviation for the counts at blank samples. The value needs to be squared ($2.5281 = 1.590 \cdot 1.590$) to get a variance and multiplied with the residual variance $\hat{\sigma}^2 = 0.946$ from the weighted regression analysis manually to obtain the final variance at the zero concentration for the AMM/RMM:

$$\hat{\sigma}_0^2 = 2.5281 \cdot 0.946 = 2.3916.$$

To obtain the total variability at a zero concentration, also taking into account the uncertainty in estimating the intercept of the concentration-response curve, the squared uncertainty ($SE_{\alpha}^2 = 0.881^2$ or 0.7762) of the estimated intercept 0.829 must be added to the variance $\hat{\sigma}_0^2$:

$$SE_{\alpha}^2 + \hat{\sigma}_0^2 = 3.1677.$$

By taking the square root of this variance 3.1677, the standard deviation is then 1.7798. The upper quantile is then determined at three times the standard deviation (approximately at the significance level of 0.1%) away from the intercept 0.829 (upwards). Thus, the upper quantile (represented as the vertical axis in **Figure 10.3-1**) is then calculated at:

$$UQ = 0.829 + 3 \cdot 1.7798 = 6.168.$$

Translating this upper quantile to a concentration level (as illustrated in **Figure 10.1.1-2**) using the weighted regression line $0.829 + 0.778\lambda$ from **Section 10.2.1**, gives an estimated LoQ equal to:

$$\lambda_{\text{LoQ}} = \frac{[6.168 - 0.829]}{0.778} = 6.863$$

represented at the horizontal axis in **Figure 10.3-1**.

Thus, contaminated samples with a concentration level ≥ 6.86 CFU most likely will not produce typical counts at blank samples when background noise is present in the AMM/RMM (for blank samples), indicating that 6.86 CFU is the LoQ and that the sample cannot be confused with a blank sample.

10.3.2 Microbiological Methods with Zero Counts at Blank Samples

When the blank sample does not generate any signal for the AMM/RMM, an approach can be used that is aligned with the calculation of the detection limit for qualitative microbiological methods. Assuming that the concentration-response curve for the AMM/RMM is given by:

$$m_A(\lambda),$$

where λ is the concentration level and $m_A(0) = 0$, the concentration level $\text{LoQ} = \lambda_{\text{LoQ}}$ may be found for which it is (highly) likely to obtain enumerations that are away from zero (the result at a blank sample). In this way, a zero result would not be obtained when the test sample contains more than the LoQ, and the test sample is no longer believed to be sterile. In mathematical terms, the LoQ is defined by:

$$\min_{\lambda} P(Y > 0 | m_A(\lambda)) \geq 1 - \alpha,$$

with $1 - \alpha$ being the level of certainty to be chosen.

In other words, the search is for the minimal concentration that does not produce zero counts with a predefined confidence level of $1 - \alpha$. As discussed for qualitative methods, the LoD was selected with a confidence level equal to 0.95 (or 95%). Taking a confidence level of 99.9% for the LoQ, as selected in **Section 10.3.1**, there would be only 0.1% risk of producing a result that would fit with a result of a blank sample. The Poisson regression analysis of the data in **Table 10.2.3-1** for the AMM/RMM in **Section 10.2.3** has been used to illustrate this approach.

The Poisson regression analysis described in **Section 10.2.3** demonstrated that the log-linear concentration-response curve was an appropriate curve that described the relationship. This relationship forces the concentration-response curve through the origin, indicating a zero mean for blank samples (and no noise or signal expected at blank samples). The coefficients were estimated at slope $\beta_2 = 1.5924$ and power $\beta_3 = 0.8552$ (see **Section 10.2** for the notation and formulation of the log-linear concentration-response curve), leading to the concentration-response curve equal to:

$$m_A(\lambda) = 1.5924 \cdot \lambda^{0.8552}$$

The probability of obtaining a zero count when the mean value of the Poisson distribution is $m_A(\lambda)$ is equal to $\exp\{-m_A(\lambda)\}$. With some simple algebra, the maximum value λ_{LoQ} can be obtained that would give $\exp\{-m_A(\lambda_{\text{LoQ}})\} \leq 0.001$. This value is:

$$\lambda_{\text{LoQ}} = \exp\left\{\frac{\log\left(\frac{-\log(0.001)}{1.5924}\right)}{0.8552}\right\} = 5.56$$

Thus, for concentration levels ≥ 5.56 CFU, it is unlikely to obtain zero enumerations, indicating that test samples will not be confused with blank test samples in routine practice. Note that this value is not so different from the estimated LoQ with weighted linear regression, but the value in **Section 10.3.1** is higher due, at least in part, to the expected positive mean signal at blank samples.

10.4 Precision

For the validation of analytical measurement systems, precision is quantified by two measures: a relative standard deviation for repeatability and a relative standard deviation for intermediate precision. Repeatability quantifies within run variability (i.e., variability in enumerations under identical circumstances), and intermediate precision quantifies the sum of within and between run variability (i.e., variation in enumerations under routine varying conditions).

The analytical run is determined by factors that can change (e.g., analysts, test days, different instruments) every time a new set of enumerations is collected. The standard data-analytic approach to calculate these two relative standard deviations for the measures of precision is analysis of variance (ANOVA). For the estimation of the measures of precision, ANOVA does not make any distributional assumptions, but it often requires normality assumptions if additional features (e.g., confidence intervals, hypothesis testing for run variability) are being used with ANOVA. For validation of microbiological methods, it is common to follow the experimental setup of analytical measurement systems and apply ANOVA to quantify the within and between run variability. This approach is demonstrated in **Section 10.4.1** using the data from **Table 10.4-1**. The run structure of the precision experiment is formed by the combinations of analysts and days (thus eight (8) runs in total).

Table 10.4-1 Precision Experiment for the Alternative/Rapid Microbiological Method Using Multiple Runs and Replicates

Replicate	Analyst 1				Analyst 2			
	Day 1	Day 2	Day 3	Day 4	Day 1	Day 2	Day 3	Day 4
1	65	71	72	60	68	75	53	62
2	67	76	69	76	73	74	70	62
3	79	64	63	68	67	68	61	53
4	71	69	66	76	68	87	66	69

Although ANOVA can quantify variability within and between runs without making any distributional assumptions, there are considerations when the counts would follow a Poisson distribution. The Poisson distribution has only one parameter λ that quantifies both the mean value and the variance of the counts, and this fixes the value for the relative standard deviation to $100\%/\sqrt{\lambda}$ (when replicated counts are collected at the same concentration). This value of relative standard deviation of the Poisson distribution is depicted in **Figure 10.4-1**, which shows that the relative standard deviation becomes small only when the mean value gets large. The relative standard deviation is never smaller than 10% for mean counts below 100 CFU. Thus, for mean counts less than 100 CFU, repeatability that is (much) less than 10% is not expected to be found when the counts follow a Poisson distribution and replicates are collected from different samples.

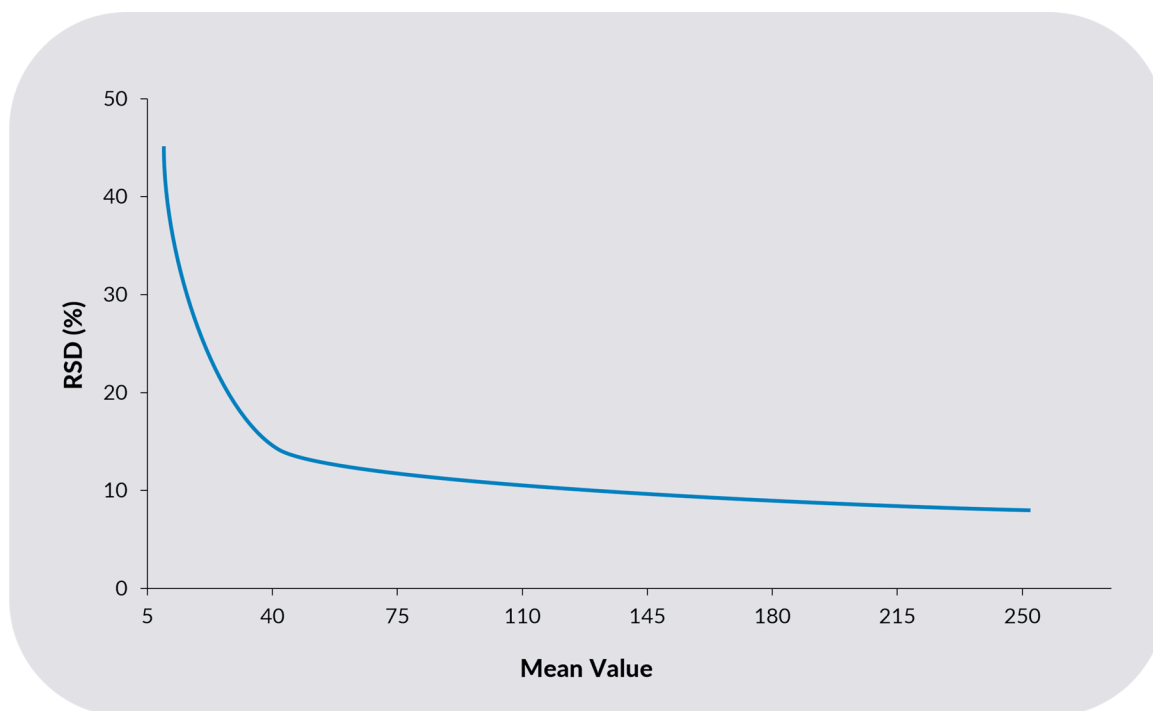


Figure 10.4-1 Visualization of the Relative Standard Deviation of a Poisson Distribution

In case the counts within runs (i.e., multiple enumerations from the same diluted suspension) follow a Poisson distribution, the relative standard deviation can be estimated by using the mean value of the counts instead of using ANOVA, which would estimate a separate variance parameter. In such situations, where the counts within runs follow a Poisson distribution, it would be better to analyze the precision experiment with a generalized linear mixed-effects model (GLMM) using the Poisson distribution. The repeatability is then determined by the estimated Poisson distribution, while the run-to-run variability is separately modeled with a latent variable that varies with the runs. This approach is illustrated in **Section 10.4.2**.

In the validation experiment for precision represented in **Table 10.4-1**, samples have been created at approximately the same spiked level for each day and each analyst. Precision experiments are often conducted this way, but typically at two or more spiked levels, to be able to quantify the precision at different microbial concentrations. At each spiked level, the data could be analyzed as demonstrated here, but there would also be options to analyze the full data simultaneously with ANOVA and GLMM.

The choice between ANOVA and GLMM depends mostly on a violation of using the Poisson distribution. If it appears that the counts at each run follow a Poisson distribution, the GLMM is preferred; however, when the variances of the counts within runs deviates (either larger or smaller) from the mean counts, the ANOVA approach may be preferred over the GLMM approach based on Poisson. This is similar to the approach for calculating accuracy at each concentration, where the choice was between a Poisson or a normal distribution (see **Section 10.1.1** and **Section 10.1.3**).

10.4.1 Analysis of Variance for Precision

The ANOVA approach calculates the mean squares for within and between runs, denoted here by MS_W and MS_B , respectively. Any general-purpose statistical software package can determine these statistics from the data presented in **Table 10.4-1**. The mean square for the within-run variability (MS_W) is a direct estimate of

the variance component (σ_W^2) for the replicates within each run (as was used for linearity and the LoQ from the weighted linear regression analysis). The assumption is that this variance is the same for each run produced with the microbiological method. The mean square for the between-run variability (MS_B) is an estimate of:

$$n\sigma_B^2 + \sigma_W^2$$

Where:

σ_B^2 = variance component for the between-run variability

σ_W^2 = variance component for the within-run variability

n = number of replicates in each run

When the number of replicates vary from run-to-run, the MS_B estimates:

$$c \cdot \sigma_B^2 + \sigma_W^2$$

Where:

c = a function of the different sample sizes in the runs (17).

Furthermore, if the software package has the option to choose a random effects model, the two variance components will be estimated directly.

The two measures of precision, repeatability and intermediate precision, are now defined by:

$$\textbf{Repeatability: } RSD_R(\%) = 100\% \cdot \frac{\sigma_W}{\mu}$$

$$\textbf{Intermediate Precision: } RSD_{IP}(\%) = 100\% \cdot \frac{\sqrt{\sigma_B^2 + \sigma_W^2}}{\mu}$$

with μ estimated by the average value of the counts.

Repeatability and intermediate precision are divided by the mean μ to obtain a measure of precision as a percentage of the average value. These measures of precision are typically not provided by standard software packages and should be calculated manually when the two variance components of σ_W^2 and σ_B^2 have been determined.

Sometimes, it is better to analyze the precision experiment on the logarithm of the counts, when the distribution in the replicated counts within a run show skewness, as counts often show positive skewness (see **Section 10.1**).

If the skewness suggests transforming the counts to the logarithmic scale, the measures of precision for the counts itself are estimated in the following way:

$$\textbf{Repeatability: } RSD_R(\%) = 100\% \cdot \sqrt{\exp\{\sigma_W^2\} - 1}$$

$$\textbf{Intermediate Precision: } RSD_{IP}(\%) = 100\% \cdot \sqrt{\exp\{\sigma_W^2 + \sigma_B^2\} - 1}$$

This calculation assumes that the logarithmically transformed counts are approximately normally distributed. The advantage is that the mean value of the counts no longer plays a role in the calculation of the measures of precision. However, to compare these measures of precision with what can be expected

from a Poisson distribution, the mean count is still required to be able to determine the relative standard deviation depicted in **Figure 10.4-1**.

10.4.1.1 Analysis of the Counts

Using the count data from **Table 10.4-1**, the mean value is estimated at 68.375 CFUs and the mean squares (rounded to two decimals) are determined at $MS_B = 86.21$ and $MS_W = 38.00$. The variance component for the between-run variability is then equal to $\sigma_B^2 = 12.0536$ (not using rounded mean squares). This results in the following values for the measures of precision:

$$RSD_R(\%) = 9.02\%$$

$$RSD_{IP}(\%) = 10.35\%$$

10.4.1.2 Analysis of the Log Counts

Using the logarithmically transformed count data from the data in **Table 10.4-1**, the mean value is estimated at 4.2199, which translates into a median count of 68.03 ($= \exp \{4.2199\}$). The mean squares are estimated at $MS_B = 0.018993$ and $MS_W = 0.008339$ using a general-purpose statistical software package. The variance component for the between-run variability is then estimated at $\hat{\sigma}_B^2 = 0.002664$. This results in the following values for the measures of precision:

$$RSD_R(\%) = 9.15\%$$

$$RSD_{IP}(\%) = 10.52\%$$

These measures of precision are very close to the measures of precision calculated from the counts directly. To make a choice between the precision measurement, the residuals of the ANOVA model could be evaluated (see **Figure 10.4.1.2-1**; most statistical software packages can generate this type of probability plot of the residuals as an output of the calculations). The probability plots show that both analyses (on counts and logarithmically transformed counts) would make sense, but the plot from the counts seems slightly closer to normality than the plot from the log counts (i.e., the data points appear visually closer to the lines), which is a marginal difference. Thus, based on this small distinction, the measures of precision calculated from the counts would be reported.

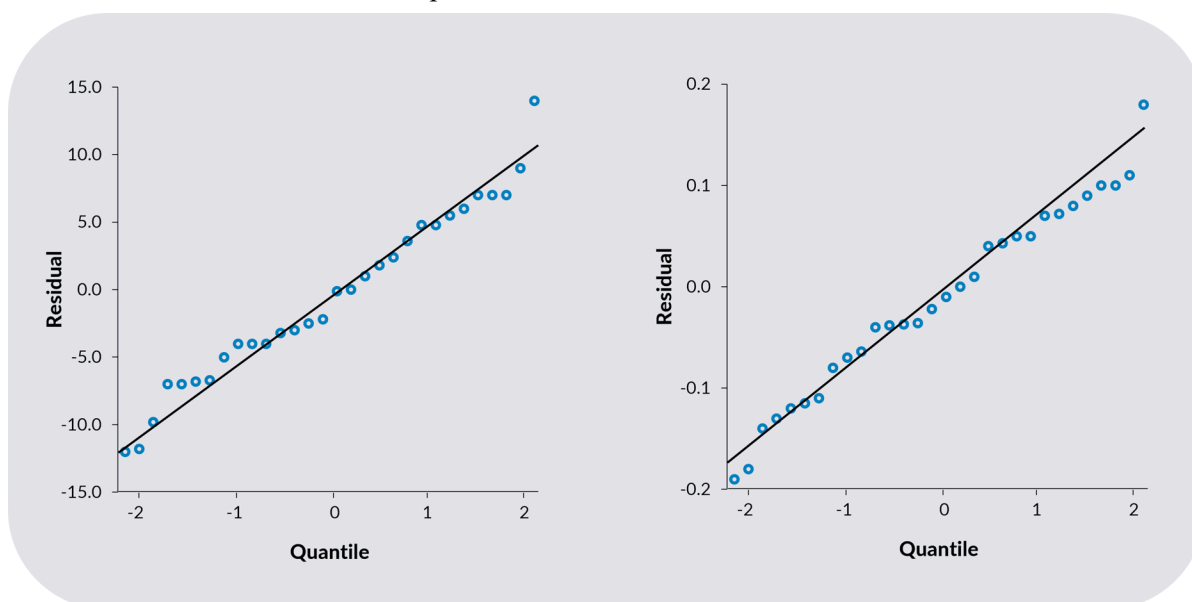


Figure 10.4.1.2-1 Probability Plot of the Residuals (Analysis of the Counts (left); Analysis of the Log Counts (right))

10.4.2 Generalized Linear Mixed Effects Model

Assuming the counts in **Table 10.4-1** within a run would follow the Poisson distribution, a generalized linear mixed effects analysis could be used. In such a model, it is assumed that there exists a latent variable z_i for run i that affects the mean count in the run. The expected count given this latent variable z_i is then formulated as:

$$E(y_{ij}|z_i) = \exp\{\alpha + z_i\}$$

Where:

λ_α = $\exp\{\alpha\}$: the median count over all possible runs for the spike level used in the precision experiment

z_i = latent variable for run i often assumed normally distributed with zero mean and variance τ^2

α = intercept

The variance τ^2 for the latent variable z_i would represent the variability between the runs in the logarithmic scale.

Based on this Poisson model, the measures of precision for the counts in their original scale can be calculated as follows:

$$\text{Repeatability: } RSD_R(\%) = 100\% \cdot \exp\left\{\frac{-\alpha}{2}\right\}$$

$$\text{Intermediate Precision: } RSD_{IP}(\%) = 100\% \cdot \left[\exp\left\{-\alpha - \frac{\tau^2}{2}\right\} + \exp\{\tau^2\} - 1 \right]^{\frac{1}{2}}$$

Fitting the GLMM may not be common in all general-purpose statistical software packages, but it could be determined with [R] and SAS®. In SAS®, the analysis can be determined by the procedure GLIMMIX. Such packages provide the estimates for α and τ^2 , from which the measures of precision can be manually determined (by using the formulas above). The TriMSA software provides these measures of precision directly in the output of the analysis.

The GLMM analysis on the data of **Table 10.4-1**, gives an estimate of α that is equal to 4.2248 and an estimate of τ^2 equal to 0.000378. In case the variance component τ^2 is equal to zero, the intermediate precision becomes equal to the repeatability, but here the intermediate precision is slightly larger than the repeatability. Thus, the two measures of precision are now (manually) estimated at:

$$\text{Repeatability: } RSD_R(\%) = 12.09\%$$

$$\text{Intermediate Precision: } RSD_{IP}(\%) = 12.25\%$$

These values are slightly larger than the calculations from ANOVA, because the Poisson distribution fixes the repeatability to $100\%/\sqrt{\lambda}$, with λ as the median value. Thus, the repeatability is now exactly equal to the relative standard deviation of the Poisson distribution for an average of $\lambda = 68.36 (= \exp\{4.2248\})$, since a Poisson distribution was assumed for the replicated counts conditionally on the executed run. Thus, the repeatability is what we can be expected from Poisson distribution, and the intermediate precision is only slightly higher than the repeatability. Overall, the measures of precision are satisfactory.

10.4.3 Comparing Precision of the Alternative/Rapid Microbiological Method with the Conventional Method

In addition to the data collection for the AMM/RMM in **Table 10.4-1**, precision data was also collected for the CMM so the AMM/RMM could be compared with the CMM. It is assumed here that the data was collected on other days and possibly with analysts other than those for the AMM/RMM. This means that the runs for the CMM are runs other than those for the AMM/RMM. The data for the CMM are presented in **Table 10.4.3-1**.

Table 10.4.3-1 Precision Experiment for the CMM Using Multiple Runs and Replicates

Replicates	Analyst 1				Analyst 2			
	Day 1	Day 2	Day 3	Day 4	Day 1	Day 2	Day 3	Day 4
1	70	70	60	78	69	69	69	71
2	82	75	54	55	84	65	65	74
3	69	73	69	59	88	74	76	76
4	64	89	65	68	81	55	57	61

It should be noted that comparing precision measures between the AMM/RMM and CMM often requires an unrealistic large number of test samples. **Table 10.4.3-2** shows the sample size per microbiological method (i.e., replicated counts when all data is collected within a single run) for traditional hypothesis-testing, assuming a specific ratio between the variance components σ_A^2/σ_C^2 (type 1 error rate is 0.05 and the power is 0.80). **Table 10.4.3-2** clearly shows that an increase in the variance of less than 50% from the CMM to the AMM/RMM is difficult to conduct for each species separately. Thus, most of the time, traditional hypothesis-testing would show that there is no difference in variance components. On the other hand, non-inferiority testing would typically lead to a wide confidence interval that would hardly ever satisfy a non-inferiority margins of 0.7 when the variance of the CMM is compared to the variance of the AMM/RMM or, alternatively, margins of 1.3 if the variance of the AMM/RMM is divided by the variance of the CMM. It is therefore recommended to either accept less strict non-inferiority margins on the ratio of variances or to compare the two microbiological methods more qualitatively. A non-inferiority margin of 2 on the ratio σ_A^2/σ_C^2 (or 0.5 on the ratio σ_C^2/σ_A^2) may be considered reasonable, since it leads to a non-inferiority margin of approximately 0.7 on the ratio of precisions σ_C/σ_A and seems practical considering the information in **Table 10.4.3-2**.

Table 10.4.3-2 Minimum Sample Size for Two Variances

Ratio Variances	Sample Size
1.25	633
1.50	193
1.75	103
2.00	68
2.25	50
2.50	40

It is common to compare the precision estimates of the AMM/RMM and CMM in a ratio where the precision of the CMM is put in the denominator. This means that a non-inferiority margin becomes a value above one, like 1.3, instead of a value below one, like 0.7, which has been used for other validation parameters so far. Indeed, when the precision of the AMM/RMM gets too large, the AMM/RMM may be considered inferior to the CMM and the ratio σ_A/σ_C becomes too far beyond one. It is important to realize that 0.7 and 1.3 are not the same non-inferiority margins in case the ratio is reversed and the precision of the AMM/RMM is put in the denominator. A non-inferiority margin of 1.3 on the ratio σ_A/σ_C would lead to a non-inferiority margin of 0.769 ($=1/1.3$) on the ratio σ_C/σ_A . Finally, putting a non-inferiority margin on the ratio of the precisions is not the same as putting a non-inferiority margin on the ratio of the variances (i.e., the ratio of squared precisions) as discussed in previous paragraph. Indeed, a non-inferiority margin of 1.3 on the ratio σ_A/σ_C is the same as putting a non-inferiority margin of 1.69 on the ratio σ_A^2/σ_C^2 or 0.592 on the ratio σ_C^2/σ_A^2 .

The approach to compare the precision of the AMM/RMM with the precision of the CMM again depends on the distributional assumptions. If it is assumed that the counts per run are Poisson-distributed, the repeatability is determined by the mean count $\exp\{\alpha\}$. Comparing this parameter between the AMM/RMM and CMM is then better done in an accuracy experiment as discussed in **Section 10.1**. Thus, for the GLMM approach, it would not make sense to compare the repeatability between the microbiological methods if an accuracy study is also conducted. The only comparison between the AMM/RMM and the CMM that would possibly make sense is the comparison of the parameter τ^2 that describes the between-run variability. However, a difference in τ^2 between the two microbiological methods may not have a strong effect on the ratio of the intermediate precision, since it also contains the within-run variability. To illustrate this, the ratio $100\% \cdot RSD_{IP}^A(\%)/RSD_{IP}^C(\%)$ of the two intermediate precisions for the microbiological methods can be visualized as a function of the ratio $100\% \cdot \tau_A/\tau_C$ for different values of the mean count $\exp\{\alpha\}$. Here, it is assumed that both microbiological methods have the same mean count (or repeatability). **Figure 10.4-1** shows that when the standard deviation τ_A of the AMM/RMM is twice as large as the standard deviation τ_C of the CMM (and the variance is thus four times larger), the intermediate precision of the AMM/RMM is only 31% (or much less) larger than the intermediate precision of the CMM for enumerations less than 90 CFU. When the repeatability is large, the repeatability dominates the intermediate precision, and a strong difference in variance components (as a factor 4, as noted previously) is not picked up. Thus, to understand

the differences between the two microbiological methods when the Poisson distribution is used, comparing the two standard deviations τ_A and τ_C from the GLMM analysis is recommended.

If the replicates within the runs of the precision experiment do not demonstrate a Poisson distribution, the counts or the log counts with the ANOVA can be modelled, assuming that the counts follow a normal distribution. The decision not to use the Poisson distribution is mainly driven by a violation of the relation between the mean and the variance of the replicated counts within runs (see also **Section 10.1.3**). When the variances for the runs deviate from the means of the runs, an alternative distribution to the Poisson distribution should be used. This evaluation is not always easy to conduct due to the relatively small number of replicates per run, and it may be necessary to do it qualitatively. If the Poisson distribution is rejected and the normal distribution describes either the counts or log counts properly for each microbiological method, which can be evaluated through a residuals plot (**Figure 10.4-2**) after conducting an ANOVA, the ANOVA approach can be used to compare both the repeatability and the intermediate precision.

10.4.3.1 ANOVA Method on the Counts

To compare the measures of precision for AMM/RMM with CMM, the values for repeatability and intermediate precision for the two methods separately can be calculated as described in **Section 10.4.1** (using counts). The calculation of the two ratios $RSD_R^A(\%)/RSD_R^C(\%)$ and $RSD_{IP}^A(\%)/RSD_{IP}^C(\%)$ are then straightforward to calculate.

Applying the ANOVA approach to the count data in **Table 10.4.3-1** for the CMM results in a mean square $MS_B = 64.375$ and $MS_W = 157.125$. The variance component for the between-run variability is then estimated at $\hat{\sigma}_B^2 = 23.19$. The mean count was determined at 69.81. Thus, the measures of precision for the CMM are now estimated at $RSD_R^C(\%) = 11.49\%$ and $RSD_{IP}^C(\%) = 13.40\%$. Comparing the measures of precision of the AMM/RMM with the measures of precision of the CMM, leads to the following ratios (expressed in percentages):

$$\text{Ratio for Repeatability: } 100\% \cdot \frac{RSD_R^A(\%)}{RSD_R^C(\%)} = 78.45$$

$$\text{Ratio for Intermediate Precision: } 100\% \cdot \frac{RSD_{IP}^A(\%)}{RSD_{IP}^C(\%)} = 77.20$$

Clearly, these ratios show that the estimated precision of the AMM/RMM is less than the estimated precision of the CMM. To better understand this ratio, it is recommended to calculate an upper 95% confidence limit for these two ratios using the logarithmic scale. Unfortunately, confidence intervals on ratios of variance components are mathematically not straightforward, particularly for the intermediate precision.

The variance of a mean square MS in a random effects ANOVA model can be estimated by:

$$\frac{2MS^2}{df}$$

with df as the degrees of freedom in the ANOVA table for the specific mean square.

The degrees of freedom of the within-run mean square $m(n - 1)$ is equal to the number of runs (m) multiplied with the number of degrees of freedom of $(n - 1)$ for the n replicated counts in one run. The asymptotic variance of the logarithmically transformed standard deviation for the within-run variability $0.5 \cdot \log(MS_W)$ is then equal to $[2m(n - 1)]^{-1}$. Hence, the asymptotic upper 95% confidence limit for the ratio $RSD_R^A(\%)/RSD_R^C(\%)$ is then determined by:

$$UCL_R = \left[\frac{RSD_R^A(\%)}{RSD_R^C(\%)} \right] \exp \left\{ \frac{1.645}{\sqrt{m(n - 1)}} \right\},$$

where the uncertainty in the mean count was not taken into account.

For the example data in **Table 10.4-1** and **Table 10.4.3-1**, the degrees of freedom is equal to 24 ($= 8 \cdot (4 - 1)$) for the within-run mean square for both microbiological methods. The upper 95% confidence limit for the ratio of repeatability is now determined by:

$$UCL_R = 109.7\%$$

The estimate for the sum of the two variance components $\sigma_B^2 + \sigma_W^2$ is equal to $[MS_B + (n - 1)MS_W]/n$. The estimated variance of this estimator is equal to:

$$\frac{2 \left[\frac{MS_B^2}{df_B} + \frac{(n - 1)^2 MS_W^2}{df_W} \right]}{n^2}$$

because the two mean squares are independent for balanced data (i.e., the same number of replicates per run).

The degrees of freedom for the mean square between-runs equal the number of runs minus one ($df_B = m - 1$). The variance of the total variability of within- and between-runs, $0.5 \cdot \log([MS_B + (n - 1)MS_W]/n)$, is then approximately equal to:

$$\eta^2 = 0.5 \frac{\left[\frac{MS_B^2}{df_B} + \frac{(n - 1)^2 MS_W^2}{df_W} \right]}{[MS_B + (n - 1)MS_W]^2}$$

The upper 95% confidence limit for the ratio $RSD_{IP}^A(\%)/RSD_{IP}^C(\%)$ is now determined by:

$$UCL_{IP} = \left[\frac{RSD_{IP}^A(\%)}{RSD_{IP}^C(\%)} \right] \exp \left\{ 1.645 \sqrt{\eta_A^2 + \eta_C^2} \right\}$$

Applying these formulas to the observed results, the upper 95% confidence limit for the ratio of intermediate precisions becomes equal to $UCL_{IP} = 107.6\%$.

The formulas for the upper 95% control limits of the ratio of repeatability and intermediate precision for the two microbiological methods can be programmed in Excel. The only input that is needed for each microbiological method is the mean, the two mean squares, and their degrees of freedom.

The upper 95% confidence limits for the ratios of both the repeatability and the intermediate precision are less than the non-inferiority limit of 141% ($= 100\%\sqrt{2}$) assuming a non-inferiority margin of 2 for the

corresponding variances. Thus, the precision measures of the AMM/RMM is non-inferior to the precision of the CMM when a non-inferiority margin (e.g., 2) is used on the variance for the relative standard deviations.

10.4.3.2 GLMM Poisson Approach

Assuming that the counts within the runs are Poisson-distributed and the runs add additional variation to the counts, the same analysis can be used for the CMM as was done for the AMM/RMM. The estimates and standard error for the parameters in the Poisson model for both the AMM/RMM and CMM are shown in Table 10.4.3.2-1.

Table 10.4.3.2-1 Estimates and Standard Error for the Poisson Model

Method	α (SE_{α})	τ^2 (SE_{τ^2})
AMM/RMM	4.2248 (0.02248)	0.000378 (0.002022)
CMM	4.2441 (0.02960)	0.003408 (0.003468)

As mentioned earlier, the ratio of the two variance components τ_A^2 and τ_C^2 for the between-run variability of the two microbiological methods should be investigated. The estimator of the variance component will be approximated with a chi-square distribution and then using the F-distribution to calculate an upper confidence limit on the ratio of the variance components. This upper limit can then be used to determine non-inferiority if a non-inferiority margin, say 2, is prescribed for the precision experiment.

To approximate the estimators of the variance component with a chi-square distribution, only the degrees of freedom need to be estimated. Making use of the Satterthwaite approach (an approach that is used for t -tests with unequal variances as well), the formula for the degrees of freedom for the estimator $\hat{\tau}^2$ for the variance component τ^2 is given by:

$$df = \frac{2\hat{\tau}^4}{SE_{\hat{\tau}^2}^2},$$

with $SE_{\hat{\tau}^2}$ the standard error of the estimator $\hat{\tau}^2$.

In practice, such degrees of freedom can sometimes become small when variance components are relatively small, so degrees of freedom will not be accepted less than the degrees of freedom for the between-run variability, which is defined by the number of runs minus one (similar to ANOVA).

To first determine an upper 95% confidence limit on the ratio of the two variance components τ_A^2/τ_C^2 , the upper 95% confidence limit is given by:

$$UCL_{VC} = \left(\frac{\tau_A^2}{\tau_C^2} \right) F_{df_A, df_C}^{-1}(0.95)$$

Where:

τ_A^2 = between-run variance component for AMM/RMM

τ_C^2 = between-run variance component for CMM

df_A = degrees of freedom for AMM/RMM

df_C = degrees of freedom for CMM

F_{df_A, df_C}^{-1} = quantile function of the F-distribution with df_1 and df_2 degrees of freedom

Using the result from the analyses, a ratio of the two variance components is obtained that is equal to $\tau_A^2/\tau_C^2 = 0.1109$. The degrees of freedom for the AMM/RMM and CMM, based on Satterthwaite approach, are determined at 0.07 and 1.93, respectively, but the number of runs for each microbiological method was set at eight (8). Thus, the degrees of freedom for calculation of the upper 95% confidence limit on the ratio of the estimated variance components will be equal to seven (7). The upper limit UCL_{VC} is then determined at 0.4200. By taking the square root, the ratio of standard deviations τ_A/τ_C , expressed in percentages, is equal to 33.3%, and the corresponding upper 95% confidence limit on the ratio of standard deviations (also expressed in percentages) is then equal to 64.8%. This upper limit is lower than a non-inferiority margin of 141% (corresponding to a factor of 2 for the variance components), making the AMM/RMM non-inferior on the between-run variability (i.e., intermediate precision) compared to CMM.

11.0 Appendix 3: Validation of Alternative and Rapid Microbiological Methods: Additional Statistical Analysis-Non-CFU Signals

Several AMM/RMMs detect or enumerate microorganisms that are not based on the principle of growth, e.g., they may measure molecules such as nicotinamide adenine dinucleotide plus hydrogen (NADH) or riboflavin or other cellular targets that are indicative of viable cells. Thus, such AMM/RMMs can detect or quantify microorganisms regardless of whether they can be cultured on or in conventional growth media or if they are non-culturable, due to stress, injury, being in a dormant state and/or other physiological conditions. This results in a reported signal that is not a CFU, but rather, some other designation based on the scientific principle of the AMM/RMM. For example, biofluorescent particle counters (BFPCs) will report the recovery of microorganisms in terms of an autofluorescent unit. Additionally, some non-CFU methods operate continuously, generating significantly more data points than a single, growth-based result. Accordingly, it may be more complicated to compare the results from non-CFU AMM/RMMs with the results from CFU-based methods (18).

As just stated, a non-CFU method may report non-culturable signals in addition to what would be expected to be reported in a CFU-based method (i.e., culturable signals). Indeed, in its simplest statistical form, the *total number of microorganisms* X_M in a sample can be conceptually defined as a sum of two components, i.e., $X_M = X_{CFU} + X_{NCFU}$, with X_{CFU} being the *number of viable and culturable* microorganisms present in the test sample and X_{NCFU} the *viable and non-culturable number* of microorganisms in the test sample. The CMM, which relies on microorganism growth, would only respond to the X_{CFU} component and does not signal anything from the X_{NCFU} component. For quantitative methods it is expected that the CMM will produce a number that is at most equal to X_{CFU} , because the CMM quantifies microorganisms in terms of CFU's and not in the actual number of microorganisms that may be present in the test sample. Alternatively, the AMM/RMM with a non-CFU signal would typically respond to both components X_{CFU} and X_{NCFU} in the sample.

Based on this statistical formulation of the non-CFU signal and assuming that the AMM/RMM and CMM perform equivalent to the X_{CFU} component, it may be expected that the non-CFU AMM/RMM is more sensitive than the CMM in detecting and enumerating microorganisms. However, this assumption may not always be true, as this will depend on a number of factors, e.g., the scientific principle of the AMM/RMM and its resulting detection signal, and the physiological state of the microorganisms being detected. In this case, the non-CFU method may report similar recoveries as the CFU-based method. Thus, the AMM/RMM can be expected to perform at least *identical* to, or alternatively, *superior* to, the CMM, even if the AMM/RMM is performing slightly worse than the CMM solely on the X_{CFU} component. Here, the additional signal of the AMM/RMM for the X_{NCFU} component would add to the signal of the X_{CFU} component. Therefore, demonstrating *non-inferiority* of the AMM/RMM to the CMM only on the total

number of microorganisms X_M in a validation study could imply that the AMM/RMM is inferior to the CMM *on the X_{CFU} component*. On the other hand, if the validation experiments can generate test samples that only or mostly contains the X_{CFU} component, the non-inferiority analyses discussed in **Section 9.0** (Appendix 1) and **Section 10.0** (Appendix 2) may be applicable, since a comparison between the AMM/RMM and CMM is then a direct comparison on the CFU signal. However, it may not be possible to construct such an experiment with 100% certainty since there will always be some level of non-culturable cells in a microbial challenge suspension, even under the best conditions. Therefore, using a non-inferiority statistical test when comparing a non-CFU-based method with a CFU-based method may not be appropriate, hence a statistical test based on a superiority model may be more appropriate.

An additional element to the issue of generating more presence tests or enumerations than the CMM which also plays a role in the validation of the AMM/RMM with a non-CFU unit, is the method's performance on blank samples (i.e., the test sample is free of viable microorganisms). Some AMM/RMMs may detect or quantify non-viable microorganisms or non-microbial particles (X_{NM}) in addition to viable microorganism (X_{CFU} and X_{NCFU}) and respond to the total number of particles ($X_P = X_{NM} + X_M = X_{NM} + X_{CFU} + X_{NCFU}$) in the test sample. The AMM/RMM could then produce signals for test samples that are free of viable microorganisms resulting in a false positive rate for blank samples with qualitative microbiological methods or a false positive count or background signal for quantitative microbiological methods. When the AMM/RMM detects or quantifies non-viable microorganisms or non-microbial particles, a comparison with the CMM becomes more difficult since it is unclear how much of the signal for the non-viable microorganisms or non-microbial particles X_{NM} is dominating the signal from the X_{CFU} and X_{NCFU} components. It is for this reason that an understanding of the false positive rate is well understood before performing comparative testing against the CMM, which is described in the method suitability section of this technical report. For example, non-CFU AMM/RMMs that detect non-viable cells or non-microbial particles may or may not be suitable for use, depending on the extent of the recovery and whether the false positives or background noise can be subtracted from the true viable recovery counts under routine use.

Section 11.0 contains many formulas; **Table 11.0-1** provides a key of all the symbols used in the formulas and their meaning.

Table 11.0-1 Nomenclature for Symbols Used in Section 11.0 (Appendix 3)

Symbol	Meaning
X	The number of microorganisms in a test sample. We use different indices to indicate the number of microorganisms for different signals. X_{CFU} is the number of viable and culturable microorganisms, X_{NCFU} is the number of viable and non-culturable microorganisms, $X_M = X_{CFU} + X_{NCFU}$ is the number of viable microorganisms. We are also using X_{NM} the number of non-viable microorganisms and/or non-microbial particles and $X_P = X_{NM} + X_M$ the total number of particles.
$\pi(X)$	The probability of a positive result on a test sample containing X microorganisms. $\pi_A(X)$ and $\pi_C(X)$ are these probabilities for the AMM/RMM and CMM, respectively. Note: The notation $\pi_A(X_{CFU})$ and $\pi_C(X_{CFU})$ may also be used for the AMM/RMM and CMM when the test sample contains X_{CFU} viable and culturable microorganisms or $\pi_A(X_{NCFU})$ and $\pi_C(X_{NCFU})$ for viable and non-culturable microorganisms.

Symbol	Meaning
$\bar{y}$	The mean count for a set of test samples at the same concentration level for the microbiological method, i.e., $\bar{y} = \sum_{i=1}^n (y_i/n)$. The means $\bar{y}_A$ and $\bar{y}_C$ are the mean counts for test samples tested with the AMM/RMM and CMM, respectively.
λ	The mean number of microorganisms or the bacterial density in a set of test samples or in a suspension. Here we make a distinction in the bacterial density λ_{CFU} for viable and culturable microorganisms and the bacterial density λ_{NCFU} for viable and non-culturable microorganisms. We will also use $\lambda_M = \lambda_{CFU} + \lambda_{NCFU}$.
μ	The mean count of a set of test samples for the microbiological method (which may deviate from the true mean number λ). We will use notations μ_A^k and μ_C^k for the mean counts of AMM/RMM and CMM, respectively, on spiked concentration k (to avoid indicating the spiked level λ_k of microorganisms).
p	The proportion of independent test samples that were tested positively for the presence of viable microorganisms with the microbiological method, also called the positive rate. p_A and p_C are the positive rates for the AMM/RMM and the CMM, respectively. When the test samples are only blank samples, the positive rate is referred to as the false positive rate. We also use the notations p_A^0 and p_C^0 for the observed positive rates of blank samples with the AMM/RMM and CMM, respectively, and p_A^+ and p_C^+ for the observed positive rates of spiked samples with the AMM/RMM and CMM, respectively.
q	The ratio of the mean number of viable and non-culturable and viable and culturable microorganisms: $q = \lambda_{NCFU}/\lambda_{CFU}$.
θ	The detection proportion for the microbiological method, which indicates the probability for a positive test for test samples with exactly one microorganism present in the test sample. θ_A and θ_C indicate the detection proportions for the AMM/RMM and CMM, respectively.
η	The false positive rate for the microbiological method, which indicates the probability for a presence test for blank test samples. η_A and η_C indicate the detection proportions for the AMM/RMM and CMM, respectively.
ξ	The estimator for $\theta\lambda$ in a set of spiked test samples with a mean number of microorganisms λ present in the test samples obtained with the microbiological method having a detection proportion θ . ξ_A and ξ_C indicate the estimators for the AMM/RMM and CMM, respectively.
τ^2	The approximated variance of the estimator ξ . τ_A^2 and τ_C^2 are the variances for the estimators ξ_A and ξ_C , respectively.
$m(\lambda)$	The concentration-response curve for the microbiological method in (spiked) concentration level λ . $m_A(\lambda)$ and $m_C(\lambda)$ are the concentration-response curves for the AMM/RMM and CMM, respectively.
$\tau(\lambda)$	The standard deviation of the log-transformed estimated concentration-response curve for the microbiological method at concentration level λ . $\tau_A(\lambda)$ and $\tau_C(\lambda)$ are the standard deviations for the log-transformed estimated concentration-response curves for the AMM/RMM and CMM, respectively.

Symbol	Meaning
R	The ratio of a statistic for the AMM/RMM and CMM and is viewed as a measure for accuracy or recovery. The ratio R_{pr} is the ratio of positive rates p_A/p_C . R_{gMPN} is the ratio of detection proportions θ_A/θ_C when the microbiological methods do not have false positive rates and R_M is the ratio of detection proportions θ_A/θ_C when the microbiological methods do have false positive rates.
n	The number of independent test samples. We may use n_A and n_C for the number of test samples tested with the AMM/RMM and CMM, respectively. Additionally, we may use n_A^0 and n_C^0 for the number of blank test samples tested with the AMM/RMM and CMM, respectively, and use n_A^+ and n_C^+ for the number of spiked test samples tested with the AMM/RMM and CMM, respectively.
P_{AGREE}	The probability that the AMM/RMM and CMM agree on the decision of presence/absence of microorganisms in the test sample.
U	The upper limit for contamination of test samples with the microbiological method. U_A and U_C are the upper limits for test samples for the AMM/RMM and CMM, respectively.

11.1 Qualitative Microbiological Methods with Non-CFU Signals

This section will discuss two situations for non-CFU AMM/RMM methods, methods that have no or hardly any false positives, and methods that produce a relevant level of false positives. This distinction is made because false positives will improperly affect positive rates.

11.1.1 Limit of Detection when the Non-CFU Alternative/Rapid Microbiological Methods Does Not (or Hardly) Detects False Positives

In this example, a superiority analysis will be conducted using data presented in **Table 11.1.1-1** to illustrate an experiment with a single organism. The positive rates (with exact 95% confidence intervals) are given by:

$$\text{AMM/RMM: } p_A = 0.933[0.779; 0.992]$$

$$\text{CMM: } p_C = 0.767 [0.577; 0.901]$$

The observed positive rate for the AMM/RMM is clearly larger than the positive rate for the CMM, which is expected because the non-CFU AMM/RMM responds to both the X_{CFU} and X_{NCFU} components and the CMM only responds to the X_{CFU} component.

Table 11.1.1-1 Presence/Absence of 30 Samples for a Single Microorganism per Microbiological Method

Method	Test Result		
	Presence	Absence	Total
AMM/RMM	28	2	30
CMM	23	7	30
Total	51	9	60

Testing the traditional null hypothesis $H_0: p_A = p_C$ can be done with Pearson's chi-square test or Fisher's exact test as discussed in **Section 9.0** (Appendix 1). However, these two test statistics do not reject the null-hypothesis for the data in **Table 11.1.1-1** at the significance level of 0.05, because the p -values are determined at 0.071 and 0.145, respectively (not shown here but can be obtained following the guidance in **Section 9.0** (Appendix 1). These statistics test the null-hypothesis against the alternative hypothesis $H_1: p_A \neq p_C$ (e.g., the recovery of the AMM/RMM is statistically inferior or superior to the recovery of the CMM). The alternative hypothesis demonstrating that the AMM/RMM is statistically superior in detecting microorganisms (i.e., a greater microbial recovery rate) than the CMM: $H_1: p_A > p_C$ is more relevant for non-CFU AMM/RMM methods. This superiority hypothesis can be better evaluated by calculating the ratio $R_{pr} = p_A/p_C$ of the positive rates and using an appropriate lower 95% confidence limit on this ratio, similar to the non-inferiority analysis in **Section 9.0** (Appendix 1). If the Farrington and Manning confidence limit is used, a ratio of $R_{pr} = 1.217$ with a lower 95% confidence limit equal to 1.018 is obtained. This implies that the hypothesis of superiority has been demonstrated, i.e., $H_0: p_A = p_C$ is rejected in favor of $H_1: p_A > p_C$ superiority, since the value one (1) is below the lower 95% confidence limit.

For a non-inferiority analysis, as described earlier in **Section 9.2.2**, it was suggested to use the gMPN when the positive rates were high, because the ratio of positive rates would converge to one when the spiked concentration level increases, and the lower 95% confidence limit would easily become above a non-inferiority margin of, for example, 0.70, irrespective of how inferior the AMM/RMM is against the CMM. For superiority testing this is less of a concern, since superiority would require a lower 95% confidence limit above the value one (1) and this is not so easy to determine if the true positive rate of the AMM/RMM is equal to or less than the true positive rate of the CMM.

Nevertheless, the gMPN which can be used here as well and applied to the data in **Table 11.1.1-1**, estimates the ratio of detection proportions $R_{gMPN} = \theta_A/\theta_C$ equal to $R_{gMPN} = 1.861$ with a lower 95% confidence limit of 1.064. Here the formula from **Section 9.2.3** was used. Thus, the gMPN shows that the null-hypothesis of equal detection proportions $H_0: \theta_A = \theta_C$ is rejected in favor of superiority $H_1: \theta_A > \theta_C$, indicating that the AMM/RMM detects test samples with a single microorganism better than the CMM. It is appropriate to note that the calculated ratio of 1.861 actually represents a ratio of $(1 + q)(\theta_A/\theta_C)$, with $q = \lambda_{NCFU}/\lambda_{CFU}$, λ_{CFU} the mean number of viable and culturable microorganisms and λ_{NCFU} the mean number of viable and non-culturable microorganisms present in the test samples. Thus, the real ratio of detection proportions θ_A/θ_C , which would investigate the detection of viable and culturable microorganisms, may not be superior $\theta_A/\theta_C < 1$. However, if the samples in the experiment contain a ratio $q = \lambda_{NCFU}/\lambda_{CFU}$ of non-

culturable and culturable microorganisms that is natural for routine sampling, then the AMM/RMM detects better than the CMM, due to a performance on detecting non-culturable microorganisms.

Thus, both the approach on positive rates and the gMPN approach show that the AMM/RMM detects microorganisms better than the CMM.

11.1.2 Limit of Detection When the Non-CFU Alternative/Rapid Microbiological Method Demonstrates a Relevant False Positive Rate

In this situation it is assumed that the AMM/RMM will also detects non-viable signals from blank samples (e.g., false positives or background noise). Clearly, this would not be an ideal situation, but it may still be desirable to know whether the AMM/RMM detects viable microorganisms as good as or better than the CMM. A direct comparison of the microbiological methods on spiked material alone may provide an incorrect conclusion, since the false positive rate masks or increase the true positive rate. In such situations, it may be helpful to combine data from test results of blank samples with data from spiked samples. To illustrate a statistical approach for such an experiment, it will be assumed that the following data has been collected (**Table 11.1.2-1**). The same data used in **Table 11.1.1-1** was also used for **Table 11.1.2-1** for the spiked samples. The statistical approach can be programmed in Excel from the formulae provided in **Section 11.1.2**.

Table 11.1.2-1 Testing of 30 Blank Samples and Spiked Samples on Both the Alternative/Rapid Microbiological Method on Absence/Presence for a Single Microorganism

Method	Test Results: Blank Samples			Test Results: Spiked Samples		
	Presence	Absence	Total	Presence	Absence	Total
AMM/RMM	4	26	30	28	2	30
CMM	0	30	30	23	7	30
Total	4	56	60	51	9	60

A statistical model that may describe the positive rate p_A for the AMM/RMM, and extends the gMPN approach, is described by $p_A = 1 - (1 - \eta_A)\exp\{-\theta_A\lambda\}$, with θ_A the detection proportion, η_A the false positive rate, and λ the spiked concentration level (9). In case no microorganisms are present in the test samples, the positive rate p_A become equal to η_A and when the false positive rate $\eta_A = 0$, the model reduces to the gMPN approach discussed in **Section 9.2.3**. For the CMM, a similar model could be assumed but with different parameters η_C and θ_C , where the false positive rate could be equal to zero ($\eta_C = 0$).

To understand whether the non-CFU AMM/RMM detects microorganisms better than the CMM testing the null-hypothesis $H_0: \theta_A = \theta_C$ against the superiority alternative hypothesis $H_1: \theta_A > \theta_C$ may be performed, irrespective of how the false positive rates η_A and η_C compare to each other. The ratio $R_M = \theta_A/\theta_C$ can now be estimated by:

$$R_{MP} = \frac{[\log(1 - p_A^0) - \log(1 - p_C^0)]}{[\log(1 - p_A^+) - \log(1 - p_C^+)]}$$

Where:

- p_A^0 = positive rate for blank samples with AMM/RMM
- p_C^0 = positive rate for blank samples with CMM
- p_A^+ = positive rate for spiked samples with AMM/RMM
- p_C^+ = positive rate for spiked samples with CMM

Calculating these positive rates with their exact 95% confidence intervals from the data in **Table 11.1.2-1**, results in the data seen in **Table 11.1.2-2**.

Table 11.1.2-2 Positive Rates with Exact 95% Confidence Intervals

Method	Test Results: Blank Samples	Test Results: Spiked Samples
AMM/RMM	0.133 (0.038; 0.307)	0.900 (0.735; 0.979)
CMM	0 [0.000; 0.116]	0.700 (0.506; 0.853)

Using the positive rates, the ratio of detection proportions can be determined, using the formula for R_M above. The ratio of detection proportions then becomes $R_{MP} = 1.763$. Note that this ratio is lower than the ratio $R_{gMPN} = 1.861$ calculated with the gMPN method on the data of **Table 11.1.2-1** in **Section 11.1.1**, because the false positive rates for the AMM/RMM is subtracted in some form from the positives obtained at the spiked ratio to compensate for the masking of false positives obtained at the spiked samples. Similar to the analysis of the data without false positives, the ratio represents $(1 + q)(\theta_A/\theta_C)$, again with $q = \lambda_{NCFU}/\lambda_{CFU}$, λ_{CFU} the mean number of viable and culturable microorganisms and λ_{NCFU} the mean number of viable and non-culturable microorganisms present in the test samples.

A lower 95% confidence limit on the ratio R_M can then be calculated by:

$$LCL_M = R_M \cdot \exp \left\{ -1.645 \sqrt{\frac{\tau_A^2}{\xi_A^2} + \frac{\tau_C^2}{\xi_C^2}} \right\}$$

Where:

$$\xi_A = \log(1 - p_A^0) - \log(1 - p_A^+)$$

$$\xi_C = \log(1 - p_C^0) - \log(1 - p_C^+)$$

$$\tau_A^2 = p_A^0/(n_A^0(1 - p_A^0)) + p_A^+/(n_A^+(1 - p_A^+))$$

$$\tau_C^2 = p_C^0/(n_C^0(1 - p_C^0)) + p_C^+/(n_C^+(1 - p_C^+))$$

$$n_A^0 = \text{number of test samples for the AMM/RMM at the blank samples}$$

$$n_A^+ = \text{number of test samples for the AMM/RMM at the spiked samples}$$

$$n_C^0 = \text{number of test samples for the CMM at the blank samples}$$

$$n_C^+ = \text{number of test samples for the CMM at the spiked samples}$$

The lower 95% confidence limit, using the data in **Table 11.1.2-1**, is then determined by $LCL_{MP} = 0.989$. Based on this lower 95% confidence limit, it can no longer be concluded that the non-CFU AMM/RMM detects microorganisms better than the CMM.

However, it could be concluded that the non-CFU AMM/RMM is non-inferior to the CMM when detecting microorganisms at a non-inferiority margin of 0.7 (or at even higher values), but this does not mean that the AMM/RMM is non-inferior on detecting culturable microorganisms (i.e., detecting the X_{CFU} part of the test sample), since the test samples may have contained several X_{NCFU} components and the AMM/RMM could be inferior on the CFU part of the test samples. This issue may also be the case when the lower 95% confidence limit is above the value one (1) (demonstrating superiority), but this may be less likely since the non-CFU AMM/RMM should be more sensitive than the CMM. Finally, showing non-inferiority with an estimate of the ratio of detection proportion above the value one (1) may still provide enough evidence that the non-CFU AMM/RMM is sensitive enough.

11.1.3 Discussion on Fit for Purpose

Superiority in detecting microorganisms with the non-CFU AMM/RMM compared to the CMM may create challenges in making GMP decisions in daily practice when a positive result of the non-CFU AMM/RMM is being compared with the established CFU-based acceptance levels for microbial recoveries. Such a positive result may be indicated in four situations. First, the sample may only contain viable and culturable microorganisms (X_{CFU}) and the non-CFU AMM/RMM would be expected to detect these. This outcome would most likely be confirmed by the CMM and the use of AMM/RMM or CMM would typically not result in a different decision (i.e., decision equivalence).

Second, a positive test in the AMM/RMM is due to the presence of only viable but non-culturable microorganisms (X_{NCFU}). The CMM would most likely not confirm the presence of these microorganisms, indicating a possible difference in decision making (i.e., non-equivalent decision making). In this case, the

non-CFU AMM/RMM would be correct in detecting these microorganisms, which would not be detected in the CMM as they would not be expected to produce a growth-based signal.

Third, a positive test in the AMM/RMM is due to the presence of both viable and culturable and viable but non-culturable microorganisms ($X_{CFU} + X_{NCFU}$). The CMM may not confirm the presence of the culturable organisms if the culturable signal X_{CFU} is far below the LoD, also indicating a possible difference in decision making (i.e., non-equivalent decision making), since the AMM/RMM would be expected to detect when culturable and non-culturable organisms are beyond the LoD. Here, the non-CFU AMM/RMM would be correct in detecting all of these microorganisms, but would not be detected in the CMM due to a too small culturable signal.

The fourth situation is that the non-CFU AMM/RMM provided a false positive and no viable microorganisms are actually present, in which case the CMM would most likely not confirm the presence of microorganisms either (i.e., again non-equivalent decision making). However, as previously discussed, care must be taken if false positives are detected in the AMM/RMM, as decisions should only be made based on the presence of viable cells and not non-viable cells or non-microbial particles. Accordingly, superiority of the non-CFU AMM/RMM in this case may lead to non-equivalent and more importantly, inappropriate decisions on the presence of supposed microorganisms in the test sample.

How well the non-CFU AMM/RMM and CMM agree in their decision-making on the same set of samples is not so easy to determine, because the agreement between the AMM/RMM and CMM would depend on many factors (e.g., the number of microorganisms in the sample, the LoD of the microbiological methods, the false positive rates, the ratio of culturable and non-culturable microorganisms, and the dependency between X_{CFU} and X_{NCFU}). However, for specific settings and under certain conditions, the degree of agreement between the AMM/RMM and CMM could be determined. Note that agreement is defined here as the expected probability of obtaining the same decision on the set of test samples.

In the scenario that a test sample does not contain microorganisms, the agreement is equal to: $P_{AGREE} = p_A^0 p_C^0 + (1 - p_A^0)(1 - p_C^0)$. For the data in **Table 11.1.2-1**, the agreement (expressed in percentages) is equal to 86.7%, indicating decision equivalence between the AMM/RMM and CMM for blank samples. Note that good or strong agreement levels are 80% or more for both inter-rater agreement and questionnaire reliability (19, 20).

For settings with samples containing only viable but non-culturable microorganisms, agreement between the AMM/RMM and CMM is determined by the probability:

$$P_{AGREE} = \pi_A(X_{NCFU})p_C^0 + (1 - \pi_A(X_{NCFU}))(1 - p_C^0)$$

with $\pi_A(X_{NCFU})$ the probability of a presence result of the AMM/RMM for a sample containing X_{NCFU} microorganisms. When the number X_{NCFU} in a sample is large enough, the AMM/RMM is likely to detect this and the agreement P_{AGREE} drops or converges to the false positive rate $P_{AGREE} = p_C^0$ of the CMM, implying a very low agreement and emphasizing the arguments above that superiority induced by non-CFU AMM/RMM-detectable cellular targets (other than culturable microorganisms) causes disagreement in decisions on samples without culturable microorganisms.

Finally, for samples containing both viable and culturable, and viable but non-culturable microorganisms ($X_{MP} = X_{CFU} + X_{NCFU}$), the agreement is given by:

$$P_{AGREE} = \pi_A(X_{NCFU} + X_{CFU})\pi_C(X_{CFU}) + (1 - \pi_A(X_{NCFU} + X_{CFU}))(1 - \pi_C(X_{CFU}))$$

with $\pi_A(X_{CFU} + X_{NCFU})$ the probability of a presence result of the AMM/RMM for a sample containing $X_{CFU} + X_{NCFU}$ microorganisms and with $\pi_C(X_{CFU})$ the probability of a presence result of the CMM for a sample containing X_{CFU} culturable microorganisms. Clearly, when the non-CFU AMM/RMM is superior to the CMM and the number of viable and culturable microorganisms is above the LoD of the CMM, the agreement would be at least $P_{AGREE} = 90\%$. This indicates that decision equivalence between the AMM/RMM and CMM is attained when the number of culturable microorganisms is above the LoD of the CMM. On the other hand, the agreement could also become substantially lower than 90%, particularly when $\pi_A(X_{NCFU} + X_{CFU}) \approx 1$ and $\pi_C(X_{CFU}) \approx 0$ (the culturable signals are below the LoD); the agreement would then be close to zero again.

To obtain an agreement level for sample from routine practice, there is a need to balance the three different levels of agreement as just described above, since these situations could possibly occur in different frequencies. The expected agreement for the statistical models that were used for the data in **Table 11.1.1-1** and **Table 11.1.2-1** have been calculated, under the assumption that the two components X_{CFU} and X_{NCFU} are Poisson distributed and that they would enter the samples independently. The formula for agreement is then provided by:

$$P_{AGREE} = 1 - (1 - p_A^0) \exp\{-\lambda_{MP}\theta_A\} - (1 - p_C^0) \exp\{-\lambda_{CFU}\theta_C\} + 2(1 - p_A^0)(1 - p_C^0) \exp\{-\lambda_M\theta_A - \lambda_{CFU}(1 - \theta_A)\theta_C\}$$

Where:

- θ_C = detection proportion of the CMM
- θ_A = detection proportion of the AMM/RMM
- p_C^0 = false positive rates of the CMM
- p_A^0 = false positive rates of the AMM/RMM
- λ_{CFU} = mean number of viable and culturable microorganisms
- λ_M = mean number of viable microorganisms (given by the sum of culturable and non-culturable viable microorganisms, $\lambda_{CFU} + \lambda_{NCFU}$) present in the samples

For overdispersed numbers of microorganisms and dependent components X_{CFU} and X_{NCFU} , extension of the agreement probability is needed. **Figure 11.1.3-1** shows a few examples of the level of agreement between the AMM/RMM and CMM as function of λ_{CFU} : i.e., the mean number of viable and culturable microorganisms in the samples.

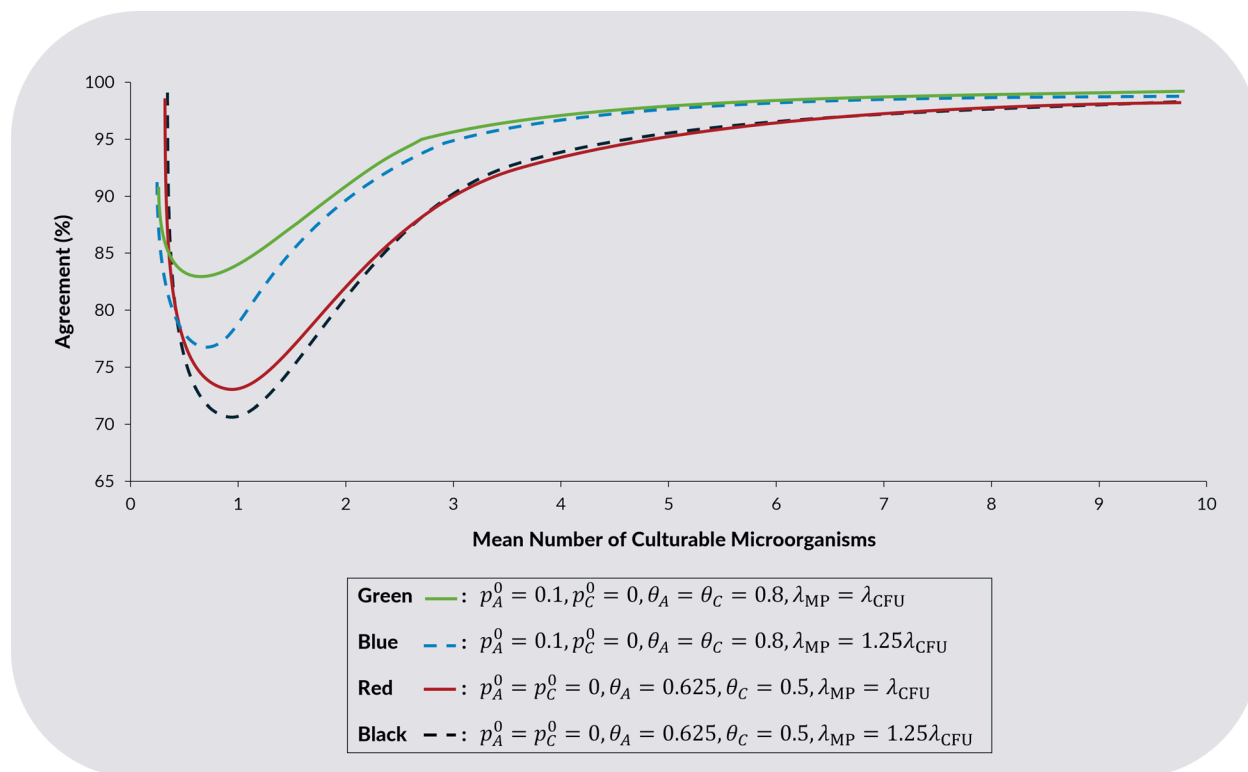


Figure 11.1.3-1 Agreement (%) Between the Alternative/Rapid Microbiological Method and the Conventional Microbiological Method as a Function of the Mean Number of Viable and Culturable Microorganisms

When there is hardly any contamination (e.g., below a mean of 0.5 CFU/sample) or there is substantial contamination (e.g., above a mean of 3 CFU/sample), agreement between the non-CFU AMM/RMM and the CMM is considered high: i.e., more than 80% agreement. However, for some practical settings, agreement can drop substantially below 80%, particularly when the mean level of contamination in practice is in the range of 0.5 to 3 CFU/sample. When one (or both) of the detection proportions θ_C and θ_A is low (red and black lines), the level of agreement is immediately reduced compared to agreement for microbiological methods with higher values of detection proportions (green and blue lines). Note that these detection proportions are directly related to the LoD of the microbiological methods. Additionally, higher ratios between the two components X_{NCFU} and X_{CFU} affects the agreement negatively (green versus blue and red versus black). Thus, depending on the detection performance of the non-CFU AMM/RMM and the ratio $q = \lambda_{NCFU}/\lambda_{CFU}$ of non-culturable and culturable viable microorganisms, decision equivalence may or may not be attained across the full range of microorganism levels.

11.2 Quantitative Microbiological Methods with Non-CFU Signals

The statistical methods described in **Section 10.0** (Appendix 2) are all applicable to AMM/RMMs with non-CFU signals when it is known that the AMM/RMM enumerates test samples with almost only culturable microorganisms. In case these conditions are not satisfied, it can be expected that AMM/RMMs would enumerate more than CMMs when confronted with the same samples, similar to the discussion on qualitative microbiological methods. Thus, for these situations it would be fair to test for superiority of the AMM/RMM against the CMM. **Section 11.2** will discuss some examples of how a non-CFU AMM/RMM

can be compared with a CMM when it is expected that the AMM/RMM would enumerate both culturable and non-culturable microorganisms.

11.2.1 Accuracy

Section 11.2.1 will analyze data of an accuracy experiment, where the number of microorganisms has been enumerated at different spiked concentration levels (Table 11.2.1-1). Note that both methods tested different samples from the same spiked suspension. The data is visualized in Figure 11.2.1-1.

Table 11.2.1-1 Counts for a Specific Microorganism for the Alternative/Rapid Microbiological Method and Conventional Method at Different Concentration Levels

Replicate	Non-CFU AMM/RMM						CMM					
	2	3	8	16	32	64	2	3	8	16	32	64
1	4	6	14	13	37	64	1	0	10	10	21	57
2	1	7	5	12	26	59	0	4	5	13	25	51
3	3	5	15	15	29	63	2	6	9	17	33	54
4	3	4	5	19	47	72	3	3	9	12	27	47
5	8	4	9	12	29	68	1	3	6	14	20	66
6	2	8	7	15	32	52	1	3	8	15	37	61
Mean	3.5	5.7	9.2	14.3	33.3	63.0	1.3	3.2	7.8	13.5	27.2	56.0
Variance	5.9	2.7	19.4	7.1	58.7	48.8	1.1	3.8	3.8	5.9	45.0	47.2

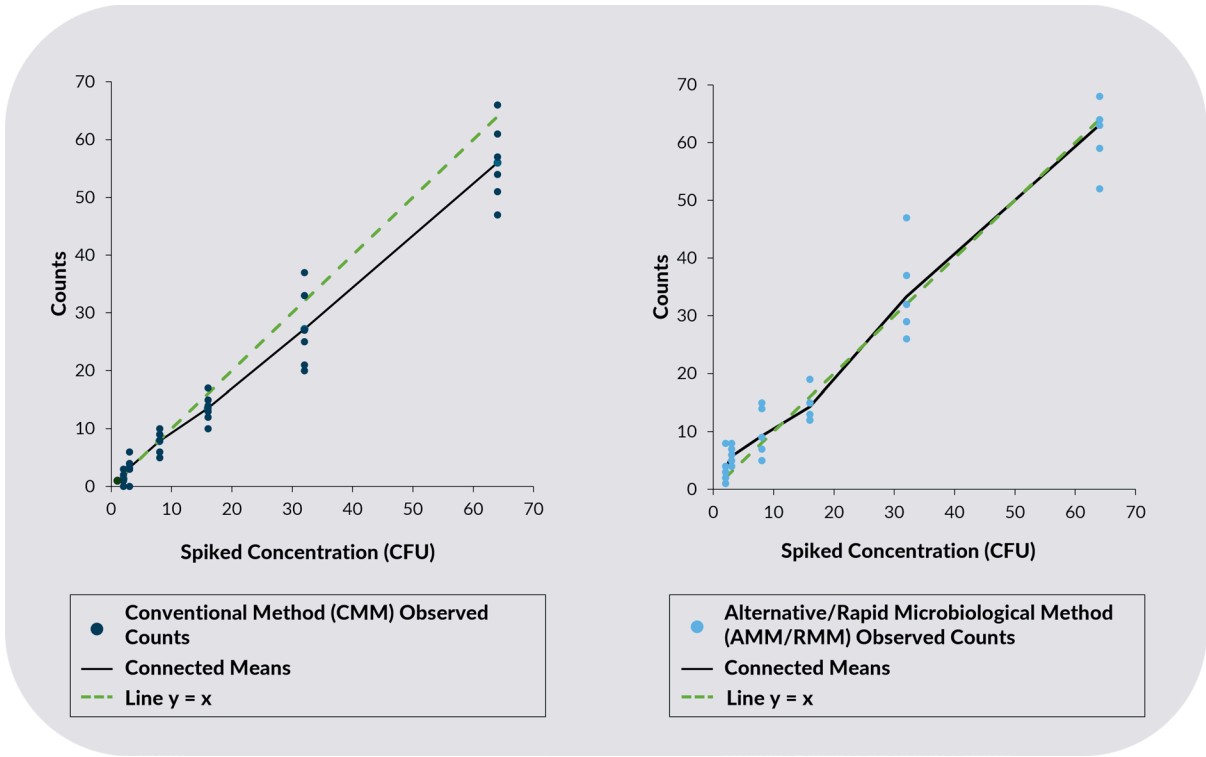


Figure 11.2.1-1 Visualization of the Enumerations of the Alternative/Rapid Microbiological Method and the Conventional Method for an Accuracy Experiment

Calculating the ratio or recovery of the mean counts of the non-CFU AMM/RMM with the mean counts of the CMM with their lower 95% confidence limit, using a Poisson distribution (as already demonstrated in **Section 10.2.3**), shows that superiority of the non-CFU AMM/RMM against the CMM cannot be demonstrated on all spiked concentration levels (see **Table 11.2.1-2**). For three spiked concentration levels, the lower 95% confidence limit is below the value one (1) (or 100% when percentages for the recovery are used).

Table 11.2.1-2 Recoveries of the Alternative/Rapid Microbiological Method versus Conventional Method per Concentration Level Using the Poisson Distribution

Spike	Mean % Recovery	LCL
2	262.5	132.5
4	179.0	111.7
8	117.0	84.4
16	106.2	82.3
32	122.7	103.2
64	112.5	99.5

On the other hand, the mean values of the AMM/RMM at the different spiked concentration levels are all consistently above the mean values of the CMM. Under the null-hypothesis that the mean counts between the two microbiological methods are equal at each spiked concentration level, i.e., $H_0: \mu_A^k = \mu_C^k, k = 1, 2, \dots, 6$ (with μ_A^k and μ_C^k the mean counts at spiked concentration level k for the AMM/RMM and CMM, respectively), the probability that this consistent ordering of means occurs is only equal to 0.016 (one-sided testing). This p -value is obtained with the sign test (i.e., using the Binomial distribution). An alternative statistical test procedure, which is often seen as a more powerful test than the sign test, is the stratified Wilcoxon ranked sign test (see chapter 6 in reference) or the Van Elteren test, where the spiked concentration level is treated as a stratification variable (21, 22). The Van Elteren test provides a value of 21.65 for the test statistic and a p -value of 0.002. Unfortunately, not all general-purpose statistical software packages have implemented the stratified Wilcoxon ranked sign test or the Van Elteren test, but the result presented here is obtained with procedure NPAR1WAY of SAS.

Although the comparison of mean counts per spiked concentration level could not demonstrate superiority in enumeration, the sign test and Van Elteren test indicate that enumeration with the AMM/RMM is on average systematically or consistently higher than the CMM, implying that the AMM/RMM is superior to the CMM on those spiked concentration levels. Note that there was no assumption made on the distribution of the counts, which is often preferred over test statistics which require additional assumptions. However, an alternative analysis is to use a Poisson regression analysis with the identify link function, where it is assumed that the expected mean counts is a straight line of the spiked concentration level. For the non-CFU AMM/RMM and the CMM it would then be assumed that:

$$\begin{aligned}\text{AMM/RMM: } E(y|\lambda_M) &= \alpha_A + \tilde{\beta}_A(\lambda_{CFU} + \lambda_{NCFU}) = \alpha_A + \tilde{\beta}_A(1 + q)\lambda_{CFU} \equiv \alpha_A + \beta_A\lambda_{CFU} \\ &\equiv m_A(\lambda_{CFU})\end{aligned}$$

$$\text{CMM: } E(y|\lambda_M) = \alpha_C + \beta_C\lambda_{CFU} \equiv m_C(\lambda_{CFU})$$

Where:

$E(y|\lambda_M)$ = expected count of a quantitative microbiological method of an arbitrary test sample that is tested at the true concentration level $\lambda_M = \lambda_{CFU} + \lambda_{NCFU}$ of viable microorganisms

λ_{CFU} = true concentration level of viable and culturable microorganisms

λ_{NCFU} = true concentration level of viable but non-culturable microorganisms

α_A = intercept of the linear concentration-response curve of the AMM/RMM

α_C = intercept of the linear concentration-response curve of the CMM

β_A = slope of the linear concentration-response curve of the AMM/RMM

β_C = slope of the linear concentration-response curve of the CMM

q = the ratio $\lambda_{NCFU}/\lambda_{CFU}$ when $\lambda_{CFU} > 0$ (as defined in Section 11.1.1 and 11.1.2)

Estimating the parameters of the linear concentration-response curves for both microbiological methods lead to the following estimates with their 95% confidence intervals shown in Table 11.2.1-3.

Table 11.2.1-3 Estimated Parameters of the Linear Concentration-Response Curve

Method	Intercept	Slope
AMM/RMM	1.517 (0.338, 2.696)	0.952 (0.863, 1.041)
CMM	-0.280 (-1.176, 0.617)	0.878 (0.797, 0.960)

Both the intercept and the slope of the AMM/RMM are higher than the intercept and slope of the CMM, indicating better enumeration across the range of spiked concentration levels and the difference in enumeration increases with higher spiked concentration levels. To determine superiority using the Poisson regression models, we can calculate the difference between the AMM/RMM and CMM and then calculate a lower 95% confidence interval on this difference (similar to what was described in **Section 10.2.3**). The difference between AMM/RMM and CMM has been calculated in this instance, and not the ratio, because a difference fits better with the additive relation $\alpha + \beta\lambda$, while a ratio fits better with the multiplicative relation $\alpha\lambda^\beta$ used in earlier analyses (where the default log link function was used).

The lower 95% confidence limit on the difference is now defined by:

$$\hat{m}_A(\lambda) - \hat{m}_C(\lambda) - 1.645 \sqrt{\hat{\tau}_A^2(\lambda) + \hat{\tau}_C^2(\lambda)}$$

Where:

$\hat{\tau}_A^2(\lambda)$ = estimated variance of the estimated mean count of the AMM/RMM

$\hat{\tau}_C^2(\lambda)$ = estimated variance of the estimated mean count of the CMM

The variance of the estimated mean count is defined by:

$$\tau^2(\lambda) = \text{VAR}(\hat{\alpha}) + \lambda_{\text{CFU}}^2 \text{VAR}(\hat{\beta}) + 2\lambda_{\text{CFU}} \text{COV}(\hat{\alpha}, \hat{\beta})$$

Figure 11.2.1-2 shows the estimated mean difference between the two microbiological methods with the lower 95% confidence limit. It demonstrates that the mean difference between the non-CFU AMM/RMM and CMM increases with spiked concentration level, as we already mentioned, and that the lower 95% confidence limit stays above the value “0” across the full range of spiked concentration levels. This last result implies that the AMM/RMM is superior to the CMM in enumerating microorganisms across the range of 2 to 64 CFUs.

In **Figure 11.2.1-2** the left graph is a visualization of the mean difference in enumerations between the non-CFU AMM/RMM and CMM with a lower 95% confidence limit as function of the spiked concentration. The right graph is a prediction of CFUs based on the average count of the non-CFU AMM/RMM together with the observed counts of the CMM and 95% prediction limits (which is explained in **Section 11.2.2**).

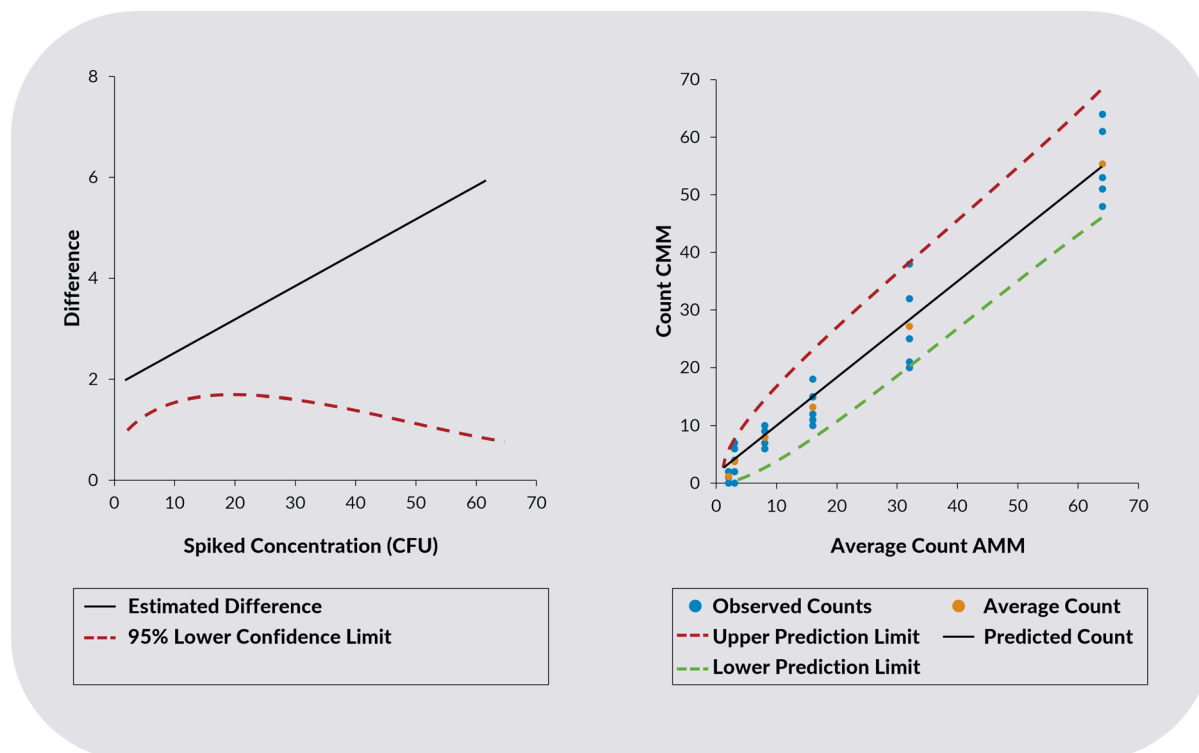


Figure 11.2.1-2 Estimated Mean Difference Between the Two Methods with Lower 95% Confidence Limit

11.2.2 Predicting the Counts of the Conventional Method Based on the Non-CFU Alternative/Rapid Microbiological Method Signal

Based on the linear relations, the mean count of the CMM as function of the mean count of the AMM/RMM may be predicted, assuming that the ratio of viable and culturable microorganisms with respect to the non-culturable microorganisms is the same in practice as in the validation experiment. The prediction equation for the range of spiked concentration levels 2 to 64 CFU's is equal to:

$$\hat{\alpha}_C - \left(\frac{\hat{\beta}_C}{\hat{\beta}_A} \right) \hat{\alpha}_A + \left(\frac{\hat{\beta}_C}{\hat{\beta}_A} \right) \bar{y}_A,$$

with $\bar{y}_A$ an average count of multiple samples for the non-CFU AMM/RMM. Based on the estimates from the previous Poisson regression models, we obtain the following equation:

$$-1.679 + 0.923\bar{y}_A$$

This prediction is visualized in **Figure 11.2.1-2** on the right-hand side, together with the observed counts of the CMM. Note that this Poisson regression model could also be directly obtained by considering the counts of the CMM as Poisson distributed and considering the average counts of the non-CFU AMM/RMM as independent variable.

Based on the Poisson distribution, 95% prediction intervals can be calculated. Such intervals would indicate a range of individual counts around the prediction function that would be obtained for the CMM with 95% confidence given the average value of the non-CFU AMM/RMM.

The limits of the prediction interval can be approximated by using:

$$-1.679 + 0.923 \bar{y}_A \pm 2\sqrt{-1.679 + 0.923 \bar{y}_A}$$

but at lower average values of the non-CFU AMM/RMM, the approximation may not be as accurate and it would be better to use the quantile of the Poisson distribution.

The 95% prediction interval is depicted in **Figure 11.2.1-2** as well. It has a stepwise shape, since the Poisson distribution is a discrete distribution, resulting in integer values for the prediction limits. Thus, when the average number of microorganisms with the non-CFU AMM/RMM is equal to 6, the expected average number of CFUs in the test samples is equal to 3.86 CFU. Furthermore, 95% of the test samples will have a CFU value that is within the lower 97.5% prediction limit of 1 and an upper 97.5% prediction limit of 8 CFU.

11.2.3 Translating a CFU Upper Limit to an Upper Limit for the Non-CFU Alternative/Rapid Microbiological Method

To determine the most likely range of possible counts for the AMM/RMM, conditionally on a given *test result* of the CMM for the same sample, would be ideal. With this information, a CFU upper limit with the CMM could be translated to an upper limit for the non-CFU AMM/RMM. This would guarantee decision equivalence when using the results from the non-CFU AMM/RMM compared to the CMM. However, such conditional probabilities can only be conducted by making (potentially) untestable and strict assumptions. Indeed, when the CMM obtains a specific count for a sample, the AMM/RMM can potentially enumerate any unlimited value, because the signal for viable but non-culturable microorganisms in the sample is not determined by the results of the CMM. Thus, how the viable culturable signal is related to the viable but non-culturable signal needs to be known, which often is not the case. Additionally, how well the two microbiological methods enumerate microorganisms also needs to be known, which is not so trivial either. For instance, a count of one CFU with the CMM, does not mean that the sample contains one microorganism or even one CFU because the CMM may be imperfect and could miss enumerating viable culturable microorganisms. Finally, to connect an observed count for a sample from the AMM/RMM given the result of the CMM on the same sample, samples with both microbiological methods in the validation study would need to be tested, which is in most situations not possible (i.e., the same test sample replicate must be analyzed in both methods; independent replicates from a test sample composite would not work).

Therefore, the only practical manner that can be considered when translating a CFU upper limit to an upper limit for signals from the non-CFU AMM/RMM is to apply the approach from **Section 11.2.2** and:

1. predict results for AMM/RMM based on an average number of the CMM
2. understand how the probability of enumeration within the upper limit for the CMM and AMM/RMM align to arrive at decision equivalence. This is not the same as providing the range of counts for the AMM/RMM given the CMM has produced a single count of a test sample, but it is a good alternative for this conditional probability.

Assuming that the ratio of the non-culturable signal and culturable signal in routine practice is the same as in the validation experiment, the prediction equation for the average count of the AMM/RMM given an average count of the CMM is given by:

$$\hat{\alpha}_A - \left(\frac{\hat{\beta}_A}{\hat{\beta}_C}\right) \hat{\alpha}_C + \left(\frac{\hat{\beta}_A}{\hat{\beta}_C}\right) \bar{y}_C = 1.213 + 1.084 \bar{y}_C$$

with $\bar{y}_C$ a given average count. Now the same approach can be used as in the previous section, since the equation can be used to translate the upper limit for the CFU signal to an upper limit for the signal of the non-CFU AMM/RMM when the *average results* from many test samples are used. Thus, when the upper limit for the CFU is equal to 1 for the average of many test samples, the upper limit for the non-CFU AMM/RMM becomes 2.297 ($= 1.213 + 1.084 \cdot 1$) for an average of many samples with the non-CFU AMM/RMM. Or if the upper limit on CFU is 5, the upper limit for the non-CFU AMM/RMM becomes 6.633 ($= 1.213 + 1.084 \cdot 5$), again when averages can be used. Thus, the calibration equation easily translates the CFU upper limit to an upper limit for non-CFU microbiological methods.

It is less easy to translate the CFU upper limit to a non-CFU signal when a decision must be made on a *single enumeration*. The probability of enumeration below the upper limit for the CMM to the probability of enumeration below the translated upper limit for the non-CFU AMM/RMM should then be mapped. If the upper limit for a single count with the CMM is 1, it needs to be decided if the upper limit for the non-CFU AMM/RMM is going to be equal to 2 (rounding the value 2.297 down) or 3 (rounding the value 2.297 up), because a value of 2.297 cannot be used as an upper limit for a single count. Similarly, if the upper limit for the CMM is 5, it needs to be decided if 6 or 7 is used as the upper limit for the AMM/RMM (rounding 6.633 down or up). In more general terms, if the upper limit for the CMM is U_C , it needs to be determined if the upper limit for non-CFU AMM/RMM is equal to the largest integer U_A below the calibration equation $1.213 + 1.084U_{CMM}$ or the lowest integer $U_A + 1$ value above the calibration equation $1.213 + 1.084U_C$.

To make this choice, both the probability that the CMM tests below the upper limit U_C and the non-CFU AMM/RMM tests below the upper limit U_A or below the alternative upper limit $U_A + 1$ given a specific average contamination needs to be calculated. This can be done for any level of contamination prior to comparing the probability curves. **Figure 11.2.3-1A** and **Figure 11.2.3-1B** below shows the probability curves for U_C equal to 1, 5, 10, and 100 assuming that the calibration model is given by $1.213 + 1.084\bar{y}_C$. The dotted line in **Figure 11.2.3-1A** and **Figure 11.2.3-1B** is the probability that the CMM tests below the upper limit U_C . The ideal case is that the probability curve for the AMM/RMM to test below the translated upper limit (either U_A or $U_A + 1$) would fall exactly on top of the probability curve for the CMM.

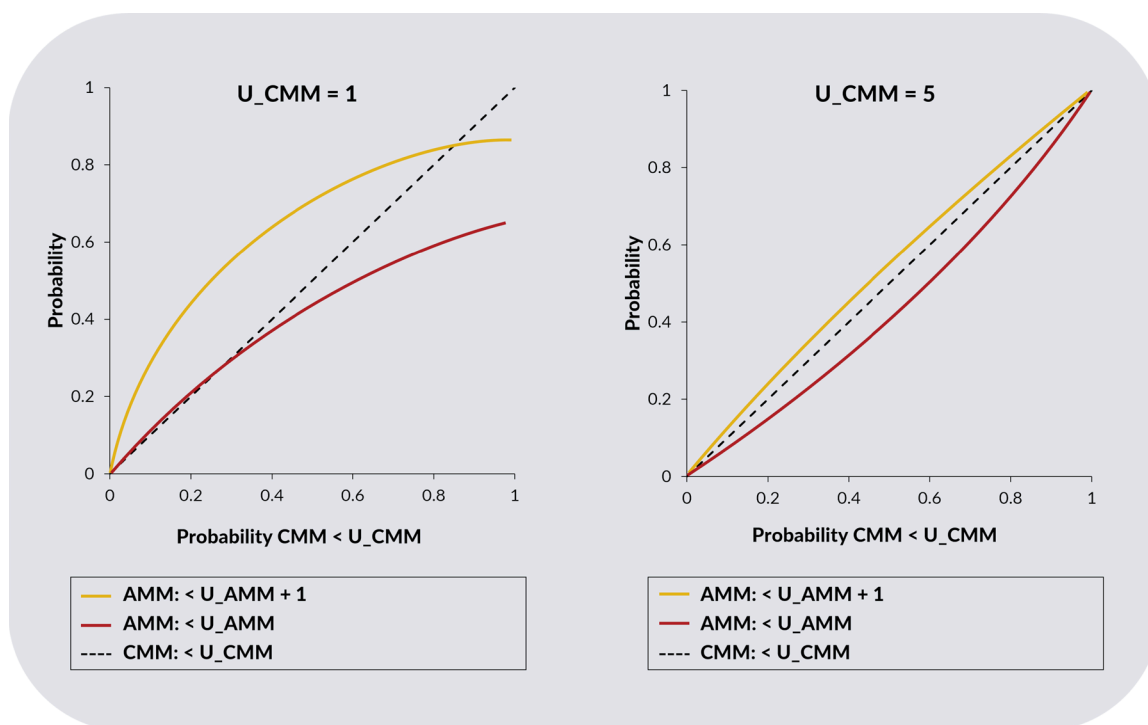


Figure 11.2.3-1A Probability Curves for the Alternative/Rapid Microbiological Methods for Testing a Single Sample Below the Translated Upper Confidence Limit ($U_{CMM} = 1$ (left); $U_{CMM} = 5$ (right))

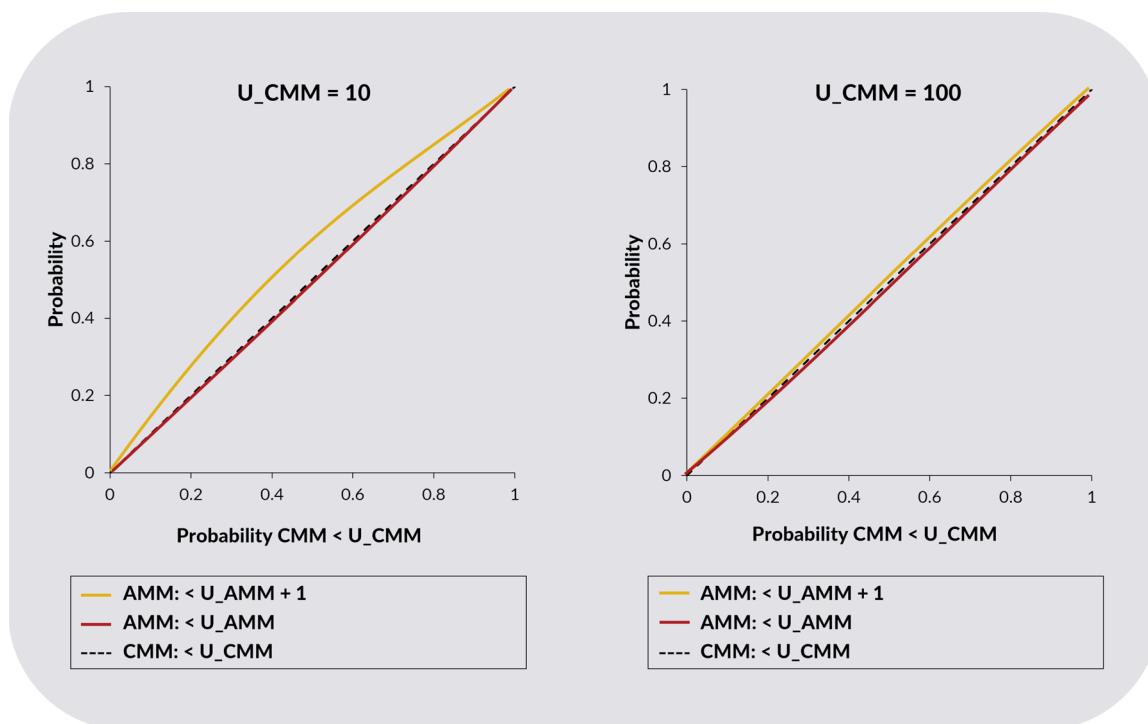


Figure 11.2.3-1B Probability Curves for the Alternative/Rapid Microbiological Methods for Testing a Single Sample Below the Translated Upper Confidence Limit ($U_{CMM} = 10$ (left); $U_{CMM} = 100$ (right))

For the highest CFU upper limit ($U_C = 100$) it can be seen that the two probability curves for the non-CFU AMM/RMM testing below $U_A = 109$ or $U_A = 110$ are both very close to the probability curve that the CMM tests below $U_C = 100$, but using the upper limit of $U_A = 110$ for the non-CFU AMM/RMM seems slightly closer to the probability curve of the CMM testing below $U_C = 100$. This means that there is decision equivalence between a CMM and non-CFU AMM/RMM. The probability of testing a single test sample below 100 with the CMM is almost identical to the probability of testing within 110 with the non-CFU AMM/RMM.

When the CFU upper limit is $U_C = 10$, it is better to choose an upper limit for the non-CFU AMM/RMM of $U_A = 12$, a value below the calibration of 12.053 ($= 1.213 + 1.084 \cdot 10$) for the non-CFU AMM/RMM than an upper limit of 13, which is just above the calibration value of 12.053.

For a CFU upper limit of $U_C = 5$ with the CMM, it would be better to round the calibration value of 6.633 up to 7 to obtain the best, or more conservative, decision equivalence. Using an upper limit of 7 for the non-CFU AMM/RMM provide reasonably similar acceptance probabilities.

When the CFU upper limit for a single enumeration is equal to $U_C = 1$, the two probability curves for the non-CFU AMM/RMM using either an upper limit of 2 or 3, do not resemble the probability curve of the CMM with upper limit of $U_C = 1$ (the two colored curves are away from the dotted curve in the top left figure in **Figure 11.2.3-1**). This implies that there is most likely going to be a lack of decision equivalence between the non-CFU AMM/RMM (with either upper limit 2 or 3) and the CMM (with upper limit 1) in making decisions against the upper limit.

If an upper limit of 2 with the AMM/RMM (red probability curve) is used, it can be seen from the **Figure 11.2.3-1** that when the probability of the CMM testing below the upper limit of one is high (say 90% on the horizontal axis), the probability of testing below the upper limit of 2 with the non-CFU AMM/RMM remains low (around 60%). This indicates that the non-CFU AMM/RMM will more likely obtain a single count at the upper limit of 2 or higher, while the CMM will hardly ever obtain a count at the upper limit of one or higher when testing the same set of samples. This means that samples that would be within the upper limit with the CMM may not be within the upper limit for the non-CFU AMM/RMM and the sponsor may unnecessary test too frequently above the upper limit of 2. When the upper limit of 3 is used with the non-CFU AMM/RMM, an opposite issue occurs. When it is likely that the CMM would provide test results at the upper limit of one or higher, the non-CFU AMM/RMM will not likely detect single counts at the upper limit of 3 or higher. Indeed, if the probability of testing below the upper limit with the CMM on the horizontal axis is, for example, 20%, the probability that the CMM will enumerate at the upper limit of 1 or higher is 80%. The probability that the non-CFU AMM/RMM will test at three or higher is only 50%, and much less likely to detect values outside the upper limit. Thus, translating the upper limit of one for the CMM to an upper limit for the non-CFU (either two or three) would most likely not result in decision equivalence. If a choice has to be made, the sponsor needs to choose between too frequent incorrect single counts beyond the upper limit of 2 or too frequent incorrect single counts within the upper limit of 3 with the non-CFU AMM/RMM.

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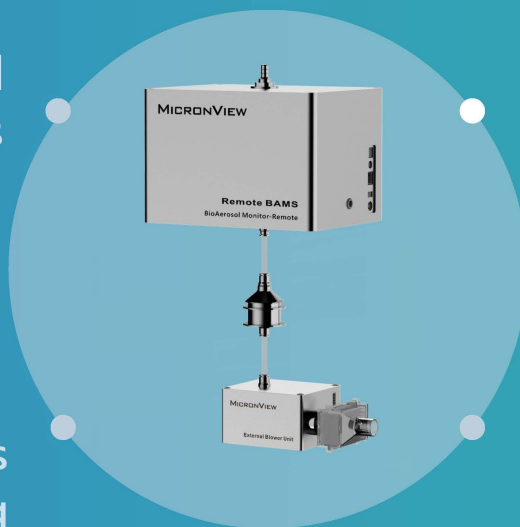


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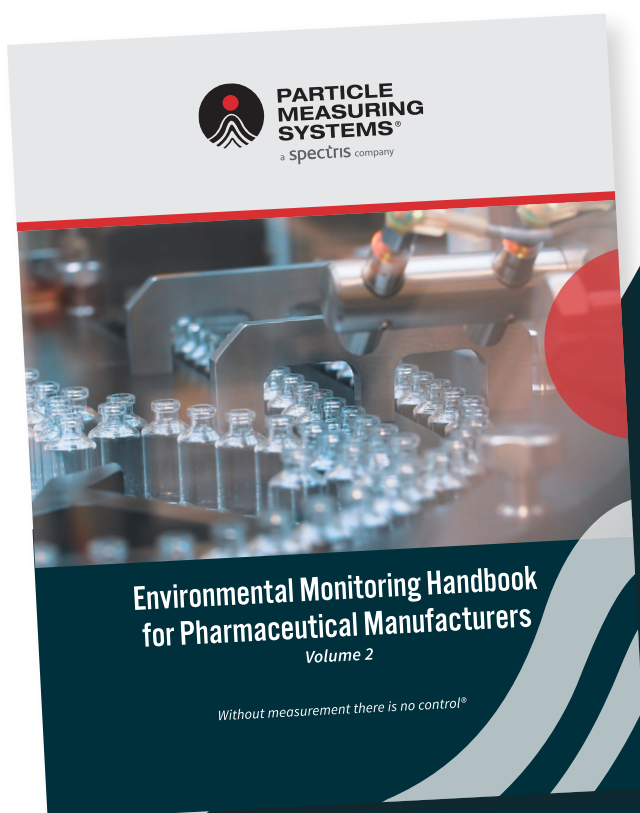
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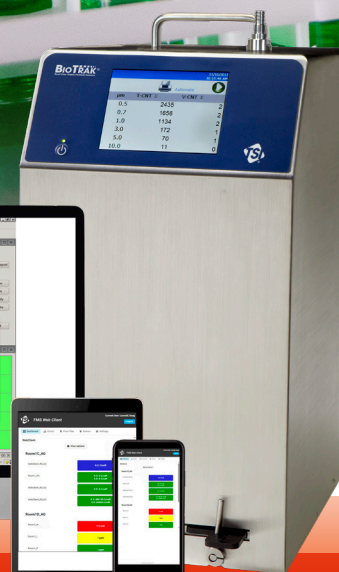
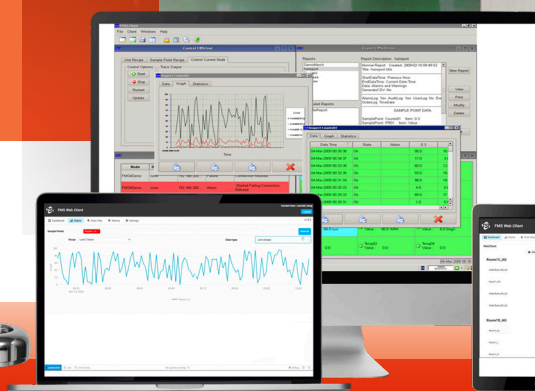
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